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DEPARTMENT OF DEFENSE STANDARD PRACTICE

ELECTRIC PLANT INSTALLATION STANDARD METHODS FOR SURFACE SHIPS AND SUBMARINES (PENETRATIONS)



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FOREWORD

1. This standard is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

2. This standard disseminates up-to-date information detailing requirements for standard installation methods for electrical penetrations employed for submarine and surface ship electrical distribution systems.

3. These criteria are for application to new construction, conversion, and alteration of existing ships.

4. Comments, suggestions, or questions on this document should be addressed to Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

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1. SCOPE

1.1 Scope. This standard covers standard methods for all types of electrical penetrations, including, but not restricted to, swage tubes, stuffing tubes, kickpipes, and multiple cable penetrations on surface ships and submarines.

1.2 Application. The installation methods in this document are intended to be used by all installing activities as required by contract, ship specifications, or similar implementing documents. These methods do not normally identify ship or type, but do establish standards for penetration installations in naval ships. The methods in this document are for new construction as well as for conversions, alterations, and repairs. It is the responsibility of the user activity to determine which method satisfies their requirements.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this standard. This section does not include documents cited in other sections of this standard or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4, or 5 of this standard, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

COMMERCIAL ITEM DESCRIPTIONS

A-A-3041	-	Wrench, Open End Ratchet (TAC Pattern) for Tube Fitting, Electrical Cable Terminals, and Stuffing Tube Gland Nuts
A-A-3042	-	Socket, Open Detachable, Standard Wall, Octagon
A-A-59004	-	Anti-Galling Compound, Thread Lubricating, Seizing Resistant, and Calcium Hydroxide Containing
A-A-59313	-	Thread Compound; Antiseize, Zinc Dust-Petrolatum

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-I-3064	-	Insulation, Electrical, Plastic-Sealer
MIL-PRF-15624	-	Gasket Material, Rubber, 50 Durometer Hardness (Maximum)
MIL-DTL-16685	-	Packing, Material and Packing Preformed (Stuffing Tube for Electric Cables)
MIL-S-24235	-	Stuffing Tubes, Metal, and Packing Assemblies for Electric Cables, General Specification for
MIL-S-24235/1	-	Stuffing Tubes, Metal, and Packing Assemblies for Electric Cables, Bulkhead, Pressureproof
MIL-DTL-24705	-	Penetrators, Multiple Cable, Electric Cable, General Specification for

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-22	-	Welded Joint Design
MIL-STD-1689	-	Fabrication, Welding, and Inspection of Ships Structure
MIL-STD-2003-2	-	Electric Plant Installation Standard Methods for Surface Ships and Submarines (Equipment)

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MIL-STD-2003-4	-	Electric Plant Installation Standard Methods for Surface Ships and Submarines (Cableways)
MIL-STD-2042	-	Fiber Optic Cable Topology Installation Standard Methods for Surface Ships and Submarines
MIL-STD-3020	-	Fire Resistance of U.S. Naval Surface Ships

(Copies of these documents are available online at https://quicksearch.dla.mil.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

NAVAL SEA SYSTEMS COMMAND (NAVSEA) DRAWINGS

803-5184182	-	Insulation, Passive Fire Protection – Installation Details

(Copies of this document are available from the applicable repositories listed in S0005-AE-PRO-010/EDM, which can be obtained online via Technical Data Management Information System (TDMIS) at https://mercury.tdmis.navy.mil/. Copies of this document may also be obtained from the Naval Ships Engineering Drawing Repository (NSEDR) online at https://199.208.213.105/webjedmics/index.jsp. To request an NSEDR account for drawing access, send an email to NNSY_JEDMICS_NSEDR_HELP_DESK@navy.mil.)

NAVAL SEA SYSTEMS COMMAND (NAVSEA) PUBLICATIONS

0924-LP-062-0010	-	Submarine Safety (SUBSAFE) Requirements Manual
S9074-AR-GIB-010/278	-	Requirements for Fabrication Welding and Inspection, and Casting Inspection and Repair for Machinery, Piping, and Pressure Vessels
S9407-AE-ASY-010/CSGA	-	Cable/Conduit Shield Ground Adapter (CSGA) Installation Procedures and Technical Specifications
T9070-AJ-DPC-030/072-3	-	Conventional Weapons Protection (Fragments)
T9070-AL-DPC-020/077-2	-	NAVSEA Hazardous Material Avoidance Process
T9070-AN-DPC-010/100-1	-	Reinforcement of Openings in Structure of Surface Ships, Other Than in Protective Plating
T9070-AN-DPC-020/100-2	-	Openings in Decks and Bulkheads for Stuffing Tubes and Pipe
T9074-AD-GIB-010/1688	-	Requirements for Fabrication, Welding, and Inspection of Submarine Structure

(Copies of these documents are available online via Technical Data Management Information System (TDMIS) at https://mercury.tdmis.navy.mil/ by searching for the document number without the suffix. Refer questions, inquiries, or problems to: DSN 296-0669, Commercial (805) 228-0669. These documents are available for ordering (hard copy) via the Naval Logistics Library (NLL) at https://nll.navsup.navy.mil. For questions regarding the NLL, contact the NLL Customer Service at nllhelpdesk@navy.mil, (866) 817-3130, or (215) 697-2626/DSN 442-2626.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

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ASTM INTERNATIONAL

ASTM F1836M - Standard Specification for Stuffing Tubes, Nylon, and Packing Assemblies (Metric)

(Copies of this document are available online at www.astm.org.)

2.3 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. DEFINITIONS

3.1 Collective protective system (CPS). A system designed to inhibit the entry of chemical, biological, and radiological contaminants into collective protection zones onboard ships. A collective protection zone is a section of the ship, which is defined by a physical boundary that inhibits the entry of chemical, biological, and radiological (CBR) contaminants into the zone. A total protection zone is pressurized and its supply ventilation air is continuously filtered to remove chemical vapors and CBR particulate and aerosols.

3.2 Community stuffing tube for bulkheads. A system of passing multiple cables through penetrations.

3.3 Kickpipe. A pipe welded into the deck with a stuffing tube attached. Kickpipes provide protection of electrical cable at deck penetrations and are used to clear an obstruction or preserve alignment. Kickpipes may be aluminum, steel, stainless steel, or brass to suit the installation or standard pipe sizes to fit the required stuffing tube.

3.4 Metal stuffing tube. A system of passing single electrical cable through decks and bulkheads and entering enclosed equipment on naval ships. Metal stuffing tubes are manufactured in accordance with MIL-S-24235. Stainless steel is an alternative material that should be considered for use in applications highly susceptible to corrosion.

3.4.1 Nylon stuffing tube. A system of passing single electrical cable through electrical equipment on naval ships. Nylon stuffing tubes are manufactured in accordance with ASTM F1836M.

3.5 Multiple cable penetrator (MCP) or multi-cable transits (MCT). A system of passing multiple cables through watertight and non-watertight bulkheads and decks in order to provide watertight, airtight, and firetight penetration of electrical cable.

3.6 RISE expanded sleeve multiple cable penetration sealing system. A trademark-named surface ship penetration and structure sealing system that uses rubber insert sleeves around and between cables and sealant to both sides of the penetration to provide a watertight seal. The sleeves will expand when exposed to heat or flame to seal the MCP to provide a fire stop and to provide a fluid seal.

3.7 Swage tube. A system of passing single cables through decks on naval ships that combines the features of the stuffing tube and kickpipe.

4. GENERAL REQUIREMENTS

4.1 Cable penetrations. Cable penetrations of pressure hulls, pressure-proof bulkheads, shielded bulkheads, ballistic bulkheads, false decks, riser boxes, decks, bulkheads and beams, and other integral parts of the hull shall be in accordance with figures 3A1 through 3E6 and the requirements of T9070-AN-DPC-010/100-1 and T9070-AN-DPC-020/100-2. Unless otherwise approved by NAVSEA, metal stuffing tubes, nylon stuffing tubes, and multiple cable penetrators for cable penetrations shall meet the design and conformance testing requirements of MIL-S-24235, ASTM F1836M, and MIL-DTL-24705, respectively. Cable penetrations that require passive fire protection at fire protection boundaries shall meet the requirements of Drawing 803-5184182 and shall be tested in accordance with MIL-STD-3020. For single cable stuffing tube penetrations that require internal cable shield bonding and grounding within the tube, see S9407-AE-ASY-010/CSGA for installation procedures.

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4.1.1 Installation welding requirements. Unless otherwise specified on the individual figure, the welding of stuffing tubes, kickpipes, swage tubes, and multiple cable penetrators shall be in accordance with the requirements of S9074-AR-GIB-010/278, T9074-AD-GIB-010/1688, MIL-STD-22, and MIL-STD-1689.

4.1.2 Cable penetration of structure. Cable penetration of decks, bulkheads, beams, and other integral parts of the hull shall conform to T9070-AN-DPC-010/100-1 and T9070-AN-DPC-020/100-2. Stuffing tubes in accordance with MIL-S-24235/1 shall be installed for cable penetrations of pressure-proof submarine bulkheads and sonar domes, which are filled with water under normal operating conditions. Stuffing tubes in accordance with MIL-S-24235/1 may be cut in half and each half used for a separate penetration. Penetrations installed, repaired, or modified in accordance with the instructions in this standard shall be restored to the original integrity requirements.

Metal stuffing tubes or multiple cable penetrators shall be used for cable penetrations of the following (Note: Multiple cable penetrators shall not be used in tanks or for cable penetrations exposed to weather (except for electro-magnetic compatibility (EMC) type penetrations covered in figure 3B63):

a. CPS boundaries

b. Watertight cable trunks

c. Watertight bulkheads and decks.

d. Bulkheads designed to withstand a waterhead

e. The portion of bulkheads specified to be watertight to a certain height

f. That portion of bulkheads below the height of the sill or the coaming of compartment accesses

g. Bulkheads surrounding compartments subject to flooding by sprinkling (if watertight):

(1) Garbage disposal rooms

(2) Battery shops

(3) Medical operating rooms

(4) Medical wards

4.1.3 Cable penetration of protective plate bulkheads and decks and bulkheads and decks forming boundaries of spaces containing volatile combustible or explosive materials. Back-to-back multiple cable penetrations consist of two single MCPs and shall be installed in accordance with figures 3B59-3B61. The definitions of a type A protective plating (ballistic) penetration and of a type B protective plating (fragmentation) penetration are in T9070-AJ-DPC-030/072-3 and in applicable ship specifications.Fig_3B59Fig_3B61

a. Cable penetration of type A protective plating (ballistic) bulkheads: Only metal stuffing tubes or back-to-back multiple cable penetrations in accordance with MIL-DTL-24705 shall be installed.

b. Cable penetrations of type A protective plating (ballistic) decks below the main deck. The following installation methods are allowed:

(1) Metal stuffing tubes.

(2) Back-to-back multiple cable penetrators in accordance with MIL-DTL-24705.

(3) Armored riser boxes with either metal stuffing tubes or single multiple cable penetrators in accordance with MIL-DTL-24705.

c. Cable penetration of type A protective plating (ballistic) bulkheads and decks at main deck and above and all type B protective plating (fragmentation) bulkheads and decks: Only metal stuffing tubes or single multiple cable penetrators in accordance with MIL-DTL-24705 shall be used.

d. Cable penetration of bulkheads and decks forming the boundaries of non-magazine storage spaces containing volatile combustible or explosive materials (e.g., flammable liquid storerooms). Only metal stuffing tubes or back-to-back multiple cable penetrators in accordance with MIL-DTL-24705 shall be installed.

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4.1.4 Cable penetration of decks, structural bulkheads, airtight bulkheads, and fume-tight bulkheads. Cable penetration of decks, structural bulkheads, airtight bulkheads, and fume-tight bulkheads shall employ one of the following:

a. Airtight metal stuffing tubes or multiple cable penetrators.

b. Fume-tight chafing collars (for multiple cable penetrations) or nipples (for single cable penetrations) having a minimum collar length of 3 inches. The entire void area within the collar (this includes the area between the collar and the cable and the area between the cables) shall be packed with plastic sealer. The minimum clearance space around cables before packing material is applied shall be ¼ inch for a single cable and 1 inch for multiple cables. See figures 4C40-4C47 of MIL-STD-2003-4.

NOTE: All surface ship stuffing tubes installed shall show a minimum of two threads and a maximum of five threads on the gland nut after it is tightened. Stuffing tubes shall have sufficient intermediate packing to ensure a tight joint when the gland nut is projecting approximately three threads out of the tube to allow for future tightening.

4.1.5 Multiple (two or more) penetrations of nonstructural steel bulkheads (other than wire mesh or expanded metal), bents, web frames, transverse girders, and longitudinal girders. Multiple cable penetrations of nonstructural steel bulkheads, bents, web frames, transverse girders, and longitudinal girders shall employ one of the following:

a. Metal stuffing tubes, multiple cable penetrators, or nipples. Nipples only apply to single cable penetrations and shall have a minimum length of 2 inches with a minimum annular area between the cable and the nipple of ¼ inch packed with plastic sealer.

b. Banding collars (for multiple cable penetrations) having a minimum collar length of 3 inches. The entire void area within the collar (this includes the area between the collar and the cable and the area between the cables) shall be packed with plastic sealer. The minimum clearance space around cables before packing material is applied shall be ¼ inch for a single cable and 1 inch for multiple cables. See figures 4C40-4C47 of MIL-STD-2003-4.

Cable penetrations of vertical non-tight structures within a compartment need not be sealed. However, all chafing collars of the structures selected for sealing shall be sealed.

4.1.6 Plastic sealer. After the cables are properly secured, plastic sealer electrical insulation in accordance with MIL-I-3064, Type HF shall be used to seal the space around the cable as follows:

a. In cable clamps and bushings, entering the top of an electrical enclosure and the side of an enclosure without a drip loop.

b. In bushings or nipples used for passing cables through light-tight and fume-tight bulkheads and decks and to seal around cables as they enter stuffing tubes, kickpipes, and swage tubes, as shown on the individual figures, except that plastic sealer is not required when silicone (red or white) grommets are used. Where compartment air tests are required, it is recommended that plastic sealer be installed after the air test has been satisfactorily performed.

4.1.7 Cable penetrations spacing. The size of stuffing tube groups shall be limited to permit tightening of gland nuts in the group using stuffing tube wrench set in accordance with A-A-3041 and A-A-3042. Penetration spacing is specified in T9070-AN-DPC-020/100-2. Spacing for stuffing tubes, swage tubes, and kickpipes in surface ships shall be as specified in figures 3B1-3B4.

4.1.8 Stuffing tube packing. Stuffing tube packing shall be in accordance with MIL-DTL-16685 for metal stuffing tubes, either the preformed (coil) class 2 or bulk class 1. When bulk packing is used, the first and last turns shall be composition “A” (retainer) and the intermediate turns shall be composition “B” (sealing) of class 1. Reinforced neoprene packing, in accordance with MIL-PRF-15624, class I, may be used as an alternate, asbestos-free, packing material (see figure 3B38). Stuffing tube packing shall be in accordance with ASTM F1836M for nylon stuffing tubes.

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4.1.9 Kickpipes. Kickpipes (aluminum, brass, stainless steel, or steel) shall be standard pipe sizes. Ends of pipe shall be chamfered and burrs existing on the inside wall shall be removed to prevent chafing of cable. For kickpipes, there shall be nine inches minimum between the deck or deck covering to the top of the kickpipe penetration for kick protection. This minimum does not apply in cases where kick protection is not required, such as entrance into a transformer.

4.1.10 Exterior application. Stuffing tubes and swage tubes used for exterior application shall have a corrosion inhibitor in accordance with applicable ship specifications applied prior to installation. The corrosion inhibitor shall be intact after the stuffing tube is installed.

4.1.11 Stuffing tube clearance. Stuffing tubes shall be installed within a minimum clearance from adjacent bulkheads or decks of 2 inches to allow for welding and cleaning.

4.1.12 Swage tube, kickpipe, and stuffing tube welding. Swage tubes, kickpipes, and stuffing tubes may be welded on one side only. Swage tubes, kickpipes, and stuffing tubes adjacent to wet and humid spaces shall be welded on both sides, except in areas where both sides cannot be accessed (such as sound dampening structure). Wet and humid spaces are those that, under normal operating conditions, may produce excessive moisture that may cause corrosion in the unwelded joint (i.e., sculleries, laundries, washrooms, and galleys). Swage tubes and stuffing tubes shall also be welded on both sides for ballistic applications, shell plating, torpedo defense system bulkheads, and penetrations exposed to weather. Kickpipe welding through ballistic decks shall be in accordance with Appendix 3D, figures 3D5 and 3D6.

4.2 Or equal. MIL-STD-2003 uses the term “or equal” to permit the use of parts, components, or tools that are equivalent and can perform the same function as the specified products. The use of the equivalent product is allowed as long as the same functional characteristics, performance, equipment safety, personnel safety, suitability for marine service, life cycle cost, maintenance cost, and supportability are attained, and agreement is obtained from NAVSEA. The request for agreement for the use of “equal” products shall include data that supports that functional and performance equivalence is retained.

4.3 Inactive for new design documents. Some of the documents referenced in MIL-STD-2003-3 have been declared as inactive for new design by decision of the Department of Defense. Where replacement documents are designated, the replacement document should be substituted for the inactive for new design document. Methods that reference inactive for new design documents shall not be used for new ship design unless approved by NAVSEA. Components and products produced in accordance with the inactive for new design documents shall not be used for new design of systems or equipment; however, this does not prohibit use for maintenance, repair, or resupply purposes.

4.4 Installation of fiber optic cables in electrical penetrations. Installation of fiber optic cables and blown fiber cables in electrical penetrations shall be in accordance with requirements of MIL-STD-2042 and this standard. For fiber optic cable installation, if there are conflicts between this standard and MIL-STD-2042, the requirements of MIL-STD-2042 take precedence.

4.5 Fastener requirements. Refer to the general fastener requirements in MIL-STD-2003-2.

4.6 Hazardous materials. Materials and products utilized to execute installation methods in this standard should avoid chemicals listed on the NAVSEA List of Targeted Chemicals (N-LTC) contained within T9070-AL-DPC-020/077-2. These chemicals pose significant risk to the user, environment, or both, and are deemed both undesirable and unsustainable by NAVSEA. NAVSEA is minimizing the use of hazardous materials in procedures such as those covered by this standard. Alternative materials should be considered for applications covered in this document to minimize the use of targeted chemicals. NAVSEA should be informed of the need for the use of any of the targeted chemicals prior to procedure execution.

4.7 Installation and servicing of electrical penetrations within the submarine SUBSAFE boundary. Installation and servicing of electrical penetrations within the submarine SUBSAFE boundary as defined in NAVSEA 0924-LP-062-0010 shall be in accordance with the requirements of NAVSEA 0924-LP-062-0010 and this standard. If there are conflicts between this standard and NAVSEA 0924-LP-062-0010, the requirements of NAVSEA 0924-LP-062-0010 take precedence.

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4.8 Thread lubricant. A thread lubricant should be used on all threads used for applying controlled torque for stuffing tube gland nuts and MCP components. Zinc dust-petrolatum in accordance with A-A-59313 shall be used on aluminum threads unless otherwise specified in the applicable ship specification or in ship drawings. Thread lubricant paste in accordance with A-A-59004 shall be used for all other applications under 1000 °F on carbon alloy steels or under 670 °F in all other applications.

a. When NO-OX-ID is used as a lubricant, no other thread lubricants or sealants are allowed to be used.

b. Thread lubricants containing molybdenum disulfide shall not be used.

4.8.1 Thread lubricant application. To be effective, the lubricant should be applied to the complete thread area minus the 2 threads closest to the surface of the attached part. A light coat shall be applied over all the external threads, and then the first one or two internal threads shall be filled. The lubricant in the first threads will then be distributed over the remaining internal threads as the fastener is assembled. Just enough lubricant shall be used to coat all of the threads.

4.9 The terms agreement or approval. Wherever such terms as “approved” or “agreement” are used without further qualification, it is the approval or agreement of the Supervising Authority or NAVSEA, as applicable, that is intended. In these cases, the approval request and the response shall be formal, in writing, and traceable. Existing methods allowed by applicable ship specifications, such as departures from specifications, are acceptable. The Supervising Authority, also referred to as the Supervisor, is defined in the applicable ship specifications.

5. DETAILED REQUIREMENTS

(See figures.)

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. This standard specifies the requirements for swage tubes, stuffing tubes, and kickpipe methods to be employed both on surface ships and submarines. Standard methods identified for electric plant installation are intended for new construction, conversion, and alteration of existing ships. The use of this standard will be specified in the contract, ship specifications, and similar implementing documents (such as COMUSFLTFORCOMINST 4790.3, Joint Fleet Maintenance Manual (JFMM) and S9086-KC-STM-010/300, Naval Ships’ Technical Manual Chapter 300, Electric Plant-General).

6.2 Acquisition requirements. Acquisition documents should specify the following:

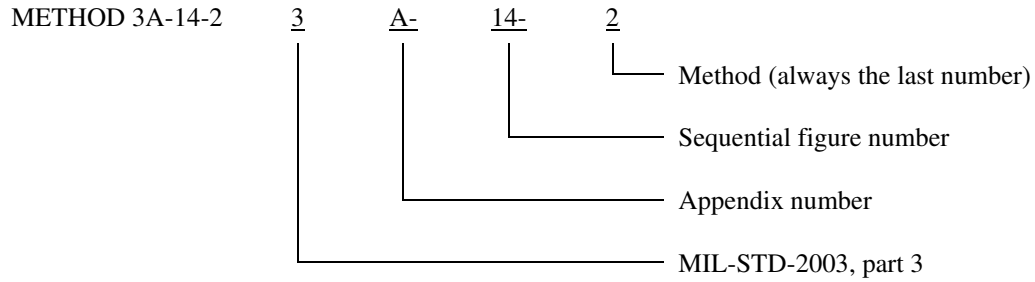
a. Title, number, and date of this standard.

6.3 Designation of electric plant installation standard methods figures. The electric plant installation standard method MIL-STD-2003-3 contains drawings that depict standard methods that are applicable for general electric plant installation on both surface ships and submarines. Each drawing has been assigned a figure number. The methods shown on the figures are grouped together in the following appendices to this standard. Each appendix provides requirements for similar functions.

MIL-STD-2003-3 (Penetrations)	A. Penetrations - Stuffing Tubes, Submarines (Appendix 3A)
	B. Penetrations - Surface Ships (Appendix 3B)
	C. Penetrations - Stuffing Tubes, General (Appendix 3C)
	D. Penetrations - Kickpipes (Appendix 3D)
	E. Penetrations - Pressure Hulls, Submarines (Appendix 3E)

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The methods shown on the figures are identified by the following alphanumeric designation system:



Thus, method 3A-14-2 identifies method 2, sequential number 14 in Appendix 3A of MIL-STD-2003-3.

6.4 Subject term (key word) listing.

Back-to-back
Banding
Bulkhead
Cable penetrations
Chafing collar
Community stuffing tube
Deck
Electro-magnetic compatibility
Fiber optic
Gland nut
Hull fitting
Kickpipe
Multi-cable transit
Multiple cable penetrator
Packing
Penetration sealing
Penetration spacing
Pressure hull
Riser
Sealer
Sealing plug
Spacing
Stuffing tube
Swage tube

6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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APPENDIX 3A – PENETRATIONS – STUFFING TUBES, SUBMARINES

A.1 SCOPE

A.1.1 Scope. This appendix describes the installation standard methods for stuffing tubes in submarines. This Appendix is a mandatory part of the standard. The information contained herein is intended for compliance.

A.2. APPLICABLE DOCUMENTS

A.2.1 General. The documents listed in this section are specified in this appendix. This section does not include documents cited in other sections of this standard or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in this appendix, whether or not they are listed.

A.2.2 Government documents.

A.2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

FEDERAL SPECIFICATIONS

MMM-A-101 - Adhesive, HAPs-Free, Bonding Vulcanized Synthetics Rubber to Steel

COMMERCIAL ITEM DESCRIPTIONS

A-A-3041 - Wrench, Open End Ratchet (TAC Pattern) for Tube Fitting, Electrical Cable Terminals, and Stuffing Tube Gland Nuts

A-A-3042 - Socket, Open Detachable, Standard Wall, Octagon

A-A-59588 - Rubber, Silicone

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-915 - Cable, Electrical, for Shipboard Use, General Specification for

MIL-DTL-24231 - Connectors, Plugs, Receptacles, Adapters, Hull Inserts, and Hull Insert Plugs, Pressure-Proof, General Specification for

MIL-S-24235 - Stuffing Tubes, Metal, and Packing Assemblies for Electric Cables, General Specification for

MIL-S-24235/1 - Stuffing Tubes, Metal, and Packing Assemblies for Electric Cables, Bulkhead, Pressureproof

MIL-DTL-24235/2 - Stuffing Tube, Packing Assemblies for Pressureproof Bulkhead

MIL-S-24235/5 - Stuffing Tubes, Metal, and Packing Assemblies for Electric Cables, Ballast Tank

MIL-S-24235/7 - Stuffing Tubes, Metal, and Packing Assemblies for Electric Cables, for Cast Enclosures, Pressureproof

MIL-S-24235/8 - Stuffing Tubes, Metal, and Packing Assemblies for Electric Cables, for Sheet Metal Enclosures, Pressureproof

MIL-P-24338 - Pipe, Carbon Steel, Seamless

MIL-DTL-24640 - Cables, Lightweight, Low Smoke, Electric, for Shipboard Use, General Specification for

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MIL-DTL-24643 - Cables, Electric, Low Smoke Halogen-Free, for Shipboard Use General Specification for

(Copies of these documents are available online at https://quicksearch.dla.mil.)

A.2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

NAVAL SEA SYSTEMS COMMAND (NAVSEA) DRAWINGS

9000-S6202-73241 - Cable Clamp - Type "C", Assembly & Details

9000-S6202-1197101 - Stuffing Tube - PP for SS & TSP Type Cables

(Copies of these documents are available from the applicable repositories listed in S0005-AE-PRO-010/EDM, which can be obtained online via Technical Data Management Information System (TDMIS) at https://mercury.tdmis.navy.mil/. Copies of these documents may also be obtained from the Naval Ships Engineering Drawing Repository (NSEDR) online at https://199.208.213.105/webjedmics/index.jsp. To request an NSEDR account for drawing access, send an email to NNSY_JEDMICS_NSEDR_HELP_DESK@navy.mil.)

NAVAL SEA SYSTEMS COMMAND (NAVSEA) PUBLICATIONS

T9074-AD-GIB-010/1688 - Requirements for Fabrication, Welding, and Inspection of Submarine Structure

(Copies of these documents are available online via Technical Data Management Information System (TDMIS) at https://mercury.tdmis.navy.mil/ by searching for the document number without the suffix. Refer questions, inquiries, or problems to: DSN 296-0669, Commercial (805) 228-0669. These documents are available for ordering (hard copy) via the Naval Logistics Library (NLL) at https://nll.navsup.navy.mil. For questions regarding the NLL, contact the NLL Customer Service at nllhelpdesk@navy.mil, (866) 817-3130, or (215) 697-2626/DSN 442-2626.)

A.2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME Y14.38 - Abbreviations and Acronyms for Use in Product Definition and Related Documents

(Copies of this document are available online at www.asme.org.)

ASTM INTERNATIONAL

ASTM A53/A53M - Standard Specification for Pipe, Steel, Black and Hot-Dipped, Zinc-Coated, Welded and Seamless

ASTM B21/B21M - Standard Specification for Naval Brass Rod, Bar, and Shapes

ASTM B150/B150M - Standard Specification for Aluminum Bronze Rod, Bar, and Shapes

ASTM B169/B169M - Standard Specification for Aluminum Bronze Sheet, Strip, and Rolled Bar

ASTM F1836M - Standard Specification for Stuffing Tubes, Nylon, and Packing Assemblies (Metric)

(Copies of these documents are available online at www.astm.org.)

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SAE INTERNATIONAL

SAE-AMS-P-83461	-	Packing, Preformed, Petroleum Hydraulic Fluid Resistant, Improved Performance at 275 °F (135 °C)
SAE-AMS-S-8802	-	Sealing Compound, Fuel Resistant, Integral Fuel Tanks and Fuel Cell Cavities
SAE-AS568	-	Aerospace Size Standard for O-Rings

(Copies of these documents are available online at www.sae.org.)

A.2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

A.3 REQUIRED EQUIPMENT AND MATERIALS

A.3.1 Required equipment and materials. The required equipment and materials for the proper installation of submarine stuffing tubes are as shown in the Appendix 3A methods.

A.4 NOTES AND PROCEDURES

A.4.1 Dimensions. For figures and tables in this appendix, all dimensions are in inches unless otherwise noted.

A.4.2 Figures. Table 3AI provides information for the figures in this appendix.

TABLE 3AI. Figures for submarine stuffing tubes installation.

Figure number	Submarine stuffing tubes installation	Page
3A1	MIL-S-24235/1 stuffing tubes for passing cables through pressureproof bulkheads (submarines)	13
3A2	MIL-S-24235/5 stuffing tube for passing cable through ballast tank partitions (submarines)	16
3A3	Passing cable through tanks (pipe extension) (submarines)	17
3A4	MIL-S-24235 stuffing tube for passing cables through shielded bulkheads (submarines)	18
3A5	Cables through elliptical bulkheads (submarines)	19
3A6	Sealing plugs for blanking steel stuffing tubes (submarines)	21
3A7	Temporary plugging of steel stuffing tubes (submarines)	23
3A8	Size and details of adapter used for changing size of installed steel bulkhead MIL-S-24235 stuffing tubes (submarines)	25
3A9	Bushing sleeves for MIL-S-24235 stuffing tubes through steel shielded bulkheads	28
3A10	Cable assignment to steel, aluminum, and nylon stuffing tubes, MCP blocks, and box connectors	30
3A11	Cable assignment to steel, aluminum, and nylon stuffing tubes, MCP blocks, and box connectors	31
3A12	Cable assignment to steel, aluminum, and nylon stuffing tubes, MCP blocks, and box connectors	32
3A13	Cable assignment to steel, aluminum, and nylon stuffing tubes, MCP blocks, and box connectors	33

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TABLE 3AI. Figures for submarine stuffing tubes installation – Continued.

Figure number	Submarine stuffing tubes installation	Page
3A14	Cable assignment to steel, aluminum, and nylon stuffing tubes, MCP blocks, and box connectors	34
3A15	Cable assignment to steel, aluminum, and nylon stuffing tubes, MCP blocks, and box connectors	35
3A16	Cable assignment to steel, aluminum, and nylon stuffing tubes, MCP blocks, and box connectors	36
3A17	Cable assignment to steel, aluminum, and nylon stuffing tubes, MCP blocks, and box connectors	37
3A18	Cable assignment to steel, aluminum, and nylon stuffing tubes, MCP blocks, and box connectors	38
3A19	Cable assignment to steel, aluminum, and nylon stuffing tubes, MCP blocks, and box connectors	39
3A20	Cable assignment to steel, aluminum, and nylon stuffing tubes, MCP blocks, and box connectors	40
3A21	Cable assignment to steel, aluminum, and nylon stuffing tubes, MCP blocks, and box connectors	41
3A22	Cable assignment to steel, aluminum, and nylon stuffing tubes, MCP blocks, and box connectors	42
3A23	Cable assignment to steel, aluminum, and nylon stuffing tubes, MCP blocks, and box connectors	43
3A24	Cable assignment to steel, aluminum, and nylon stuffing tubes, MCP blocks, and box connectors	44
3A25	Cable assignment to steel, aluminum, and nylon stuffing tubes, MCP blocks, and box connectors	45

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METHOD 3A-1-1

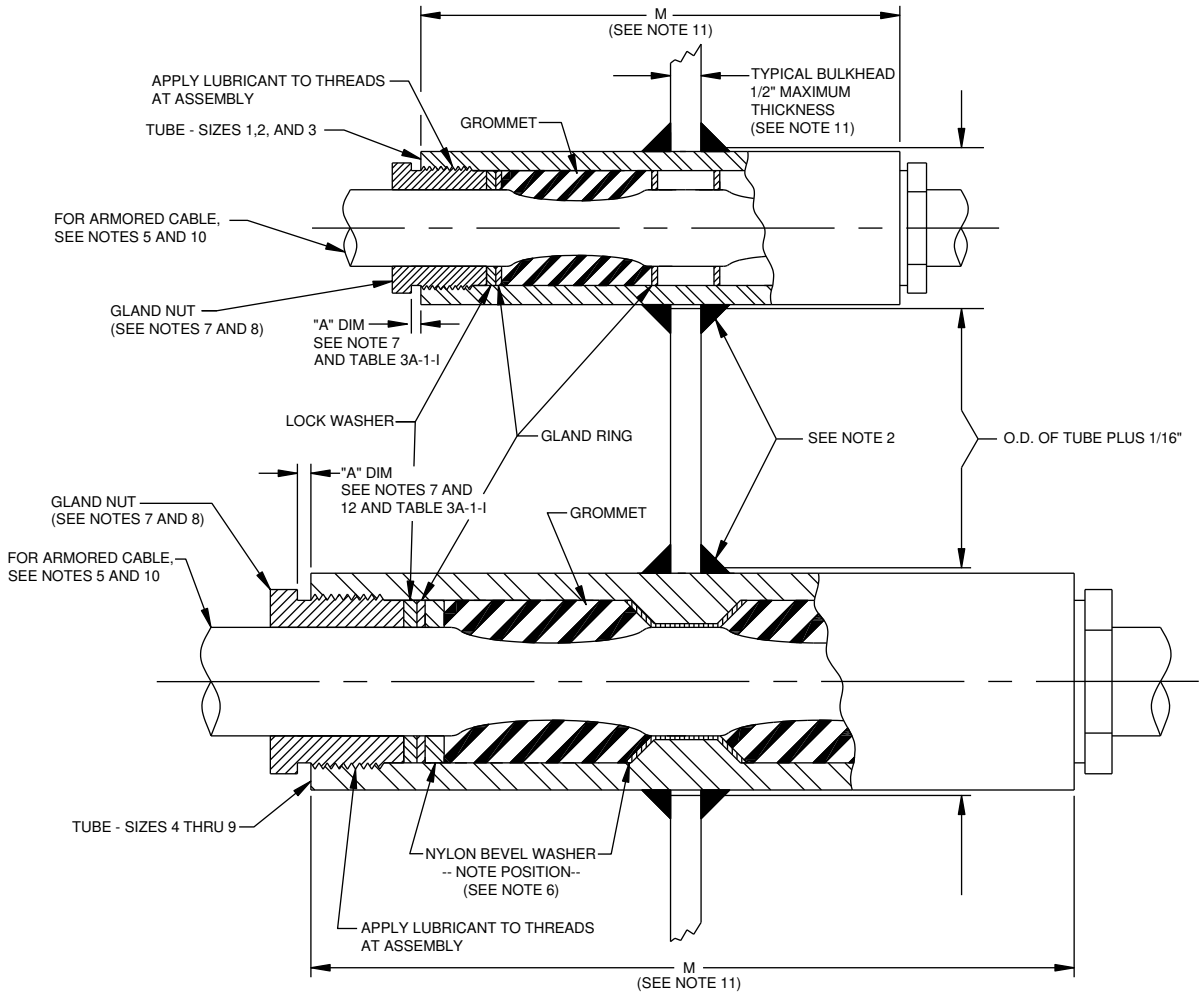


FIGURE 3A1. MIL-S-24235/1 stuffing tubes for passing cables through pressureproof bulkheads (submarines).

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NOTES:

1. Stuffing tubes are designed for cables in accordance with the following specifications (latest revisions): MIL-DTL-915, MIL-DTL-24640, and MIL-DTL-24643.
2. All welding shall comply with approved methods. All welding to or on HY-80 shall comply with T9074-AD-GIB-010/1688.
3. Tubes may be thrown out of line by welding. This is permissible up to 1/16 inch.
4. For non-nuclear bulkheads, grommets may be of neoprene (black). For nuclear bulkheads, grommets shall be of silicone (red or white).
5. To prevent water seepage with armored cable, use a mixture consisting of 3 to 4 parts by weight of silicone grease (DC-4 or DC-111; Dow Corning, or equal [see 4.2]) and 1 part by weight of milled fiberglass (Owens Corning, Composite Materials, Fiberglas, or equal [see 4.2]). See note 10 for methods of applying this mixture to grommets or cable. In applications where one side of the penetration is submerged, when cables have an armored layer under the outer jacket, such as LSMDY type, the outer jacket and armored layer shall be stripped and removed from within 1-2 inches of the bulkhead penetrations and through the penetration. This ensures that any damage that may occur to the cable outer jacket does not permit water to traverse or wick along the underlying armor layer and through the penetration.
6. Care shall be taken to assemble the bevel washers in the position indicated hereon.
7. All gland nuts shall be tightened with a torque wrench using the following procedure:
   a. Tighten gland nuts to an "A" dimension of approximately 1/4 inch (two threads visible on gland nut) whenever possible without exceeding the maximum torque indicated in table 3A1-I for a particular tube size (this maximum torque restriction applies only to tube sizes 1 through 4 inclusive). If maximum torque is obtained before the 1/4 inch spacing requirement is achieved, the "A" dimension shall not exceed 3/8 inch.
   b. In the event minimum torque (as indicated in table 3A1-I) is not obtained at 1/8 inch, continue to tighten the gland nut until this torque is reached. If metal-to-metal contact is achieved without obtaining minimum torque, the gland nut shall be backed off, an additional split gland ring inserted in the stuffing tube, and the gland nut retightened. No more than two additional gland rings, inserted one at a time, shall be installed in each end of a tube to raise the torque value to the minimum specified in table 3A1-I.

TABLE 3A1-I. Gland nut torque values (see notes 7, 12, and 14).

Tube size	Min. (ft-lbs)	Max. (ft-lbs)	Max. "A" dimension (inches)
1	17	30	3/8
2	20	35	3/8
3	40	80	3/8
4	50	100	3/8
5	60	No max.	1/8
6	70	No max.	1/8
7	100	No max.	1/8
8	110	No max.	1/8
9	125	No max.	1/8

FIGURE 3A1. MIL-S-24235/1 stuffing tubes for passing cables through pressureproof bulkheads (submarines) – Continued.

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NOTES (continued):

8. It is recommended that sockets (type II, class I, style A, form B, standard wall octagon) and wrench heads (type I, class I, style A, open ratchet, crowfoot, multipurpose) manufactured in accordance with A-A-3041 and A-A-3042 be used with a suitable torque wrench (preferably a sensory signaling type) to tighten gland nuts. Torque values listed in table 3A1-I were obtained using sockets and wrench heads listed above.
9. Existing gland nuts manufactured to the canceled Drawing 815-1197030 may be used, providing care is taken to ensure a good fit between the gland nut and the applicable socket. Where possible, use gland nuts in accordance with MIL-S-24235/1 and MIL-DTL-24235/2. This is necessary to prevent deforming the gland nut corners and wedging of the socket on the gland nut, causing erroneous torque readings.
10. Methods for applying glass flock mixture:
    a. Coat entire inner surface and leading edge of split grommet with mixture just prior to attachment to cable. Grommet should be attached just outside of tube to minimize loss of mixture.
    b. Coat that portion of the cable that will eventually be under grommet (this shall be done just prior to pulling final few inches of cable into tube).
11. For bulkheads in excess of ½ inch but not greater than 1 inch, tube body dimensions shall be modified as shown in table 3A1-II.

TABLE 3A1-II. Length modifications (inches).

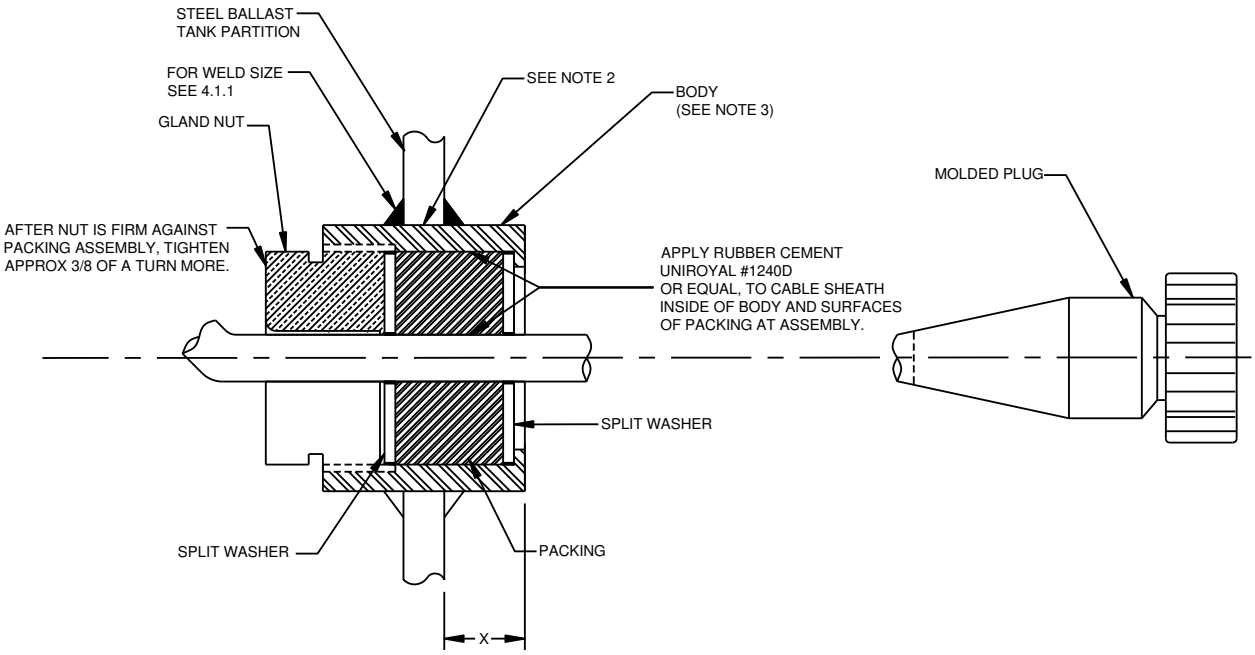
Tube size	Center section dimension	Tube length "M" dimension
1	1¾	6½
2	1¾	7⅛
3	1¾	8
4	1¾	9⅜
5	1¾	10⅞
6	1¾	11½
7	1¾	12¼
8	1¾	12½
9	1¾	12¾

12. The "A" dimension for tube sizes 5 through 9 shown in table 3A1-I are desired dimensions. Variations from this dimension, up to a maximum "A" dimension of 5/16 inch, may be approved locally at the discretion of the Naval Electrical Design Representatives, provided all stuffing tube parts and the installed cable are first inspected to determine compliance with applicable specifications.
13. This method may be employed on surface ship sonar dome applications. Welding shall be in accordance with 4.1.1.
14. See 4.8 for thread lubrication guidance.

FIGURE 3A1. MIL-S-24235/1 stuffing tubes for passing cables through pressureproof bulkheads (submarines) – Continued.

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METHOD 3A-2-1



NOTES:

1. Tubes may be thrown out of line by welding. This is permissible up to 1/16 inch.
2. Hole diameter shall be outside diameter of tube plus 1/16 inch.
3. Position stuffing tube so that gland nut is on the most accessible side of ballast tank partition.
4. See 4.8 for thread lubrication guidance.
5. Note that 3/8 of a turn is equivalent to two to three wrench flats.

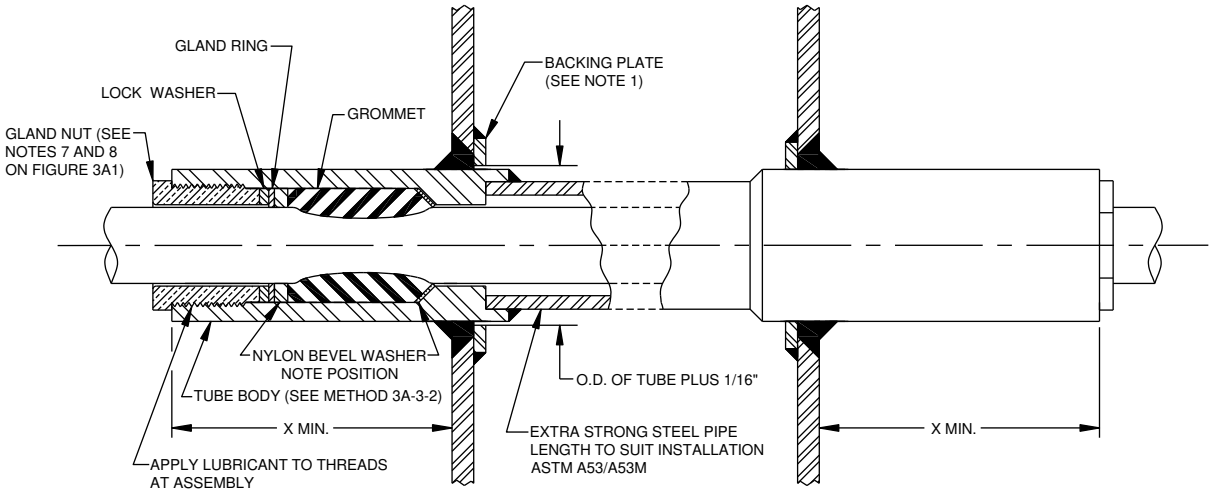
TABLE 3A2-I. Application table.

Stuffing tube sym. no.	Cable type	Molded plug MIL-DTL-24231	Dimension "X"
2425	TSS-4 DSS-3 DSS-4	713	9/16
2425.1	TSS-4 FSS-2	713.1	3/4
2425.2	FSS-4	713.1	3/4

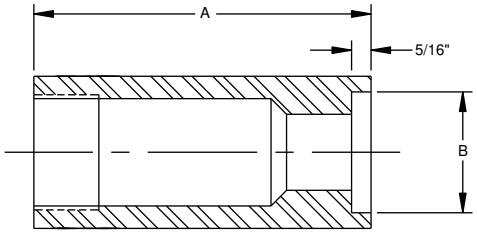
FIGURE 3A2. MIL-S-24235/5 stuffing tube for passing cable through ballast tank partitions (submarines).

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METHOD 3A-3-1



METHOD 3A-3-2
SAME AS 1197030 EXCEPT AS SHOWN



NOTES:

1. When backing plate is used, it shall be the diameter of the tube plus 1⅜ inches thick.
2. Notes on figure 3A1 apply.
3. Pipe shall be in accordance with ASTM A53/A53M.

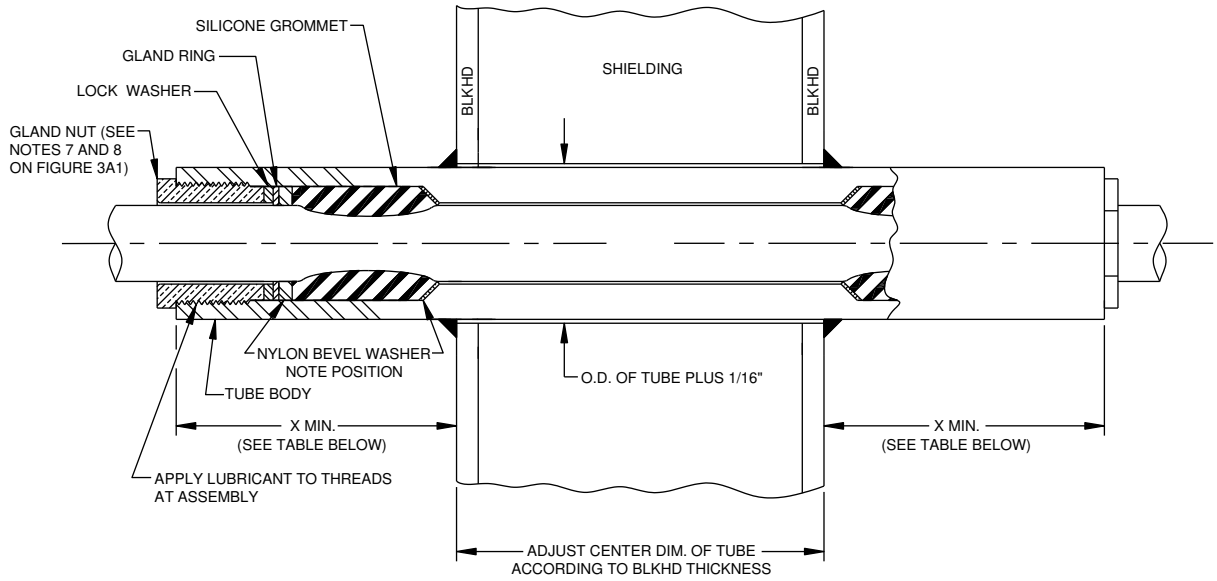
TABLE 3A3-I. Table of dimensions.

Tube size	Extra strong pipe length	"X"	"A"	"B"
1	¾	3	4	1.065
2	1	3⅝	4⅝	1.330
3	1¼	3¾	4¾	1.675
4	1½	4⅞	5⅞	1.915
5	2	5⅜	6⅜	2.400
6	2	5½	6½	2.400
7	2½	5⅞	6⅞	2.906
8	3	6	7	3.648
9	3	6⅛	7⅛	3.546

FIGURE 3A3. Passing cable through tanks (pipe extension) (submarines).

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METHOD 3A-4-1



NOTE:

1. Notes on figure 3A1 apply.

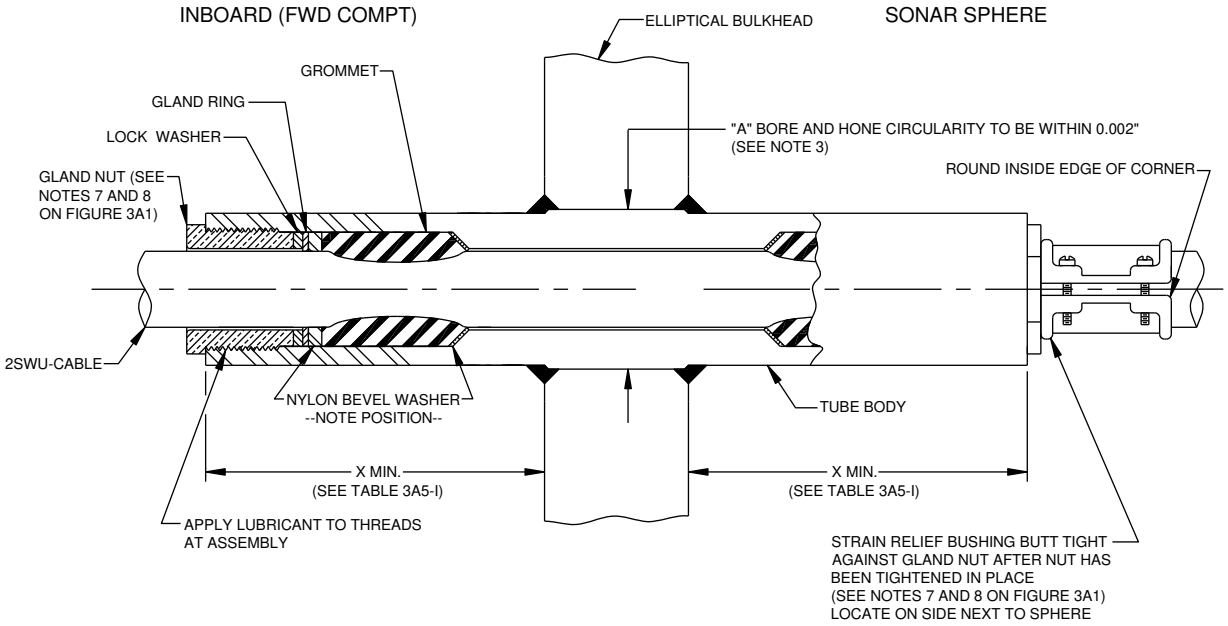
TABLE 3A4-I. Table of dimensions.

Tube size	Dimension "X"
1	3
2	3 5/16
3	3 3/4
4	4 7/16
5	5 3/16
6	5 1/2
7	5 7/8
8	6
9	6 1/8

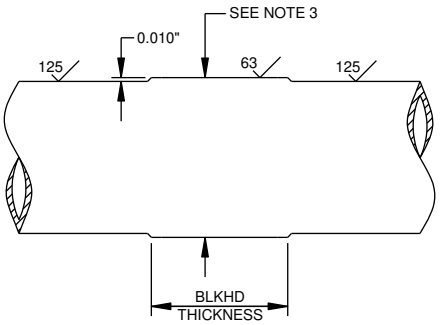
FIGURE 3A4. MIL-S-24235 stuffing tube for passing cables through shielded bulkheads (submarines).

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METHOD 3A-5-1



TUBE BODY



NOTES:

1. Notes on figure 3A1 apply. For size 9 stuffing tubes, the nylon bevel washer to the left of the grommet shown in the above illustration may be replaced by a flat washer to avoid the washer cutting into cable when the gland nut is torqued.
2. Installation shall be in accordance with note 2 on figure 3A1. Tube spacing and weld joint efficiency shall be in accordance with T9074-AD-GIB-010/1688.
3. Provide an interface fit of 0.000 inch and 0.001 inch between “A” bore and tube body.
4. Bore and length dimensions (A and X) are based on 2⅝-inch thick elliptical bulkhead.
5. For correct strain relief bushing, see table 3A5-I below and 9000-S6202-73241.

FIGURE 3A5. Cables through elliptical bulkheads (submarines).

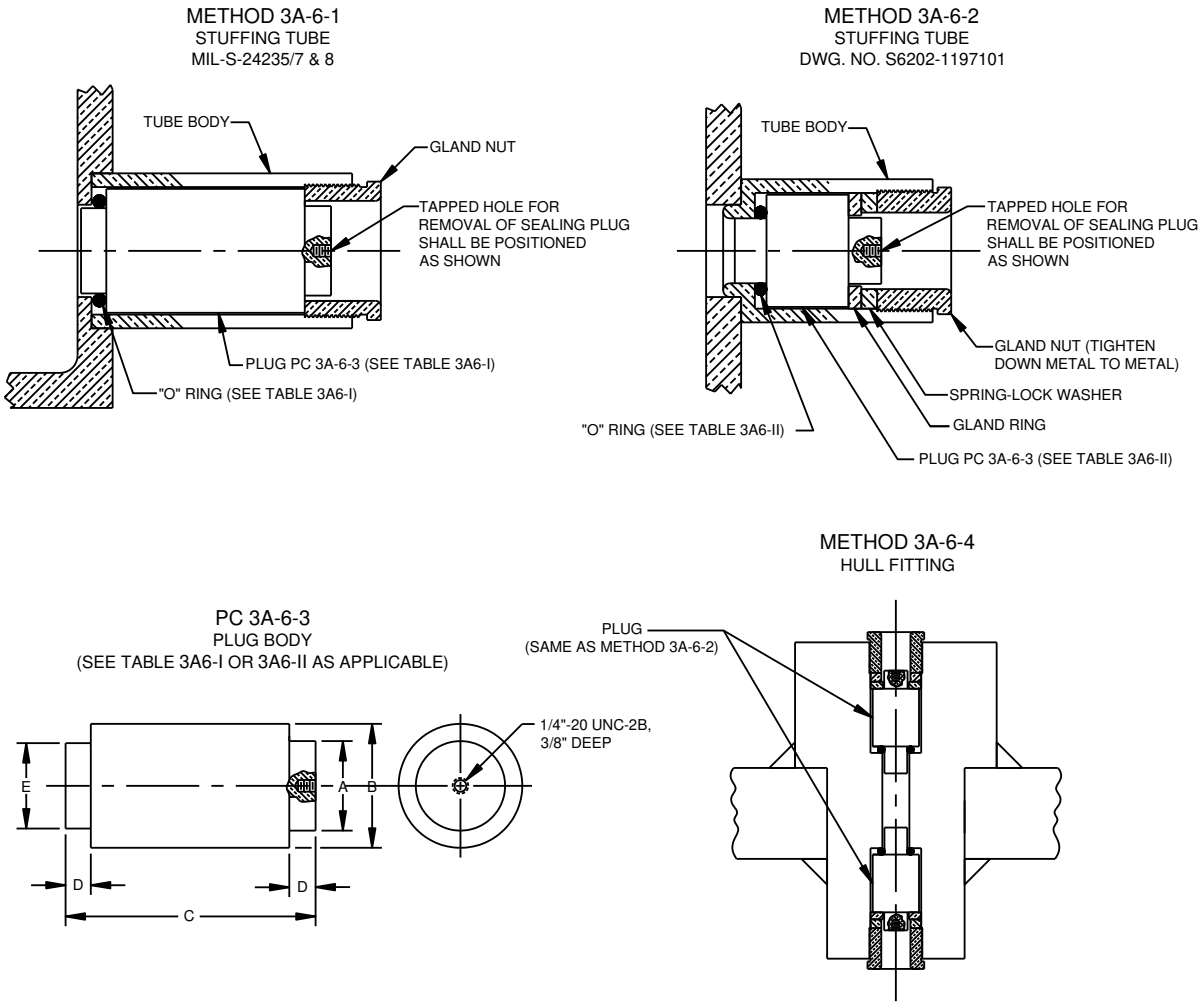
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TABLE 3A5-I. Bore and length dimensions and strain relief bushings.

Tube size	Strain relief bushing size	Dimension "X" (inches)	Dimension "A" (inches)
1		3¼	1.488 1.484
2		3 9/16	1.669 1.684
3		4 3/16	2.109 2.103
4		4⅞	2.299 2.292
5		5 13/16	2.669 2.662
6	5	6 3/16	2.799 2.792
7	5	6 9/16	3.299 3.291
8	6a	7⅛	3.924 3.916
9	6a	7⅛	3.924 3.916

FIGURE 3A5. Cables through elliptical bulkheads (submarines) – Continued.

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NOTES:

1. Sealing plugs covered by this figure are intended for temporary blanking of stuffing tubes installed prior to installation of cables.
2. Material for plug body to be brass in accordance with ASTM B21/B21M.
3. “O” rings shall be synthetic rubber in accordance with SAE-AMS-P-83461. “O” ring sizes shall be in accordance with SAE-AS568. See tables 3A6-I and 3A6-II.
4. See 4.8 for thread lubrication guidance.
5. See figure 3A1 for gland nut tightening methods.

FIGURE 3A6. Sealing plugs for blanking steel stuffing tubes (submarines).

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TABLE 3A6-I. Table of dimensions – sealing plugs, method 3A-6-1.

M24235/7 and 8 tube size	A	B	C	D	E	"O" ring size (see note 3)
A	17/32	53/64	2 1/16	1/4	17/32	-113
B	23/32	1 1/32	2 7/16	5/16	23/32	-116
C	31/32	1 11/32	2 13/16	5/16	31/32	-214
D	31/32	1 11/32	3 7/32	5/16	31/32	-214
E	1 3/16	1 37/64	2 13/16	5/16	1 3/16	-218
F	1 17/32	2	2 13/16	5/16	1 17/32	-223

TABLE 3A6-II. Table of dimensions – sealing plugs, method 3A-6-2.

S6202-1197101 tube sym. no.	A	B	C	D	E	"O" ring PC no. (see note 3)
2446.1	0.500	0.867	2.077	0.625	0.500	-112
2446.2	0.621	1.084	2.267	0.625	0.646	-115
2446.3	0.725	1.302	2.449	0.625	0.711	-211
2446.4	1.100	1.867	2.782	0.750	1.100	-216

FIGURE 3A6. Sealing plugs for blanking steel stuffing tubes (submarines) – Continued.

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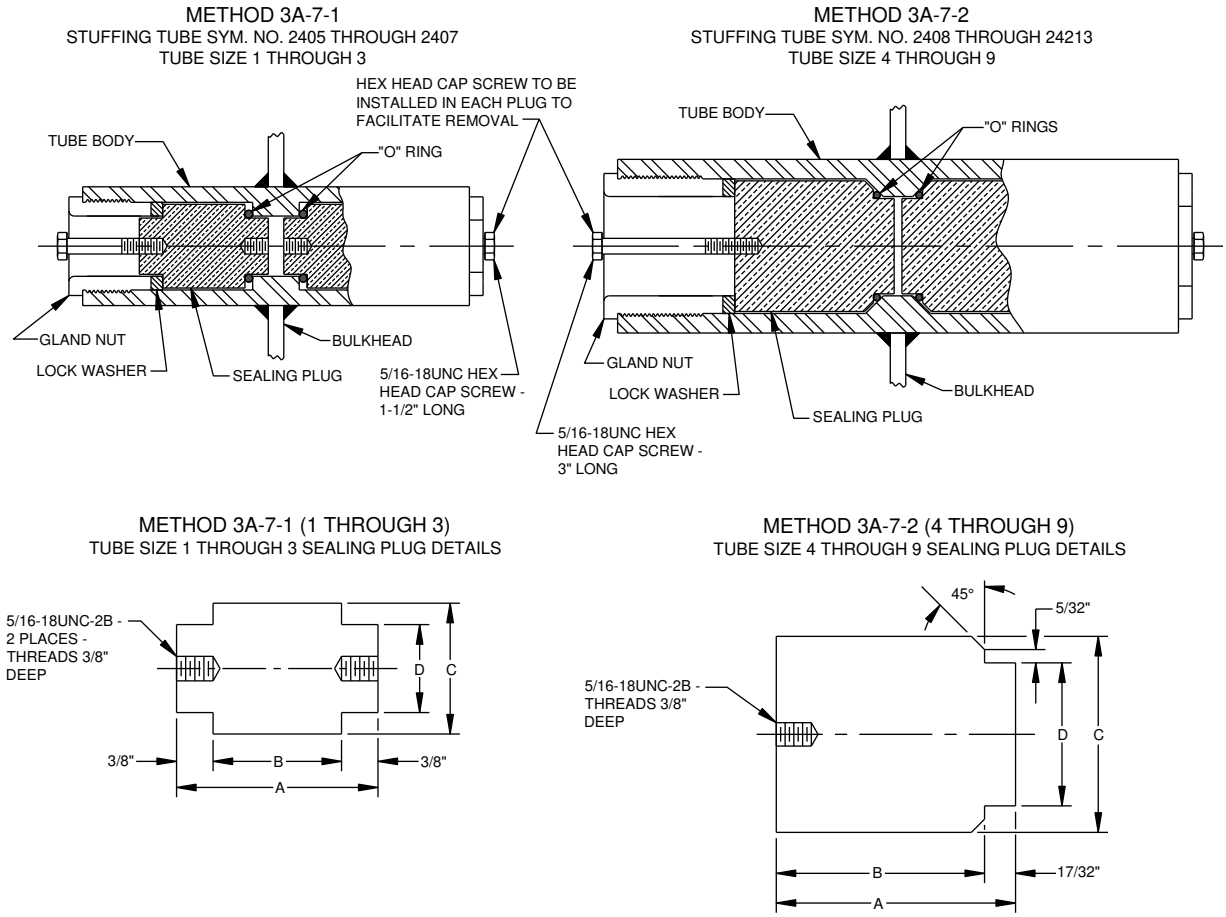


TABLE 3A7-I. Associated data and sealing plug dimensions.

Tube size	"O" ring size (2 required)	Sealing plug (2 required)	A	B	C	D
1	-113	3A-7-1-1	1 53/64	1 5/64	0.986	0.594
2	-116	3A-7-1-2	2 9/64	1 25/64	1.173	0.797
3	-214	3A-7-1-3	2 1/2	1 3/4	1.486	1.047
4	-218	3A-7-2-4	2 1/2	1 31/32	1.673	1.250
5	-220	3A-7-2-5	3 1/8	2 19/32	1.923	1.422
6	-223	3A-7-2-6	3 5/16	2 25/32	2.173	1.610
7	-225	3A-7-2-7	3 1/2	2 31/32	2.548	1.860
8	-227	3A-7-2-8	3 5/8	3 3/32	2.861	2.172
9	-228	3A-7-2-9	3 11/16	3 5/32	2.988	2.297

FIGURE 3A7. Temporary plugging of steel stuffing tubes (submarines).

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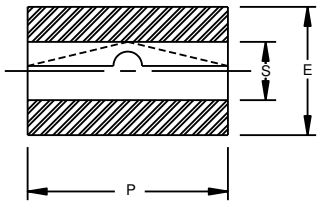
NOTES:

1. The pieces and methods shown on this figure are means of temporarily plugging an electrical fitting which, due to design changes, no longer has a specific cable assigned to it. Temporary plugging shall be used for any of the following conditions:
   a. Fitting is so inaccessible as to prevent accomplishing a permanent patch.
   b. Welding heat involved in permanent patching is liable to damage adjacent fittings and cable.
   c. Immediate reuse is likely on another system.
2. At the time of final preparation for delivery of the ship, all unused cable fittings not in categories 1.a, 1.b, or 1.c shall be removed and patched permanently, as indicated by the applicable hull structural plans for permanent patching of bulkhead and hull penetrations.
3. "O" rings shall be synthetic rubber in accordance with SAE-AMS-P-83461. "O" ring sizes shall be in accordance with SAE-AS568. See table 3A7-I.
4. Use existing lockwashers and gland nuts.
5. Material for sealing plugs shall be brass.
6. Hex head cap screws shall remain installed until plug is removed.
7. See 4.8 for thread lubrication guidance.
8. See figure 3A1 for gland nut tightening methods.
9. See 4.5 for fastener and fastener material requirements.

FIGURE 3A7. Temporary plugging of steel stuffing tubes (submarines) – Continued.

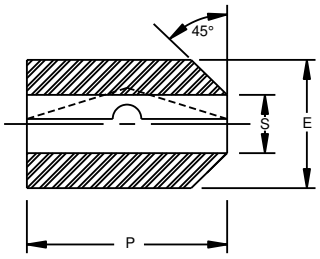
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PIECE NO. (PC) 3A-8-1



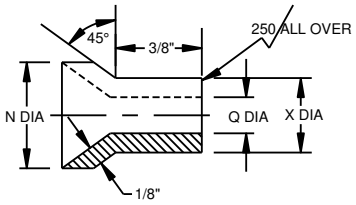
GROMMET FOR TUBE SIZE 2 AND 3
1. SIMILAR TO M24235/2-061 THROUGH -076 EXCEPT FOR DIMENSIONS.

PIECE NO. (PC) 3A-8-2



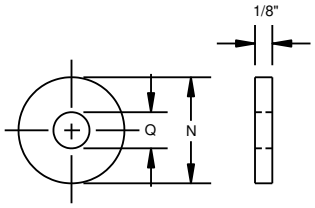
GROMMET FOR TUBE SIZE 4 THROUGH 9
1. SIMILAR TO M24235/2-090 THROUGH -110 EXCEPT FOR DIMENSIONS.

PIECE NO. (PC) 3A-8-3



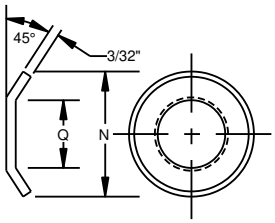
BUSHING
1. MATL. AL. BRZ., ASTM B150/B150M AND ASTM B169/B169M.
2. BREAK ALL SHARP CORNERS.

PIECE NO. (PC) 3A-8-4



GLAND RING
1. SIMILAR TO M24235/2-120 THROUGH -144 EXCEPT FOR DIMENSIONS.

PIECE NO. (PC) 3A-8-5



BEVEL WASHER
1. SIMILAR TO M24235/2-160 THROUGH 180 EXCEPT FOR DIMENSIONS.

NOTES:

1. This figure provides a method for adapting existing MIL-S-24235 stuffing tubes to accept cables assigned to smaller tube sizes. To determine dimensions required to adapt a stuffing tube for a particular cable, obtain the grommet number for the cable from the tables on figures 3A10 through 3A25. Enter table 3A8-I with the existing tube size and packing symbol number. A dash indicates that particular stuffing tube size/packing assembly combination is either not applicable or not allowed.
2. Bushing, gland ring, bevel washer, and grommet shall be manufactured to the requirements of MIL-S-24235, except as shown on this drawing. Each item shall be marked with a piece number consisting of symbol number and "(MOD)" (for example, "2405.3 (MOD)").
3. Adapter assembly for stuffing tubes size 2 and 3 consists of four gland rings, PC 3A-8-4, and two grommets, PC 3A-8-1. Adapter assembly for stuffing tubes size 4 through 9 consists of two each: gland ring, PC 3A-8-4, grommet, PC 3A-8-2, bushing, PC 3A-8-3, and bevel washer, PC 3A-8-5.

FIGURE 3A8. Size and details of adapter used for changing size of installed steel bulkhead MIL-S-24235 stuffing tubes (submarines).

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NOTES (continued):

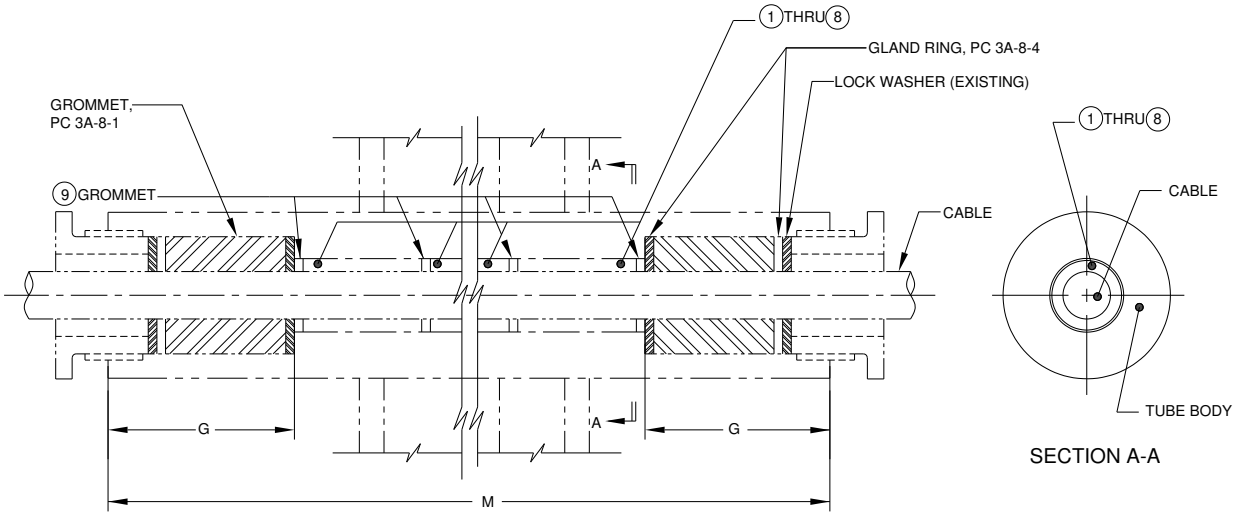
4. Adapter assembly shall be installed in accordance with figure 3A1 for non-nuclear bulkheads.
5. Use of adapter assemblies on nuclear bulkheads shall be in accordance with figure 3A9.
6. On non-nuclear bulkheads, step-down of two sizes is permitted without NAVSEA approval. Step-down of three or more sizes requires specific NAVSEA approval.

FIGURE 3A8. Size and details of adapter used for changing size of installed steel bulkhead MIL-S-24235 stuffing tubes (submarines) – Continued.

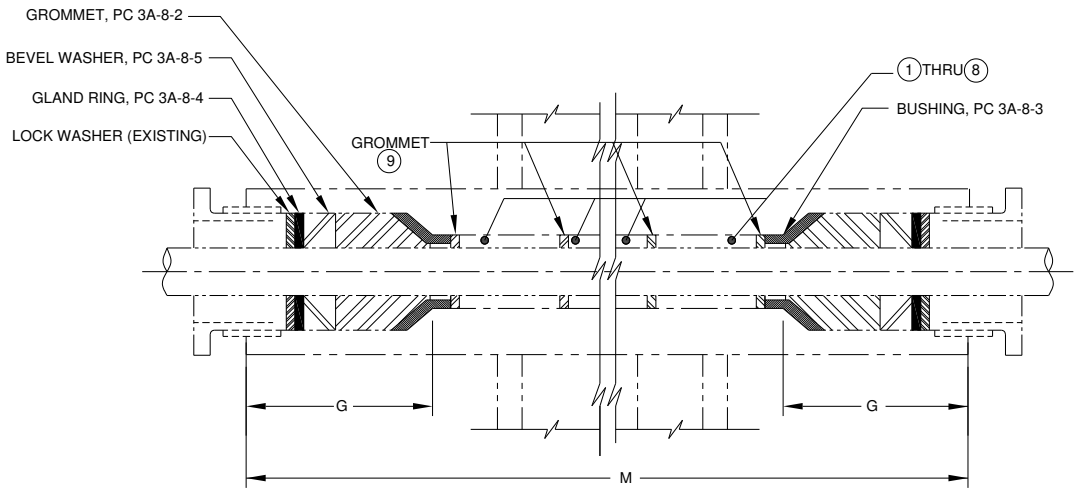
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TABLE 3A8-I. Table of dimensions.

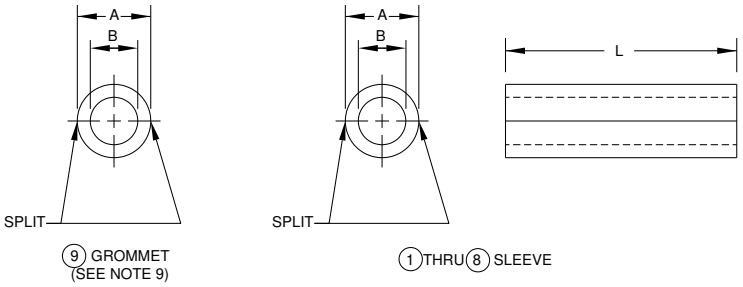
| Tube size | | | | | MIL-C-24235 tube modified packing assemblies symbol numbers | | | | |



ASSEMBLY 1



ASSEMBLY 2



NOTES:

1. This method and the drawings called out were applicable to S5W submarines only and are included in this document for historical purposes only. Any current use of this method shall be expressly approved by NAVSEA 08.
2. Any deviations from this figure are not permitted without prior written approval of NAVSEA 08 or the NAVSEA-designated reactor plant design yard.

FIGURE 3A9. Bushing sleeves for MIL-S-24235 stuffing tubes through steel shielded bulkheads.

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NOTES (continued):

3. Abbreviations are in accordance with ASME Y14.38.
4. The requirements of this figure are not retroactive for existing bushed stuffing tubes.
5. Stuffing tube bushing sleeves in accordance with this figure shall be installed in bushed stuffing tubes through shielded bulkheads when the difference between the inner diameter of the stuffing tube throat and the outer diameter of the cable is ¼ inch or greater.
6. Stuffing tube bushing sleeves items 1 through 8 can be installed in one, two, or three sections as required to facilitate installation.
7. Stuffing tube bushing sleeves items 1 through 8 and grommets item 9 shall be installed in stuffing tube, and tube shall be packed in accordance with instructions on figure 3A1. The "A" and "B" dimensions of item 9 shall duplicate the "A" and "B" dimensions of items 1-8.
8. Material for items 1 through 8 shall be of steel bar, round with cold finish, or seamless carbon steel pipe in accordance with MIL-P-24338 may be substituted to facilitate machining, provided that no deviations from the tolerances of table 3A9-I are made.
9. Grommet item 9 shall be manufactured from silicon sheet rubber 0.25 inch thick in accordance with A-A-59588, Class 2B, Grade 50 and shall be cemented to bushing items 1 through 8 prior to installation using adhesive in accordance with MMM-A-101.
10. For dimensions not detailed herein, see MIL-S-24235 and figure 3A8.

TABLE 3A9-I. Sleeve dimensions (see note 7).

Assy	Stuffing tube size	Sleeve item no.	Dim. "A"	Dim. "B"	Dim. "L" (note 5)
1	1	-	-	"B" = cable OD+ 0.062	(M-2G)-0.50 = L When sleeve is installed in one section.
	2	1	0.750		$\frac{\text{(M-2G)-0.75}}{2}$ = L When sleeve is installed in two sections.
	3	2	1.000		$\frac{\text{(M-2G)-1.00}}{3}$ = L When sleeve is installed in three sections.
2	4	3	1.203		(M-2G)-1.25 = L When sleeve is installed in one section.
	5	4	1.375		$\frac{\text{(M-2G)-1.50}}{2}$ = L When sleeve is installed in two sections.
	6	5	1.563		
	7	6	1.813		$\frac{\text{(M-2G)-1.75}}{3}$ = L When sleeve is installed in three sections.
	8	7	2.125		
	9	8	2.250		

TOLERANCES

Fractional	±	1/32"
Decimal	±	0.005"
Angles	±	0°-30′
Unless otherwise noted		

FIGURE 3A9. Bushing sleeves for MIL-S-24235 stuffing tubes through steel shielded bulkheads – Continued.

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TABLE 3A10-I. Cable assignment to steel, aluminum, and nylon stuffing tubes, MCP blocks, and box connectors.

Cable			Nylon tube		MCP [2/]		Stuffing tube			Box conn.	Cable	
Type [1/]	Size OD		Size	Packing assembly, ASTM F1836M	Insert block	Insert block	Surface size	Submarine			MIL-Spec	Slant sheet
	Min.	Max.						Size	Grommet			
DSS-2	0.370	0.390	2	17-0003	15/9		B	1	2-062	½	MIL-DTL-915	8
DSS-3	0.480	0.500	4T	19-0001	20/12	30/12	B	1	2-065	½	MIL-DTL-915	8
DSS-4	0.480	0.500	4T	19-0001	20/12	30/12	B	1	2-065	½	MIL-DTL-915	8
DSWS-4	0.770	0.800	5	20-0001	30/20		E	2	2-071	1	MIL-DTL-915	7
FSS-2	0.480	0.500	4T	19-0001	20/12	30/12	B	1	2-065	½	MIL-DTL-915	8
FSS-3	0.480	0.500	4T	19-0001	20/12	30/12	B	1	2-065	½	MIL-DTL-915	8
FSS-4	0.600	0.625	4T	19-0004	20/15	30/15	C	2	2-068	¾	MIL-DTL-915	8
MCSF-4	1.388	1.500	7	22-0002	60/37		P	6	2-099	1½	MIL-DTL-915	10
MSP	1.565	1.635	7	22-0003	60/41		R	7	2-101	2	MIL-DTL-915	67
MSPW	1.565	1.635	7	22-0003	60/41		R	7	2-101	2	MIL-DTL-915	66
MWF-7	0.480	0.500	3	18-0018	20/12	30/12	C	1	2-065	½	MIL-DTL-915	58
MWF-10	0.605	0.635	4T	19-0004	20/16	30/16	D	2	2-068	¾	MIL-DTL-915	58
MWF-14	0.605	0.635	4T	19-0005	20/16	30/16	D	2	2-068	¾	MIL-DTL-915	58
MWF-19	0.710	0.745	4T	19-0007	30/18		E	2	2-071	¾	MIL-DTL-915	58
MWF-24	0.800	0.836	5	20-0002	30/21		G	3	2-072	1	MIL-DTL-915	58
MWF-30	0.905	0.945	5	20-0005	30/23	40/23	J	3	2-074	1	MIL-DTL-915	58
MWF-37	1.005	1.045	5	20-0008	40/26		J	3	2-076	1¼	MIL-DTL-915	58
THOF-42	1.200	1.250	6	21-0004	40/30		L	4	2-094	1¼	MIL-DTL-24643	3
THOF-400	2.680	2.800	9	24-0008	90/71		AA			3	MIL-DTL-915	6
THOF-500	2.920	3.100			120/78		BB			3½	MIL-DTL-915	6
TPUM-6	0.856	0.925	5	20-0004	30/23		G	3	2-074	1	MIL-DTL-915	79
TSP-11	0.680	0.730	4T	19-0007	30/17		D	2	2-070	¾	MIL-DTL-915	22
TSP-31	0.982	1.062	5	20-0008	40/26		J	4	2-090	1¼	MIL-DTL-915	22
TSPA-11	0.680	0.73	4T	19-0007	30/18		E	2	2-071	1	MIL-DTL-915	22
TSPA-31	0.982	1.062	5	20-0008	40/26		K	4	2-091	1¼	MIL-DTL-915	22
TSS-2	0.385	0.400	2	17-0003	20/10		A	1	2-062	½	MIL-DTL-915	8
TSS-3	0.480	0.500	4T	19-0001	20/12	30/12	B	1	2-065	½	MIL-DTL-915	8
TSS-4	0.480	0.500	4T	19-0001	20/12	30/12	B	1	2-065	½	MIL-DTL-915	8
1PR-A20E	0.28	0.29	1	16-0005	15/7	20/7		1	2-061		MIL-DTL-915	81
1PR-16	0.380	0.400	2	17-0003	20/10			1	2-062		MIL-DTL-915	82
1O-16	0.380	0.400	2	17-0003	20/10			1	2-062		MIL-DTL-915	86
1SPR-16S	0.380	0.400	2	17-0003	20/10			1	2-062		MIL-DTL-915	89
1SWF-2	0.600	0.625	4T	19-0004	20/15	30/15	C	2	2-068	¾	MIL-DTL-915	47
1TR-16	0.380	0.400	2	17-0003	20/10			1	2-062		MIL-DTL-915	87
2SPR-16	0.630	0.650	4T	19-0004	20/16	30/16		2	2-039		MIL-DTL-915	84
2SWF-3	0.600	0.625	4T	19-0004	20/15	30/15	D	2	2-068	¾	MIL-DTL-915	48
2SWF-4	0.600	0.625	4T	19-0004	20/15	30/15	D	2	2-068	¾	MIL-DTL-915	48
2SWF-7	0.780	0.815	5	20-0002	30/20		F	3	2-072	¾	MIL-DTL-915	48
2SWF-24	1.190	1.250	6	21-0004	40/30			4	2-094		MIL-DTL-915	48
3PR-16	0.530	0.550	4T	19-0002	20/13	30/13		1	2-066		MIL-DTL-915	85
7PR-16	0.630	0.650	4T	19-0004	20/16	30/16		2	2-069		MIL-DTL-915	83
7SPR-16S	0.920	0.950	5	20-0005	30/24	40/24		3	2-074		MIL-DTL-915	88
7SS-2	0.600	0.625	4T	19-0004	20/15	30/15	C	2	2-068	¾	MIL-DTL-915	8
DX-3	0.223	0.241	1	16-0005	15/6	20/6	A	1	2-061	½	MIL-DTL-24640	1
DX-4	0.266	0.286	1	16-0005	15/7	20/7	A	1	2-061	½	MIL-DTL-24640	1

NOTES:

1. Cable assignment to steel, aluminum, and nylon stuffing tubes, MCP blocks, and BX connectors shown on figures 3A10 through 3A25 are for MIL-DTL-915, MIL-DTL-24640, and MIL-DTL-24643 cable.
2. MCP block size and stuffing tube designations are provided for information and guidance. Some of the MCP block and stuffing tube sizes shown in this table may not accommodate the full range of allowed cable diameters for a given cable type. In those cases, different MCP block sizes or stuffing tubes should be selected to ensure the required block gap and retest requirements are met. See figure 3B22 for MCPs.

FIGURE 3A10. Cable assignment to steel, aluminum, and nylon stuffing tubes, MCP blocks, and box connectors.

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TABLE 3A11-I. Cable assignment to steel, aluminum, and nylon stuffing tubes, MCP blocks, and box connectors.

Cable			Nylon tube		MCP [2/]		Stuffing tube			Box conn.	Cable	
Type [1/]	Size OD		Size	Packing assembly, ASTM F1836M	Insert block	Insert block	Surface size	Submarine			MIL-Spec	Slant sheet
	Min.	Max.						Size	Grommet			
DXOW-3	0.294	0.316	2	17-0001	15/7	20/7	A	1	2-062	½	MIL-DTL-24640	19
DXOW-4	0.328	0.354	2	17-0002	15/8	20/8	A	1	2-062	½	MIL-DTL-24640	19
DXW-3	0.239	0.257	1	16-0005	15/6	20/6	A	1	2-061	½	MIL-DTL-24640	19
DXW-4	0.281	0.303	1	16-0006	15/7	20/7	A	1	2-061	½	MIL-DTL-24640	19
FX-3	0.262	0.282	1	16-0006	15/7	20/7	A	1	2-061	½	MIL-DTL-24640	3
FX-4	0.311	0.335	2	17-0002	15/8	20/8	A	1	2-062	½	MIL-DTL-24640	3
FXOW-3	0.324	0.350	1	16-0006	15/8	20/8		1	2-062	½	MIL-DTL-24640	21
FXOW-4	0.366	0.394	2	17-0003	20/9			1	2-062	½	MIL-DTL-24640	21
FXW-3	0.266	0.286	1	16-0005	15/7	20/7	A	1	2-061	½	MIL-DTL-24640	21
FXW-4	0.315	0.339	2	17-0002	15/8	20/8	A	1	2-062	½	MIL-DTL-24640	21
MXCOW-7	0.340	0.366	2	17-0002	15/9	20/9	A	1	2-062	⅜	MIL-DTL-24640	23
MXCOW-10	0.415	0.447	2	17-0004	20/11		B	1	2-063	½	MIL-DTL-24640	23
MXCOW-14	0.440	0.474	3	18-0018	20/11	30/11	B	1	2-064	½	MIL-DTL-24640	23
MXCOW-19	0.477	0.515	4T	19-0001	20/12	30/12	C	1	2-065	½	MIL-DTL-24640	23
MXCOW-24	0.557	0.601	4T	19-0003	20/15	30/15	C	2	2-068	¾	MIL-DTL-24640	23
MXCOW-30	0.584	0.630	4T	19-0005	20/15	30/15	C	2	2-068	¾	MIL-DTL-24640	23
MXCOW-37	0.622	0.670	4T	19-0005	30/16		D	2	2-069	¾	MIL-DTL-24640	23
MXCOW-44	0.697	0.751	4T	20-0001	30/18		E	2	2-071	1	MIL-DTL-24640	23
MXCOW-61	0.757	0.817	5	20-0002	30/20		F	3	2-072	1	MIL-DTL-24640	23
MXCW-7	0.295	0.319	2	17-0001	15/8	20/8	A	1	2-062	⅜	MIL-DTL-24640	23
MXCW-10	0.375	0.405	2	17-0004	20/10		B	1	2-062	½	MIL-DTL-24640	23
MXCW-14	0.402	0.434	3	18-0018	20/10		B	1	2-063	½	MIL-DTL-24640	23
MXCW-19	0.440	0.474	3	18-0018	20/11	30/11	B	1	2-064	½	MIL-DTL-24640	23
MXCW-24	0.520	0.560	4T	19-0003	20/14	30/14	C	1	2-067	¾	MIL-DTL-24640	23
MXCW-30	0.547	0.589	4T	19-0003	20/14	30/14	C	1	2-067	¾	MIL-DTL-24640	23
MXCW-37	0.584	0.630	4T	19-0005	20/15	30/5	C	2	2-068	¾	MIL-DTL-24640	23
MXCW-44	0.656	0.708	4T	19-0006	30/17		D	2	2-070	¾	MIL-DTL-24640	23
MXCW-61	0.729	0.785	5	20-0001	30/19		E	2	2-071	1	MIL-DTL-24640	23
MXO-10	0.349	0.377	3	18-0018	15/9	20/9	A	1	2-062	⅜	MIL-DTL-24640	8
MXO-14	0.380	0.410	2	17-0004	20/10		B	1	2-063	½	MIL-DTL-24640	8
MXSO-2	0.282	0.304	1	16-0005	15/7	20/7	A	1	2-061	⅜	MIL-DTL-24640	10
MXSO-9	0.424	0.458	3	18-0018	20/11		B	1	2-064	½	MIL-DTL-24640	10
MXSO-21	0.552	0.596	4T	19-0004	20/15	30/15	C	2	2-068	½	MIL-DTL-24640	10
MXSO-37	0.644	0.694	4T	19-0007	30/17		D	2	2-070	½	MIL-DTL-24640	10
MXSOW-2	0.282	0.304	1	16-0005	15/7	20/7	A	1	2-061	⅜	MIL-DTL-24640	10
MXSOW-9	0.424	0.458	3	18-0018	20/11		B	1	2-064	½	MIL-DTL-24640	10
MXSOW-21	0.552	0.596	4T	19-0004	20/15	30/15	C	2	2-068	½	MIL-DTL-24640	10
MXSOW-37	0.644	0.694	4T	19-0007	30/17		D	2	2-070	½	MIL-DTL-24640	10
TTX-3	0.296	0.320	2	17-0002	15/8	20/8	A	1	2-062	⅜	MIL-DTL-24640	4
TTX-15	0.549	0.591	4T	19-0005	20/14	30/14	C	2	2-068	½	MIL-DTL-24640	4
TTXOW-1½	0.253	0.273	1	16-0004	15/6	20/6	A	1	2-061	⅜	MIL-DTL-24640	24
TTXOW-3	0.333	0.359	2	17-0002	15/9	20/9	A	1	2-062	⅜	MIL-DTL-24640	24
TTXOW-5	0.376	0.406	2	17-0003	20/10		B	1	2-063	½	MIL-DTL-24640	24

NOTES:

1. Cable assignment to steel, aluminum, and nylon stuffing tubes, MCP blocks, and BX connectors shown on figures 3A10 through 3A25 are for MIL-DTL-915, MIL-DTL-24640, and MIL-DTL-24643 cable.
2. MCP block size and stuffing tube designations are provided for information and guidance. Some of the MCP block and stuffing tube sizes shown in this table may not accommodate the full range of allowed cable diameters for a given cable type. In those cases, different MCP block sizes or stuffing tubes should be selected to ensure the required block gap and retest requirements are met. See figure 3B22 for MCPs.

FIGURE 3A11. Cable assignment to steel, aluminum, and nylon stuffing tubes, MCP blocks, and box connectors.

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TABLE 3A12-I. Cable assignment to steel, aluminum, and nylon stuffing tubes, MCP blocks, and box connectors.

Cable			Nylon tube		MCP [2/]		Stuffing tube			Box conn.	Cable	
Type [1/]	Size OD		Size	Packing assembly, ASTM F1836M	Insert block	Insert block	Surface size	Submarine			MIL-Spec	Slant sheet
	Min.	Max.						Size	Grommet			
TTXOW-15	0.556	0.600	4T	19-0004	20/15	30/15	C	2	2-068	½	MIL-DTL-24640	24
TTXOW-20	0.614	0.662	4T	19-0005	20/16	30/16	D	2	2-069	½	MIL-DTL-24640	24
TTXOW-30	0.717	0.772	5	20-0001	30/19		E	2	2-071	1	MIL-DTL-24640	24
TTXOW-40	0.823	0.887	5	20-0004	30/22	40/22	G	3	2-073	1	MIL-DTL-24640	24
TTXS-2	0.335	0.361	2	17-0003	15/9	20/9	A	1	2-062	⅜	MIL-DTL-24640	5
TTXS-4	0.393	0.424	2	17-0004	20/10		B	1	2-063	⅜	MIL-DTL-24640	5
TTXSO-2	0.386	0.416	2	17-0004	20/10		B	1	2-063	½	MIL-DTL-24640	5
TTXSO-6	0.506	0.546	4T	19-0003	20/13	30/13	C	1	2-066	¾	MIL-DTL-24640	5
TTXSO-8	0.587	0.633	4T	19-0005	20/15	30/15	D	2	2-068	¾	MIL-DTL-24640	5
TTXSO-10	0.627	0.675	4T	19-0006	30/17		D	2	2-069	¾	MIL-DTL-24640	5
TTXW-1½	0.181	0.195	1	16-0004	15/4	20/4	A	1	2-061	⅜	MIL-DTL-24640	24
TTXW-3	0.285	0.307	2	17-0001	15/7	20/7	A	1	2-061	⅜	MIL-DTL-24640	24
TTXW-5	0.331	0.357	2	17-0002	15/8	20/8	A	1	2-062	⅜	MIL-DTL-24640	24
TTXW-10	0. 456	0.492	4T	19-0001	20/12	30/12	B	1	2-065	½	MIL-DTL-24640	24
TTXW-15	0.527	0.569	4T	19-0003	20/14	30/14	B	1	2-067	¾	MIL-DTL-24640	24
TTXW-20	0.577	0.621	4T	19-0005	20/15	30/15	C	2	2-068	¾	MIL-DTL-24640	24
TTXW-30	0.684	0.738	4T	19-0007	30/18		D	2	2-071	¾	MIL-DTL-24640	24
TTXW-40	0.790	0.852	5	20-0003	30/21		G	3	2-073	1	MIL-DTL-24640	24
TX-3	0.243	0.261	1	16-0005	15/6	20/6	A	1	2-061	⅜	MIL-DTL-24640	2
TX-4	0.288	0.310	2	17-0001	15/7	20/7	A	1	2-061	⅜	MIL-DTL-24640	2
TXOW-3	0.305	0.329	2	17-0002	15/8	20/8	A	1	2-062	⅜	MIL-DTL-24640	20
TXOW-4	0.343	0.369	2	17-0002	15/9	20/9	A	1	2-062	⅜	MIL-DTL-24640	20
TXW-3	0.246	0.266	1	16-0005	15/6	20/6	A	1	2-061	⅜	MIL-DTL-24640	20
TXW-4	0.292	0.314	2	17-0002	15/7	20/7	A	1	2-062	⅜	MIL-DTL-24640	20
1XMSO-7	0.344	0.370	2	17-0002	15/9	20/9	A	1	2-062	⅜	MIL-DTL-24640	7
1XMSO-16	0.471	0.507	4T	19-0002	20/12	30/12	C	1	2-065	½	MIL-DTL-24640	7
1XMSO-70	0.859	0.925	5	20-0006	30/23		J	3	2-074	1	MIL-DTL-24640	7
1XSOW-2	0.292	0.314	2	17-0001	15/7	20/7	A	1	2-062	⅜	MIL-DTL-24640	14
1XSOW-14	0.470	0.506	4T	19-0001	20/12	30/12	C	1	2-065	½	MIL-DTL-24640	14
1XSOW-20	0.542	0.584	4T	19-0003	20/14	30/14	C	1	2-067	½	MIL-DTL-24640	14
1XSOW-30	0.614	0.662	4T	19-0005	20/16	30/16	D	2	2-069	½	MIL-DTL-24640	14
2XAO-2	0.307	0.331	2	17-0001	15/8	20/8	A	1	2-062	⅜	MIL-DTL-24640	6
2XAO-7	0.392	0.422	2	17-0004	20/10		B	1	2-063	½	MIL-DTL-24640	6
2XAO-10	0.475	0.511	4T	19-0002	20/12	30/12	C	1	2-065	½	MIL-DTL-24640	6
2XAO-18	0.567	0.611	4T	19-0004	20/15	30/15	C	2	2-068	½	MIL-DTL-24640	6
2XAO-40	0.796	0.858	5	20-0004	30/21		G	3	2-073	¾	MIL-DTL-24640	6
2XO-6	0.305	0.329	2	17-0001	15/8	20/8	A	1	2-062	⅜	MIL-DTL-24640	12
2XO-18	0.417	0.449	3	18-0018	20/11		B	1	2-064	½	MIL-DTL-24640	12
2XO-24	0.473	0.509	4T	19-0002	20/12	30/12	C	1	2-065	½	MIL-DTL-24640	12
2XO-42	0.565	0.609	4T	19-0004	20/15	30/15	C	2	2-068	½	MIL-DTL-24640	12
2XO-60	0.641	0.691	4T	19-0006	30/17		D	2	2-070	¾	MIL-DTL-24640	12
2XO-77	0.728	0.785	5	20-0002	30/19		E	2	2-071	¾	MIL-DTL-24640	12
2XOW-6	0.336	0.363	2	17-0002	15/9	20/9	A	1	2-062	⅜	MIL-DTL-24640	17

NOTES:

1. Cable assignment to steel, aluminum, and nylon stuffing tubes, MCP blocks, and BX connectors shown on figures 3A10 through 3A25 are for MIL-DTL-915, MIL-DTL-24640, and MIL-DTL-24643 cable.
2. MCP block size and stuffing tube designations are provided for information and guidance. Some of the MCP block and stuffing tube sizes shown in this table may not accommodate the full range of allowed cable diameters for a given cable type. In those cases, different MCP block sizes or stuffing tubes should be selected to ensure the required block gap and retest requirements are met. See figure 3B22 for MCPs.

FIGURE 3A12. Cable assignment to steel, aluminum, and nylon stuffing tubes, MCP blocks, and box connectors.

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TABLE 3A13-I. Cable assignment to steel, aluminum, and nylon stuffing tubes, MCP blocks, and box connectors.

Cable			Nylon tube		MCP [2/]		Stuffing tube			Box conn.	Cable	
Type [1/]	Size OD		Size	Packing assembly, ASTM F1836M	Insert block	Insert block	Surface size	Submarine			MIL-Spec	Slant sheet
	Min.	Max.						Size	Grommet			
2XOW-18	0.468	0.504	4T	19-0002	20/12	30/12	C	1	2-065	½	MIL-DTL-24640	17
2XOW-24	0.546	0.588	4T	19-0004	20/14	30/14	C	1	2-067	½	MIL-DTL-24640	17
2XOW-42	0.646	0.686	4T	19-0006	30/17		D	2	2-070	½	MIL-DTL-24640	17
2XOW-60	0.744	0.802	5	20-0002	30/20		F	3	2-072	¾	MIL-DTL-24640	17
2XOW-77	0.840	0.906	5	20-0005	30/22		G	3	2-074	¾	MIL-DTL-24640	17
2XS-2	0.308	0.332	2	17-0002	15/8	20/8	A	1	2-062	⅜	MIL-DTL-24640	9
2XS-3	0.325	0.350	2	17-0002	15/8	20/8	A	1	2-062	½	MIL-DTL-24640	9
2XS-7	0.423	0.455	3	18-0018	20/11		B	1	2-063	½	MIL-DTL-24640	9
2XS-10	0.537	0.579	4T	19-0004	20/14	30/14	C	1	2-067	½	MIL-DTL-24640	9
2XS-14	0.582	0.627	4T	19-0005	20/15	30/15	D	2	2-068	½	MIL-DTL-24640	9
2XS-19	0.644	0.694	4T	19-0007	30/17		D	2	2-070	½	MIL-DTL-24640	9
2XS-24	0.758	0.818	5	20-0003	30/20		G	3	2-072	¾	MIL-DTL-24640	9
2XS-30	0.804	0.866	5	20-0004	30/21		G	3	2-073	¾	MIL-DTL-24640	9
2XSAOW-3	0.405	0.437	2	17-0004	20/10		B	1	2-063	½	MIL-DTL-24640	15
2XSAOW-7	0.510	0.550	4T	19-0002	20/13	30/13	C	1	2-066	½	MIL-DTL-24640	15
2XSAOW-10	0.631	0.681	4T	19-0006	30/17		D	2	2-070	½	MIL-DTL-24640	15
2XSAOW-14	0.689	0.743	5	20-0001	30/18		D	2	2-071	½	MIL-DTL-24640	15
2XSAOW-19	0.757	0.817	5	20-0003	30/20		F	3	2-072	¾	MIL-DTL-24640	15
2XSAOW-24	0.884	0.952	5	20-0006	30/24	40/24	J	3	2-075	¾	MIL-DTL-24640	15
2XSAOW-30	0.941	1.020	5	20-0007	30/25	40/25	J	3	2-076	1⅜	MIL-DTL-24640	15
2XSAOW-37	1.010	1.090	5	20-0010	40/27		K	4	2-090	1⅜	MIL-DTL-24640	15
2XSAW-3	0.368	0.396	2	17-0004	20/9		B	1	2-062	½	MIL-DTL-24640	15
2XSAW-7	0.461	0.497	4T	19-0001	20/12	30/12	B	1	2-065	½	MIL-DTL-24640	15
2XSAW-14	0.641	0.691	4T	19-0006	30/17		D	2	2-069	½	MIL-DTL-24640	15
2XSAWA-3	0.418	0.456	2	17-0004	20/11		B	1	2-063	½	MIL-DTL-24640	15
2XSAWA-7	0.511	0.557	4T	19-0001	20/14	30/14	C	1	2-066	½	MIL-DTL-24640	15
2XSAWA-14	0.691	0.751	4	19-0006	30/18		E	2	2-071	½	MIL-DTL-24640	15
2XSO-3	0.380	0.410	2	17-0004	20/10		B	1	2-063	½	MIL-DTL-24640	9
2XSO-7	0.474	0.510	4T	19-0002	20/12	30/12	C	1	2-065	½	MIL-DTL-24640	9
2XSO-10	0.594	0.640	4T	19-0005	20/16	30/16	D	2	2-069	½	MIL-DTL-24640	9
2XSO-14	0.636	0.686	4T	19-0006	30/17		D	2	2-070	½	MIL-DTL-24640	9
2XSO-19	0.709	0.765	5	20-0001	30/19		E	2	2-071	¾	MIL-DTL-24640	9
2XSO-30	0.869	0.937	5	20-0005	30/23		G	3	2-074	¾	MIL-DTL-24640	9
2XSOW-3	0.487	0.525	4T	19-0002	20/13	30/13	C	1	2-065	½	MIL-DTL-24640	16
2XSOW-7	0.606	0.656	4T	19-0005	20/16	30/16	D	2	2-069	½	MIL-DTL-24640	16
2XSOW-12	0.802	0.864	5	20-0004	30/21		G	3	2-073	¾	MIL-DTL-24640	16
2XSOW-19	0.938	1.010	5	20-0007	30/25	40/25	J	3	2-076	1⅜	MIL-DTL-24640	16
2XSOW-30	1.180	1.270	6	21-0005	40/32	60/32	M	5	2-095	1⅜	MIL-DTL-24640	16
2XSW-1	0.240	0.258	1	16-0005	15/6	20/6	A	1	2-051	⅜	MIL-DTL-24640	16
2XSW-3	0.436	0.470	3	18-0018	20/11	30/11	B	1	2-064	⅜	MIL-DTL-24640	16
2XSW-7	0.573	0.617	4T	19-0005	20/15	30/15	C	2	2-068	½	MIL-DTL-24640	16
2XSXO-4	0.333	0.359	2	17-0002	15/9	20/9	A	1	2-062	⅜	MIL-DTL-24640	13
3XS-7	0.601	0.647	4T	19-0005	20/16	30/16	D	2	2-069	½	MIL-DTL-24640	11
3XSOW-3	0.519	0.559	4T	19-0003	20/14	30/14	C	1	2-067	½	MIL-DTL-24640	18
3XSOW-7	0.659	0.711	4T	19-0007	30/17		D	2	2-070	½	MIL-DTL-24640	18
3XSOW-10	0.835	0.901	5	20-0005	30/22	40/22	G	3	2-074	¾	MIL-DTL-24640	18

NOTES:

1. Cable assignment to steel, aluminum, and nylon stuffing tubes, MCP blocks, and BX connectors shown on figures 3A10 through 3A25 are for MIL-DTL-915, MIL-DTL-24640, and MIL-DTL-24643 cable.
2. MCP block size and stuffing tube designations are provided for information and guidance. Some of the MCP block and stuffing tube sizes shown in this table may not accommodate the full range of allowed cable diameters for a given cable type. In those cases, different MCP block sizes or stuffing tubes should be selected to ensure the required block gap and retest requirements are met. See figure 3B22 for MCPs.

FIGURE 3A13. Cable assignment to steel, aluminum, and nylon stuffing tubes, MCP blocks, and box connectors.

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TABLE 3A14-I. Cable assignment to steel, aluminum, and nylon stuffing tubes, MCP blocks, and box connectors.

Cable			Nylon tube		MCP [2/]		Stuffing tube			Box conn.	Cable	
Type [1/]	Size OD		Size	Packing assembly, ASTM F1836M	Insert block	Insert block	Surface size	Submarine			MIL-Spec	Slant sheet
	Min.	Max.						Size	Grommet			
3XSOW-14	0.898	0.968	5	20-0006	30/24	40/24	J	3	2-075	¾	MIL-DTL-24640	18
3XSOW-19	1.010	1.090	5	20-0010	40/27		K	4	2-091	1⅜	MIL-DTL-24640	18
3XSOW-24	1.200	1.300	6	21-0005	40/32	60/32	M	5	2-095	1⅜	MIL-DTL-24640	18
3XSW-3	0.472	0.508	4T	19-0001	20/11		B	1	2-065	½	MIL-DTL-24640	18
3XSW-7	0.620	0.668	4T	19-0005	20/16	30/16	D	2	2-069	½	MIL-DTL-24640	18
3XSW-10	0.803	0.865	5	20-0004	30/21		G	3	2-073	¾	MIL-DTL-24640	18
3XSW-14	0.873	0.941	5	20-0005	30/23		G	3	2-074	¾	MIL-DTL-24640	18
7XW-3	0.315	0.339	2	17-0002	15/8	20/8	A	1	2-062	⅜	MIL-DTL-24640	22
7XW-4	0.374	0.404	2	17-0004	15/9		B	1	2-062	½	MIL-DTL-24640	22
LS1SAU-44	0.910	0.99	5	20-0006	30/25	40/25	J	3	2-074	1	MIL-DTL-24643	41
LS1SMU-5	0.465	0.500	4T	19-0001	20/12	30/12	B	1	2-065	½	MIL-DTL-24643	40
LS1SMWU-70	1.465	1.555	7	22-0008	60/39		P	6	2-100	2	MIL-DTL-24643	47
LS1SU-36	0.910	0.985	5	20-0005	30/24	40/24	J			1	MIL-DTL-24643	42
LS1SU-60	1.210	1.310	6	21-0004	40/33	60/33	M	5	2-096	1¼	MIL-DTL-24643	42
LS1SWU-2	0.430	0.455	2	17-0004	20/11		B	1	2-063	½	MIL-DTL-24643	30
LS1SWU-14	0.825	0.870	5	20-0003	30/21	40/21	G	3	2-073	1	MIL-DTL-24643	30
LS1SWU-20	0.970	1.030	5	20-0007	40/26		J	3	2-076	1¼	MIL-DTL-24643	30
LS1SWU-30	1.135	1.2	6	21-0003	40/30		L	4	2-094	1¼	MIL-DTL-24643	30
LS1S50MU-16	0.760	0.825	5	20-0002	30/20		G	3	2-072	¾	MIL-DTL-24643	28
LS1S50MU-20	0.835	0.905	5	20-0004	30/22	40/22	G	3	2-074	¾	MIL-DTL-24643	28
LS1S50MU-40	1.095	1.185	6	21-0001	40/29		L	4	2-094	1	MIL-DTL-24643	28
LS1S50MU-70	1.465	1.555	7	22-0001	60/39		P	6	2-100	1½	MIL-DTL-24643	28
LS1S50MUS-16	0.820	0.885	5	20-0004	30/22	40/22	G	3	2-073	¾	MIL-DTL-24643	28
LS1S50MUS-20	0.895	0.965	5	20-0005	30/24	40/24	J	3	2-075	¾	MIL-DTL-24643	28
LS1S50MUS-40	1.155	1.245	6	21-0003	40/31		L	4	2-094	1	MIL-DTL-24643	28
LS1S50MUS-70	1.525	1.615	7	22-0003	60/40		P	7	2-101	1½	MIL-DTL-24643	28
LS1S75MU-8	0.950	1.030	5	20-0007	40/26		J	3	2-076	1	MIL-DTL-24643	39
LS2AU-40	1.320	1.370	6	21-0005	40/34	60/34	M	5	2-096	1½	MIL-DTL-24643	27
LS2AUS-40	1.380	1.430	7	22-0001	60/36		N	6	2-098	1½	MIL-DTL-24643	27
LS2CS-6	0.400	0.430	2	17-0004	20/10		B	1	2-063	⅜	MIL-DTL-24643	58
LS2CS-18	0.550	0.590	4T	19-0003	20/14	30/14	C	1	2-067	½	MIL-DTL-24643	58

NOTES:

1. Cable assignment to steel, aluminum, and nylon stuffing tubes, MCP blocks, and BX connectors shown on figures 3A10 through 3A25 are for MIL-DTL-915, MIL-DTL-24640, and MIL-DTL-24643 cable.
2. MCP block size and stuffing tube designations are provided for information and guidance. Some of the MCP block and stuffing tube sizes shown in this table may not accommodate the full range of allowed cable diameters for a given cable type. In those cases, different MCP block sizes or stuffing tubes should be selected to ensure the required block gap and retest requirements are met. See figure 3B22 for MCPs.

FIGURE 3A14. Cable assignment to steel, aluminum, and nylon stuffing tubes, MCP blocks, and box connectors.

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TABLE 3A15-I. Cable assignment to steel, aluminum, and nylon stuffing tubes, MCP blocks, and box connectors.

Cable			Nylon tube		MCP [2/]		Stuffing tube			Box conn.	Cable	
Type [1/]	Size OD		Size	Packing assembly, ASTM F1836M	Insert block	Insert block	Surface size	Submarine			MIL-Spec	Slant sheet
	Min.	Max.						Size	Grommet			
LS2CS-42	0.750	0.800	5	20-0002	30/20		E	2	2-071	¾	MIL-DTL-24643	58
LS2CS-60	0.880	0.930	5	20-0005	30/23		G	3	2-074	¾	MIL-DTL-24643	58
LS2CS-77	1.000	1.070	5	20-0009	40/27		K	4	2-090	1	MIL-DTL-24643	58
LS2SJ-7	0.600	0.615	4T	19-0003	20/15	30/15	C	2	2-068	¾	MIL-DTL-24643	43
LS2SJ-9	0.525	0.545	4T	19-0001	20/13	30/13	C	1	2-066	½	MIL-DTL-24643	43
LS2SJ-11	0.447	0.460	2	17-0004	20/11		B	1	2-064	½	MIL-DTL-24643	43
LS2SJ-12	0.417	0.430	2	17-0002	20/10		A	1	2-063	½	MIL-DTL-24643	43
LS2SJ-14	0.337	0.350	1	16-0006	15/8	20/8	A	1	2-062	½	MIL-DTL-24643	43
LS2SJ-16	0.309	0.325	1	16-0004	15/8	20/8	A	1	2-062	½	MIL-DTL-24643	43
LS2SJ-18	0.295	0.310	1	16-0004	15/7	20/7	A	1	2-061	½	MIL-DTL-24643	43
LS2SJ-20	0.273	0.290	1	16-0001	15/7	20/7	A	1	2-061	½	MIL-DTL-24643	43
LS2SJ-22	0.261	0.275	1	16-0001	15/6	20/6	A	1	2-061	½	MIL-DTL-24643	43
LS2SU-3	0.480	0.520	4T	19-0001	20/13	30/13	C	1	2-065	½	MIL-DTL-24643	31
LS2SU-7	0.610	0.660	4T	19-0005	20/16	30/16	D	2	2-069	¾	MIL-DTL-24643	31
LS2SU-10	0.770	0.830	5	20-0002	30/20		F	3	2-072	1	MIL-DTL-24643	31
LS2SU-14	0.860	0.930	5	20-0004	30/23		G	3	2-074	1	MIL-DTL-24643	31
LS2SU-19	0.970	1.040	5	20-0007	40/26		J	3	2-076	1¼	MIL-DTL-24643	31
LS2SU-24	1.120	1.210	6	21-0001	40/30		L	4	2-094	1¼	MIL-DTL-24643	31
LS2SU-30	1.190	1.260	6	21-0004	40/32	60/32	M	5	2-095	1¼	MIL-DTL-24643	31
LS2SU-37	1.290	1.380	6	21-0004	40/34	60/34	M	5	2-097	1½	MIL-DTL-24643	31
LS2SU-44	1.460	1.550	7	22-0002	60/39		P	6	2-100	2	MIL-DTL-24643	31
LS2SU-61	1.660	1.740	8	23-0003	60/44		R	7	2-103	2	MIL-DTL-24643	31
LS2SUS-3	0.540	0.580	4T	19-0002	20/14	30/14	C	1	2-067	½	MIL-DTL-24643	31
LS2SUS-7	0.670	0.720	4T	19-0006	30/18		D	2	2-070	½	MIL-DTL-24643	31
LS2SUS-10	0.830	0.890	5	20-0004	30/22	40/22	G	3	2-073	¾	MIL-DTL-24643	31
LS2SUS-14	0.920	0.990	5	20-0006	30/25	40/25	J	3	2-075	¾	MIL-DTL-24643	31
LS2SUS-19	1.030	1.100	5	20-0008	40/26		K	4	2-091	1	MIL-DTL-24643	31
LS2SUS-24	1.180	1.270	6	21-0003	40/32	60/32	M	5	2-095	1	MIL-DTL-24643	31
LS2SUS-30	1.250	1.340	6	21-0004	40/33	60/33	M	5	2-096	1¼	MIL-DTL-24643	31
LS2SUS-37	1.350	1.440	6	21-0007	60/36		N	6	2-098	1¼	MIL-DTL-24643	31
LS2SUS-44	1.520	1.610	7	22-0003	60/40		P	7	2-101	1¼	MIL-DTL-24643	31
LS2SUS-61	1.720	1.800	8	23-0003	50/45		S	7	2-105	1½	MIL-DTL-24643	31

NOTES:

1. Cable assignment to steel, aluminum, and nylon stuffing tubes, MCP blocks, and BX connectors shown on figures 3A10 through 3A25 are for MIL-DTL-915, MIL-DTL-24640, and MIL-DTL-24643 cable.
2. MCP block size and stuffing tube designations are provided for information and guidance. Some of the MCP block and stuffing tube sizes shown in this table may not accommodate the full range of allowed cable diameters for a given cable type. In those cases, different MCP block sizes or stuffing tubes should be selected to ensure the required block gap and retest requirements are met. See figure 3B22 for MCPs.

FIGURE 3A15. Cable assignment to steel, aluminum, and nylon stuffing tubes, MCP blocks, and box connectors.

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TABLE 3A16-I. Cable assignment to steel, aluminum, and nylon stuffing tubes, MCP blocks, and box connectors.

Cable			Nylon tube		MCP [2/]		Stuffing tube			Box conn.	Cable	
Type [1/]	Size OD		Size	Packing assembly, ASTM F1836M	Insert block	Insert block	Surface size	Submarine			MIL-Spec	Slant sheet
	Min.	Max.						Size	Grommet			
LS2SWAU-3	0.480	0.520	4T	19-0001	20/13	30/13	C	1	2-065	½	MIL-DTL-24643	32
LS2SWAU-7	0.610	0.660	4T	19-0005	20/16	30/16	D	2	2-069	¾	MIL-DTL-24643	32
LS2SWAU-10	0.770	0.830	5	20-0002	30/20		F	3	2-072	1	MIL-DTL-24643	32
LS2SWAU-14	0.860	0.930	5	20-0005	30/23		G	3	2-074	1	MIL-DTL-24643	32
LS2SWAU-19	0.970	1.040	5	20-0007	40/26		J	3	2-076	1¼	MIL-DTL-24643	32
LS2SWAU-24	1.120	1.210	6	21-0003	40/30		L	4	2-094	1¼	MIL-DTL-24643	32
LS2SWAU-30	1.190	1.280	6	21-0004	40/32	60/32	M	5	2-095	1¼	MIL-DTL-24643	32
LS2SWAU-37	1.290	1.380	6	21-0006	40/34	60/34	M	5	2-097	1½	MIL-DTL-24643	32
LS2SWAU-44	1.460	1.550	7	22-0002	60/39		P	6	2-100	2	MIL-DTL-24643	32
LS2SWAU-61	1.660	1.740	8	23-0003	60/44		R	7	2-103	2	MIL-DTL-24643	32
LS2SWL-7	0.860	0.910	5	20-0004	30/23		G	3	2-074	¾	MIL-DTL-24643	56
LS2SWU-1	0.240	0.255	1	16-0004	15/6	20/6	A	1	2-061	⅜	MIL-DTL-24643	33
LS2SWU-3	0.670	0.710	4T	19-0005	30/18		D	2	2-070	½	MIL-DTL-24643	33
LS2SWU-7	0.860	0.910	5	20-0004	30/23		G	3	2-074	¾	MIL-DTL-24643	33
LS2SWU-12	1.130	1.200	6	21-0001	40/30		L	4	2-094	1⅜	MIL-DTL-24643	33
LS2SWU-19	1.292	1.380	6	21-0005	40/34	60/34	M	5	2-097	1⅜	MIL-DTL-24643	33
LS2SWU-24	1.500	1.590	7	22-0002	60/39		P	7	2-101	1½	MIL-DTL-24643	33
LS2SWU-30	1.670	1.760	8	23-0002	60/44		S	7	2-104	1¾	MIL-DTL-24643	33
LS2SWU-37	1.785	1.870	8	23-0003	60/46		S	8	2-106	1¾	MIL-DTL-24643	33
LS2SWU-61	2.205	2.300	9	24-0004	90/55		W			2⅜	MIL-DTL-24643	33
LS2U-10	0.450	0.480	3	18-0018	20/12	30/12	B	1	2-064	½	MIL-DTL-24643	45
LS2U-15	0.530	0.560	4T	19-0002	20/14	30/14	C	1	2-067	½	MIL-DTL-24643	45
LS2U-19	0.550	0.580	4T	19-0002	20/14	30/14	C	1	2-067	½	MIL-DTL-24643	45
LS2U-30	0.670	0.700	4T	19-0006	30/17		D	2	2-070	¾	MIL-DTL-24643	45
LS2U-45	0.830	0.870	5	20-0004	30/22	40/22	G	3	2-073	1	MIL-DTL-24643	45
LS2U-60	0.920	0.960	5	20-0005	30/24	40/24	J	3	2-075	1	MIL-DTL-24643	45
LS2UW-42	0.750	0.790	4T	19-0007	30/19		E	2	2-071	¾	MIL-DTL-24643	57
LS2UWS-42	0.785	0.825	5	20-0002	30/20		F	3	2-073	1	MIL-DTL-24643	57
LS2WAU-40	1.320	1.370	6	21-0005	40/34	60/34	M	5	2-096	1½	MIL-DTL-24643	46

NOTES:

1. Cable assignment to steel, aluminum, and nylon stuffing tubes, MCP blocks, and BX connectors shown on figures 3A10 through 3A25 are for MIL-DTL-915, MIL-DTL-24640, and MIL-DTL-24643 cable.
2. MCP block size and stuffing tube designations are provided for information and guidance. Some of the MCP block and stuffing tube sizes shown in this table may not accommodate the full range of allowed cable diameters for a given cable type. In those cases, different MCP block sizes or stuffing tubes should be selected to ensure the required block gap and retest requirements are met. See figure 3B22 for MCPs.

FIGURE 3A16. Cable assignment to steel, aluminum, and nylon stuffing tubes, MCP blocks, and box connectors.

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TABLE 3A17-I. Cable assignment to steel, aluminum, and nylon stuffing tubes, MCP blocks, and box connectors.

Cable			Nylon tube		MCP [2/]		Stuffing tube			Box conn.	Cable	
Type [1/]	Size OD		Size	Packing assembly, ASTM F1836M	Insert block	Insert block	Surface size	Submarine			MIL-Spec	Slant sheet
	Min.	Max.						Size	Grommet			
LS3SF-7	1.000	1.040	5	20-0007	40/26		J	3	2-076	1	MIL-DTL-24643	44
LS3SJ-9	0.594	0.620	4T	19-0003	20/15	30/15	C	2	2-068	½	MIL-DTL-24643	43
LS3SJ-12	0.440	0.455	2	17-0003	20/11	0	B	1	2-063	⅜	MIL-DTL-24643	43
LS3SJ-14	0.353	0.370	2	17-0001	15/9	20/9	A	1	2-062	⅜	MIL-DTL-24643	43
LS3SJ-16	0.323	0.340	1	16-0005	15/8	20/8	A	1	2-062	⅜	MIL-DTL-24643	43
LS3SJ-18	0.308	0.325	1	16-0004	15/8	20/8	A	1	2-062	⅜	MIL-DTL-24643	43
LS3SJ-20	0.284	0.300	1	16-0004	15/7	20/7	A	1	2-061	⅜	MIL-DTL-24643	43
LS3SJ-22	0.271	0.285	1	16-0001	15/7	20/7	A	1	2-061	⅜	MIL-DTL-24643	43
LS3SU-3	0.650	0.700	4T	19-0005	30/17		D	2	2-070	¾	MIL-DTL-24643	35
LS3SU-7	0.840	0.910	5	20-0004	30/23	40/22	G	3	2-074	1	MIL-DTL-24643	35
LS3SU-10	1.100	1.190	5	20-0010	40/30		L	4	2-094	1¼	MIL-DTL-24643	35
LS3SU-14	1.200	1.290	6	21-0003	40/32	60/32	M	5	2-095	1¼	MIL-DTL-24643	35
LS3SU-19	1.340	1.430	6	21-0007	60/36		N	6	2-098	1½	MIL-DTL-24643	35
LS3SU-24	1.580	1.670	7	22-0003	60/42		R	7	2-102	2	MIL-DTL-24643	35
LS3SU-30	1.680	1.770	8	23-0002	60/44		S	7	2-103	2	MIL-DTL-24643	35
LS3SU-37	1.840	1.930	8	23-0005	60/48		T	8	2-107	2	MIL-DTL-24643	35
LS3SU-44	2.060	2.150	9	24-0001	60/53	90/54	V	9	2-109	2½	MIL-DTL-24643	35
LS3SUS-3	0.710	0.760	4T	19-0005	30/19		E	2	2-071	¾	MIL-DTL-24643	35
LS3SUS-7	0.900	0.970	5	20-0004	30/24	40/24	J	3	2-075	1	MIL-DTL-24643	35
LS3SUS-10	1.160	1.250	5	20-0010	40/31		L	4	2-094	1¼	MIL-DTL-24643	35
LS3SUS-14	1.260	1.350	6	21-0003	40/33	60/34	M	5	2-096	1¼	MIL-DTL-24643	35
LS3SUS-19	1.400	1.490	6	21-0007	60/37		P	6	2-099	1½	MIL-DTL-24643	35
LS3SUS-24	1.640	1.730	7	22-0003	60/43		R	7	2-103	2	MIL-DTL-24643	35
LS3SUS-30	1.740	1.830	8	23-0002	60/46		S	7	2-105	2	MIL-DTL-24643	35
LS3SUS-37	1.900	1.990	9	23-0005	60/50		T	8	2-107	2	MIL-DTL-24643	35
LS3SUS-44	2.120	2.210	9	24-0001	60/56	90/55	W	9	2-110	2½	MIL-DTL-24643	35
LS3SWU-3	0.61	0.655	4T	19-0006	20/16	30/16	D	2	2-069	¾	MIL-DTL-24643	36
LS3SWU-7	0.880	0.940	5	20-0004	30/23		G	3	2-074	1	MIL-DTL-24643	36
LS3SWU-10	1.100	1.180	6	21-0002	40/29		L	4	2-094	1¼	MIL-DTL-24643	36
LS3SWU-14	1.200	1.280	6	21-0004	40/32	60/32	M	5	2-095	1¼	MIL-DTL-24643	36
LS3SWU-19	1.370	1.450	6	21-0007	60/36		N	6	2-098	1½	MIL-DTL-24643	36
LS3SWU-24	1.640	1.76	8	23-0001	60/44		R	7	2-104	2	MIL-DTL-24643	36
LS3SWU-30	1.760	1.860	8	23-0003	60/47		S	8	2-106	2	MIL-DTL-24643	36
LS3SWU-37	1.890	1.94	8	23-0006	60/49	90/49	T	8	2-107	2	MIL-DTL-24643	36
LS3SWU-44	2.140	2.240	9	24-0003	90/56		V	9	2-110	2½	MIL-DTL-24643	36
LS3SWUS-3	0.675	0.71	4T	19-0005	30/17		D	2	2-070	¾	MIL-DTL-24643	36
LS3SWUS-7	0.940	1.000	5	20-0005	30/25	40/24	J	3	2-075	1	MIL-DTL-24643	36
LS3SWUS-10	1.160	1.240	5	20-0010	40/31		L	4	2-094	1⅜	MIL-DTL-24643	36
LS3SWUS-14	1.260	1.340	6	21-0004	40/33	60/33	M	5	2-096	1⅜	MIL-DTL-24643	36
LS3SWUS-19	1.430	1.510	6	21-0007	60/38		N	6	2-099	1¾	MIL-DTL-24643	36

NOTES:

1. Cable assignment to steel, aluminum, and nylon stuffing tubes, MCP blocks, and BX connectors shown on figures 3A10 through 3A25 are for MIL-DTL-915, MIL-DTL-24640, and MIL-DTL-24643 cable.
2. MCP block size and stuffing tube designations are provided for information and guidance. Some of the MCP block and stuffing tube sizes shown in this table may not accommodate the full range of allowed cable diameters for a given cable type. In those cases, different MCP block sizes or stuffing tubes should be selected to ensure the required block gap and retest requirements are met. See figure 3B22 for MCPs.

FIGURE 3A17. Cable assignment to steel, aluminum, and nylon stuffing tubes, MCP blocks, and box connectors.

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TABLE 3A18-I. Cable assignment to steel, aluminum, and nylon stuffing tubes, MCP blocks, and box connectors.

Cable			Nylon tube		MCP [2/]		Stuffing tube			Box conn.	Cable	
Type [1/]	Size OD		Size	Packing assembly, ASTM F1836M	Insert block	Insert block	Surface size	Submarine			MIL-Spec	Slant sheet
	Min.	Max.						Size	Grommet			
LS3SWUS-24	1.700	1.82	7	22-0003	60/46		S	7	2-105	2	MIL-DTL-24643	36
LS3SWUS-30	1.820	1.920	8	23-0004	60/48		T	8	2-107	2	MIL-DTL-24643	36
LS3SWUS-37	1.950	2.050	8	23-0006	60/51		T	8	2-108	2⅜	MIL-DTL-24643	36
LS3SWUS-44	2.200	2.300	9	24-0003	90/55		W			2½	MIL-DTL-24643	36
LS3U-3	0.580	0.620	4T	19-0004	20/15	30/15	C	2	2-068	¾	MIL-DTL-24643	37
LS3U-7	0.760	0.810	5	20-0002	30/20		E	3	2-072	1	MIL-DTL-24643	37
LS3U-12	1.030	1.090	5	20-0009	40/26		K	4	2-091	1¼	MIL-DTL-24643	37
LS4NW-8	0.684	0.740	4T	19-0008	30/18		D	2	2-070	¾	MIL-DTL-24643	55
LS4SJ-14	0.380	0.395	2	17-0002	20/9		A	1	2-062	¾	MIL-DTL-24643	43
LS4SJ-16	0.346	0.360	2	17-0002	15/9	20/9	A	1	2-062	¾	MIL-DTL-24643	43
LS4SJ-20	0.303	0.320	1	16-0004	15/8	20/8	A	1	2-062	¾	MIL-DTL-24643	43
LS5KVTSGU-100	1.656	1.74	8	23-0002	60/44		S	7	2-105	2	MIL-DTL-24643	22
LS5KVTSGU-150	1.804	1.950	8	23-0005	60/49		T	8	2-107	2	MIL-DTL-24643	22
LS5KVTSGU-250	2.054	2.220	9	24-0002	90/56		W	9	2-110	2½	MIL-DTL-24643	22
LS5KVTSGU-350	2.266	2.450	9	24-0005	90/62		X			3	MIL-DTL-24643	22
LS5KVTSGU-400	2.405	2.600	9	24-0007	90/65		Y			3	MIL-DTL-24643	22
LS6SGU-100	1.480	1.600	8	23-0001	60/40		P	7	2-101	2	MIL-DTL-24643	19
LS6SGU-125	1.656	1.790	8	23-0004	60/45		S	7	2-105	2	MIL-DTL-24643	19
LS6SGU-150	1.813	1.960	8	23-0006	60/49		T	8	2-107	2	MIL-DTL-24643	19
LS6SGU-200	2.035	2.200	9	24-0003	60/55		W	9	2-109	2½	MIL-DTL-24643	19
LS7PS-6	0.717	0.775	4T	19-0007	30/19		E	2	2-071	1	MIL-DTL-24643	26
LS7SGU-3	0.504	0.545	4T	19-0002	20/13	30/13	C	1	2-066	½	MIL-DTL-24643	20
LS7SGU-4	0.550	0.595	4T	19-0003	20/15	30/15	C	2	2-068	¾	MIL-DTL-24643	20
LS8NW-6	0.620	0.670	4T	19-0004	30/16		D	2	2-069	¾	MIL-DTL-24643	54
LSDCOP-1	0.235	0.250	1	16-0004	15/6	20/6	A	1	2-061	½	MIL-DTL-24643	2
LSDCOP-1½	0.300	0.315	1	16-0006	15/7	20/7	A	1	2-062	½	MIL-DTL-24643	2
LSDCOP-2	0.310	0.330	2	17-0001	15/8	20/8	A	1	2-062	½	MIL-DTL-24643	2
LSDHOF-3	0.405	0.425	2	17-0004	20/10		B	1	2-063	½	MIL-DTL-24643	3
LSDHOF-4	0.440	0.460	3	18-0018	20/11		B	1	2-064	½	MIL-DTL-24643	3
LSDHOF-6	0.490	0.510	4T	19-0001	20/12	30/12	B	1	2-065	½	MIL-DTL-24643	3
LSDHOF-9	0.540	0.570	4T	19-0003	20/14	30/14	C	1	2-067	½	MIL-DTL-24643	3
LSDHOF-14	0.675	0.705	4T	19-0007	30/17		D	2	2-070	¾	MIL-DTL-24643	3
LSDHOF-23	0.820	0.860	5	20-0003	30/21		G	3	2-073	1	MIL-DTL-24643	3
LSDHOF-30	0.920	0.960	5	20-0005	30/24	40/24	J	3	2-075	1	MIL-DTL-24643	3
LSDHOF-83	1.390	1.450	6	21-0007	60/36		N	6	2-098	1½	MIL-DTL-24643	3
LSDHOF-250	2.000	2.100	9	24-0002	60/53		V	8	2-108	2½	MIL-DTL-24643	3
LSDHOF-400	2.400	2.500	9	24-0006	90/60		Y			3	MIL-DTL-24643	3
LSDNW-3	0.361	0.390	2	17-0003	15/9		A	1	2-062	½	MIL-DTL-24643	48

NOTES:

1. Cable assignment to steel, aluminum, and nylon stuffing tubes, MCP blocks, and BX connectors shown on figures 3A10 through 3A25 are for MIL-DTL-915, MIL-DTL-24640, and MIL-DTL-24643 cable.
2. MCP block size and stuffing tube designations are provided for information and guidance. Some of the MCP block and stuffing tube sizes shown in this table may not accommodate the full range of allowed cable diameters for a given cable type. In those cases, different MCP block sizes or stuffing tubes should be selected to ensure the required block gap and retest requirements are met. See figure 3B22 for MCPs.

FIGURE 3A18. Cable assignment to steel, aluminum, and nylon stuffing tubes, MCP blocks, and box connectors.

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TABLE 3A19-I. Cable assignment to steel, aluminum, and nylon stuffing tubes, MCP blocks, and box connectors.

Cable			Nylon tube		MCP [2/]		Stuffing tube			Box conn.	Cable	
Type [1/]	Size OD		Size	Packing assembly, ASTM F1836M	Insert block	Insert block	Surface size	Submarine			MIL-Spec	Slant sheet
	Min.	Max.						Size	Grommet			
LSDNW-4	0.398	0.430	2	17-0004	20/10		B	1	2-063	½	MIL-DTL-24643	48
LSDNW-9	0.504	0.545	4T	19-0002	20/13	30/13	C	1	2-066	½	MIL-DTL-24643	48
LSDNW-14	0.564	0.610	4T	19-0004	20/15	30/15	C	2	2-068	½	MIL-DTL-24643	48
LSDNW-23	0.638	0.690	4T	19-0005	30/17		D	2	2-070	¾	MIL-DTL-24643	48
LSDNW-50	0.842	0.910	5	20-0004	30/23		G	3	2-074	1	MIL-DTL-24643	48
LSDNW-75	1.000	1.080	5	20-0009	40/27		K	4	2-091	1¼	MIL-DTL-24643	48
LSDNW-100	1.082	1.170	6	21-0002	40/29		L	4	2-093	1¼	MIL-DTL-24643	48
LSDPS-3	0.421	0.455	2	17-0003	20/11		B	1	2-063	½	MIL-DTL-24643	26
LSDPS-4	0.452	0.489	2	17-0004	20/12	30/12	B	1	2-064	½	MIL-DTL-24643	26
LSDPS-6	0.541	0.585	4T	19-0001	20/14	30/14	C	1	2-067	¾	MIL-DTL-24643	26
LSDPS-9	0.581	0.628	4T	19-0002	20/15	30/15	C	2	2-068	¾	MIL-DTL-24643	26
LSDPS-14	0.675	0.730	4T	19-0007	30/17		D	2	2-070	¾	MIL-DTL-24643	26
LSDRW	0.620	0.670	4T	19-0005	30/16		D	2	2-069	¾	MIL-DTL-24643	53
LSDSGU-3	0.362	0.391	2	17-0003	15/9		A	1	2-062	½	MIL-DTL-24643	15
LSDSGU-4	0.395	0.427	2	17-0004	20/10		B	1	2-063	½	MIL-DTL-24643	15
LSDSGU-9	0.503	0.544	4T	19-0002	20/13	30/13	C	1	2-066	¾	MIL-DTL-24643	15
LSDSGU-14	0.620	0.670	4T	19-0006	30/16		D	2	2-069	¾	MIL-DTL-24643	15
LSDSGU-23	0.722	0.781	5	20-0002	30/19		E	2	2-071	1	MIL-DTL-24643	15
LSDSGU-50	0.843	0.911	5	20-0004	30/23		G	3	2-074	1	MIL-DTL-24643	15
LSDSGU-75	0.993	1.074	5	20-0009	40/27		K	4	2-090	1¼	MIL-DTL-24643	15
LSDSGU-100	1.079	1.167	6	21-0001	40/29		K	4	2-093	1¼	MIL-DTL-24643	15
LSDSGU-200	1.464	1.583	7	22-0003	60/40		P	7	2-101	2	MIL-DTL-24643	15
LSDSGU-300	1.703	1.841	8	23-0004	60/46		S	8	2-106	2	MIL-DTL-24643	15
LSDSGU-400	1.914	2.069	9	24-0001	60/52		V	8	2-108	2½	MIL-DTL-24643	15
LSECM	1.290	1.370	6	21-0006	40/34	60/34	M	5	2-096	1½	MIL-DTL-24643	38
LSFHOF-3	0.460	0.480	3	18-0018	20/12	30/12	B	1	2-064	½	MIL-DTL-24643	3
LSFHOF-4	0.520	0.550	4T	19-0002	20/13	30/13	C	1	2-066	¾	MIL-DTL-24643	3
LSFHOF-9	0.630	0.660	4T	19-0005	20/16	30/16	D	2	2-069	¾	MIL-DTL-24643	3
LSFHOF-42	1.300	1.380	6	21-0006	40/34	60/34	M	5	2-097	1½	MIL-DTL-24643	3
LSFHOF-60	1.430	1.510	7	22-0002	50/38		N	6	2-099	2	MIL-DTL-24643	3
LFHOF-133	1.920	2.000	8	23-0006	60/50	90/50	T	8	2-107	2½	MIL-DTL-24643	3
LSFNW-3	0.413	0.447	3	18-0018	20/11		B	1	2-063	½	MIL-DTL-24643	50
LSFNW-4	0.475	0.513	4T	19-0001	20/12	30/12	B	1	2-065	¾	MIL-DTL-24643	50
LSFNW-9	0.583	0.630	4T	19-0004	20/15	30/15	C	2	2-069	¾	MIL-DTL-24643	50
LSFNW-23	0.768	0.830	5	20-0002	30/20		F	3	2-072	1	MIL-DTL-24643	50

NOTES:

1. Cable assignment to steel, aluminum, and nylon stuffing tubes, MCP blocks, and BX connectors shown on figures 3A10 through 3A25 are for MIL-DTL-915, MIL-DTL-24640, and MIL-DTL-24643 cable.
2. MCP block size and stuffing tube designations are provided for information and guidance. Some of the MCP block and stuffing tube sizes shown in this table may not accommodate the full range of allowed cable diameters for a given cable type. In those cases, different MCP block sizes or stuffing tubes should be selected to ensure the required block gap and retest requirements are met. See figure 3B22 for MCPs.

FIGURE 3A19. Cable assignment to steel, aluminum, and nylon stuffing tubes, MCP blocks, and box connectors.

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TABLE 3A20-I. Cable assignment to steel, aluminum, and nylon stuffing tubes, MCP blocks, and box connectors.

Cable			Nylon tube		MCP [2/]		Stuffing tube			Box conn.	Cable	
Type [1/]	Size OD		Size	Packing assembly, ASTM F1836M	Insert block	Insert block	Surface size	Submarine			MIL-Spec	Slant sheet
	Min.	Max.						Size	Grommet			
LSFPS-14	0.754	0.815	5	20-0002	30/20		F	3	2-072	1	MIL-DTL-24643	26
LSFSGU-3	0.413	0.447	2	17-0004	20/11		B	1	2-063	½	MIL-DTL-24643	17
LSFSGU-4	0.475	0.513	4T	19-0001	20/13	30/12	B	1	2-065	½	MIL-DTL-24643	17
LSFSGU-9	0.583	0.630	4T	19-0004	20/16	30/15	C	2	2-068	¾	MIL-DTL-24643	17
LSFSGU-23	0.823	0.890	5	20-0004	30/22	40/22	G	3	2-073	1	MIL-DTL-24643	17
LSFSGU-50	0.971	1.050	5	20-0008	40/26		J	4	2-090	1¼	MIL-DTL-24643	17
LSFSGU-75	1.147	1.240	6	21-0004	40/31		L	4	2-094	1¼	MIL-DTL-24643	17
LSFSGU-100	1.256	1.358	6	21-0006	40/34	60/34	M	5	2-096	1½	MIL-DTL-24643	17
LSFSGU-150	1.503	1.625	8	23-0001	60/41		R	7	2-101	2	MIL-DTL-24643	17
LSFSGU-200	1.684	1.820	8	23-0004	60/46		S	7	2-105	2	MIL-DTL-24643	17
LSMCOS-2	0.440	0.460	2	17-0004	20/11		B	1	2-064	½	MIL-DTL-24643	4
LSMCOS-4	0.490	0.510	4T	19-0001	20/12	30/12	B	1	2-065	½	MIL-DTL-24643	4
LSMCOS-5	0.375	0.390	2	17-0003	15/9		A	1	2-062	½	MIL-DTL-24643	4
LSMCOS-6	0.460	0.480	3	18-0018	20/12	30/12	B	1	2-064	½	MIL-DTL-24643	4
LSMCOS-7	0.575	0.595	4T	19-0004	20/15	30/15	C	2	2-068	¾	MIL-DTL-24643	4
LSMDU-6	0.925	1.000	5	20-0006	30/24	40/25	J	4	2-091	1¼	MIL-DTL-24643	5
LSMDU-14	1.290	1.395	6	21-0007	40/34	60/35	M	5	2-097	1½	MIL-DTL-24643	5
LSMDU-23	1.633	1.765	8	23-0003	60/44		S	7	2-104	2	MIL-DTL-24643	5
LSMDU-40	1.887	2.040	9	24-0001	60/51	90/51	T	8	2-108	2½	MIL-DTL-24643	5
LSMDU-60	2.155	2.330	9	24-0005	90/59		X			2½	MIL-DTL-24643	5
LSMHOF-7	0.465	0.500	4T	19-0001	20/12	30/12	B	1	2-065	½	MIL-DTL-24643	7
LSMHOF-10	0.540	0.585	4T	19-0003	20/14	30/14	C	1	2-067	¾	MIL-DTL-24643	7
LSMHOF-14	0.585	0.635	4T	19-0004	20/15	30/16	C	2	2-068	¾	MIL-DTL-24643	7
LSMHOF-19	0.650	0.705	4T	19-0006	30/17		D	2	2-070	¾	MIL-DTL-24643	7
LSMHOF-24	0.735	0.795	5	20-0001	30/20		E	2	2-071	1	MIL-DTL-24643	7
LSMHOF-30	0.775	0.835	5	20-0002	30/21		F	3	2-072	1	MIL-DTL-24643	7
LSMHOF-37	0.855	0.925	5	20-0004	30/23		G	3	2-074	1	MIL-DTL-24643	7
LSMHOF-44	0.925	1.000	5	20-0006	30/25	40/24	J	3	2-075	1¼	MIL-DTL-24643	7
LSMHOF-61	1.100	1.175	6	21-0001	40/29		L	4	2-093	1¼	MIL-DTL-24643	7
LSMMOP-5	0.282	0.305	1	16-0006	15/7	20/7	A	1	2-061	½	MIL-DTL-24643	8
LSMNW-7	0.370	0.400	2	17-0003	20/10		A	1	2-062	½	MIL-DTL-24643	51
LSMNW-10	0.457	0.495	3	18-0018	20/12	30/12	B	1	2-065	½	MIL-DTL-24643	51
LSMNW-14	0.494	0.535	4T	19-0001	20/13	30/13	C	1	2-066	½	MIL-DTL-24643	51
LSMNW-19	0.545	0.590	4T	19-0003	20/14	30/14	C	1	2-067	½	MIL-DTL-24643	51
LSMNW-24	0.633	0.685	4T	19-0005	30/17		D	2	2-070	¾	MIL-DTL-24643	51
LSMNW-30	0.670	0.725	4T	19-0007	30/18		D	2	2-070	¾	MIL-DTL-24643	51
LSMNW-37	0.726	0.785	5	20-0001	30/19		E	2	2-071	¾	MIL-DTL-24643	51
LSMNW-44	0.823	0.890	5	20-0004	30/22	40/22	G	3	2-073	1	MIL-DTL-24643	51

NOTES:

1. Cable assignment to steel, aluminum, and nylon stuffing tubes, MCP blocks, and BX connectors shown on figures 3A10 through 3A25 are for MIL-DTL-915, MIL-DTL-24640, and MIL-DTL-24643 cable.
2. MCP block size and stuffing tube designations are provided for information and guidance. Some of the MCP block and stuffing tube sizes shown in this table may not accommodate the full range of allowed cable diameters for a given cable type. In those cases, different MCP block sizes or stuffing tubes should be selected to ensure the required block gap and retest requirements are met. See figure 3B22 for MCPs.

FIGURE 3A20. Cable assignment to steel, aluminum, and nylon stuffing tubes, MCP blocks, and box connectors.

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TABLE 3A21-I. Cable assignment to steel, aluminum, and nylon stuffing tubes, MCP blocks, and box connectors.

Cable			Nylon tube		MCP [2/]		Stuffing tube			Box conn.	Cable	
Type [1/]	Size OD		Size	Packing assembly, ASTM F1836M	Insert block	Insert block	Surface size	Submarine			MIL-Spec	Slant sheet
	Min.	Max.						Size	Grommet			
LSMRI-D-1	0.069	0.075	1	16-0001			A	1	2-061		MIL-DTL-24643	9
LSMRI-D-2½	0.094	0.102	1	16-0001			A	1	2-061		MIL-DTL-24643	9
LSMRI-T-2½	0.094	0.102	1	16-0001			A	1	2-061	⅜	MIL-DTL-24643	9
LSMS-37	0.740	0.800	5	20-0001	30/20		E	2	2-071	1	MIL-DTL-24643	34
LSMSCS-7	0.507	0.544	4T	19-0002	20/13	30/13	C	1	2-066	½	MIL-DTL-24643	18
LSMSCS-10	0.635	0.682	4T	19-0005	30/17		D	2	2-070	¾	MIL-DTL-24643	18
LSMSCS-14	0.677	0.728	4T	19-0007	30/17		E	2	2-071	1	MIL-DTL-24643	18
LSMSCS-19	0.742	0.798	5	20-0002	30/20		F	2	2-071	1	MIL-DTL-24643	18
LSMSCS-24	0.850	0.915	5	20-0004	30/23		G	3	2-074	1	MIL-DTL-24643	18
LSMSCS-30	0.893	0.961	5	20-0005	30/24	40/24	J	2	2-069	1¼	MIL-DTL-24643	18
LSMSCS-37	0.986	1.065	5	20-0009	40/26		K	4	2-090	1¼	MIL-DTL-24643	18
LSMSCS-44	1.090	1.174	6	21-0002	40/29		L	4	2-093	1¼	MIL-DTL-24643	18
LSMSCS-61	1.216	1.310	6	21-0005	40/33	60/33	M	5	2-096	1½	MIL-DTL-24643	18
LSMSCS-91	1.429	1.540	7	22-0002	60/39		P	6	2-100	2	MIL-DTL-24643	18
LSMSCU-7	0.447	0.484	3	18-0018	20/12	30/12	B	1	2-064	½	MIL-DTL-24643	18
LSMSCU-10	0.575	0.622	4T	19-0004	20/15	30/15	C	2	2-068	¾	MIL-DTL-24643	18
LSMSCU-14	0.617	0.668	4T	19-0005	20/16	30/16	D	2	2-069	¾	MIL-DTL-24643	18
LSMSCU-19	0.682	0.738	4T	19-0007	30/18		D	2	2-071	¾	MIL-DTL-24643	18
LSMSCU-24	0.790	0.855	5	20-0003	30/21		G	3	2-073	1	MIL-DTL-24643	18
LSMSCU-30	0.833	0.901	5	20-0004	30/22	40/22	G	3	2-074	1	MIL-DTL-24643	18
LSMSCU-37	0.926	1.002	5	20-0006	30/25	40/25	J	3	2-076	1	MIL-DTL-24643	18
LSMSCU-44	1.030	1.114	5	20-0009	40/28		K	4	2-091	1¼	MIL-DTL-24643	18
LSMSCU-61	1.156	1.250	6	21-0003	40/31		L	4	2-094	1¼	MIL-DTL-24643	18
LSMSCU-91	1.369	1.480	7	22-0001	60/37		N	6	2-099	1½	MIL-DTL-24643	18
LSMU-14	0.365	0.400	2	17-0003	20/10		A	1	2-062	½	MIL-DTL-24643	29
LSMUS-14	0.425	0.460	2	17-0004	20/11		B	1	2-024	½	MIL-DTL-24643	29
LSPBTMU-5	0.499	0.540	4T	19-0002	20/13	30/13	C	1	2-066	½	MIL-DTL-24643	10
LSPBTMU-15	0.694	0.750	4T	19-0007	30/18		D	2	2-071	1	MIL-DTL-24643	10
LSPBTMU-30	0.907	0.980	5	20-0006	30/24	40/24	J	3	2-075	1	MIL-DTL-24643	10
LSPI-3	0.634	0.685	4T	19-0004	30/17		D	2	2-070	¾	MIL-DTL-24643	25
LSPI-7	0.833	0.900	5	20-0002	30/22	40/22	G	3	2-073	1	MIL-DTL-24643	25
LSPI-12	1.068	1.155	5	20-0009	40/29		K	4	2-093	1¼	MIL-DTL-24643	25
LSSHOF-3	0.195	0.210	1	16-0001	15/5	20/5	A	1	2-061	½	MIL-DTL-24643	3
LSSHOF-23	0.440	0.460	3	18-0018	20/11		B	1	2-064	½	MIL-DTL-24643	3
LSSHOF-60	0.570	0.600	4T	19-0004	20/15	30/15	C	2	2-068	¾	MIL-DTL-24643	3
LSSHOF-150	0.830	0.870	5	20-0003	30/21	40/22	F	3	2-073	1	MIL-DTL-24643	3
LSSHOF-200	0.940	0.980	5	20-0006	30/24	40/24	J	3	3-075	1	MIL-DTL-24643	3
LSSHOF-250	1.035	1.085	5	20-0009	40/26		K	4	2-091	1¼	MIL-DTL-24643	3

NOTES:

1. Cable assignment to steel, aluminum, and nylon stuffing tubes, MCP blocks, and BX connectors shown on figures 3A10 through 3A25 are for MIL-DTL-915, MIL-DTL-24640, and MIL-DTL-24643 cable.
2. MCP block size and stuffing tube designations are provided for information and guidance. Some of the MCP block and stuffing tube sizes shown in this table may not accommodate the full range of allowed cable diameters for a given cable type. In those cases, different MCP block sizes or stuffing tubes should be selected to ensure the required block gap and retest requirements are met. See figure 3B22 for MCPs.

FIGURE 3A21. Cable assignment to steel, aluminum, and nylon stuffing tubes, MCP blocks, and box connectors.

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TABLE 3A22-I. Cable assignment to steel, aluminum, and nylon stuffing tubes, MCP blocks, and box connectors.

Cable			Nylon tube		MCP [2/]		Stuffing tube			Box conn.	Cable	
Type [1/]	Size OD		Size	Packing assembly, ASTM F1836M	Insert block	Insert block	Surface size	Submarine			MIL-Spec	Slant sheet
	Min.	Max.						Size	Grommet			
LSSHOF-500	1.380	1.450	7	22-0001	60/36		N	6	2-098	½	MIL-DTL-24643	3
LSSHOF-650	1.540	1.610	7	22-0003	60/40		P	7	2-101	2	MIL-DTL-24643	3
LSSHOF-800	1.600	1.670	8	23-0001	60/42		R	7	2-101	2	MIL-DTL-24643	3
LSSRW	0.370	0.400	2	17-0003	20/10		A	1	2-062	½	MIL-DTL-24643	53
LSSSF-7	1.020	1.100	5	20-0009	40/27		K	4	2-091	1	MIL-DTL-24643	11
LSSSGU-50	0.481	0.520	4T	19-0001	20/13	30/13	C	1	2-065	½	MIL-DTL-24643	14
LSSSGU-75	0.557	0.602	4T	19-0004	20/15	30/15	C	2	2-068	¾	MIL-DTL-24643	14
LSSSGU-100	0.619	0.669	4T	19-0005	20/16	30/16	D	2	2-069	¾	MIL-DTL-24643	14
LSSSGU-200	0.807	0.872	5	20-0003	30/22	40/22	G	3	2-073	1	MIL-DTL-24643	14
LSSSGU-300	0.926	1.001	5	20-0006	30/25	40/25	J	3	2-076	1¼	MIL-DTL-24643	14
LSSSGU-400	1.034	1.118	5	20-0010	40/28		K	4	2-092	1¼	MIL-DTL-24643	14
LSSSGU-650	1.268	1.371	6	21-0006	40/34	60/34	M	5	2-097	1½	MIL-DTL-24643	14
LSSSGU-800	1.374	1.485	7	22-0001	60/37		N	6	2-099	1½	MIL-DTL-24643	14
LSSSGU-1000	1.499	1.620	8	23-0001	60/41		P	7	2-101	2	MIL-DTL-24643	14
LSSSGU-1600	1.859	2.010	9	24-0001	60/50		T	8	2-108	2½	MIL-DTL-24643	14
LSSSGU-2000	2.044	2.210	9	24-0004	90/56		W	9	2-110	2½	MIL-DTL-24643	14
LSTCJU-4	0.400	0.430	2	17-0004	20/10		B	1	2-063	½	MIL-DTL-24643	21
LSTCJX-3	0.686	0.742	4T	19-0005	30/18		D	2	2-071	¾	MIL-DTL-24643	24
LSTCJX-7	0.909	0.983	5	20-0005	30/24	40/24	J	3	2-075	1	MIL-DTL-24643	24
LSTCJX-12	1.174	1.269	6	21-0003	40/32		M	5	2-095	1¼	MIL-DTL-24643	24
LSTCKX-1	0.422	0.456	2	17-0003	20/11		B	1	2-064	½	MIL-DTL-24643	24
LSTCKX-3	0.686	0.742	4T	19-0005	30/18		D	2	2-071	¾	MIL-DTL-24643	24
LSTCKX-7	0.909	0.983	5	20-0005	30/24	40/24	J	3	2-075	1	MIL-DTL-24643	24
LSTCKX-12	1.174	1.269	6	21-0003	40/32		M	5	2-095	1¼	MIL-DTL-24643	24
LSTCOP-2	0.325	0.345	2	17-0001	15/8	20/8	A	1	2-062	½	MIL-DTL-24643	2
LSTCTU-4	0.400	0.430	2	17-0004	20/10		B	1	2-063	½	MIL-DTL-24643	21
LSTCTX-1	0.324	0.350	2	17-0001	15/8	20/8	A	1	2-062	½	MIL-DTL-24643	24
LSTCTX-3	0.511	0.552	4T	19-0001	20/13	30/13	C	1	2-066	½	MIL-DTL-24643	24
LSTCTX-7	0.676	0.731	4T	19-0005	30/17		D	2	2-071	¾	MIL-DTL-24643	24
LSTCTX-12	0.892	0.964	5	20-0004	20/24	30/24	J	3	2-075	1	MIL-DTL-24643	24
LSTHOF-3	0.430	0.450	2	17-0004	20/11		B	1	2-064	½	MIL-DTL-24643	3
LSTHOF-4	0.460	0.480	3	18-0018	20/12	30/12	B	1	2-064	½	MIL-DTL-24643	3
LSTHOF-6	0.520	0.550	4T	19-0002	20/13	30/13	C	1	2-066	¾	MIL-DTL-24643	3
LSTHOF-9	0.570	0.600	4T	19-0004	20/15	30/15	C	2	2-068	¾	MIL-DTL-24643	3
LSTHOF-14	0.720	0.750	4T	19-0007	30/18		E	2	2-070	1	MIL-DTL-24643	3
LSTHOF-23	0.860	0.900	5	20-0004	30/22	40/22	G	3	2-073	1	MIL-DTL-24643	3
LSTHOF-42	1.200	1.250	6	21-0003	40/30		L	4	2-094	1¼	MIL-DTL-24643	3
LSTHOF-150	1.740	1.820	8	23-0004	60/46		S	7	2-105	2	MIL-DTL-24643	3
LSTHOF-250	2.140	2.240	9	24-0003	90/56		W	9	2-100	2½	MIL-DTL-24643	3
LSTHOF-400	2.680	2.800	9	24-0008	90/71		AA			3	MIL-DTL-24643	3
LSTHOF-500	2.920	3.100			120/78		BB			3½	MIL-DTL-24643	3
LSTHOF-600	2.980	3.150			120/79		BB			3½	MIL-DTL-24643	3
LSTNW-3	0.380	0.411	2	17-0003	20/10		B	1	2-063	½	MIL-DTL-24643	49
LSTNW-4	0.415	0.449	2	17-0004	20/11		B	1	2-063	½	MIL-DTL-24643	49

NOTES:

1. Cable assignment to steel, aluminum, and nylon stuffing tubes, MCP blocks, and BX connectors shown on figures 3A10 through 3A25 are for MIL-DTL-915, MIL-DTL-24640, and MIL-DTL-24643 cable.
2. MCP block size and stuffing tube designations are provided for information and guidance. Some of the MCP block and stuffing tube sizes shown in this table may not accommodate the full range of allowed cable diameters for a given cable type. In those cases, different MCP block sizes or stuffing tubes should be selected to ensure the required block gap and retest requirements are met. See figure 3B22 for MCPs.

FIGURE 3A22. Cable assignment to steel, aluminum, and nylon stuffing tubes, MCP blocks, and box connectors.

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TABLE 3A23-I. Cable assignment to steel, aluminum, and nylon stuffing tubes, MCP blocks, and box connectors.

Cable			Nylon tube		MCP [2/]		Stuffing tube			Box conn.	Cable	
Type [1/]	Size OD		Size	Packing assembly, ASTM F1836M	Insert block	Insert block	Surface size	Submarine			MIL-Spec	Slant sheet
	Min.	Max.						Size	Grommet			
LSTNW-9	0.578	0.625	4T	19-0004	20/15	30/15	C	2	2-068	¾	MIL-DTL-24643	49
LSTNW-14	0.620	0.670	4T	19-0005	30/16		D	2	2-069	¾	MIL-DTL-24643	49
LSTNW-23	0.703	0.760	4T	19-0008	30/19		E	2	2-071	1	MIL-DTL-24643	49
LSTNW-50	0.869	0.969	5	20-0005	30/24	40/24	J	3	2-075	1	MIL-DTL-24643	49
LSTNW-75	1.048	1.134	5	20-0010	40/28		K	4	2-093	1¼	MIL-DTL-24643	49
LSTNW-100	1.171	1.266	6	21-0004	40/32		M	5	2-095	1¼	MIL-DTL-24643	49
LSTNW-150	1.401	1.515	7	22-0001	60/38		N	6	2-099	1½	MIL-DTL-24643	49
LSTPNW-1½	0.217	0.235	1	16-0004	15/5	20/5	A	1	2-061	½	MIL-DTL-24643	52
LSTPNW-3	0.287	0.310	1	16-0006	15/7	20/7	A	1	2-061	½	MIL-DTL-24643	52
LSTPNW-5	0.338	0.365	2	17-0001	15/9	20/9	A	1	2-062	½	MIL-DTL-24643	52
LSTPNW-10	0.435	0.470	3	18-0018	20/11		B	1	2-064	½	MIL-DTL-24643	52
LSTPNW-15	0.490	0.530	4T	19-0001	20/13	30/13	C	1	2-065	½	MIL-DTL-24643	52
LSTPNW-20	0.532	0.575	4T	19-0002	20/14	30/14	C	1	2-067	¾	MIL-DTL-24643	52
LSTPNW-30	0.629	0.680	4T	19-0005	30/17		D	2	2-069	¾	MIL-DTL-24643	52
LSTPNW-40	0.708	0.765	4T	19-0008	30/19		E	2	2-071	1	MIL-DTL-24643	52
LSTPS-3	0.439	0.475	2	17-0004	20/11	30/11	B	1	2-064	½	MIL-DTL-24643	26
LSTPS-4	0.512	0.553	3	18-0018	20/13	30/13	C	1	2-066	½	MIL-DTL-24643	26
LSTPS-6	0.574	0.620	4T	19-0002	20/15	30/15	C	2	2-068	¾	MIL-DTL-24643	26
LSTPS-9	0.608	0.657	4T	19-0003	20/16	30/16	D	2	2-069	¾	MIL-DTL-24643	26
LSTPS-14	0.695	0.751	4T	19-0006	30/18		E	2	2-071	1	MIL-DTL-24643	26
LSTPS-23	0.801	0.866	5	20-0003	30/21		G	3	2-073	1	MIL-DTL-24643	26
LSTPS-30	0.915	0.989	5	20-0004	30/25	40/25	J	3	2-075	1	MIL-DTL-24643	26
LSTRW	0.567	0.710	4T	19-0005	30/18		D	2	2-070	¾	MIL-DTL-24643	53
LSTSGU-3	0.380	0.411	2	17-0003	20/10		B	1	2-063	½	MIL-DTL-24643	16
LSTSGU-4	0.415	0.449	2	17-0004	20/11		B	1	2-063	½	MIL-DTL-24643	16
LSTSGU-9	0.532	0.575	4T	19-0003	20/14	30/14	C	1	2-067	¾	MIL-DTL-24643	16
LSTSGU-14	0.664	0.718	4T	19-0007	30/18		D	2	2-070	¾	MIL-DTL-24643	16
LSTSGU-23	0.751	0.812	5	20-0003	30/20		E	3	2-072	1	MIL-DTL-24643	16

NOTES:

1. Cable assignment to steel, aluminum, and nylon stuffing tubes, MCP blocks, and BX connectors shown on figures 3A10 through 3A25 are for MIL-DTL-915, MIL-DTL-24640, and MIL-DTL-24643 cable.
2. MCP block size and stuffing tube designations are provided for information and guidance. Some of the MCP block and stuffing tube sizes shown in this table may not accommodate the full range of allowed cable diameters for a given cable type. In those cases, different MCP block sizes or stuffing tubes should be selected to ensure the required block gap and retest requirements are met. See figure 3B22 for MCPs.

FIGURE 3A23. Cable assignment to steel, aluminum, and nylon stuffing tubes, MCP blocks, and box connectors.

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TABLE 3A24-I. Cable assignment to steel, aluminum, and nylon stuffing tubes, MCP blocks, and box connectors.

Cable			Nylon tube		MCP [2/]		Stuffing tube			Box conn.	Cable	
Type [1/]	Size OD		Size	Packing assembly, ASTM F1836M	Insert block	Insert block	Surface size	Submarine			MIL-Spec	Slant sheet
	Min.	Max.						Size	Grommet			
LSTSGU-50	0.896	0.969	5	20-0005	30/24	40/24	J	3	2-075	1	MIL-DTL-24643	16
LSTSGU-75	1.049	1.134	5	20-0010	40/28		L	4	2-093	1¼	MIL-DTL-24643	16
LSTSGU-100	1.171	1.266	6	21-0004	40/32		M	5	2-095	1¼	MIL-DTL-24643	16
LSTSGU-150	1.401	1.515	7	22-0002	60/38		N	6	2-099	2	MIL-DTL-24643	16
LSTSGU-200	1.544	1.669	8	23-0001	60/42		R	7	2-101	2	MIL-DTL-24643	16
LSTSGU-300	1.810	1.957	8	24-0003	60/49		T	8	2-107	2	MIL-DTL-24643	16
LSTSGU-400	2.038	2.203	9	23-0006	60/55		W	9	2-110	2½	MIL-DTL-24643	16
LSTTOP-3	0.444	0.480	3	18-0018	20/12	30/12	B	1	2-064	½	MIL-DTL-24643	12
LSTTOP-5	0.546	0.590	4T	19-0003	20/14	30/14	C	1	2-067	¾	MIL-DTL-24643	12
LSTTOP-10	0.648	0.700	4T	19-0006	30/17		D	2	2-070	¾	MIL-DTL-24643	12
LSTTOP-15	0.768	0.830	5	20-0002	30/20		F	3	2-072	1	MIL-DTL-24643	12
LSTTRS-2	0.629	0.680	4T	19-0005	30/17		D	2	2-069	¾	MIL-DTL-24643	13
LSTTRS-4	0.685	0.740	4T	19-0007	30/18		D	2	2-071	¾	MIL-DTL-24643	13
LSTTRS-6	0.814	0.880	5	20-0003	30/22	40/22	G	3	2-073	1	MIL-DTL-24643	13
LSTTRS-8	0.916	0.990	5	20-0006	30/25	40/25	J	3	2-075	1¼	MIL-DTL-24643	13
LSTTRS-10	0.999	1.080	5	20-0009	40/27		K	4	2-091	1¼	MIL-DTL-24643	13
LSTTRS-12	1.017	1.100	5	20-0009	40/27		K	4	2-091	1¼	MIL-DTL-24643	13
LSTTRS-16	1.100	1.190	6	21-0001	40/30		L	4	2-094	1¼	MIL-DTL-24643	13
LSTTSU-1½	0.305	0.330	2	17-0001	15/8	20/8	A	1	2-062	½	MIL-DTL-24643	23
LSTTSU-3	0.416	0.450	2	17-0004	20/11		B	1	2-063	½	MIL-DTL-24643	23
LSTTSU-5	0.500	0.540	4T	19-0001	20/13	30/13	C	1	2-066	¾	MIL-DTL-24643	23
LSTTSU-10	0.624	0.675	4T	19-0004	30/17		D	2	2-069	¾	MIL-DTL-24643	23
LSTTSU-15	0.740	0.800	4T	19-0007	30/20		E	2	2-071	1	MIL-DTL-24643	23
LSTTSU-20	0.805	0.870	5	20-0002	30/21	40/21	G	3	2-073	1	MIL-DTL-24643	23
LSTTSU-30	0.999	1.080	5	20-0006	40/27		K	4	2-091	1¼	MIL-DTL-24643	23
LSTTSU-40	1.110	1.200	6	21-0002	40/30		L	4	2-094	1¼	MIL-DTL-24643	23
LSTTSU-50	1.295	1.400	6	21-0007	40/35	60/35	M	5	2-097	1½	MIL-DTL-24643	23
LSTTSU-60	1.341	1.450	7	22-0001	60/36		N	6	2-098	1½	MIL-DTL-24643	23

NOTES:

1. Cable assignment to steel, aluminum, and nylon stuffing tubes, MCP blocks, and BX connectors shown on figures 3A10 through 3A25 are for MIL-DTL-915, MIL-DTL-24640, and MIL-DTL-24643 cable.
2. MCP block size and stuffing tube designations are provided for information and guidance. Some of the MCP block and stuffing tube sizes shown in this table may not accommodate the full range of allowed cable diameters for a given cable type. In those cases, different MCP block sizes or stuffing tubes should be selected to ensure the required block gap and retest requirements are met. See figure 3B22 for MCPs.

FIGURE 3A24. Cable assignment to steel, aluminum, and nylon stuffing tubes, MCP blocks, and box connectors.

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TABLE 3A25-I. Cable assignment to steel, aluminum, and nylon stuffing tubes, MCP blocks, and box connectors.

Cable			Nylon tube		MCP [2/]		Stuffing tube			Box conn.	Cable	
Type [1/]	Size OD		Size	Packing assembly, ASTM F1836M	Insert block	Insert block	Surface size	Submarine			MIL-spec	Slant sheet
	Min.	Max.						Size	Grommet			
LSC5FS-4	0.240	0.360	1	16-0006	15/9	20/9	A	1	2-062	⅜	MIL-DTL-24643	59
LSC5FSW-4	0.240	0.380	1	16-0006	15/9	20/9	A	1	2-062	⅜	MIL-DTL-24643	59
LSC5OS-4	0.290	0.400	1	16-0006	20/10		A	1	2-062	⅜	MIL-DTL-24643	59
LSC5OSW-4	0.300	0.420	2	17-0002	20/10		A	1	2-063	½	MIL-DTL-24643	59
LSC5-4	0.220	0.270	1	16-0004	15/6	20/6	A	1	2-061	⅜	MIL-DTL-24643	60
LSC5W-4	0.230	0.290	1	16-0005	15/7	20/7	A	1	2-061	⅜	MIL-DTL-24643	60
LSC5P-4	0.230	0.280	1	16-0005	15/7	20/7	A	1	2-061	⅜	MIL-DTL-24643	61
LSC5POS-4	0.290	0.370	2	17-0002	15/9	20/9	A	1	2-062	⅜	MIL-DTL-24643	61
LSC5POSR-4	0.230	0.280	1	16-0005	15/7	20/7	A	1	2-061	⅜	MIL-DTL-24643	61
LSPB2SD-1	0.330	0.390	2	17-0002	15/9		A	1	2-062	½	MIL-DTL-24643	62
LSPB2SDW-1	0.345	0.405	2	17-0003	20/10		B	1	20-63	½	MIL-DTL-24643	62
LSPB2SDOS-1	0.440	0.500	4T	19-0001	20/12	30/12	B	1	2-065	½	MIL-DTL-24643	62
LSPB2SDOSW-1	0.450	0.530	3	18-0018	20/13	30/13	C	1	2-065	¾	MIL-DTL-24643	62
LSYSGU-2½		0.535	4T	19-0001	20/13	30/13	C	1	2-066	¾	MIL-DTL-24643	63
LSYSGU-4		0.580	4T	19-0001	20/14	30/14	C	1	2-067	¾	MIL-DTL-24643	63
LSYSGU-7		0.640	4T	19-0003	20/16	30/16	C	2	2-069	¾	MIL-DTL-24643	63
LSYSGU-10		0.730	4T	19-0007	30/18		D	2	2-070	¾	MIL-DTL-24643	63
LSYSGU-16		1.000	5	20-0006	30/25	40/25	J	4	2-091	1½	MIL-DTL-24643	63
LSYSGU-42		1. 206	6	21-0001	40/30		L	4	2-093	1½	MIL-DTL-24643	63
LSYSGU-65		1.426	6	21-0005	40/36	60/36	M	5	2-096	1½	MIL-DTL-24643	63
LSYSGU-85		1.541	6	21-0007	60/39		N	6	2-098	1½	MIL-DTL-24643	63
LSYSGU-100		1.669	8	23-0001	60/42		P	7	2-101	2	MIL-DTL-24643	63
LSYSGU-133		1.828	8	23-0002	60/46		S	7	2-105	2	MIL-DTL-24643	63
LSYSGU-165		1.99	8	23-0004	60/50		S	8	2-106	2	MIL-DTL-24643	63
LSTPSJ-1-24	0.154	0.186			15/4						MIL-DTL-24643	64
LSTPSJ-1-22	0.166	0.198			15/4						MIL-DTL-24643	64
LSTPSJ-1-20	0.184	0.216	1	16-0001	15/5	20/5	A	1	2-061	½	MIL-DTL-24643	64
LSTPSJ-1-18	0.204	0.236	1	16-0004	15/5	20/5	A	1	2-061	½	MIL-DTL-24643	64
LSTPSJ-1-16	0.218	0.250	1	16-0004	15/6	20/6	A	1	2-061	½	MIL-DTL-24643	64
LSTPSJ-1-14	0.246	0.278	1	16-0004	15/6	20/6	A	1	2-061	½	MIL-DTL-24643	64
LSTPSJ-2-24	0.230	0.290	1	16-0005	15/7	20/7	A	1	2-061	½	MIL-DTL-24643	64
LSTPSJ-2-22	0.255	0.315	2	17-0001	15/7	20/7	A	1	2-062	½	MIL-DTL-24643	64
LSTPSJ-2-20	0.295	0.355	2	17-0004	20/8		B	1	2-063	½	MIL-DTL-24643	64
LSTPSJ-2-18	0.360	0.420	2	17-0004	20/10		B	1	2-063	½	MIL-DTL-24643	64
LSTPSJ-2-16	0.390	0.450	3	18-0018	20/11		B	1	2-064	½	MIL-DTL-24643	64
LSTPSJ-2-14	0.435	0.495	4T	19-0001	20/12	30/12	B	1	2-065	½	MIL-DTL-24643	64
LSTPSJ-3-24	0.245	0.305	1	16-0005	15/7	20/7	A	1	2-061	⅜	MIL-DTL-24643	64
LSTPSJ-3-22	0.255	0.315	2	17-0001	15/7	20/7	A	1	2-062	½	MIL-DTL-24643	64
LSTPSJ-3-20	0.295	0.355	1	16-0006	15/8	20/8	A	1	2-062	½	MIL-DTL-24643	64
LSTPSJ-3-18	0.360	0.420	2	17-0004	20/10		B	1	2-063	½	MIL-DTL-24643	64
LSTPSJ-3-16	0.420	0.480	2	17-0002	20/12	30/12	B	1	2-064	½	MIL-DTL-24643	64
LS3C179DT	0.335	0.365	2	17-0002	15/9	20/9	A	1	2-062	½	MIL-DTL-24643	65
LS6C176DT	0.425	0.455	2	17-0004	20/11		B	1	2-063	½	MIL-DTL-24643	65
LS2OW-3	0.412	0.442	2	17-0002	15/11	20/11	A	1	2-062	½	MIL-DTL-24643	66
LS2OW-4	0.429	0.46	2	17-0002	15/11		A	1	2-062	½	MIL-DTL-24643	66
LS2OW-9	0.513	0.551	4T	19-0001	20/12	30/12	B	1	2-065	½	MIL-DTL-24643	66
LS2OW-14	0.639	0.686	4T	19-0006	30/17		D	2	2-070	¾	MIL-DTL-24643	66
LS2OW-23	0.743	0.797	5	20-0002	30/20		F	2	2-071	1	MIL-DTL-24643	66

FIGURE 3A25. Cable assignment to steel, aluminum, and nylon stuffing tubes, MCP blocks, and box connectors.

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TABLE 3A25-I. Cable assignment to steel, aluminum, and nylon stuffing tubes, MCP blocks, and box connectors – Continued.

Cable			Nylon tube		MCP 2/		Stuffing tube			Box conn.	Cable	
Type 1/	Size OD		Size	Packing assembly, ASTM F1836M	Insert block	Insert block	Surface size	Submarine			MIL-spec	Slant sheet
	Min.	Max.						Size	Grommet			
LS2OW-50	0.894	0.959	5	20-0004	30/24	40/24	J	3	2-075	¾	MIL-DTL-24643	66
LS2OW-75	1.033	1.108	5	20-0008	40/28		K	4	2-091	1¼	MIL-DTL-24643	66
LS2OW-100	1.149	1.232	6	21-0001	40/31		L	4	2-094	1¼	MIL-DTL-24643	66
LS2OW-200	1.519	1.630	7	22-0003	60/41		R	7	2-101	2	MIL-DTL-24643	66
LS3OW-3	0.431	0.463	2	17-0002	15/11		A	1	2-062	½	MIL-DTL-24643	67
LS3OW-4	0.449	0.482	2	17-0004	20/12		B	1	2-063	½	MIL-DTL-24643	67
LS3OW-9	0.54	0.58	3	18-0018	20/13	30/13	C	1	2-065	½	MIL-DTL-24643	67
LS3OW-14	0.684	0.733	4T	19-0005	30/17		D	2	2-071	¾	MIL-DTL-24643	67
LS3OW-23	0.769	0.853	5	20-0003	30/21		G	3	2-073	1	MIL-DTL-24643	67
LS3OW-50	0.957	1.027	5	20-0006	40/25		J	3	2-076	1	MIL-DTL-24643	67
LS3OW-75	1.107	1.187	6	21-0001	40/30		L	4	2-094	1¼	MIL-DTL-24643	67
LS3OW-100	1.231	1.321	6	21-0004	40/33	60/33	M	5	2-096	1½	MIL-DTL-24643	67
LS3OW-200	1.630	1.748	8	23-0002	60/44		S	7	2-103	2	MIL-DTL-24643	67
LS3OW-300	1.916	2.055	3	23-0006	60/52		T	8	2-108	2⅜	MIL-DTL-24643	67
LS3OW-400	2.190	2.349	9	24-0004	90/55		X			2¼	MIL-DTL-24643	67
LS4OW-3	0.463	0.497	2	17-0002	20/11		B	1	2-063	½	MIL-DTL-24643	68
LS4OW-4	0.485	0.521	3	18-0018	20/12	30/12	B	1	2-064	½	MIL-DTL-24643	68
LS4OW-9	0.588	0.631	4T	19-0003	20/15	30/15	C	2	2-068	¾	MIL-DTL-24643	68
LS4OW-14	0.772	0.828	5	20-0002	30/20		G	3	2-072	¾	MIL-DTL-24643	68
LS4OW-23	0.900	0.965	5	20-0006	30/24	40/24	J	3	2-075	1	MIL-DTL-24643	68
LS4OW-50	1.084	1.163	6	21-0001	40/29		K	4	2-093	1¼	MIL-DTL-24643	68
LS4OW-75	1.254	1.345	6	21-0003	40/34	60/34	M	5	2-096	1⅜	MIL-DTL-24643	68
LS4OW-100	1.396	1.498	7	22-0001	60/37		N	6	2-099	1½	MIL-DTL-24643	68
LS4OW-200	1.850	1.984	8	23-0006	60/50	90/50	T	8	2-107	2½	MIL-DTL-24643	68
LSC264-4	0.735	0.765	4T	19-0005	30/19		E	2	2-071	¾	MIL-DTL-24643	69

NOTES:

1. Cable assignment to steel, aluminum, and nylon stuffing tubes, MCP blocks, and BX connectors shown on figures 3A10 through 3A25 are for MIL-DTL-915, MIL-DTL-24640, and MIL-DTL-24643 cable.
2. MCP block size and stuffing tube designations are provided for information and guidance. Some of the MCP block and stuffing tube sizes shown in this table may not accommodate the full range of allowed cable diameters for a given cable type. In those cases, different MCP block sizes or stuffing tubes should be selected to ensure the required block gap and retest requirements are met. See figure 3B22 for MCPs.

FIGURE 3A25. Cable assignment to steel, aluminum, and nylon stuffing tubes, MCP blocks, and box connectors – Continued.

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APPENDIX 3B – PENETRATIONS –SURFACE SHIPS

B.1 SCOPE

B.1.1 Scope. This appendix describes the installation standard methods for electrical penetrations in surface ships. This Appendix is a mandatory part of the standard. The information contained herein is intended for compliance.

B.2. APPLICABLE DOCUMENTS

B.2.1 General. The documents listed in this section are specified in this appendix. This section does not include documents cited in other sections of this standard or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in this appendix, whether or not they are listed.

B.2.2 Government documents.

B.2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

COMMERCIAL ITEM DESCRIPTIONS

A-A-3041	-	Wrench, Open End Ratchet (TAC Pattern) for Tube Fitting, Electrical Cable Terminals, and Stuffing Tube Gland Nuts
A-A-3042	-	Socket, Open Detachable, Standard Wall, Octagon

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-917	-	Electric Power Equipment, Basic Requirements for
MIL-DTL-2726/68	-	Receptacle, Plug, Connector, Electrical, 10-Ampere, 125-Volt, 3-Phase, Grounded (Symbol No. 706.1) (see 4.3)
MIL-I-3064	-	Insulation, Electrical, Plastic-Sealer
MIL-PRF-15624	-	Gasket Material, Rubber, 50 Durometer Hardness (Maximum)
MIL-DTL-16685	-	Packing, Material and Packing Preformed (Stuffing Tube for Electric Cables)
MIL-R-17882	-	Repair Kits, Metallic Pipe and General Purpose, Damage Control
MIL-S-22698	-	Steel Plate, Shapes and Bars, Weldable Ordinary Strength and Higher Strength: Structural
MIL-S-24235	-	Stuffing Tubes, Metal, and Packing Assemblies for Electric Cables, General Specification for
MIL-S-24235/1	-	Stuffing Tubes, Metal, and Packing Assemblies for Electric Cables, Bulkhead, Pressureproof
MIL-S-24235/9	-	Stuffing Tubes, Metal, and Packing Assemblies for Electric Cables, Brass and Steel, for Decks and Bulkheads with Pipe Protection
MIL-S-24235/10	-	Stuffing Tubes, Metal, and Packing Assemblies for Electric Cables, Steel, for Decks and Bulkheads Without Pipe Protection
MIL-S-24235/17	-	Stuffing Tubes, Metal, and Packing Assemblies for Electric Cables, Swage Type, Steel and Aluminum, for Deck and Bulkheads with Pipe Protection

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MIL-S-24235/18	-	Stuffing Tubes, Metal, and Packing Assemblies for Electric Cables, Swage Type, Reduced Diameter, Steel and Aluminum, for Deck and Bulkheads with Pipe Protection
MIL-S-24235/19	-	Stuffing Tubes, Metal, and Packing Assemblies for Electric Cables, Sealing Plug for Unused Stuffing Tubes
MIL-DTL-24705	-	Penetrators, Multiple Cable, Electric Cable, General Specification for
MIL-DTL-24705/1	-	Penetrators, Multiple Cable, Electric Cable, Rectangular Metal Frame, Square-Faced Blocks
MIL-DTL-24705/3	-	Penetrators, Multiple Cable, Electric Cable, Round Frame, Square-Faced Blocks

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-22	-	Welded Joint Design
MIL-STD-2003-2	-	Electric Plant Installation Standard Methods for Surface Ships and Submarines (Equipment)
MIL-STD-2003-4	-	Electric Plant Installation Standard Methods for Surface Ships and Submarines (Cableways)

(Copies of these documents are available online at https://quicksearch.dla.mil.)

B.2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

NAVAL SEA SYSTEMS COMMAND (NAVSEA) DRAWINGS

302-7583331	-	Standard Rise Installation Procedure, Penetration Install

(Copies of this document are available from the applicable repositories listed in S0005-AE-PRO-010/EDM, which can be obtained online via Technical Data Management Information System (TDMIS) at https://mercury.tdmis.navy.mil/. Copies of this document may also be obtained from the Naval Ships Engineering Drawing Repository (NSEDR) online at https://199.208.213.105/webjedmics/index.jsp. To request an NSEDR account for drawing access, send an email to NNSY_JEDMICS_NSEDR_HELP_DESK@navy.mil.)

NAVAL SEA SYSTEMS COMMAND (NAVSEA) PUBLICATIONS

S9086-CJ-STM-010/075	-	NSTM Chapter 075, Fasteners
SE000-01-IMB-010	-	Navy Installation and Maintenance Book
T9070-AN-DPC-010/100-1	-	Reinforcement of Openings in Structure of Surface Ships, Other Than in Protective Plating
T9070-AN-DPC-020/100-2	-	Openings in Decks and Bulkheads for Stuffing Tubes and Pipe

(Copies of these documents are available online via Technical Data Management Information System (TDMIS) at https://mercury.tdmis.navy.mil/ by searching for the document number without the suffix. Refer questions, inquiries, or problems to: DSN 296-0669, Commercial (805) 228-0669. These documents are available for ordering (hard copy) via the Naval Logistics Library (NLL) at https://nll.navsup.navy.mil. For questions regarding the NLL, contact the NLL Customer Service at nllhelpdesk@navy.mil, (866) 817-3130, or (215) 697-2626/DSN 442-2626.)

B.2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

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ASTM INTERNATIONAL

ASTM A36/A36M - Standard Specification for Carbon Structural Steel

ASTM A53/A53M - Standard Specification for Pipe, Steel, Black and Hot-Dipped, Zinc-Coated, Welded and Seamless

ASTM A106/A106M - Standard Specification for Seamless Carbon Steel Pipe for High-Temperature Service

ASTM A1008/A1008M - Standard Specification for Steel, Sheet, Cold-Rolled, Carbon, Structural, High-Strength Low-Alloy, High-Strength Low-Alloy with Improved Formability, Solution Hardened, and Bake Hardenable

ASTM A1011/A1011M - Standard Specification for Steel, Sheet and Strip, Hot-Rolled, Carbon, Structural, High-Strength Low-Alloy, High-Strength Low-Alloy with Improved Formability, and Ultra-High Strength

ASTM B209 - Standard Specification for Aluminum and Aluminum-Alloy Sheet and Plate

ASTM F1836M - Standard Specification for Stuffing Tubes, Nylon, and Packing Assemblies (Metric)

(Copies of these documents are available online at www.astm.org.)

NATIONAL AEROSPACE STANDARDS COMMITTEE (NA/NAS)

NASM 25027 - Nut, Self-Locking, 250 °F, 450 °F, and 800 °F

(Copies of this document are available online at www.aia-aerospace.org.)

B.2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

B.3 REQUIRED EQUIPMENT AND MATERIALS

B.3.1 Required equipment and materials. The required equipment and materials for the proper installation of surface ship stuffing tubes are as shown in the Appendix 3B methods.

B.4 NOTES AND PROCEDURES

B.4.1 Dimensions. For figures and tables in this appendix, all dimensions are in inches unless otherwise noted.

B.4.2 Figures. Table 3BI provides information for the figures in this appendix.

TABLE 3BI. Figures for surface ship penetration installation.

Figure number	Surface ship penetration installation	Page
3B1	MIL-S-24235 stuffing tube and kickpipes minimum spacing (surface ships)	52
3B2	MIL-S-24235 (other than /17 and /18 stuffing tube and pipe (steel) minimum spacing (surface ships)	53
3B3	MIL-S-24235/17 swage tube (steel) minimum spacing (surface ships)	54
3B4	MIL-S-24235/18 reduced diameter swage and kickpipe minimum spacing (surface ships)	55
3B5	MIL-S-24235 stuffing tubes for steel or aluminum bulkheads (surface ships)	56

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TABLE 3BI. Figures for surface ship stuffing tubes – Continued.

Figure number	Surface ship penetration installation	Page
3B6	MIL-S-24235 stuffing tubes for steel or aluminum bulkheads (surface ships)	58
3B7	MIL-S-24235 stuffing tubes through shielded bulkheads (surface ships)	59
3B8	MIL-S-24235 stuffing tubes for ballistic and type A protective plate bulkheads (surface ships)	60
3B9	MIL-S-24235 stuffing tubes through acoustical spaces (surface ships)	61
3B10	MIL-S-24235 stuffing tubes through acoustical spaces (surface ships)	63
3B11	MIL-S-24235 stuffing tubes through permanent sections of false decks (surface ships)	65
3B12	MIL-S-24235 stuffing tubes through portable sections of false decks (surface ships)	66
3B13	MIL-S-24235 stuffing tubes through air spaces (surface ships)	67
3B14	Stuffing tubes for sonar dome area (surface ships)	68
3B15	Topside stuffing tubes with riser box (surface ships)	70
3B16	MIL-S-24235 topside stuffing tubes with riser box for medium steel decks (surface ships)	72
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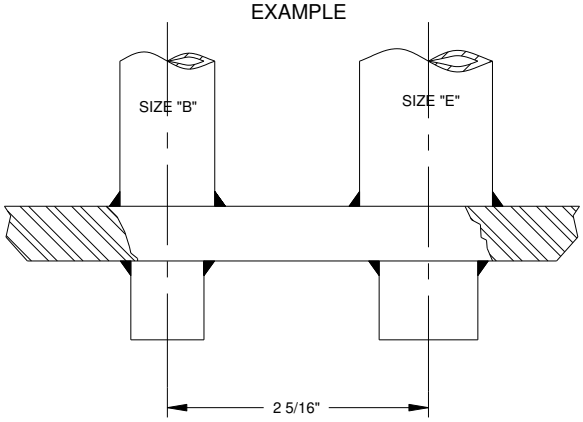
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METHOD 3B-1-1
HOLE SPACING IN DECKS AND BULKHEADS
FROM T9070-AN-DPC-020/100-2

EXAMPLE



NOTES:

1. The table referred to in the following notes is the hole-spacing table on figures 3B2, 3B3, and 3B4.
2. The dimensions shown in the hole-spacing table are suitable for the piercing of ballistic and type A protective plate, high tensile protective plating, and medium steel decks and bulkheads.
3. This table may be used for all plate thicknesses not exceeding 3 inches.
4. This table may also be used for aluminum decks and bulkheads where installation of stuffing tubes is required.
5. The values of the table are minimum permitted values and may be increased as judged to be desirable for any particular application.
6. Use of this table does not require any increase in thickness of plating or addition of doubler plates. The metal restored to a plate by the mounting of stuffing tubes or the passing of pipe is adequate compensation for the metal removed from the plate due to any drilling of holes in accordance with the table and limitations of note 2.
7. This table is applicable to the great majority of installations. Exceptions to the table are recognized when it is expedient to mount stuffing tubes on alternate sides of the plate being pierced, or as otherwise noted in this installation drawing.
8. For strength preservation of both medium and ballistic and type A protective platings, stuffing tube hole spacings shall in no case be less than 1.8 mean diameters of any two adjacent holes. For medium steel calculations for the hole diameters, use the internal diameters of the extra strong kickpipes. For ballistic and type A protective plate, use the actual drilled hole diameters given in column three of figure 3B2.
9. In locating a hole from a boundary structural member or cable support, the distance shall not be less than half the center-to-center spacing given in the table for two tubes of the required size.
10. This table does not impose any restrictions on how the holes shall be arranged as long as the restriction regarding minimum allowable spacing is not violated.
11. The spacings for tube sizes “A” through “Z” are sufficient for the use of the standard open ratchet wrenches, as specified in A-A-3041 and A-A-3042. Spacings for tube sizes “AA” and “BB” are sufficient for the use of 32-point box type crowfoot wrenches with square drive. This does not, however, exclude the use of other types of wrenches suitable for the purpose.

FIGURE 3B1. MIL-S-24235 stuffing tube and kickpipes minimum spacing (surface ships).

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TABLE 3B2-I. Hole spacing in decks and bulkheads.

(Derived from requirements of T9070-AN-DPC-020/100-2)

Spacing of holes for stuffing tubes and pipes This table also applicable to armor plate.																										
Nominal dia. of kickpipe (iron pipe size)	Drill for pipe (medium steel only)	Drill for pipe (ballistic and type A protective plate)	Tube size	A	B	C	D	E	F	G	J	K	L	M	N	P	R	S	T	V	W	X	Y	Z	AA	BB
⅜"	23/32"	0.406"	A	1 15/16"																						
½"	⅞"	0.515"	B	2"	2"																					
¾"	1 3/32"	0.640"	C	2 1/16"	2 1/16"	2 1/16"																				
¾"	1 3/32"	0.750"	D	2¼"	2 5/16"	2⅜"	2 7/16"																			
1"	1 21/64"	0.812"	E	2¼"	2 5/16"	2⅜"	2½"	2½"																		
1"	1 21/64"	0.843"	F	2¼"	2 5/16"	2⅜"	2½"	2½"	2½"																	
1"	1 21/64"	0.953"	G	2¾"	2 13/16"	2⅞"	2 15/16"	3"	3"	3 1/16"																
1¼"	1 45/64"	1.062"	J	2¾"	2 13/16"	2⅞"	2 15/16"	3"	3"	3 3/16"	3 3/16"															
1¼"	1 45/64"	1.171"	K	2¾"	2 13/16"	2⅞"	2 15/16"	3"	3"	3 3/16"	3 3/16"	3 3/16"														
1¼"	1 45/64"	1.265"	L	2¾"	2 13/16"	2⅞"	2 15/16"	3"	3"	3¼"	3¼"	3¼"	3¼"													
1½"	1 15/16"	1.406"	M	2¾"	2 13/16"	2⅞"	2 15/16"	3"	3"	3⅜"	3⅜"	3⅜"	3⅜"	3⅜"												
1½"	1 15/16"	1.515"	N	3 5/16"	3⅜"	3 7/16"	3½"	3 9/16"	3 9/16"	3 11/16"	3¾"	3 13/16"	3 13/16"	3 15/16"	4"											
2"	2 13/32"	1.625"	P	3 5/16"	3⅜"	3 7/16"	3½"	3 9/16"	3 9/16"	3 11/16"	3¾"	3 13/16"	3 13/16"	3 15/16"	4⅛"	4⅛"										
2"	2 13/32"	1.750"	R	3 5/16"	3⅜"	3 7/16"	3½"	3 9/16"	3 9/16"	3 11/16"	3¾"	3 13/16"	3 13/16"	3 15/16"	4⅛"	4⅛"	4⅛"									
2"	2 13/32"	1.875"	S	3 5/16"	3⅜"	3 7/16"	3½"	3 9/16"	3 9/16"	3 11/16"	3¾"	3 13/16"	3 13/16"	3 15/16"	4 5/16"	4 5/16"	4 5/16"	4 5/16"								
2½"	2 29/32"	2.062"	T	3 5/16"	3⅜"	3 7/16"	3½"	3 9/16"	3 9/16"	3 13/16"	3 13/16"	3 13/16"	3 13/16"	3 15/16"	4 7/16"	4 7/16"	4 7/16"	4 7/16"	4 7/16"							
2½"	2 29/32"	2.187"	V	3 11/16"	3¾"	3 13/16"	3⅞"	3 15/16"	3 15/16"	4 1/16"	4⅛"	4 3/16"	4¼"	4 5/16"	4 7/16"	4½"	4½"	4 11/16"	4¾"	4 13/16"						
2½"	2 29/32"	2.312"	W	3 11/16"	3¾"	3 13/16"	3⅞"	3 15/16"	3 15/16"	4 1/16"	4⅛"	4 3/16"	4¼"	4 5/16"	4½"	4½"	4½"	4 11/16"	4¾"	4⅞"	4⅞"					
3"	3 17/32"	2.500"	X	3 11/16"	3¾"	3 13/16"	3⅞"	3 15/16"	3 15/16"	4⅛"	4⅛"	4 3/16"	4¼"	4 5/16"	4 11/16"	4 11/16"	4 11/16"	4 11/16"	4¾"	5 1/16"	5 1/16"	5 1/16"				
3"	3 17/32"	2.609"	Y	3 11/16"	3¾"	3 13/16"	3⅞"	3 15/16"	3 15/16"	4⅛"	4⅛"	4 3/16"	4¼"	4 5/16"	4 11/16"	4 11/16"	4 11/16"	4 11/16"	4¾"	5 1/16"	5 1/16"	5 1/16"	5 1/16"			
3"	3 17/32"	2.781"	Z	3 11/16"	3¾"	3 13/16"	3⅞"	3 15/16"	3 15/16"	4⅛"	4⅛"	4 3/16"	4¼"	4 5/16"	4¾"	4¾"	4¾"	4¾"	4¾"	5⅛"	5⅛"	5⅛"	5⅛"	5 5/16"		
3"	3 7/32"	2.875"	AA	3 13/16"	3⅞"	3 15/16"	4"	4"	4 1/32"	4 5/32"	4¼"	4 9/32"	4 5/16"	4 7/16"	4 15/32"	4⅝"	4 19/32"	4 25/32"	4 31/32"	4 31/32"	5 1/16"	5 9/16"	5 9/16"	5 9/16"	5 17/32"	
3½"	4 1/32"	3.157"	BB	4"	4 1/16"	4⅛"	4 3/16"	4 3/16"	4 7/32"	4 11/32"	4 7/16"	4 15/32"	4½"	4⅝"	4 21/32"	5 1/16"	5 1/16"	5 1/16"	5 13/32"	5 13/32"	5 13/32"	5 31/32"	5 31/32"	5 31/32"	5 31/32"	6 13/32"

NOTE:

1. See figure 3B1 for notes.

FIGURE 3B2. MIL-S-24235 (other than /17 and /18 stuffing tube and pipe (steel) minimum spacing (surface ships).

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TABLE 3B3-I. Hole spacing in decks and bulkheads.

(Derived from requirements of T9070-AN-DPC-020/100-2)

Spacing of hole for swage tubes This table does not apply to armor plate.																									
Drill for swage tube M24235/17	Nominal outside dia. of swage tube	Tube size	A	B	C	D	E	F	G	J	K	L	M	N	P	R	S	T	V	W	X	Y	Z	AA	BB
1 3/32	1 1/16	A	2 3/32																						
1 3/32	1 1/16	B	2 3/32	2 3/32																					
1 11/32	1 5/16	C	2 11/32	2 11/32	2 9/16																				
1 11/32	1 5/16	D	2 11/32	2 11/32	2 9/16	2 9/16																			
1 11/32	1 5/16	E	2 11/32	2 11/32	2 9/16	2 9/16	2 9/16																		
1 11/32	1 5/16	F	2 11/32	2 11/32	2 9/16	2 9/16	2 9/16	2 9/16																	
1 11/16	1 21/32	G	2 21/32	2 21/32	2 29/32	2 29/32	2 29/32	2 29/32	3 9/32																
1 15/16	1 29/32	J	2 29/32	2 29/32	3 1/8	3 1/8	3 1/8	3 1/8	3 15/32	3 11/16															
1 15/16	1 29/32	K	2 29/32	2 29/32	3 1/8	3 1/8	3 1/8	3 1/8	3 15/32	3 11/16	3 11/16														
1 15/16	1 29/32	L	2 29/32	2 29/32	3 1/8	3 1/8	3 1/8	3 1/8	3 15/32	3 11/16	3 11/16	3 11/16													
2 13/32	2 3/8	M	3 11/32	3 11/32	3 19/32	3 19/32	3 19/32	3 19/32	3 29/32	4 1/16	4 5/32	4 5/32	4 19/32												
2 13/32	2 3/8	N	3 11/32	3 11/32	3 19/32	3 19/32	3 19/32	3 19/32	3 29/32	4 1/16	4 5/32	4 5/32	4 19/32	4 19/32											
2 13/32	2 3/8	P	3 11/32	3 11/32	3 19/32	3 19/32	3 19/32	3 19/32	3 29/32	4 1/16	4 5/32	4 5/32	4 19/32	4 19/32	4 19/32										
2 13/32	2 3/8	R	3 11/32	3 11/32	3 19/32	3 19/32	3 19/32	3 19/32	3 29/32	4 1/16	4 5/32	4 5/32	4 19/32	4 19/32	4 19/32	4 19/32									
2 29/32	2 7/8	S	3 13/16	3 13/16	4 1/16	4 1/16	4 1/16	4 1/16	4 3/8	4 5/8	4 5/8	4 5/8	5 1/16	5 1/16	5 1/16	5 1/16	5 17/32								
3 3/16	3 1/8	T	4 3/32	4 3/32	4 5/16	4 5/16	4 5/16	4 5/16	4 21/32	4 7/8	4 7/8	4 7/8	5 11/32	5 11/32	5 11/32	5 11/32	5 13/16	6 1/16							
3 5/16	3 1/4	V	4 3/16	4 3/16	4 7/16	4 7/16	4 7/16	4 7/16	4 3/4	5	5	5	5 7/16	5 7/16	5 7/16	5 7/16	5 15/16	6 3/16	6 5/16						
3 9/16	3 1/2	W	4 7/16	4 7/16	4 15/32	4 15/32	4 15/32	4 15/32	5	5 1/4	5 1/4	5 1/4	5 11/16	5 11/16	5 11/16	5 11/16	6 5/32	6 7/16	6 17/32	6 25/32					
3 9/16	3 1/2	X	4 7/16	4 7/16	4 15/32	4 15/32	4 15/32	4 15/32	5	5 1/4	5 1/4	5 1/4	5 11/16	5 11/16	5 11/16	5 11/16	6 5/32	6 7/16	6 17/32	6 25/32	6 25/32				
4 1/16	4	Y	4 29/32	4 29/32	5 5/32	5 5/32	5 5/32	5 5/32	5 15/32	5 23/32	5 23/32	5 23/32	6 5/32	6 5/32	6 5/32	6 5/32	6 5/8	6 29/32	7 1/32	7 1/4	7 1/4	7 23/32			
4 1/16	4	Z	4 29/32	4 29/32	5 5/32	5 5/32	5 5/32	5 5/32	5 15/32	5 23/32	5 23/32	5 23/32	6 5/32	6 5/32	6 5/32	6 5/32	6 5/8	6 29/32	7 1/32	7 1/4	7 1/4	7 23/32	7 23/32		
4 1/16	4	AA	4 29/32	4 29/32	5 5/32	5 5/32	5 5/32	5 5/32	5 15/32	5 23/32	5 23/32	5 23/32	6 5/32	6 5/32	6 5/32	6 5/32	6 5/8	6 29/32	7 1/32	7 1/4	7 1/4	7 23/32	7 23/32	7 23/32	
4 9/16	4 1/2	BB	5 3/8	5 3/8	5 5/8	5 5/8	5 5/8	5 5/8	5 15/16	6 3/16	6 3/16	6 3/16	6 5/8	6 5/8	6 5/8	6 5/8	7 1/8	7 3/8	7 1/2	7 23/32	7 23/32	8 7/32	8 7/32	8 7/32	8 11/32

NOTES:

1. See figure 3B1 for notes.
2. See figure 3B4 for MIL-S-24235/18 spacing.

FIGURE 3B3. MIL-S-24235/17 swage tube (steel) minimum spacing (surface ships).

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TABLE 3B4-I. Hole spacing in decks and bulkheads.

(Derived from requirements of T9070-AN-DPC-020/100-2)

Spacing of hole for swage tubes This table does not apply to armor plate.																									
Drill for swage tube M24235/18	Nominal outside dia. of swage tube	Tube size	A	B	C	D	E	F	G	J	K	L	M	N	P	R	S	T	V	W	X	Y	Z	AA	BB
7/8	0.840 (27/32)	A	1 31/32																						
7/8	0.840 (27/32)	B	1 31/32	2																					
1 3/32	1.050 (1/16)	C	1 31/32	2	2 1/16																				
1 3/32	1.050 (1/16)	D	2 7/32	2 1/4	2 5/16	2 3/8																			
1 23/64	1.315 (1 21/64)	E	2 7/32	2 1/4	2 5/16	2 3/8	2 13/32																		
1 23/64	1.315 (1 21/64)	F	2 7/32	2 1/4	2 5/16	2 3/8	2 13/32	2 1/2																	
1 23/64	1.315 (1 21/64)	G	2 21/32	2 23/32	2 25/32	2 27/32	2 7/8	2 29/32	2 31/32																
1 45/64	1.660 (1 43/64)	J	2 21/32	2 23/32	2 25/32	2 27/32	2 7/8	2 29/32	2 31/32	3 3/16															
1 45/64	1.660 (1 43/64)	K	2 21/32	2 23/32	2 25/32	2 27/32	2 7/8	2 29/32	2 31/32	3 3/16	3 3/16														
1 45/64	1.660 (1 43/64)	L	2 21/32	2 23/32	2 25/32	2 27/32	2 7/8	2 29/32	2 31/32	3 3/16	3 3/16	3 3/16													
1 15/64	1.900 (1 29/32)	M	2 21/32	2 23/32	2 25/32	2 27/32	2 7/8	2 29/32	3	3 15/32	3 15/32	3 15/32	3 5/8												
1 15/16	1.900 (1 29/32)	N	3 1/4	3 5/16	3 3/8	3 7/16	3 15/32	3 1/2	3 9/16	3 19/32	3 11/16	3 23/32	3 13/16	3 27/32											
2 13/32	2.375 (2 3/8)	P	3 1/4	3 5/16	3 3/8	3 13/32	3 15/32	3 1/2	3 9/16	3 7/8	3 7/8	3 7/8	4 3/32	4 3/32	4 17/32										
2 13/32	2.375 (2 3/8)	R	3 1/4	3 5/16	3 3/32	3 13/32	3 17/32	3 17/32	3 9/16	3 7/8	3 7/8	3 7/8	4 3/32	4 3/32	4 17/32	4 17/32									
2 13/32	2.375 (2 3/8)	S	3 1/4	3 5/16	3 11/32	3 13/32	3 17/32	3 17/32	4 1/32	3 7/8	3 7/8	3 7/8	4 3/32	4 3/32	4 17/32	4 17/32	4 17/32								
2 15/16	2.875 (2 7/8)	T	3 9/16	3 9/16	3 25/32	4 25/32	4 1/32	4 1/32	4 1/32	4 11/32	4 11/32	4 11/32	4 19/32	4 19/32	5 1/32	5 1/32	5 1/32	5 1/2							
2 15/16	2.875 (2 7/8)	V	3 5/8	3 11/16	3 25/32	3 13/16	4 1/32	4 1/32	4 1/32	4 11/32	4 11/32	4 11/32	4 19/32	4 1/32	5 1/32	5 1/32	5 1/32	5 1/2	5 1/2						
2 15/16	2.875 (2 7/8)	W	3 5/8	3 11/16	3 25/32	3 13/16	4 1/32	4 1/32	4 1/32	4 11/32	4 11/32	4 11/32	4 19/32	4 19/32	5 1/32	5 1/32	5 1/32	5 1/2	5 1/2	5 1/2					
3 9/16	3.500 (3 1/2)	X	4 5/32	4 5/32	4 3/8	4 3/8	4 5/8	4 5/8	4 5/8	4 15/16	4 15/16	4 15/16	5 3/16	5 3/16	5 9/16	5 9/16	5 9/16	6 3/32	6 3/32	6 3/32	6 11/16				
3 9/16	3.500 (3 1/2)	Y	4 5/32	4 5/32	4 3/8	4 3/8	4 5/8	4 5/8	4 5/8	4 15/16	4 15/16	4 15/16	5 3/16	5 3/16	5 9/16	5 9/16	5 9/16	6 3/32	6 3/32	6 3/32	6 11/16	6 11/16			
3 9/16	3.500 (3 1/2)	Z	4 5/32	4 5/32	4 3/8	4 3/8	4 5/8	4 5/8	4 5/8	4 15/16	4 15/16	4 15/16	5 3/16	5 3/16	5 9/16	5 9/16	5 9/16	6 3/32	6 3/32	6 3/32	6 11/16	6 11/16	6 11/16		
3 9/16	3.500 (3 1/2)	AA	4 5/32	4 5/32	4 3/8	4 3/8	4 5/8	4 5/8	4 5/8	4 5/16	4 15/16	4 15/16	5 15/16	5 3/16	5 3/16	5 5/8	5 5/8	6 3/32	6 3/32	6 3/32	6 11/16	6 11/16	6 11/16	6 11/16	
4 1/16	4.000 (4)	BB	4 19/32	4 19/32	4 27/32	4 27/32	5 3/32	5 3/32	5 3/32	5 13/32	5 13/32	5 13/32	5 21/32	5 21/32	6 3/32	6 3/32	6 3/32	6 9/16	6 9/16	6 9/16	7 5/32	7 5/32	7 5/32	7 5/32	7 21/32

NOTE:

1. See figure 3B3 for spacing of MIL-S-24235/17 for regular swage tubes.

FIGURE 3B4. MIL-S-24235/18 reduced diameter swage and kickpipe minimum spacing (surface ships).

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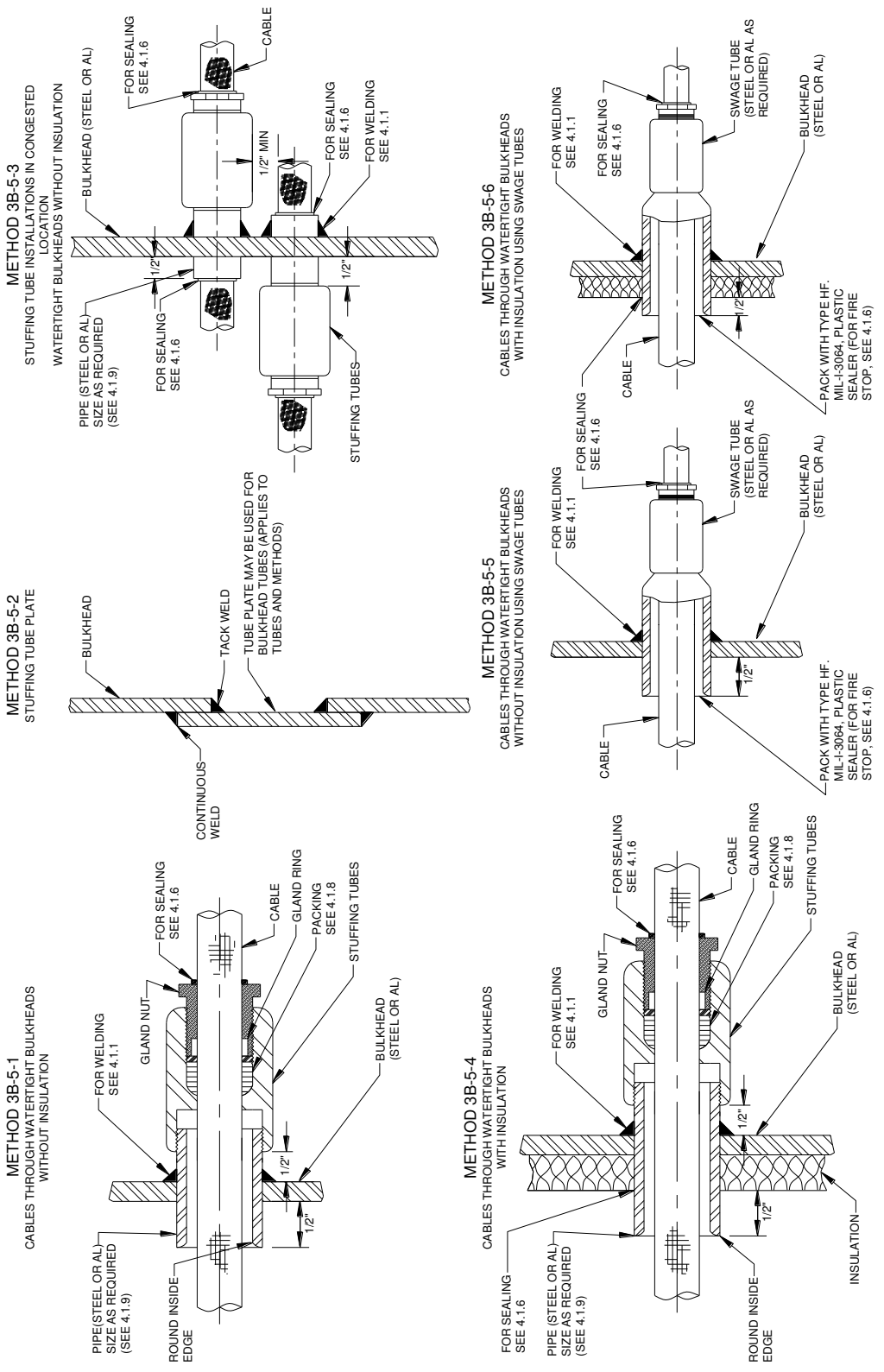
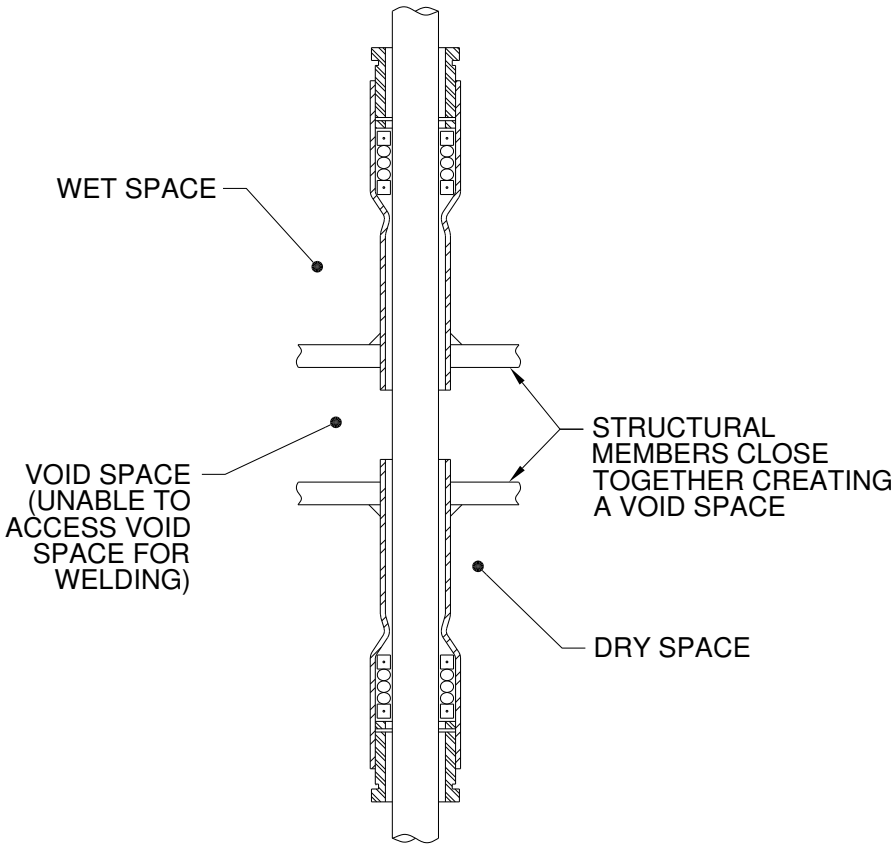


FIGURE 3B5. MIL-S-24235 stuffing tubes for steel or aluminum bulkheads (surface ships).

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Method 3B-5-7
Cables through wet spaces and spaces adjacent
to wet spaces where the tube cannot be accessed on both sides

1. This method is applicable to installation of cable penetrations to wet spaces or spaces adjacent to wet spaces where both sides of a tube cannot be accessed for welding (such as a sound dampening structure).
2. Penetrators shall be welded on the side that can be accessed.
3. Penetrator tubes shall not be touching in the void space.



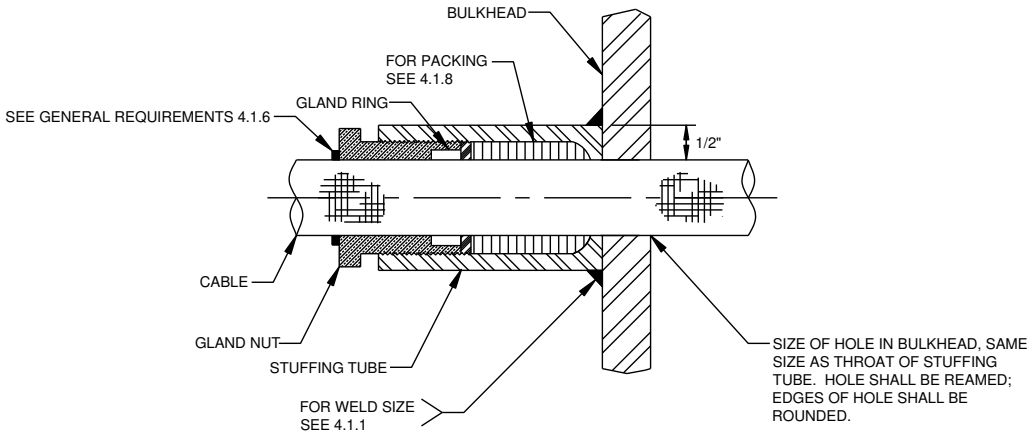
NOTES:

1. Pipes used in conjunction with stuffing tubes shall be welded on one side of the bulkhead penetrated except for the following types of bulkheads where welds on both sides are required:
   a. Ballistic bulkheads
   b. Shell plating
   c. Torpedo defense system bulkheads
   d. Penetrations exposed to weather or wet spaces
2. Plastic sealer is only required on kickpipes, stuffing tubes, etc. when exposed to the elements (penetrates the skin of the ship).
3. See 4.8 for thread lubrication guidance.
4. See figure 3A1 for gland nut tightening methods.

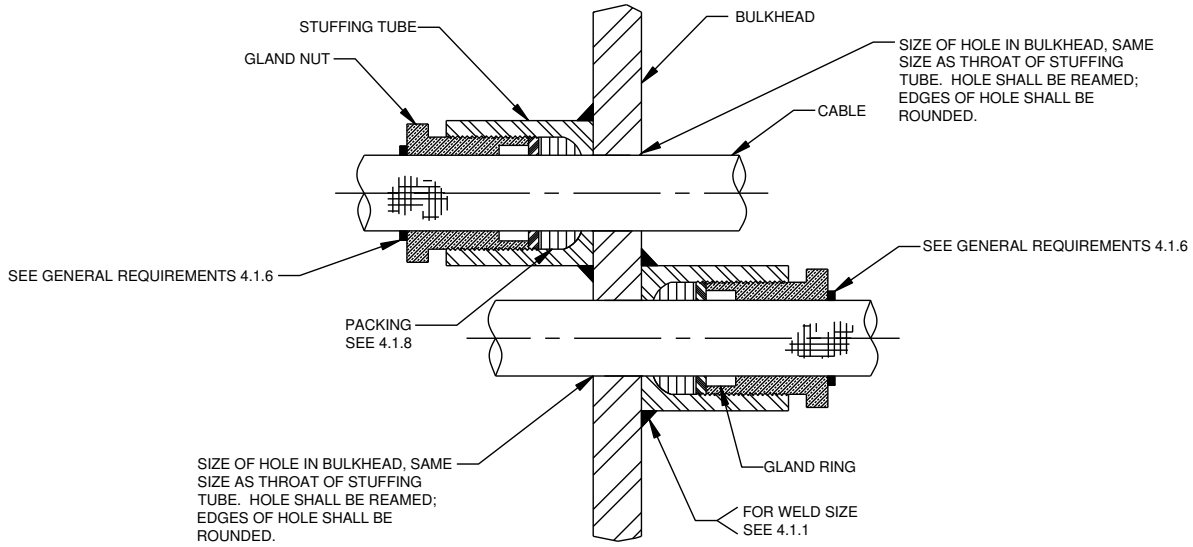
FIGURE 3B5. MIL-S-24235 stuffing tubes for steel or aluminum bulkheads (surface ships) – Continued.

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METHOD 3B-6-1
CABLES THROUGH
WATERTIGHT MEDIUM STEEL BULKHEADS
WITHOUT INSULATION



METHOD 3B-6-2
STUFFING TUBE INSTALLATION IN CONGESTED LOCATIONS
WATERTIGHT MEDIUM STEEL BULKHEAD
WITHOUT INSULATION

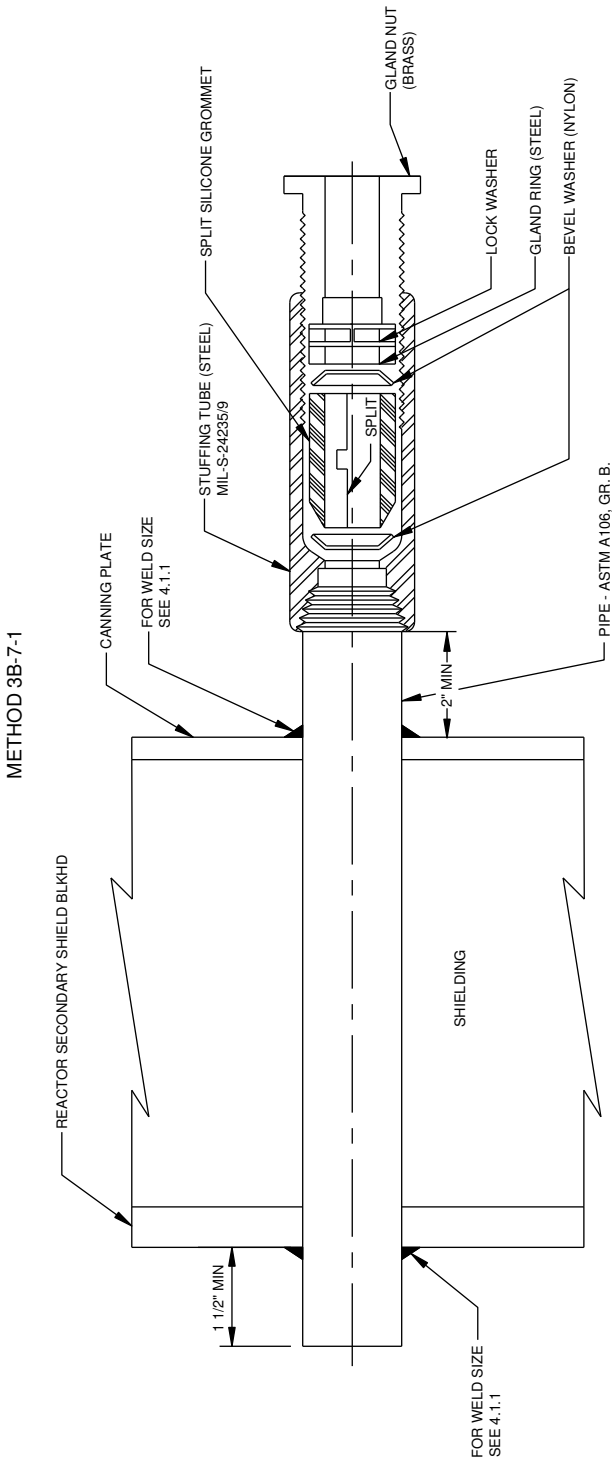


NOTES:

1. Plastic sealer is only required on kickpipes, stuffing tubes, etc. when exposed to the elements (penetrates the skin of the ship).
2. See 4.8 for thread lubrication guidance.
3. See figure 3A1 for gland nut tightening methods.

FIGURE 3B6. MIL-S-24235 stuffing tubes for steel or aluminum bulkheads (surface ships).

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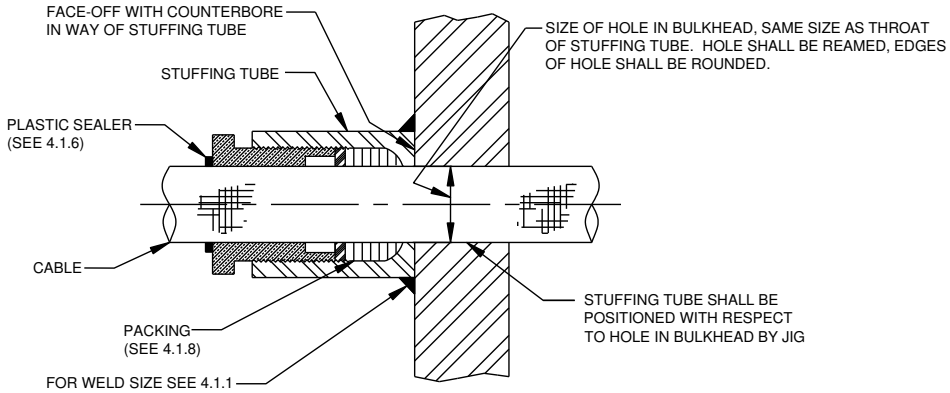
NOTES:

1. Plastic sealer is only required on kickpipes, stuffing tubes, etc. when exposed to the elements (penetrates the skin of the ship).
2. See 4.8 for thread lubrication guidance.
3. See figure 3A1 for gland nut tightening methods.

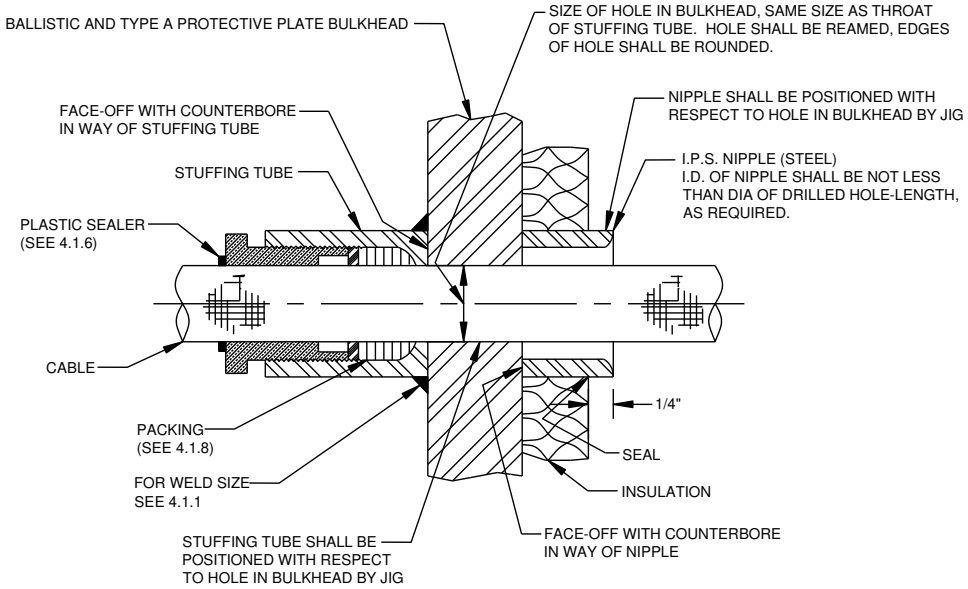
FIGURE 3B7. MIL-S-24235 stuffing tubes through shielded bulkheads (surface ships).

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METHOD 3B-8-1
CABLES THROUGH BALLISTIC AND TYPE A PROTECTIVE PLATE BULKHEADS
WITHOUT INSULATION



METHOD 3B-8-2
CABLES THROUGH BALLISTIC AND TYPE A PROTECTIVE PLATE BULKHEADS
WITH INSULATION



NOTES:

1. Plastic sealer is only required on kickpipes, stuffing tubes, etc. when exposed to the elements (penetrates the skin of the ship).
2. For information on the definition of a type A protective plate bulkhead or a ballistic bulkhead, see 4.1.3.
3. See 4.8 for thread lubrication guidance.
4. See figure 3A1 for gland nut tightening methods.

FIGURE 3B8. MIL-S-24235 stuffing tubes for ballistic and type A protective plate bulkheads (surface ships).

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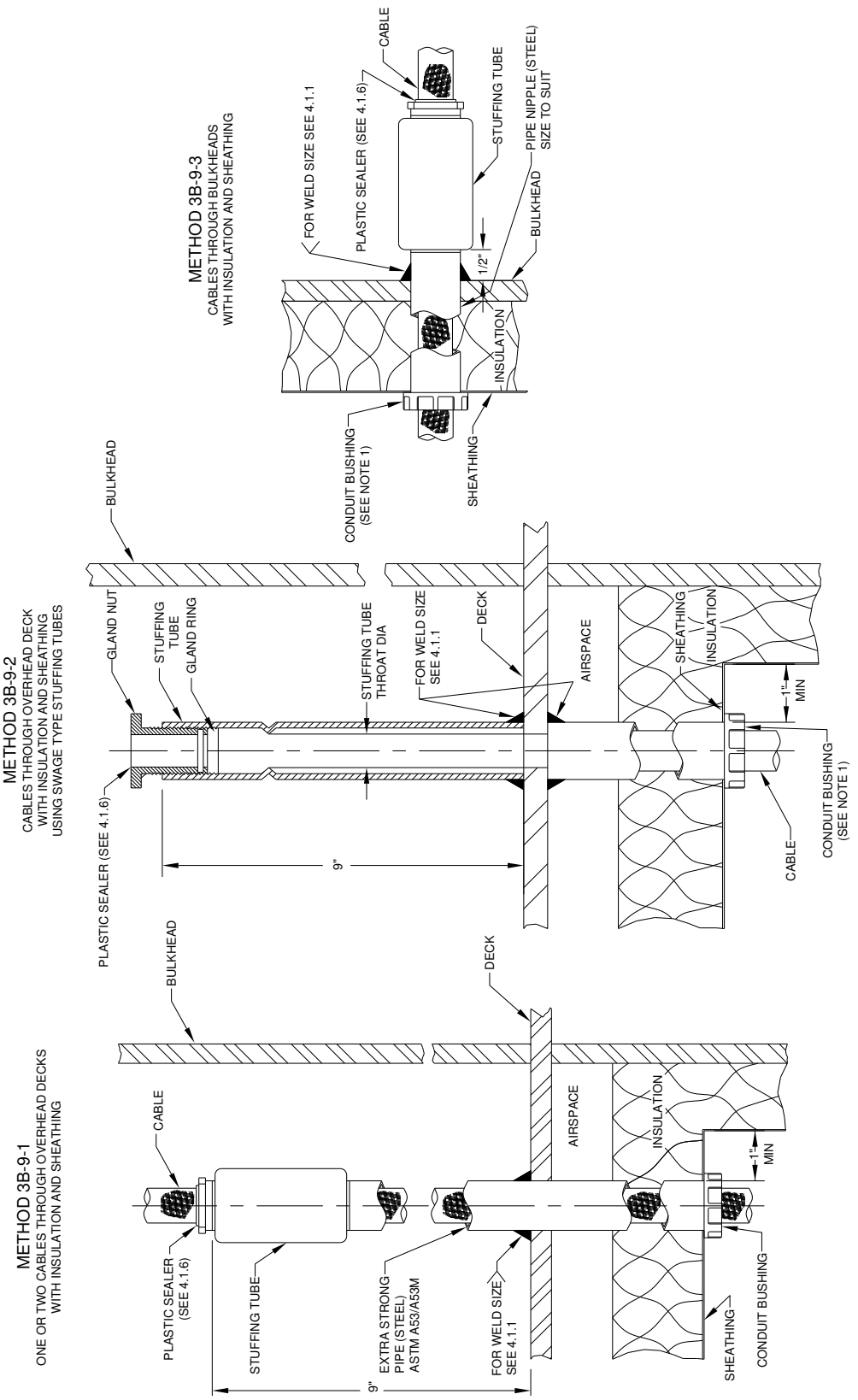


FIGURE 3B9. MIL-S-24235 stuffing tubes through acoustical spaces (surface ships).

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NOTES:

1. Pipe nipples and ends of kickpipes shall be threaded for installation of conduit bushing.
2. Plastic sealer is only required on kickpipes, stuffing tubes, etc. when exposed to the elements (penetrates the skin of the ship).
3. See 4.8 for thread lubrication guidance.
4. See figure 3A1 for gland nut tightening methods.

FIGURE 3B9. MIL-S-24235 stuffing tubes through acoustical spaces (surface ships) – Continued.

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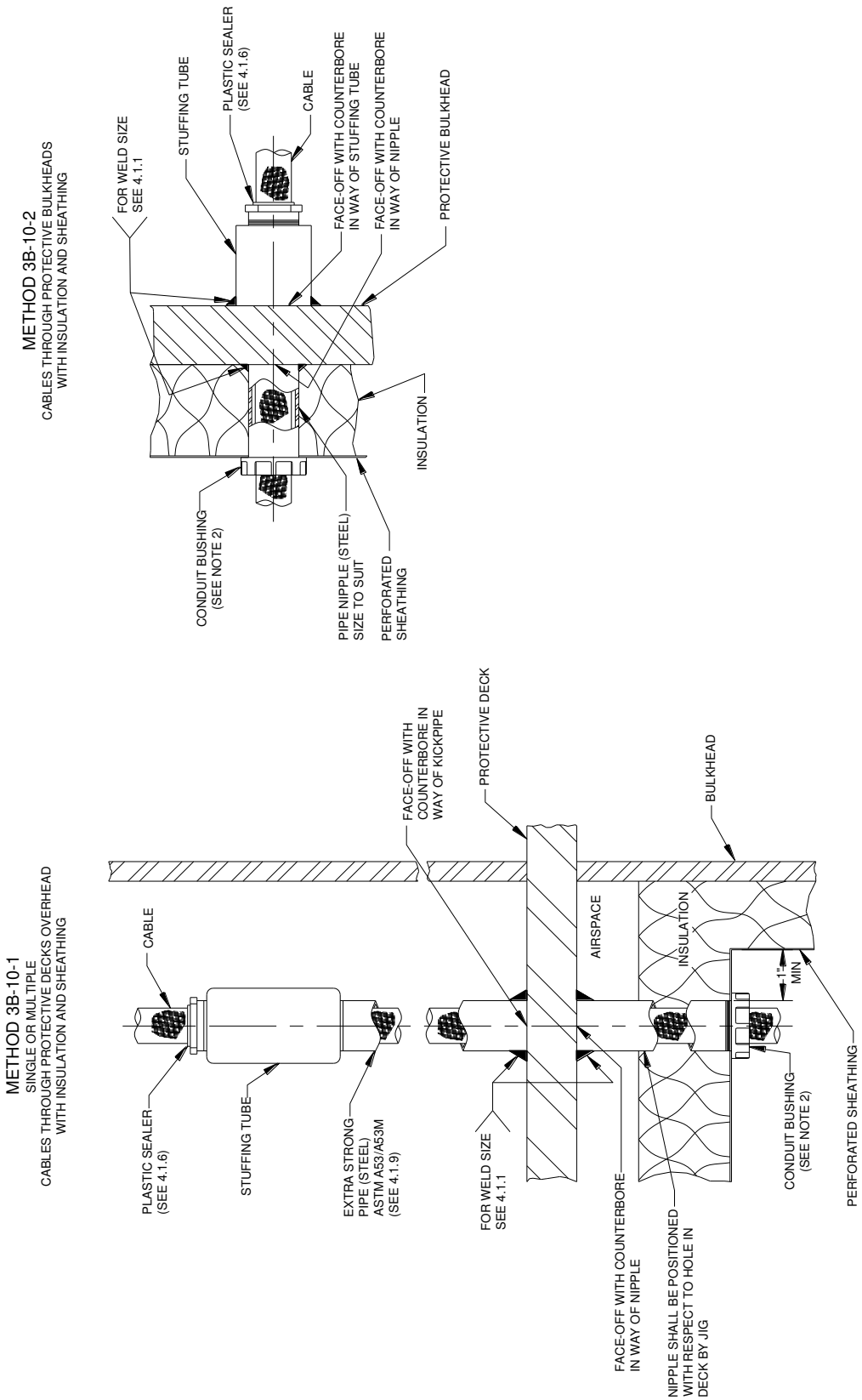


FIGURE 3B10. MIL-S-24235 stuffing tubes through acoustical spaces (surface ships).

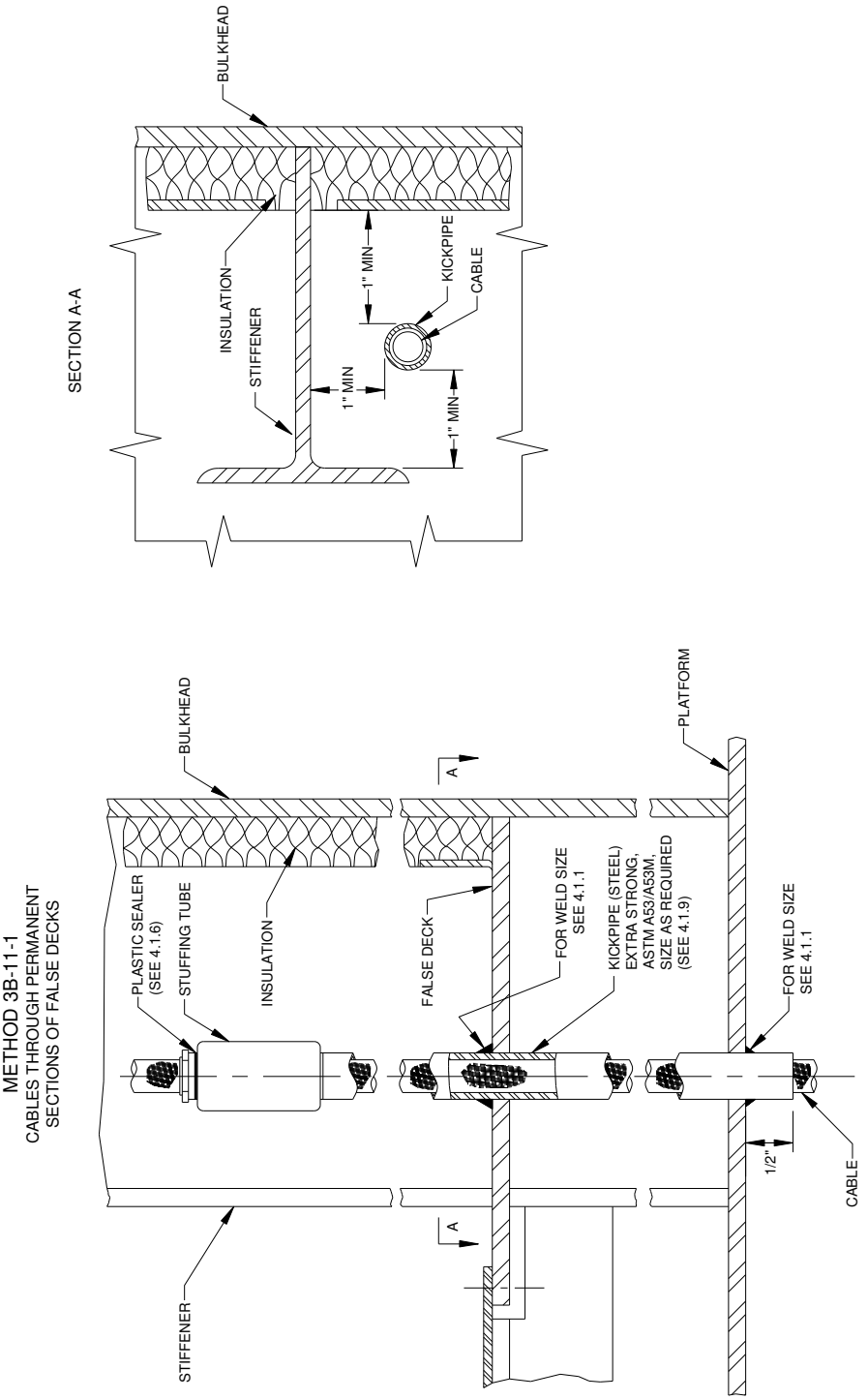
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NOTES:

1. Pipes used in conjunction with stuffing tubes shall be welded on both sides of the bulkhead for the following types of bulkheads:
   a. Ballistic bulkheads
   b. Shell plating
   c. Torpedo defense system bulkheads
   d. Penetrations exposed to weather or wet spaces
2. Pipe nipples shall be threaded for installation of conduit bushing.
3. Plastic sealer is only required on kickpipes, stuffing tubes, etc. when exposed to the elements (penetrates the skin of the ship).
4. See 4.8 for thread lubrication guidance.
5. See figure 3A1 for gland nut tightening methods.

FIGURE 3B10. MIL-S-24235 stuffing tubes through acoustical spaces (surface ships) – Continued.

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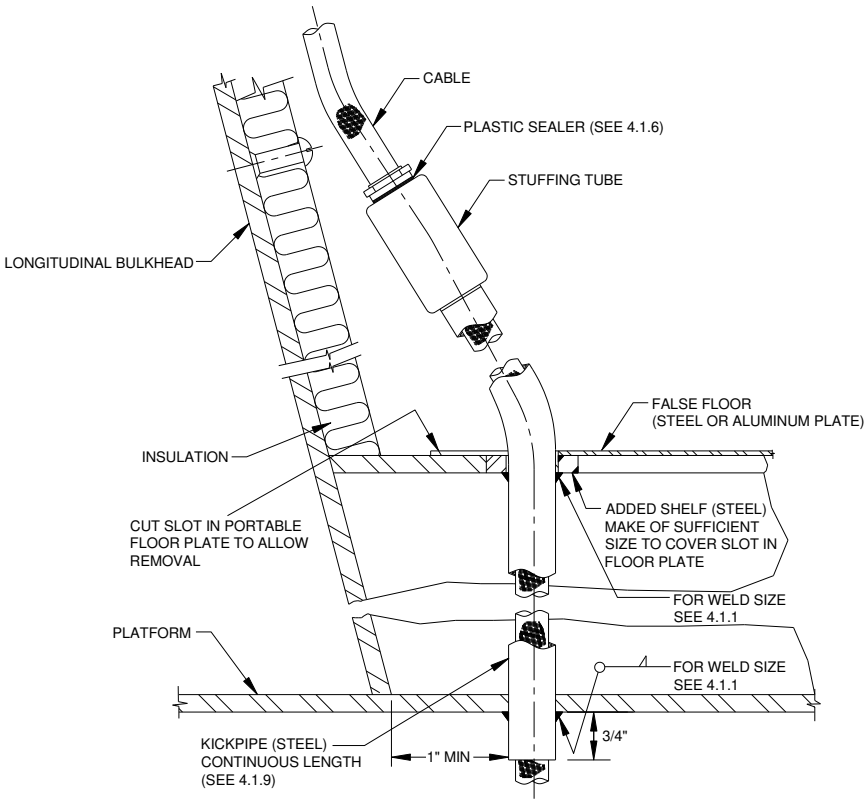
NOTES:

1. The kickpipe can either be straight or bent to suit installation requirements.
2. See 4.8 for thread lubrication guidance.
3. See figure 3A1 for gland nut tightening methods.

FIGURE 3B11. MIL-S-24235 stuffing tubes through permanent sections of false decks (surface ships).

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METHOD 3B-12-1
CABLES THROUGH PORTABLE
SECTIONS OF FALSE DECKS

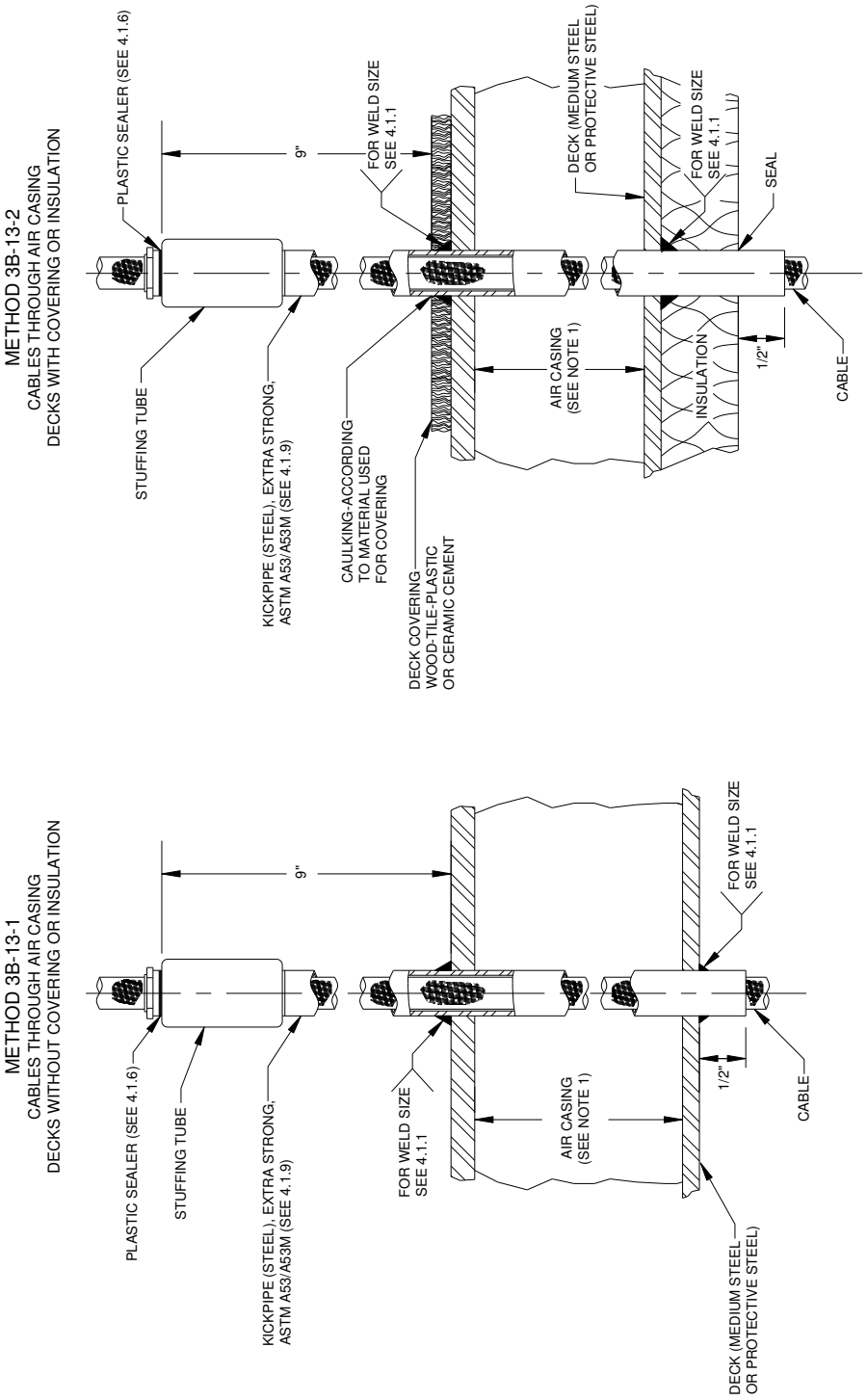


NOTES:

1. For suitable cable support, see MIL-STD-2003-4.
2. See 4.8 for thread lubrication guidance.
3. See figure 3A1 for gland nut tightening methods.

FIGURE 3B12. MIL-S-24235 stuffing tubes through portable sections of false decks (surface ships).

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NOTES:

1. For kickpipe installations through protective decks bounding air casings. The kickpipe shall extend through the air casing and both decks in one continuous piece.
2. See 4.8 for thread lubrication guidance.
3. See figure 3A1 for gland nut tightening methods.

FIGURE 3B13. MIL-S-24235 stuffing tubes through air spaces (surface ships).

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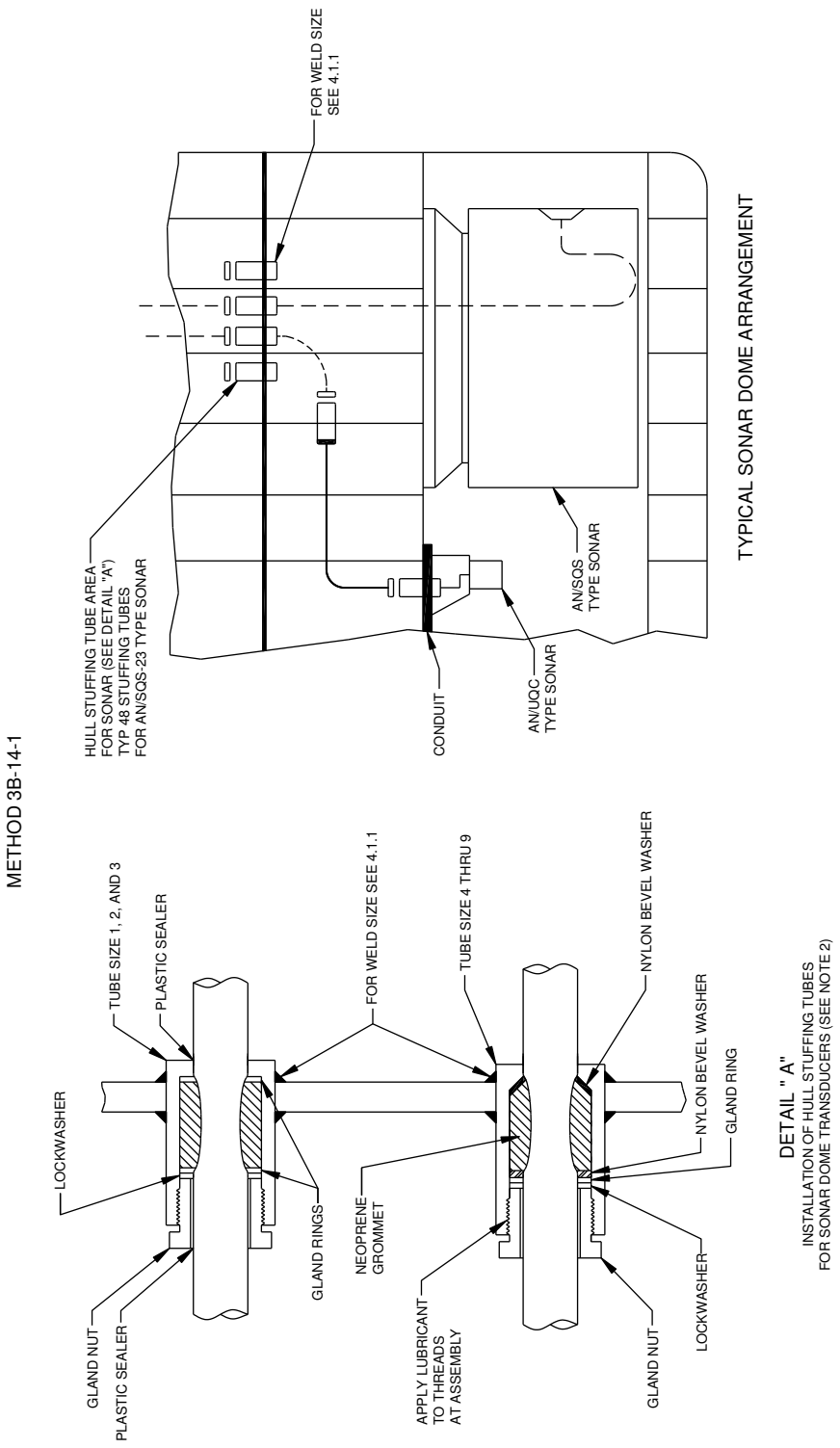


FIGURE 3B14. Stuffing tubes for sonar dome area (surface ships).

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NOTES:

1. The stuffing tube shown is equivalent to one-half of a standard submarine type double-ended stuffing tube in accordance with MIL-S-24235/1 (½ of the "M" dimension). See applicable notes on figure 3A1.
2. The arrangement of hull stuffing tube area for sonar dome transducers shall be as specified on detailed construction drawings. Installation of stuffing tubes shall be in accordance with this figure.
3. Holes for stuffing tubes shall be drilled or otherwise mechanically cut.
4. Stuffing tube spacing shall be in accordance with figures 3B1 and 3B2.
5. Plastic sealer use should be in accordance with 4.1.6.
6. See 4.8 for thread lubrication guidance.
7. For size 9 stuffing tubes, the nylon bevel washer to the left of the grommet shown in the above illustration may be replaced by a flat washer to avoid the washer cutting into cable when the gland nut is torqued.
8. In applications where one side of the penetration is submerged, when cables have an armored layer under the outer jacket, such as LSMDY type, the outer jacket and armored layer shall be stripped and removed from within 1-2 inches of the bulkhead penetrations and through the penetration. This ensures that any damage that may occur to the cable outer jacket does not permit water to traverse or wick along the underlying armor layer and through the penetration.
9. See figure 3A1 for gland nut tightening methods.

FIGURE 3B14. Stuffing tubes for sonar dome area (surface ships) – Continued.

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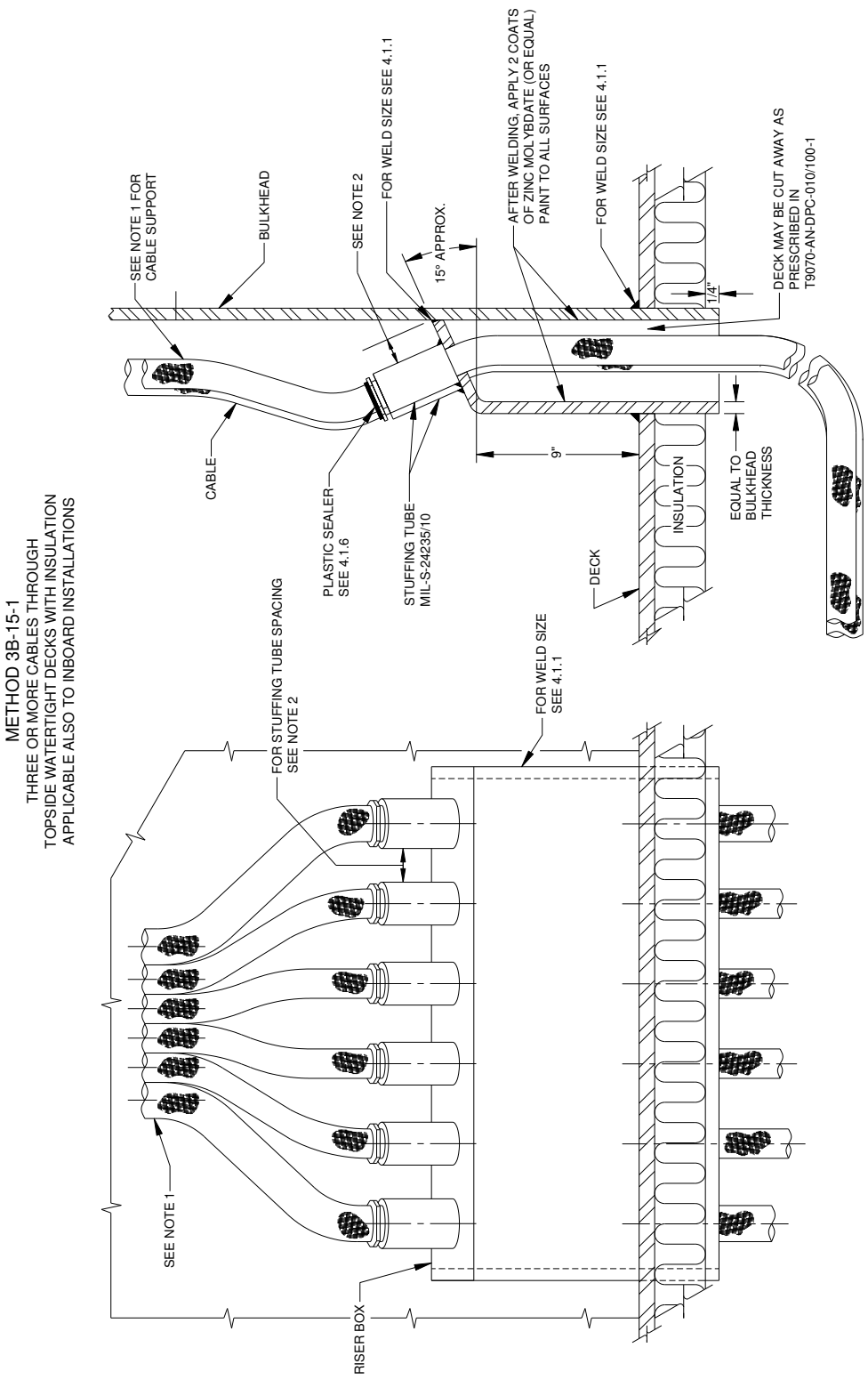


FIGURE 3B15. Topside stuffing tubes with riser box (surface ships).

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NOTES:

1. For suitable cable support, see MIL-STD-2003-4.
2. Tubes shall be spaced 1 inch minimum from bulkhead and from each other in accordance with figures 3B2 and 3B3.
3. See 4.8 for thread lubrication guidance.
4. See figure 3A1 for gland nut tightening methods.

FIGURE 3B15. Topside stuffing tubes with riser box (surface ships) – Continued.

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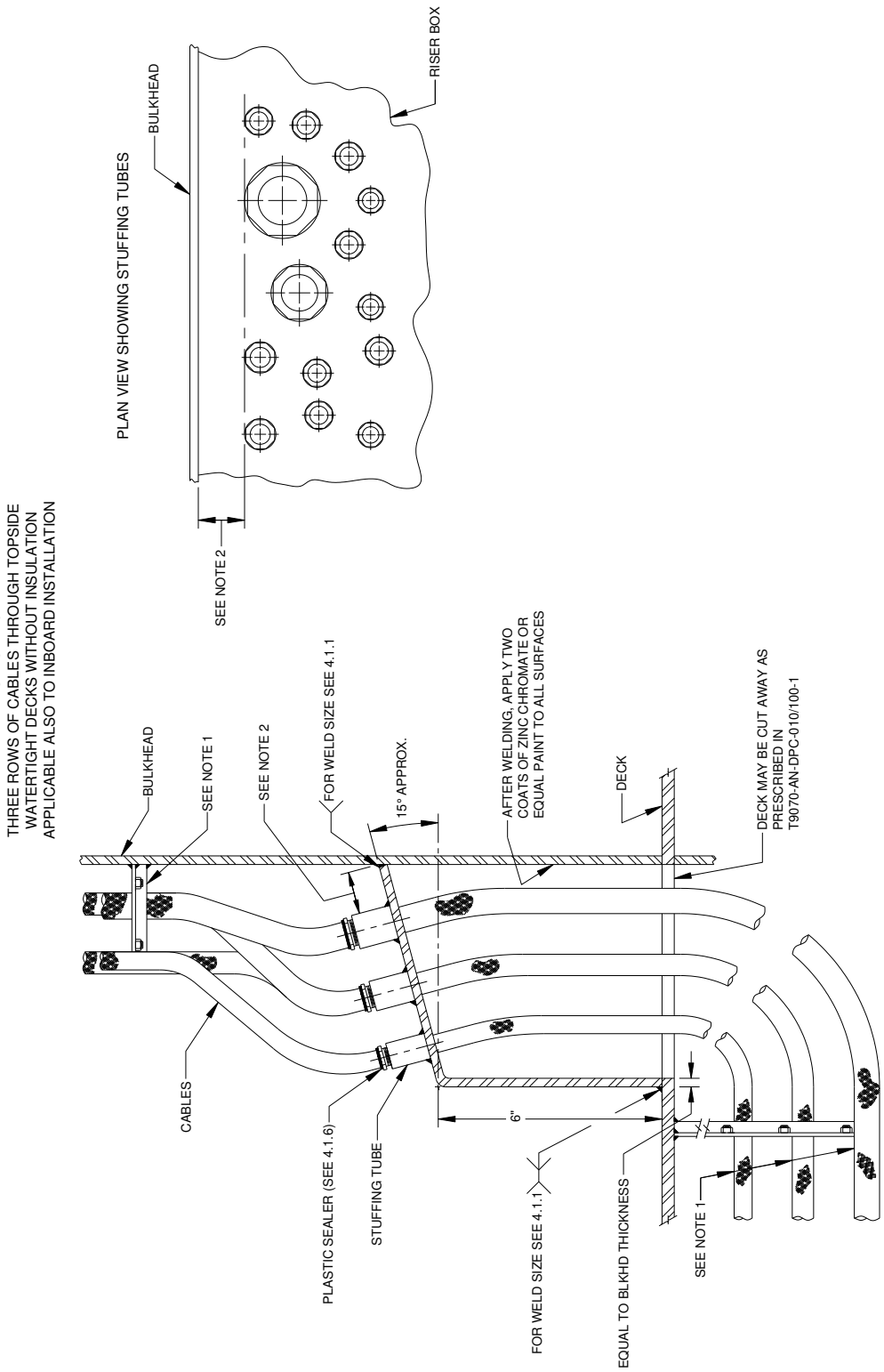


FIGURE 3B16. MIL-S-24235 topside stuffing tubes with riser box for medium steel decks (surface ships).

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NOTES:

1. For suitable cable support, see MIL-STD-2003-4.
2. Tubes shall be spaced 1 inch minimum from bulkhead and from each other in accordance with stuffing tube spacing figures 3B2 and 3B3.
3. See 4.8 for thread lubrication guidance.
4. See figure 3A1 for gland nut tightening methods.

FIGURE 3B16. MIL-S-24235 topside stuffing tubes with riser box for medium steel decks (surface ships) – Continued.

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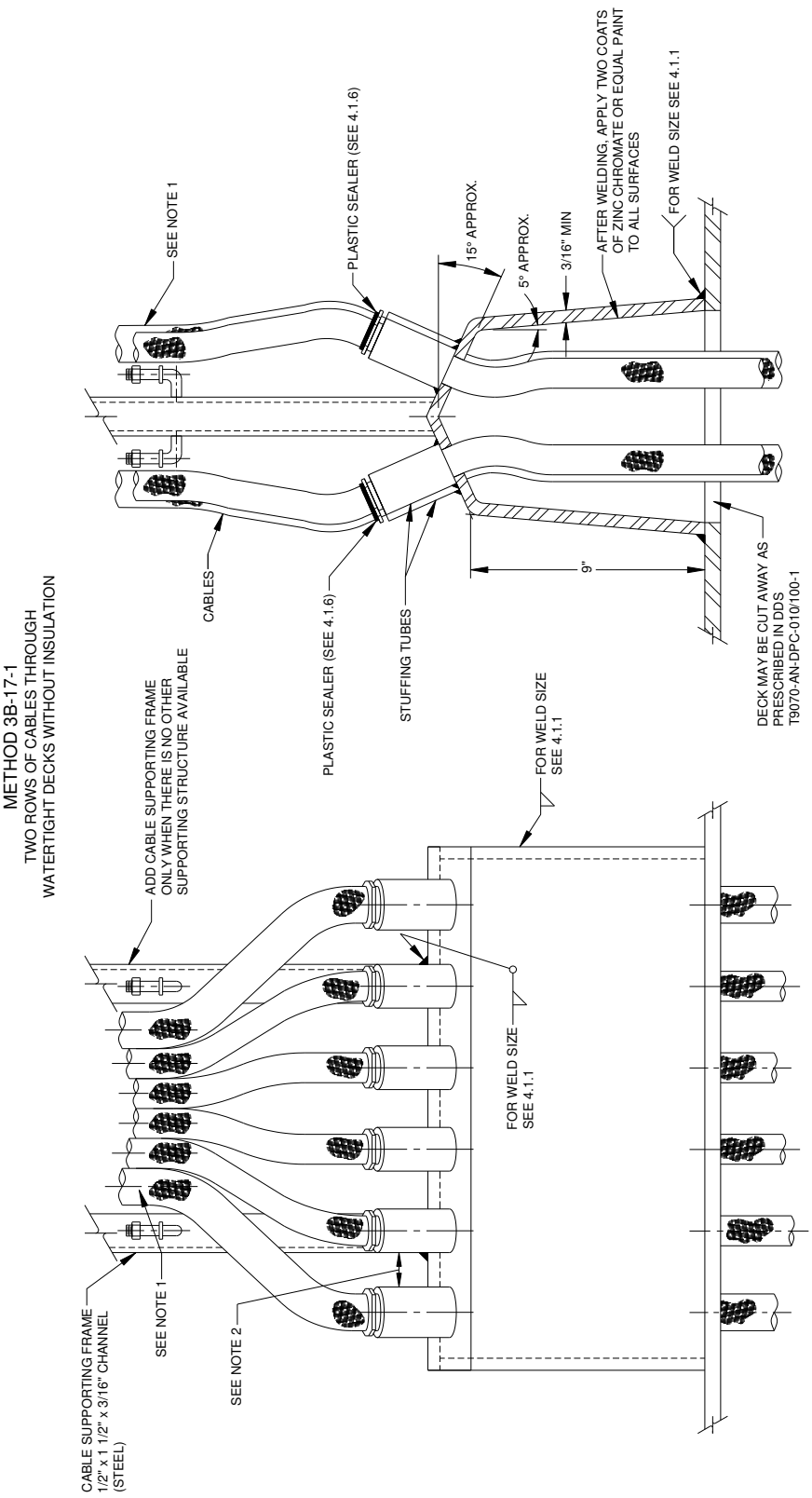


FIGURE 3B17. MIL-S-24235 topside stuffing tubes with riser box on medium steel decks not adjacent to a bulkhead (surface ships).

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NOTES:

1. For suitable cable support, see MIL-STD-2003-4.
2. Stuffing tubes shall be spaced from each other and framework as prescribed on figures 3B2 and 3B3.
3. See 4.8 for thread lubrication guidance.
4. See figure 3A1 for gland nut tightening methods.

FIGURE 3B17. MIL-S-24235 topside stuffing tubes with riser box on medium steel decks not adjacent to a bulkhead (surface ships) – Continued.

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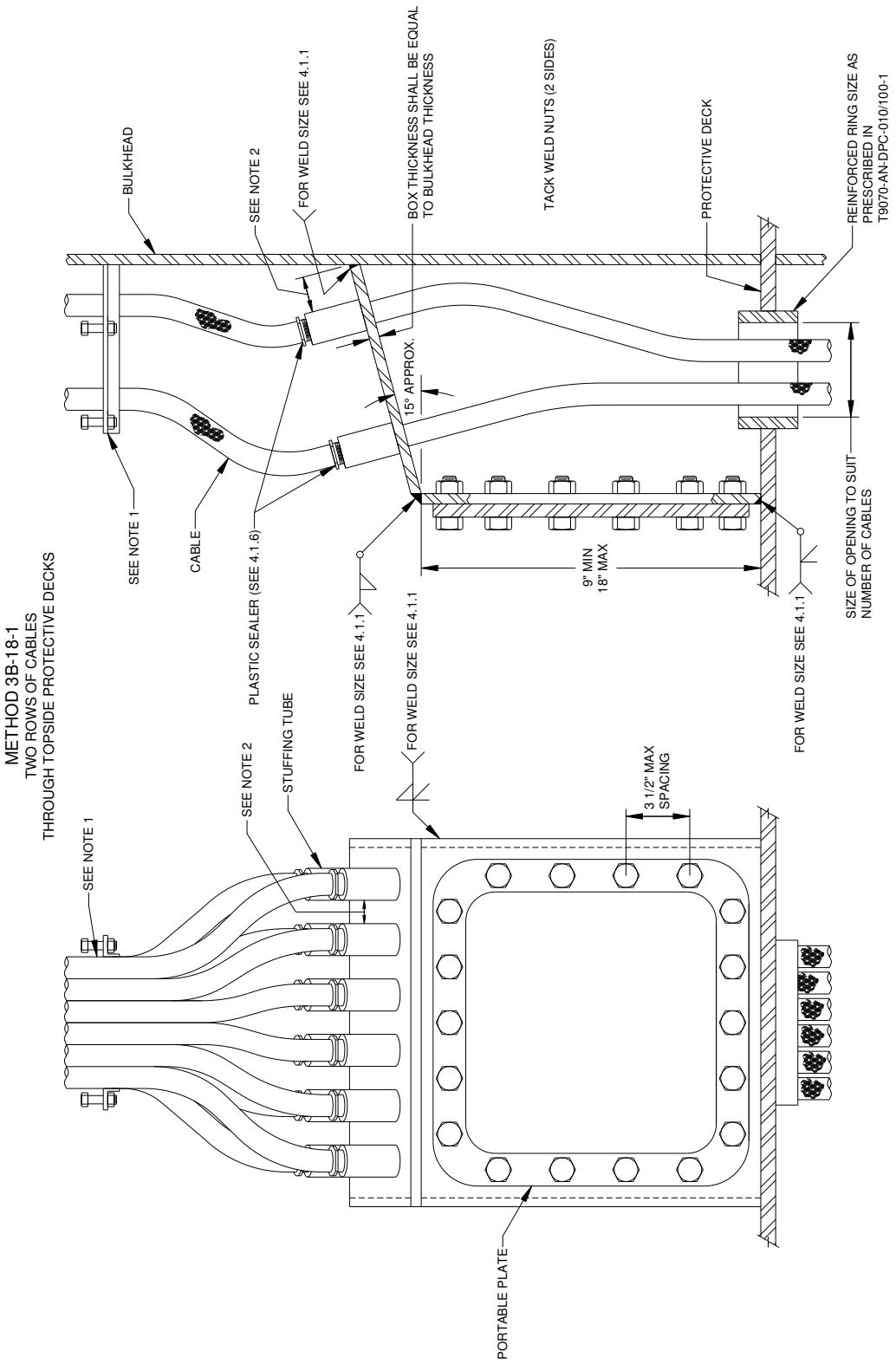


FIGURE 3B18. MIL-S-24235 topside stuffing tubes with riser box (surface ships).

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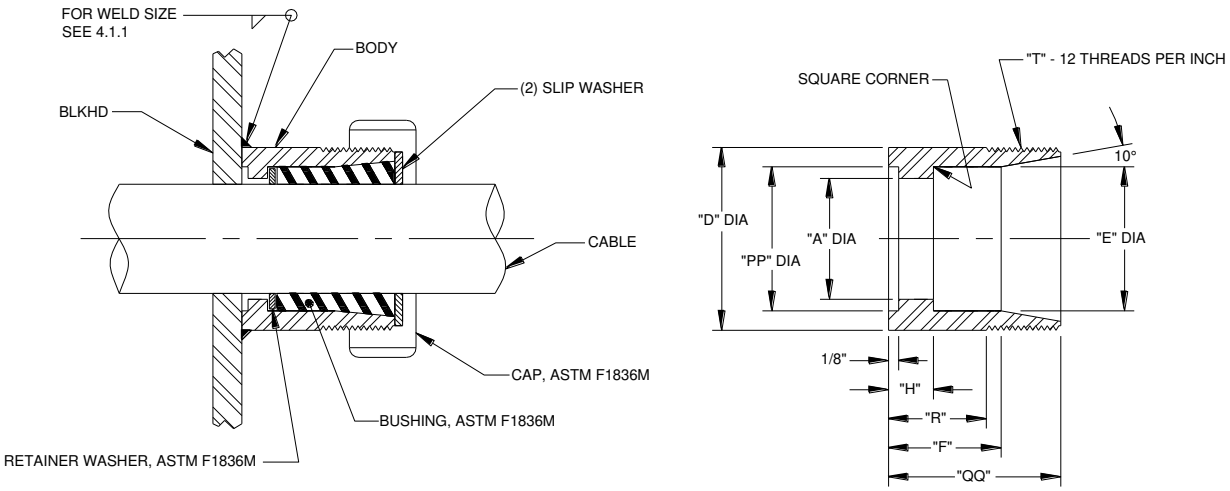
NOTES:

1. For suitable cable support, see MIL-STD-2003-4.
2. Tubes shall be spaced 1 inch minimum from bulkhead and from each other in accordance with stuffing tube spacing on figures 3B2 and 3B3.
3. See 4.8 for thread lubrication guidance.
4. See figure 3A1 for gland nut tightening methods.

FIGURE 3B18. MIL-S-24235 topside stuffing tubes with riser box (surface ships) – Continued.

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METHOD 3B-19-1
(ASSEMBLED VIEW AND BODY VIEW)



ASSEMBLED VIEW

BODY VIEW

TABLE 3B19-I. Stuffing tube body dimensions.

Tube size	Tube body										Clearance hole in bulkhead	Std. pipe size for riser
	A	D	E	F	H	R	T		PP	QQ		
1	0.3500	0.8750	0.5600	1.1870	0.7500	0.8430	0.8750		0.7100	1.437	0.750	0.375
2	0.5640	1.0000	0.6250	1.1870	0.7500	0.8440	1.0000		0.8750	1.500	0.875	0.500
3	0.6870	1.1250	0.7650	1.1870	0.7500	0.8440	1.1250		0.8750	1.500	1.000	0.500
4	0.8250	1.2500	0.9060	1.1870	0.7500	0.8440	1.2500		1.0630	1.500	1.125	0.750
5	1.1870	2.0000	1.3430	1.5000	0.7500	0.8750	2.0000		1.6880	1.875	1.625	1.125
6	1.5050	2.5000	1.7500	1.5620	0.7500	0.8750	2.5000		1.9380	2.125	1.938	1.500
7	1.6890	2.7500	1.9370	1.6250	0.7500	0.8750	2.7500		2.4380	2.250	2.125	2.000
8	2.0890	3.2500	2.3150	1.7500	0.7500	1.0000	3.1250		2.9380	2.500	2.562	2.500
9	2.8420	4.0000	3.2500	1.8430	0.7500	1.1880	4.0000		3.5630	2.750	3.312	3.000

NOTES:

1. Method 3B-19-1 is for welding to kickpipe (riser) and bulkheads.
2. These stuffing tubes are authorized for use only on weather decks in deck house bulkheads and below decks above the tightness level in surface ships.
3. See 4.8 for thread lubrication guidance.
4. See figure 3A1 for gland nut tightening methods.

FIGURE 3B19. Stuffing tube for steel and aluminum bulkheads (surface ships).

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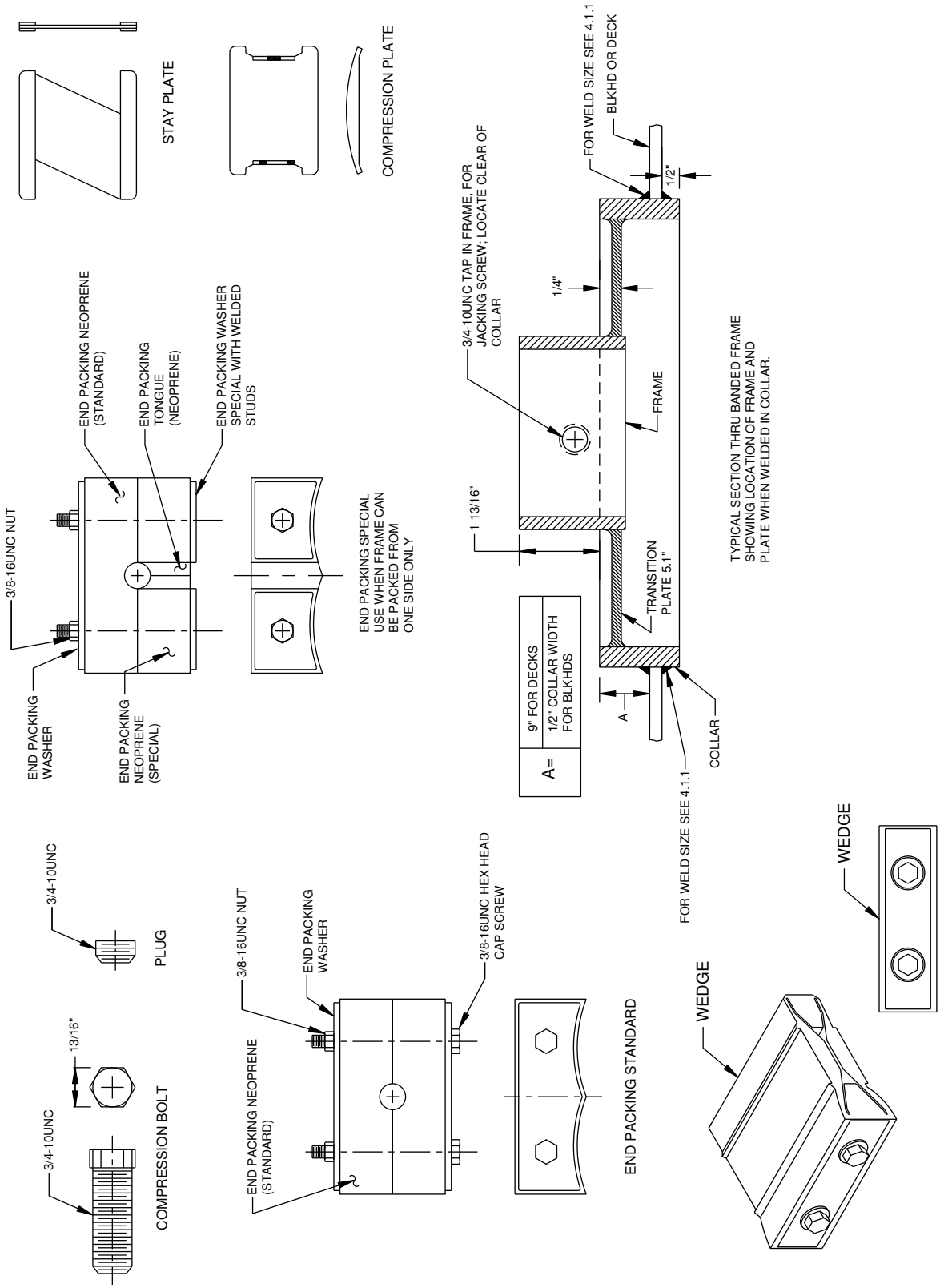
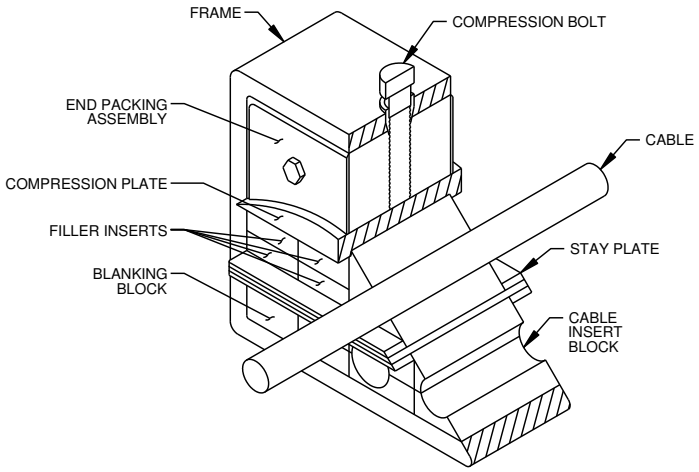


FIGURE 3B20. Multiple cable penetrator nomenclature.

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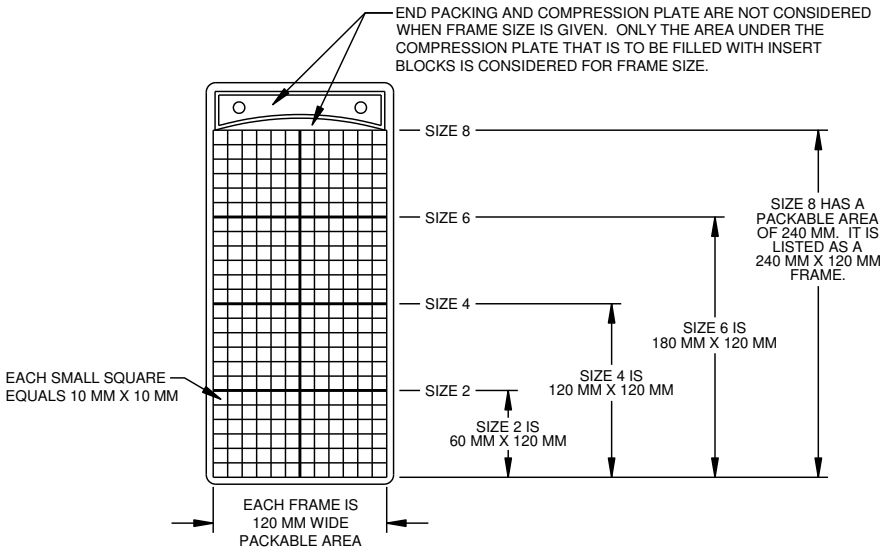


FIGURE 3B20. Multiple cable penetrator nomenclature – Continued.

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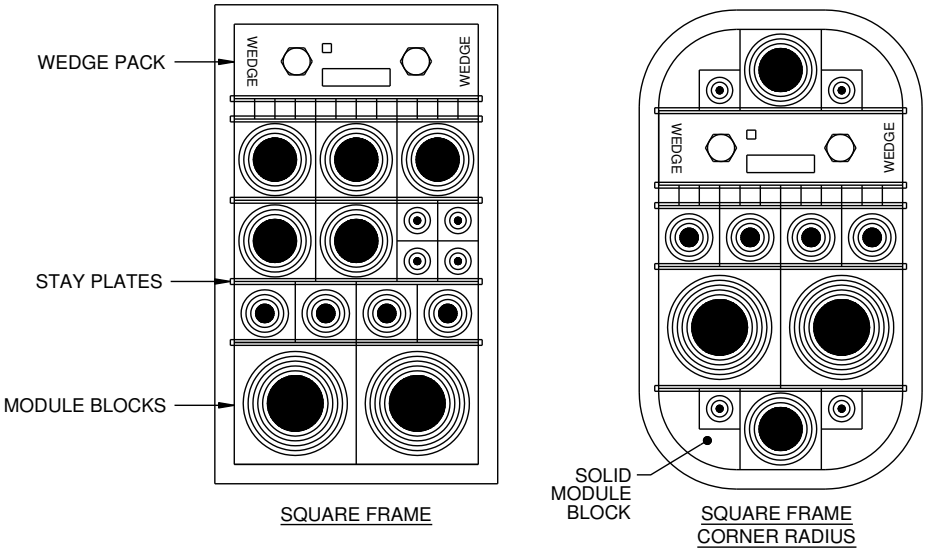


FIGURE 3B20. Multiple cable penetrator nomenclature – Continued.

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Multiple cable penetrators (MCP):

A system of passing cables through watertight and non-watertight bulkheads and decks designed to suit particular bulkhead and deck weights. MCP components shall meet the requirements of MIL-DTL-24705. In general, the MCP component views, tables, and installation instructions in figures 3B20-3B30 apply. However, there are several different MCP products that meet the requirements of MIL-DTL-24705. Therefore, to achieve satisfactory installation, manufacturer's instructions should be used in addition to the instructions in this standard.

Nomenclature:

Throughout this standard of electric plant installation drawings, the following descriptions apply:

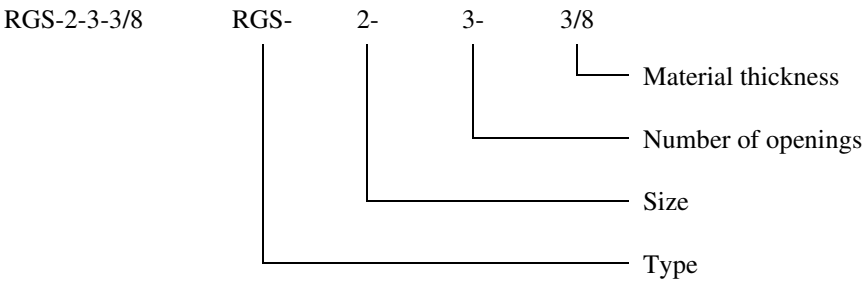
a. Rectangular frames and square frame corner radius, unbanded
   (1) Consist of four basic sizes; 2, 4, 6, and 8. All are 5.506 inches wide. Size 2 is 4.726 inches high; size 4 is 7.029 inches high; size 6 is 9.333 inches high; and size 8 is 11.660 inches high.
   (2) Material shall be either steel or aluminum, ¼ or ⅜ thick (depending on weight of bulkhead or deck) in steel or ½ thick in aluminum. All frames are 2⅜ deep.
   (3) There are four different type frames:
   RGS – Steel, four corners welded (completely closed)
   RGSO – Steel, two corners welded (open bottom)
   RGA – Aluminum, four corners welded (completely closed)
   RGAO – Aluminum, two corners welded (open bottom)
   RGSC – Steel or aluminum, square frame, rounded corners, see figure 3B64
   (4) To recap sub-paragraphs (1) through (3), an unbanded frame is identified as follows: an RGS-2 x ⅜ is a ⅜-thick size 2 steel frame with four corners welded (completely closed).

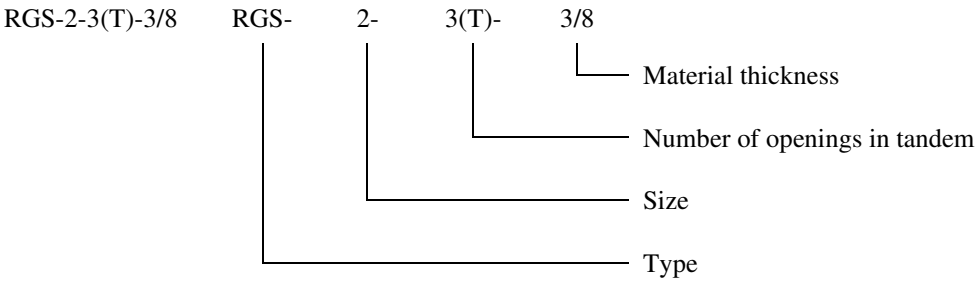
b. Rectangular frames, banded:
   (1) Consist of the four basic sizes in sub-paragraph a(1), each of which is enclosed by a reinforcing band of proper size material to suit particular bulkhead weight. In addition to the band filler, plates are employed to seal empty spaces between frame and band.
   (2) RGS and RGA banded frames are complete units ready to be installed in bulkhead. RGSO and RGAO banded frames have lower half of band and frame temporarily tack welded. At installation, this tack weld is broken, frame is wrapped around cables, and all welding is completed.

c. Physical arrangement and identification
   (1) Various arrangements of frames are available in all sizes and types. Examples and dimensions are shown on figures 3B24, 3B25, and 3B26.
   (2) Identification of various arrangements are made by the following examples:

FIGURE 3B20. Multiple cable penetrator nomenclature – Continued.

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Single or double wall, banded or unbanded shall be specified.

FIGURE 3B20. Multiple cable penetrator nomenclature – Continued.

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Nomenclature (continued):

d. Compression screw or bolt:
   (1) Stainless steel, ¾-10UNC thread, when tightened, seats the compression plate farther down into the transit frame
e. Plug:
   (1) Stainless steel, ¾-10UNC thread used to plug tapped hole in frame opposite compression screw
f. Stay plate:
   (1) Either steel or aluminum normally placed between every other row of insert blocks keeping them positioned in the transit frame.
g. Compression plate:
   (1) Either cast iron or cast aluminum seats and compresses the insert blocks so that the end packing can be inserted in the transit frame.
h. End packing – standard (see item q)
   (1) Compresses the insert blocks used when frame can be packed from either side
   (2) Assembly, which consists of two neoprene elastomer blocks and two end packing washers held together by two ⅜-16UNC hex head cap screws and hex nuts. Washers, nuts, and cap screws are steel cadmium plated. For non-magnetic applications, stainless steel is used.
i. End packing – special (see item q):
   (1) Compresses the insert blocks used when frame can only be packed from one side
   (2) Assembly, which consists of four neoprene elastomer pieces, one end packing washer, two end packing washers special, and two ⅜-16UNC hex nuts, all are steel cadmium plated. For non-magnetic applications, stainless steel is used.
j. Insert blocks:
   (1) Twin half blocks are specially formulated of a neoprene elastomer with a centered semi-circular groove. When matched around a cable, these blocks form a single block with a tight fit. Basic sizes are shown on figure 3B21 for information and guidance.
k. Spares:
   (1) Solid blocks are specially formulated of a neoprene elastomer and are used as fillers or spare provisions for addition of future cables. Sizes are shown on figure 3B21 for information and guidance. When installing new MCPs, at least 20 percent of the MCP usable area shall be reserved for future cable installations.
l. Fill-ins:
   (1) Same material as insert blocks and spares used to pack thicknesses of 5 or 10 millimeters caused by the presence of different size blocks in the same row. Also employed to increase the pressure in the penetrator frame when exceptionally soft cables are used. These fill-ins are available in two sizes 24 x 5/0 or 12 x 10/0 (see figure 3B21). Serrations have been provided to permit slicing of the fill-ins to the required length.

Example: 6 x 10/0, 8 x 5/0, etc.



m. Tallow:
   (1) Tallow is a natural lubricant that allows the blocks, stayplates, compression system and cables to move freely during the compression process. Without tallow, the friction during compression will create gaps in the system and an effective seal will not be obtained. Therefore, as stated in the installation instructions in figures 3B22, 3B39, 3B41, and 3B63, tallow must be applied for every insert block.
n. Sealer:
   (1) Liquid silicone rubber applied to cable side of each gauge block upon compression. This provides a seal between the armor of the cable and its impervious inner sheath.

FIGURE 3B20. Multiple cable penetrator nomenclature – Continued.

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Nomenclature (continued):

o. Nut:

(1) All nuts are to be in accordance with NASM 25027.

p. All dimensions are in inches unless otherwise stated.

q. Wedge: a one-piece compression system that may be used in place of compression plates and end-packing. The wedge may be installed from one side and may be installed anywhere in the frame.

NOTE:

1. Multiple cable penetrators shall not be used in bulkheads or decks which are exposed to the weather except for electro-magnetic compatibility (EMC) type penetrations covered in figure 3B63.

FIGURE 3B20. Multiple cable penetrator nomenclature – Continued.

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TABLE 3B21-I. Cable information and assignment for MCP insert blocks.

Cable type	Cable diameter in inches	Cable diameter in mm	Insert block part number size mm 3/	Insert block part number size mm 3/
SSGA-3	0.305	7.75	15/7	20/7
SSGA-4	0.323	8.20	15/8	20/8
SSGA-9	0.371	9.42	15/9	20/9
SSGA-14	0.414	10.51	20/10	
SSGA-23	0.453	11.50	20/11	
SSGA-30	0.484	12.30	20/12	30/12
SSGA-40	0.515	13.08	20/13	30/13
SSGA-150	0.844	21.44	30/21	
SSGA-500	1.290	32.77	40/30	
SSGA-1300	1.880	47.75	60/48	
DCOP-3	0.425	10.80	20/10	
DCOP-4	0.460	11.68	20/12	30/12
DCOP-6	0.510	12.95	20/13	30/13
DCOP-9	0.570	14.48	20/14	30/14
DCOP-14	0.705	17.91	30/18	
DCOP-23	0.860	21.84	30/22	
DCOP-30	0.960	24.38	30/24	40/24
DCOP-83	1.450	36.83	60/36	
DCOP-250	2.100	53.34	60/52	
DCOP-400	2.500	63.50	90/60	
DSGA-14	0.680	17.27	30/17	
DSGA-23	0.781	19.84	30/19	
DSGA-30	0.852	21.64	30/21	
DSGA-40	0.898	22.81	30/22	40/22
DSGA-50	0.961	24.41	30/24	40/24
DSGA-60	1.031	26.19	40/26	
DSGA-125	1.374	34.90	40/34	60/34
DSGA-150	1.479	37.57	60/36	
DSGA-250	1.759	34.68	60/44	

TABLE 3B21-I. Cable information and assignment for MCP insert blocks – Continued.

Cable type	Cable diameter in inches	Cable diameter in mm	Insert block part number size mm 3/	Insert block part number size mm 3/
(TTHFWA)				
TT-1-1/2	0.380	9.65	15/9	20/9
TT-3	0.500	12.70	20/12	30/12
TT-5	0.590	14.99	20/14	30/14
TT-10	0.690	17.53	30/17	
TT-15	0.800	20.32	30/20	
TT-20	0.880	22.35	30/22	40/22
TT-30	1.030	26.16	40/26	
TT-40	1.130	28.70	40/28	
TT-50	1.265	32.13	40/32	60/32
TT-60	1.350	34.29	40/34	60/34
TTRSA-2	0.740	18.80	30/18	
TTRSA-4	0.800	20.32	30/20	
TTRSA-6	0.940	23.88	30/23	
TTRSA-8	1.050	26.67	40/26	
TTRSA-10	1.1.40	28.96	40/28	
TTRSA-12	1.160	29.46	40/28	
TTRSA-16	1.250	31.75	40/30	
2SA 2SWA-3	0.570	14.48	20/14	30/14
-7	0.710	18.03	30/18	
-10	0.880	22.35	30/22	40/22
-14	0.980	24.89	30/24	40/24
-19	1.090	27.69	40/26	
-24	1.260	32.00	40/30	
-30	1.330	34.80	40/34	60/34
-37	1.430	36.32	60/36	
-44	1.600	40.64	60/40	
-61	1.790	45.47	60/44	

TABLE 3B21-I. Cable information and assignment for MCP insert blocks – Continued.

Cable type	Cable diameter in inches	Cable diameter in mm	Insert block part number size mm 3/	Insert block part number size mm 3/
RG-12A/U	0.475	12.07	20/12	30/12
RG-22B/U	0.420	10.67	20/10	
RG-27A/U	0.670	17.02	30/17	
RG-58C/U	0.195	4.96	15/4	20/4
RG-59B/U	0.242	6.15	15/6	20/6
RG-62A/U	0.242	6.15	15/6	20/6
RG-63B/U	0.405	10.29	20/10	
RG-71B/U	0.250	6.35	15/6	20/6
RG-79B/U	0.475	12.07	20/12	30/12
RG-111A/U	0.490	12.45	20/12	30/12
RG-142B/U	0.195	4.96	15/4	20/4
RG-157/U	0.725	18.42	30/18	
RG-213/U	0.405	10.29	20/10	
RG-214/U	0.425	10.80	20/10	
RG-215/U	0.475	12.07	20/12	30/12
RG-218/U	0.870	22.10	30/22	40/22
RG-219/U	0.945	24.01	30/23	
RG-224/U	0.615	15.62	30/15	
RG-227/U	0.490	12.45	20/12	30/12
RG-236/U	0.500	12.70	20/12	30/12
RG-258/U	1.886	47.90	60/46	
RG-318/U	1.085	27.56	40/26	
RG-254/U	1.090	27.69	40/26	
RG-179B/U	0.100	2.54		
RG-333/U	1.052	26.72	40/26	

TABLE 3B21-I. Cable information and assignment for MCP insert blocks – Continued.

Cable type	Cable diameter in inches	Cable diameter in mm	Insert block part number size mm 3/	Insert block part number size mm 3/
MA-14	0.450	11.43	20/11	
TCOP-42	1.250	31.75	40/30	
MDGA-19(6)	1.285	32.63	40/32	60/32
MDGA-19(14)	1.520	38.61	60/38	
MDGA-19(23)	1.820	46.23	60/46	
MDGA-19(40)	2.100	53.34	60/52	
SPCLXDUCER 7819478P1	1.200	30.48	40/30	
ISMU-5	0.500	12.70	20/12	30/12

FIGURE 3B21. Cable information and assignment for MCP insert blocks.

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TABLE 3B21-I. Cable information and assignment for MCP insert blocks – Continued.

Cable type	Cable diameter in inches	Cable diameter in mm	Insert block part number size mm 3/	Insert block part number size mm 3/
TSGA-14	0.718	18.23	30/18	
TSGA-30	0.902	22.91	30/22	40/22
TSGA-40	0.950	24.13	30/24	40/24
TSGA-60	1.110	28.19	40/28	
TSGA-125	1.458	37.03	60/36	
TSGA-250	1.844	46.84	60/46	
TSGA-350	2.123	53.92	60/52	
FSGA-60	1.205	30.61	40/30	
MDY-6	1.190	30.23	40/30	
MDY-14	1.570	39.88	60/39	
MDY-23	1.960	49.78	60/49	
MDY-40	2.240	56.90	90/55	
MDY-60	2.525	64.14	90/63	
ISA-44	1.040	26.42	40/26	
ISMWA-70	1.605	40.77	60/40	
ISWA-2	0.505	12.83	20/12	30/12
ISWA-14	0.920	23.37	30/23	
ISWA-20	1.080	27.43	40/26	
ISWA-30	1.250	31.75	40/30	

TABLE 3B21-I. Cable information and assignment for MCP insert blocks – Continued.

Cable type	Cable diameter in inches	Cable diameter in mm	Insert block part number size mm 3/	Insert block part number size mm 3/
3SA 3SWA-3	0.750	19.05	30/18	
-7	0.960	24.38	30/24	40/24
-10	1.240	31.50	40/30	
-14	1.340	34.04	40/33	60/34
-19	1.480	37.59	60/36	
-24	1.720	43.69	60/42	
-30	1.820	46.23	60/46	
-37	1.980	50.29	60/50	90/50
-44	2.200	55.88	90/55	
-61	2.360	59.94	90/55	
PBJX-4	0.480	12.19	20/12	30/12
SHFR-4	0.540	13.72	20/13	30/13
DHFR-4	0.844	21.44	30/21	
THFR-4	0.683	22.43	30/22	40/22
1S50MA-16	0.875	22.23	30/22	40/22
1S50MA-20	0.955	24.26	30/24	40/24
1S50MA-40	1.235	31.37	40/30	
1S50MA-70	1.605	40.77	60/40	
IS75MA-8	1.080	27.43	40/26	

TABLE 3B21-I. Cable information and assignment for MCP insert blocks – Continued.

Cable type	Cable diameter in inches	Cable diameter in mm	Insert block part number size mm 3/	Insert block part number size mm 3/
MHFF-2	0.460	11.68	20/11	
MHFF-4	0.520	13.21	20/13	30/13
MHFF-7	0.627	15.93	30/15	
MHFF-10	0.795	20.20	30/20	
MHFF-14	0.844	21.44	30/21	
MHFF-19	0.995	25.28	30/24	40/24
MHFF-24	1.120	28.45	40/28	
MHFF-30	1.194	30.33	40/30	
MHFF-37	1.290	32.77	40/32	60/32
MHFF-44	1.420	36.07	60/35	

TABLE 3B21-II. Insert blocks.

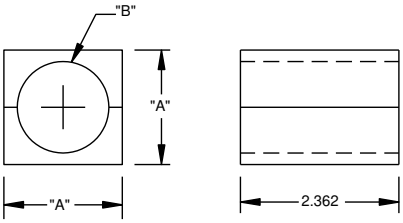
Insert block part no size mm 3/	"A" dim in.	"B" dim in.
15/4	0.590	0.157
15/5	0.590	0.197
15/6	0.590	0.236
15/7	0.590	0.276
15/8	0.590	0.315
15/9	0.590	0.354
20/4	0.787	0.157
20/5	0.787	0.197
20/6	0.787	0.236
20/7	0.787	0.276
20/8	0.787	0.315
20/9	0.787	0.354
20/10	0.787	0.394
20/11	0.787	0.433
20/12	0.787	0.472
20/13	0.787	0.512
20/14	0.787	0.551
30/12	1.181	0.472
30/13	1.181	0.512
30/14	1.181	0.551
30/15	1.181	0.591
30/16	1.181	0.630
30/17	1.181	0.669
30/18	1.181	0.709
30/19	1.181	0.748
30/20	1.181	0.787
30/21	1.181	0.827
30/22	1.181	0.866
30/23	1.181	0.906
30/24	1.181	0.945
40/22	1.575	0.866

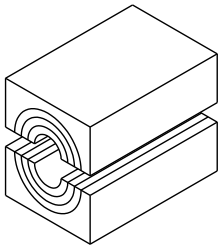
TABLE 3B21-II. Insert blocks – Continued.

Insert block part no size mm 3/	"A" dim in.	"B" dim in.
40/24	1.575	.945
40/26	1.575	1.024
40/28	1.575	1.102
40/30	1.575	1.181
40/32	1.575	1.260
40/34	1.575	1.339
60/32	2.362	1.260
60/34	2.362	1.339
60/36	2.362	1.417
60/38	2.362	1.496
60/40	2.362	1.575
60/42	2.362	1.654
60/44	2.362	1.732
60/46	2.362	1.811
60/48	2.362	1.890
60/50	2.362	1.969
60/52	2.362	2.047
60/54	2.362	2.128
90/50	3.543	1.969
90/55	3.543	2.165
90/60	3.543	2.362
90/65	3.543	2.559
90/70	3.543	2.756
120/75	4.724	2.953
120/80	4.724	3.150
120/85	4.724	3.346
120/90	4.724	3.543
120/95	4.724	3.740

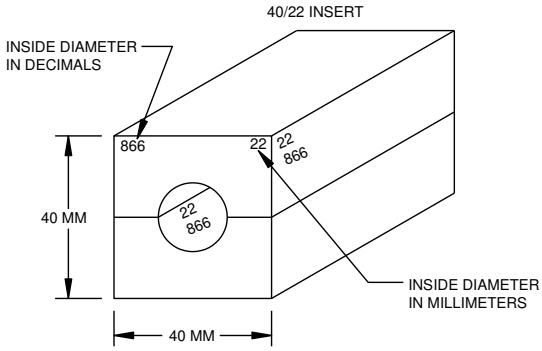
FIGURE 3B21. Cable information and assignment for MCP insert blocks – Continued.

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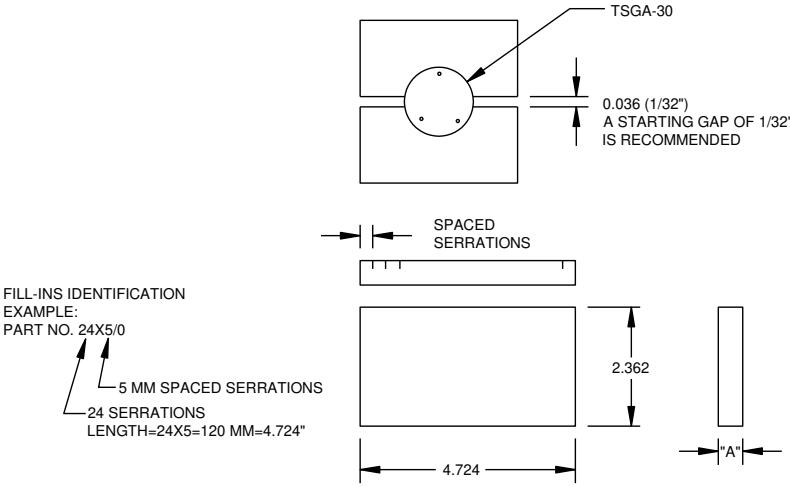


MULTI-LAYER INSERT MODULE

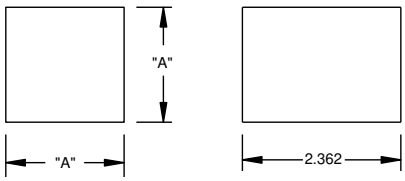


TYPICAL INSERT BLOCK IDENTIFICATION

TSGA-30 CABLE HAS AN OUTSIDE DIAMETER OF 0.902. SELECT INSERT BLOCKS CONTAINING THE NEXT INSIDE DIAMETER THAT IS LESS THAN THE OUTSIDE DIAMETER OF THE CABLE. A SIZE 22 MM HOLE IS 0.866 INCHES AND WILL PROVIDE THE FOLLOWING PRIOR TO COMPRESSION.



Fill-ins	"A" dim. (inches)
24x5/0	0.196
12x10/0	0.393



Spares part no. size mm	"A" dim. (inches)
15/0	0.590
20/0	0.787
30/0	1.181

NOTES:

1. See figures 3A10 through 3A25 and this figure for current cable assignments.
2. Approved multi-diameter insert blocks may be used, where available.
3. MCP block size designations are provided for information and guidance. Some of the MCP block sizes shown in this table may not accommodate the full range of allowed cable diameters for a given cable type. In those cases, different MCP block sizes should be selected to ensure the required block gap and retest requirements are met. See figure 3B22.

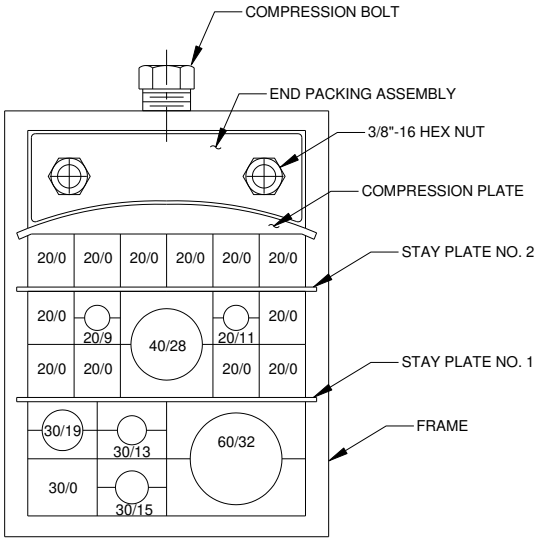
FIGURE 3B21. Cable information and assignment for MCP insert blocks – Continued.

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Installation notes:

RGS and RGA multiple cable penetrators

1. Multiple cable penetrator (MCP) is a modular system that utilizes standard units and dimensions to allow maximum flexibility and compatibility with various numbers and sizes of cables. Final sealed units will provide watertight, airtight, and fireproof bulkhead or deck penetrations. In general, the MCP component views, tables, and installation instructions in figures 3B20-3B30 apply. However, there are several different MCP products that meet the requirements of MIL-DTL-24705. Therefore, to achieve satisfactory installation, manufacturer's instructions should be used in addition to the instructions in this standard.
2. Remove any dirt from inside the frame. Pull all cable through cable penetrator, bearing in mind cables shall always approach parallel to each other and in a horizontal plane perpendicular to the penetrator. This can be accomplished if the nearest cable hanger is 16 inches minimum from bulkhead. Cables running up a bulkhead and through a cable penetrator shall have its full bending radius outside of the penetrator. While in the process of pulling cables, it is imperative that a reasonable amount of slack should be left to facilitate movement of cable within the cable penetrator to ease packing operation. See MIL-STD-2003-4 for cable slack requirements. Also, it is necessary that each cable be identified on the compression screw side of MCP by lead number and cable size. All cables through a new MCP shall be pulled prior to installing insert blocks. See 4.4 for specific restrictions and instructions for installation of fiber optic and blown fiber cables in electrical penetrations.
3. After pulling all cables through a particular MCP, it can be packed. First apply a liberal coat of cable lubricant (tallow) to all outside surfaces of the insert blocks, the inside portion of the frame, and the curved surface of the compression plate. When packing armored cable, apply Nelson Fire stop Caulk, or equal (see 4.2), sealer on the surface of each block that contacts the cable (a ⅛ inch bead every ½ inch around the circumference). Each frame shall be laid out (blocked) showing solid blocks (used as spares) and cable assignments prior to packing. A typical example of a packed RGS-4 frame with instructions for packing is shown below.



   a. Select insert blocks and spares from tables 3B21-I and 3A10-I through 3A25-I (presented for information and guidance) and prepare as above (lubricant and sealer where required). Blocks should be selected to avoid any gaps under compression. Fill-ins as explained in figure 3B20 and illustrated in figure 3B21 may be used if different size blocks are installed in the same row to ensure an even installation. Although not desirable, the fill-ins also may be used if soft cable would leave a gap under compression. For multi-diameter modules, the insert blocks should be adapted by peeling one layer at a time to achieve a 0.1- to 1-millimeter gap between the two halves when the block is held against the cable. See following illustration. The layers shall be peeled from both sides to achieve an even installation. Once a layer is peeled, the layer may not be reinserted into the block.

FIGURE 3B22. Multiple cable penetrator installation notes (type RGS and RGA).

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Installation notes (continued):

b. Starting with heaviest cable, place the lower half of the 60/32 insert block under the cable in the frame. Next, repeat the procedure with the 30/15 then add one 30/0 spare. Add the upper half of the 60/32 and the 30/15. Place lower half of 30/13 and 30/19 under cable, then add upper halves. Across the top of these, add stay plate no. 1. Use steel stay plates with steel frames and aluminum stay plates with aluminum frames. Use corrosion resistant steel (CRES) stay plates for medium or high voltage cables.

c. On top of the stay plate, place four 20/0 spares and the lower half of the 40/28 under its cable. In the next higher row, place two 20/0 spares and the lower half of 20/9 and 20/11 under their respective cables. Then add upper halves of 40/28, 20/11, and 20/9. On top of these, add stay plate no. 2.

d. Add compression plate to the frame.

e. Lift compression plate clear of stay plate no. 2. Add final row of six 20/0 spares. Lower compression plate to this level.

f. If individual conductors of a three-phase power circuit are in different cables, the individual cables shall be installed in adjacent insert blocks. There shall be no ferrous material, e.g., stay plates or frame segments between the individual cables. The purpose of this requirement is to avoid heating due to electro-magnetic induction.

g. Tighten compression screw down on compression plate to compress entire contents of frame and provide just enough clearance to insert end packing assembly. Apply a generous amount of lubricant to this cavity to assist insertion of end packing. A wedge, as described in figure 3B20, may be used instead of the end packing assembly. The wedge may be installed anywhere within the MCP assembly. The wedge shall be installed in accordance with manufacturer's instructions. Guidance for wedge installation is provided in step 28.

h. Disassemble end packing assembly and install in frame around the compression screw. Reassemble and tighten ⅜-16 UNC hex nuts evenly until end packing insert material expands in the cavity and there is a slight roll of the material around the end packing metal washers. Back off compression screw approximately ⅛ of a turn. Packing is now complete.

i. See SE000-01-IMB-010 for modifications to the MCPs to support the installation of waveguides.

4. When a multiple cable penetrator frame is installed in a structure in such a manner that the compression screw is located on the bottom, packing procedure is the same except:
   a. The compression plate is installed first.
   b. The frame is packed from the bottom up with the smallest cables being packed first.
5. To add a new cable to a packed frame, loosen the ⅜-16UNC hex nuts and remove the end packing metal washers. Tighten the compression screw until end packing insert material is loose. Remove insert material from frame, loosen compression screw, and lift compression plate clear of top six 20/0 spares. Remove these inserts. Remove compression plate and full access to the insert blocks and cable is available. If a wedge is used, follow manufacturer instructions for wedge removal.
6. Template all work from ship.
7. All welding and inspection shall be in accordance with 4.1.1.
8. All painting shall be done in accordance with MIL-DTL-917.
9. The structural reinforcement shown on this plan does not apply to flight deck bents, flight deck support structures, or other similar type structures on other ships. Such cases shall be individually resolved with structural design.
10. Material of structural reinforcement is to be similar to material of bulkhead, deck, or riser box penetrated.
11. This plan was developed from T9070-AN-DPC-010/100-1, reinforcement of openings in structure of surface ships other than in protective plating, to permit installation of multiple cable penetrator frame [Tecrona Products of Nelson Electric, or equal (see 4.2)] in watertight decks and bulkheads of surface ships.

FIGURE 3B22. Multiple cable penetrator installation notes (type RGS and RGA) – Continued.

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Installation notes (continued):

12. Fillet weld reinforcement for T2V.1 weld joints in accordance with MIL-STD-22 shall be ⅛ inch for plating up to 15.3# and ¼ inch for plating above 17.85#.
13. The bulkheads and decks that are considered to be longitudinal strength members are defined in T9070-AN-DPC-010/100-1 as components in "strength envelope (i.e., in outside plating, uppermost strength deck, inner bottom, and their associated longitudinal framing) and for other strength decks within the midship three-fifths length." If there is difficulty in definition, consult with structural design.
14. Care should be taken while installing multiple cable penetrators to ensure that all of its coaming protrudes through bulkhead sufficiently to weld in place. After welding, examine for cracks and sharp edges that may damage cable when being pulled through penetrator. If grinding is required, ensure grinding areas do not weaken the penetrator.
15. It is mandatory to maintain a minimum of ⅞ inch from bulkhead to bolt hole center for wrench clearance. Also, leave a 4-inch clearance to allow for installation of compression screw (from bolt hole side of the multiple cable penetrator to any structure).
16. Riser box to be attached to bulkhead of 7.65# or heavier. If bulkhead is less than 7.65# adjacent to riser box installation, install a back plate of 10.2# between the back of the riser box and the bulkhead.
17. After installation is complete, entire penetrator may be painted to suit surrounding structure in accordance with the applicable ship specifications. Wait 24 hours or longer (refer to manufacturer's instructions) after installation before exposing the cables/pipes to strain or pressure. The wait time shall be observed prior to the post-installation testing.
18. Post installation testing:
    a. After installation of a new MCP or work on an existing MCP, retest shall be conducted to confirm integrity of the electrical penetration.
    b. Perform a visual inspection on plastic or rubber inserts inside of the MCP to ensure there are no visible or discernible openings.
    c. For airtight, oiltight, and watertight penetrations, conduct an air blast test or a vacuum box test as follows:
        (1) Air blast test: The air hose nozzle shall be held a maximum of 3 inches from the MCP area. The distance should be as close to 3 inches as practicable to avoid distortion of the MCP blocks due to high air flow. The pressure shall be directed at the structure under test in a manner most likely to disclose leaks. The minimum nozzle diameter shall be ⅜ inch and the nozzle pressure shall be 60 to 90 pounds per square inch gage (PSIG) as monitored at the nozzle.
            (a) Obtain a pressure differential in the same direction as would occur during a full compartment, tank, or void test using the air hose. The air pressure shall be maintained for at least 3 minutes. As soon as the pressure is applied, apply a soapy solution to the opposite side of the structure and inspect for leaks.
        (2) Vacuum box test requirements: Apply a soapy solution to the joint to be tested. Install the vacuum box so the pressure differential will be in the same direction as a full compartment, tank, or void completion test. Draw 10.2"Hg vacuum (minimum). Maintain the vacuum for at least 3 minutes. Inspect the joint or fitting for leakage by observing the formation of bubbles.
    d. For penetrations through fume-tight bulkheads, conduct a visual/light test to ensure no light transmits from one side of the MCP to the other.

RGSO and RGAO multiple cable penetrators

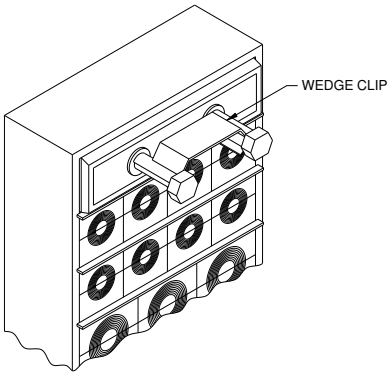
19. Temporarily insulate cable for protection against welding and burning.
20. Remove old non-watertight structure.
21. Prepare hole in bulkhead to receive multiple cable penetrator frame (designated "RGSO"). External banding is used only when required.
22. Bottom of "RGSO" unit is tack welded to other three sides – remove this piece.
23. Segregate and confine required cables within each three-sided frame.
24. Wrap welders blanket around cable. Replace bottom piece and continuously weld corners. Grind smooth same as top corners.
25. Slide unit into prepared hole in bulkhead. Position so that center line of tapped hole is ⅞" out from face of bulkhead to permit clearance for the compression screw.

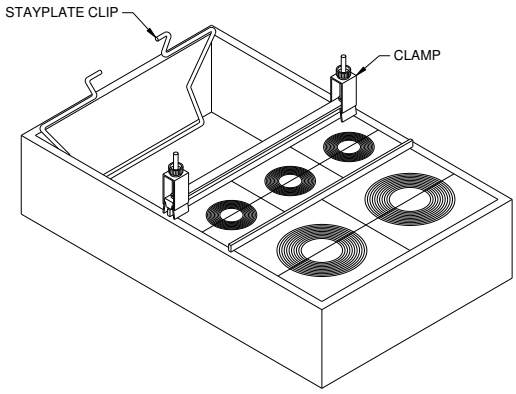
FIGURE 3B22. Multiple cable penetrator installation notes (type RGS and RGA) – Continued.

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Installation notes (continued):

26. Weld frame to bulkhead with continuous weld all around frame on both sides of bulkhead.
27. Proceed as outlined in notes 3-18.
28. Guidance for wedge installation: Pack the MCP frame up to and including step 3f above. Then, proceed as follows:
    a. Before inserting the final row of cable or spare insert blocks, insert two stayplates.
    b. Separate the two stayplates. Insert the final row between the stayplates.
    c. Place the upper stayplate on top of the modules.
    d. Lubricate all sides of the wedge, except for the front and back. Turn the wedge according to the markings and insert into the top of the frame (standard position). As an option, the wedge may be installed anywhere in the frame. The face marked with "Stayplate this side" shall always face a stayplate.
    e. Tighten the wedge fasteners until full stop, approximately 15 foot-pounds.
    f. Attach the wedge clip (see following illustration) to the screws to check that the wedge is properly tightened.
    g. Stayplate clips and clamps may be used to simplify horizontal assembly (see following illustration).
    h. Proceed to step 6 above.





NOTE:

1. The tables and dimensions shown on these sheets are for typical steel installations. Other unique situations shall be handled as a case-by-case basis through the structures department.

FIGURE 3B22. Multiple cable penetrator installation notes (type RGS and RGA) – Continued.

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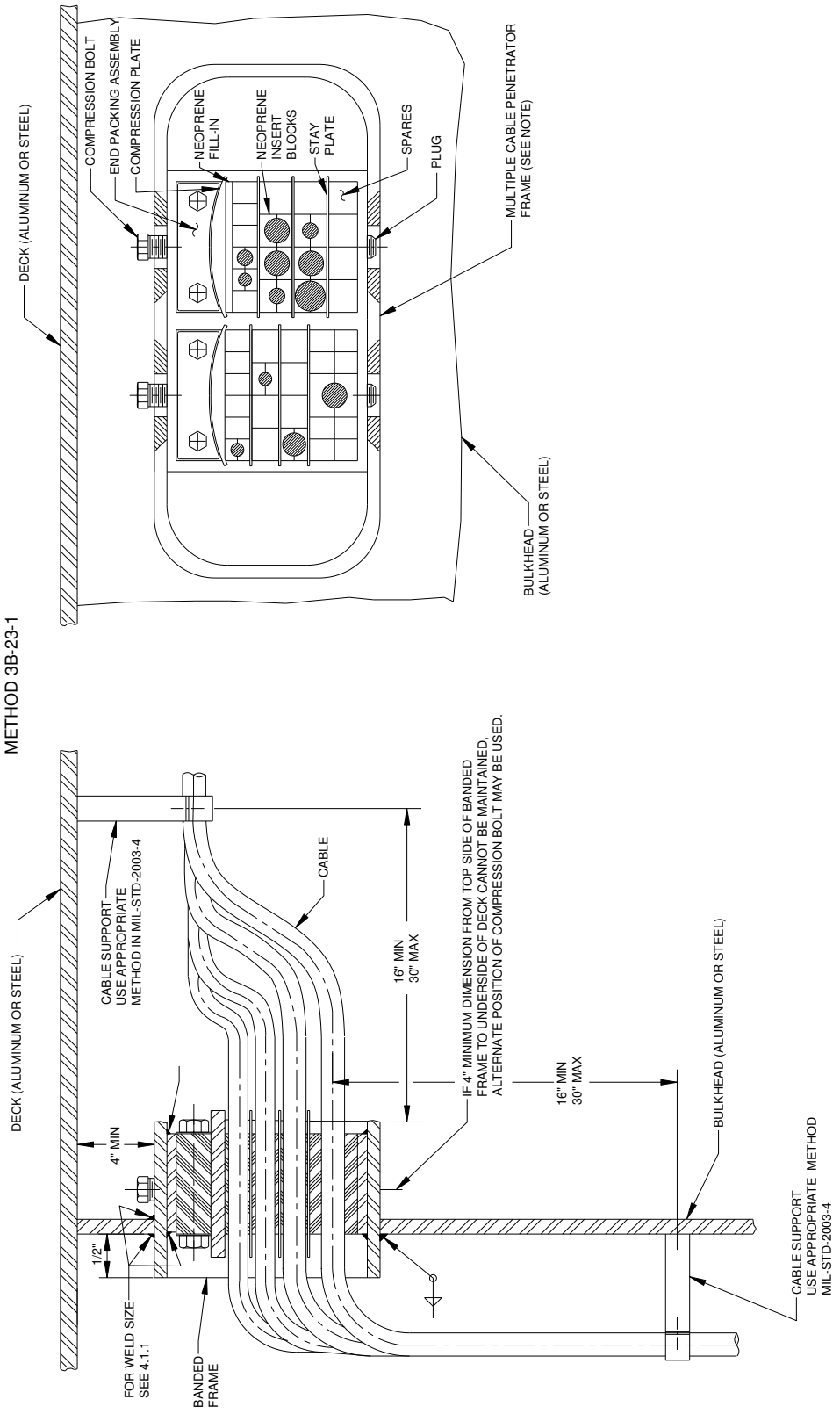


FIGURE 3B23. Multiple cable penetrator installation in steel or aluminum bulkheads using two-frame penetrator (types RGS and RGA).

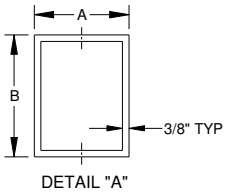
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NOTE:

1. Multiple cable penetrator frames are available in various sizes and arrangements. A two-frame penetrator is shown.

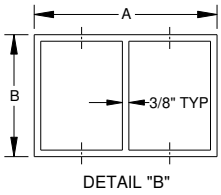
FIGURE 3B23. Multiple cable penetrator installation in steel or aluminum bulkheads using two-frame penetrator (types RGS and RGA) – Continued.

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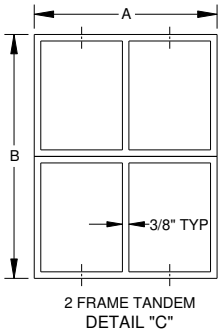
⅜ thick single frame

Type	A (see note 3)	B (see note 3)	Packable area dimensions per frame (mm) (see note 2)
RGS-2	5.5	4.726	120/60
RGS-4	5.5	7.029	120/120
RGS-6	5.5	9.333	120/180
RGS-8	5.5	11.660	120/240



⅜ thick 1 thru 6 multi-frame

Type	Dim (see note 3)	1 Frame	2 Frame	3 Frame	4 Frame	5 Frame	6 Frame
RGS-2	A	5.506	10.637	15.768	20.899	26.030	31.161
RGS-4	A	5.506	10.637	15.768	20.899	26.030	31.161
RGS-6	A	5.506	10.637	15.768	20.899	26.030	31.161
RGS-8	A	5.506	10.637	15.768	20.899	26.030	31.161
RGS-2	B	4.726	4.726	4.726	4.726	4.726	4.726
RGS-4	B	7.029	7.029	7.029	7.029	7.029	7.029
RGS-6	B	9.333	9.333	9.333	9.333	9.333	9.333
RGS-8	B	11.66	11.66	11.66	11.66	11.66	11.66

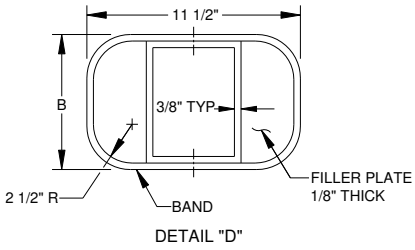


⅜ thick 1 thru 6 tandem multi-frame

Type	Dim (see note 3)	T Fr.	2T Fr.	3T Fr.	4T Fr.	5T Fr.	6T Fr.
RGS-2	A	5.506	10.637	15.768	20.899	26.030	31.161
RGS-4	A	5.506	10.637	15.768	20.899	26.030	31.161
RGS-6	A	5.506	10.637	15.768	20.899	26.030	31.161
RGS-8	A	5.506	10.637	15.768	20.899	26.030	31.161
RGS-2	B	9.452	9.452	9.452	9.452	9.452	9.452
RGS-4	B	14.058	14.058	14.058	14.058	14.058	14.058
RGS-6	B	18.666	18.666	18.666	18.666	18.666	18.666
RGS-8	B	23.31	23.31	23.31	23.31	23.31	23.31

FIGURE 3B24. Multiple cable penetrator details (types RGS and RGA).

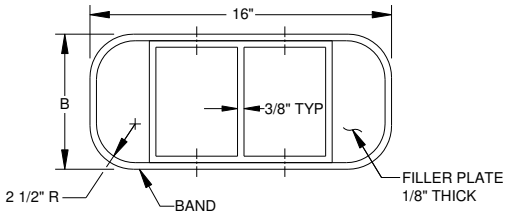
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DETAIL "D"

⅜ single frame banded

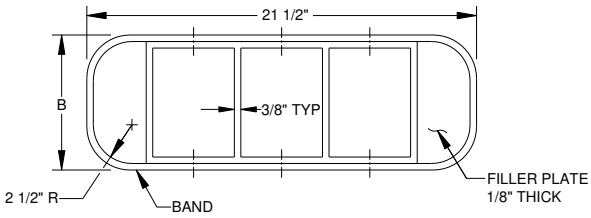
Dimension "B" (see note 3)			
BHD plate #	10.2 12.75 15.3	17.85 20.4 23.0	25.5 28.0 30.6
RGS-2	5 9/16	5 13/16	6 5/16
RGS-4	7⅞	8⅛	8⅝
RGS-6	10 3/16	10 7/16	10 15/16
RGS-8	12½	12¾	13¼
BAND	⅜ x 4"	½ x 4"	¾ x 4"



DETAIL "E"

⅜ thick 2 frame banded

Dimension "B" (see note 3)			
BHD plate #	10.2 12.75 15.3	17.85 20.4	23.0 25.5 28.0 30.6
RGS-2	5 9/16	5 13/16	6 5/16
RGS-4	7⅞	8⅛	8⅝
RGS-6	10 3/16	10 7/16	10 15/16
RGS-8	12½	12¾	13¼
BAND	⅜x4"	½x4"	¾x4"



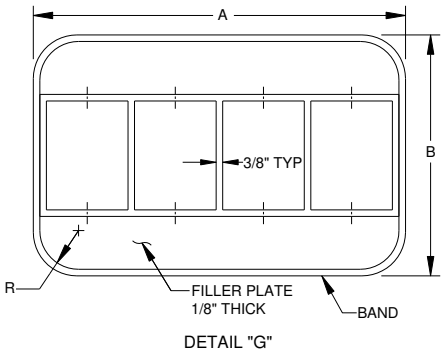
DETAIL "F"

⅜ thick multi-frame banded

Dimension "B" (see note 3)			
BHD plate #	10.2 12.75	15.3 17.85 20.4	23.0 25.5 28.0 30.6
RGS-2	5 9/16	5 13/16	6 5/16
RGS-4	7⅞	8⅛	8⅝
RGS-6	10 3/16	10 7/16	10 15/16
RGS-8	12½	12¾	13¼
BAND	⅜ x 4"	½ x 4"	¾ x 4"

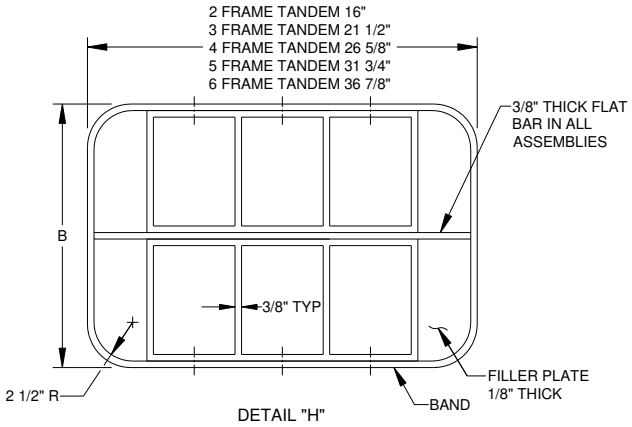
FIGURE 3B24. Multiple cable penetrator details (types RGS and RGA) – Continued.

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⅜ thick 4 frame banded

BHD plate lbs #				"B" dim (see note 3)				"A" (see note 3)	"R"	Band
				RGS-2	RGS-4	RGS-6	RGS-8			
10.2	12.75	15.3	17.85	11 1/16	13 3/8	15 11/16	18	21 3/4	3 1/8	3/8 x 4"
17.85	20.4	23.0	25.55	11 9/16	13 7/8	16 3/16	18 1/2	22 1/4	3 3/8	5/8 x 4"
25.5	28.0	30.6	30.6	11 13/16	14 1/8	16 7/16	18 3/4	22 1/2	3 1/2	3/4 x 4"



⅜ thick 2 thru 6 tandem multi-frame banded

Dimension "B" (see note 3)						
BHD plate #	7.65	10.2	12.75 15.3	17.85 20.4	25.5	30.6 35.7
RGS-2	10 5/8	10 7/8	11 1/8	11 3/8	11 5/8	11 5/8
RGS-4	15 3/16	15 7/16	15 11/16	15 15/16	16 3/16	16 3/16
RGS-6	19 13/16	20 1/16	20 5/16	20 9/16	20 13/16	20 13/16
RGS-8	24 7/16	24 11/16	24 15/16	25 3/16	25 7/16	25 7/16
BAND	3/8 x 4	1/2 x 4	5/8 x 4	3/4 x 5	7/8 x 6	7/8 x 6 1/2

FIGURE 3B24. Multiple cable penetrator details (types RGS and RGA) – Continued.

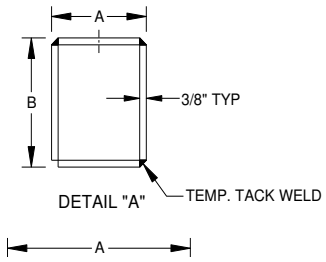
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NOTES:

1. Material RGS frames and filler plates are mild steel ASTM A36/A36M. Reinforcing bands are either mild steel (ASTM A36/A36M) or high tensile steel MIL-S-22698 in order for material to be similar to bulkhead or deck in which penetration is being made.
2. All single frames have the same packable area, whether in the single frame or multi-frame configuration.
3. For aluminum MCP frames, the frame edge thickness in accordance with MIL-DTL-24705/1 is ½ inch rather than ⅜ inch for steel frames. Therefore, the appropriate amount should be added to the dimensions shown in the tables in figure 3B24 to account for the ½-inch aluminum frame edges.
4. Dimensions in inches unless otherwise noted.

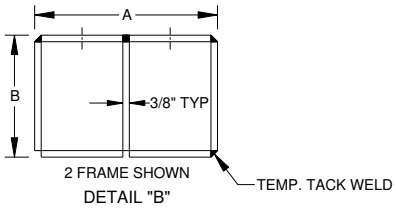
FIGURE 3B24. Multiple cable penetrator details (types RGS and RGA) – Continued.

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⅜ thick single frame

Type	A (see note 3)	B (see note 3)	Packable area dimensions per frame (mm) (see note 2)
RGSO-2	5.506	4.726	120/60
RGSO-4	5.506	7.029	120/120
RGSO-6	5.506	9.333	120/180
RGSO-8	5.506	11.660	120/240

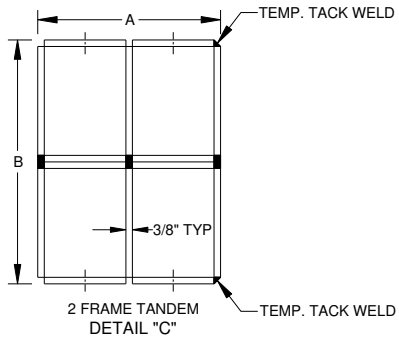


⅜ thick 2 thru 6 multi-frame

Type	Dim (see note 3)	2 Frame	3 Frame	4 Frame	5 Frame	6 Frame
RGSO-2	A	10.637	15.768	20.899	26.030	31.161
RGSO-4	A	10.637	15.768	20.899	26.030	31.161
RGSO-6	A	10.637	15.768	20.899	26.030	31.161
RGSO-8	A	10.637	15.768	20.899	26.030	31.161
RGSO-2	B	4.726	4.726	4.726	4.726	4.726
RGSO-4	B	7.029	7.029	7.029	7.029	7.029
RGSO-6	B	9.333	9.333	9.333	9.333	9.333
RGSO-8	B	11.660	11.660	11.660	11.660	11.660

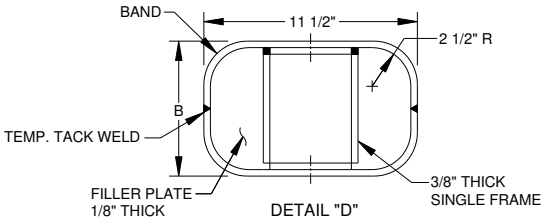
FIGURE 3B25. Multiple cable penetrator details (types RGSO and RGAO).

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⅜ thick 1 thru 6 tandem multi-frame

Type	Dim (see note 3)	T frame	2T frame	3T frame	4T frame	5T frame	6T frame
RGSO-2	A	5.506	10.637	15.768	20.899	26.030	31.161
RGSO-4	A	5.506	10.637	15.768	20.899	26.030	31.161
RGSO-6	A	5.506	10.637	15.768	20.899	26.030	31.161
RGSO-8	A	5.506	10.637	15.768	20.899	26.030	31.161
RGSO-2	B	9.452	9.452	9.452	9.452	9.452	9.452
RGSO-4	B	14.058	14.058	14.058	14.058	14.058	14.058
RGSO-6	B	18.666	18.666	18.666	18.666	18.666	18.666
RGSO-8	B	23.31	23.31	23.31	23.31	23.31	23.31



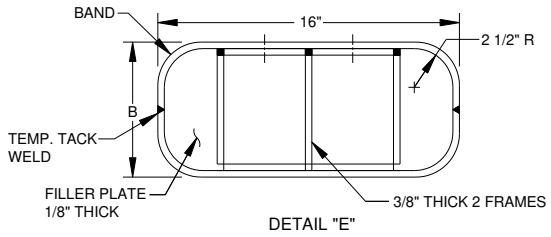
⅜ thick single frame banded

Dimension "B" (see note 3)			
BHD plate #	10.2 12.75 15.3	17.85 20.4	23.0 25.5 28.0 30.6
RGSO-2	5 9/16	5 13/16	6 5/16
RGSO-4	7⅞	8⅛	8⅝
RGSO-6	10 3/16	10 7/16	10 15/16
RGSO-8	12½	12¾	13¼
BAND	⅜ x 4"	½ x 4"	¾ x 4"

Catalog no. for RGSO-6 banded with ⅜ x 4 flat bar would be RGSO-6 4375.

FIGURE 3B25. Multiple cable penetrator details (types RGSO and RGAO) – Continued.

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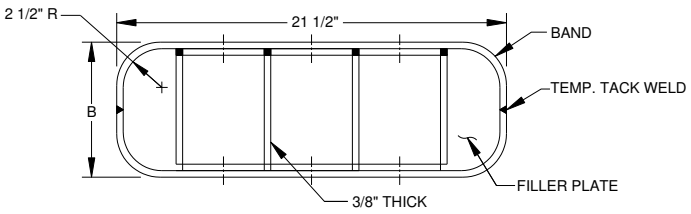


DETAIL "E"

⅜ thick 2 frame banded

Dimension "B" (see note 3)			
BHD plate #	10.2 12.75 15.3	17.85 20.4	23.0 25.5 28.0 30.6
RGSO-2	5 9/16	5 13/16	6 5/16
RGSO-4	7⅞	8⅛	8⅝
RGSO-6	10 3/16	10 7/16	10 15/16
RGSO-8	12½	12¾	13¼
BAND	⅜ x 4"	½ x 4"	¾ x 4"

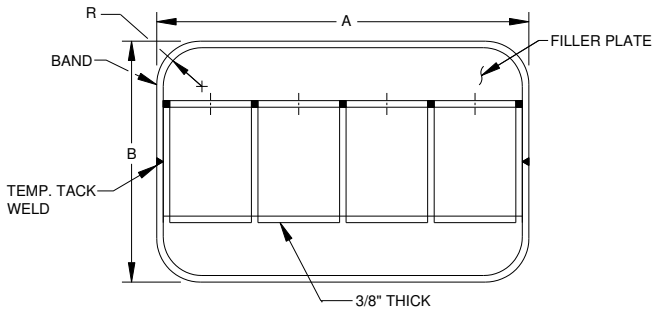
Catalog no. for RGSO-6x2 banded with ⅜ x 4 flat bar would be RGSO-6x2 4375.



DETAIL "F"

⅜ thick multi-frame banded

Dimension "B" (see note 3)			
BHD plate #	10.2 12.75	15.3 17.85 20.4	23.0 25.5 28.0 30.6
RGSO-2	5 9/16	5 13/16	6 5/16
RGSO-4	7⅞	8⅛	8⅝
RGSO-6	10 3/16	10 7/16	10 15/16
RGSO-8	12½	12¾	13¼
BAND	⅜ x 4"	½ x 4"	¾ x 4"



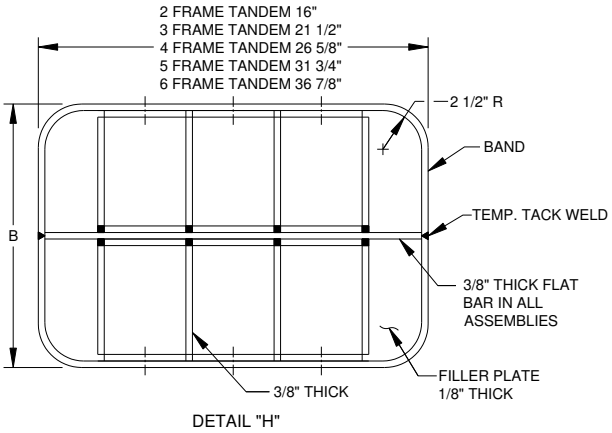
DETAIL "G"

⅜ thick 4 frame banded

BHD plate #				Dimension "B" (see note 3)				"A" (see note 3)	"R"	Band
				RGSO-2	RGSO-4	RGSO-6	RGSO-8			
10.2	12.75	15.3	17.85	11 1/16	13⅜	15 11/16	18	21¾	3⅛	⅜ x 4"
17.85	20.4	23.0	25.55	11 9/16	13⅞	16 3/16	18½	22¼	3⅜	⅝ x 4"
25.5	28.0	30.6	30.6	11 15/16	14⅛	16 7/16	18¾	22½	3½	¾ x 4"

FIGURE 3B25. Multiple cable penetrator details (types RGSO and RGAO) – Continued.

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⅜ thick tandem (T) frames banded

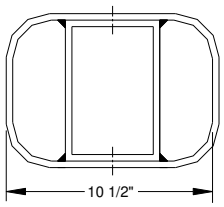
Dimension "B" (see note 3)						
BHD plate #	7.65	10.2	12.75 15.3	17.85 20.4	25.5	30.6 35.7
RGSO-2	10⅝	10⅞	11⅛	11⅜	11⅝	11⅝
RGSO-4	15 3/16	15 7/16	15 11/16	15 15/16	16 3/16	16 3/16
RGSO-6	19 13/16	20 1/16	20 5/16	20 9/16	20 13/16	20 13/16
RGSO-8	24 7/16	24 11/16	24 15/16	25 3/16	25 7/16	25 7/16
BAND	⅜ x 4	½ x 4	⅝ x 4	¾ x 5	⅞ x 6	⅞ x 6½

NOTES:

1. RGSO frames and filler plates are mild steel ASTM A36/A36M. Reinforcing bands are either mild steel (ASTM A36/A36M) or high tensile steel MIL-S-22698 in order for material to be similar to bulkhead or deck in which penetration is being made.
2. All single frames have the same packable area, whether in the single frame or multi-frame configuration.
3. For aluminum MCP frames, the frame edge thickness, in accordance with MIL-DTL-24705/1 is ½ inch rather than ⅜ inch for steel frames. Therefore, the appropriate amount should be added to the dimensions shown in the tables in figure 3B25 to account for the ½-inch aluminum frame edges.
4. Dimensions in inches unless otherwise noted.

FIGURE 3B25. Multiple cable penetrator details (types RGSO and RGAO) – Continued.

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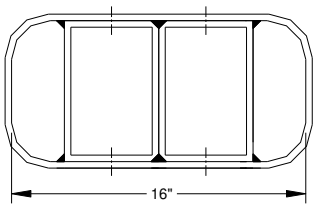


SINGLE FRAME BANDED

Type I Size of band			
Wt of structure penetrated #	**Long'l strength member**		**BHDs and decks other than long'l strength members**
	BHDs	**Decks**	
7.65	5" x 5/16	12" x 5/16	5" x 5/16
10.2	5" x 5/16	12" x 5/16	5" x 5/16
12.75	5" x 5/16	12" x 5/16	5" x 5/16
15.3	5" x 5/16	12" x 5/16	5" x 5/16
17.85	5" x 3/8	12" x 3/8	5" x 5/16
20.4	5" x 3/8	12" x 3/8	5" x 5/16
25.5	5" x 3/8	12" x 3/8	5" x 3/8
30.6	5" x ½	12" x ½	5" x ½
35.7	5" x ½	12" x ½	5" x ½

FIGURE 3B26. Multiple cable penetrator details (types RGS and RGA).

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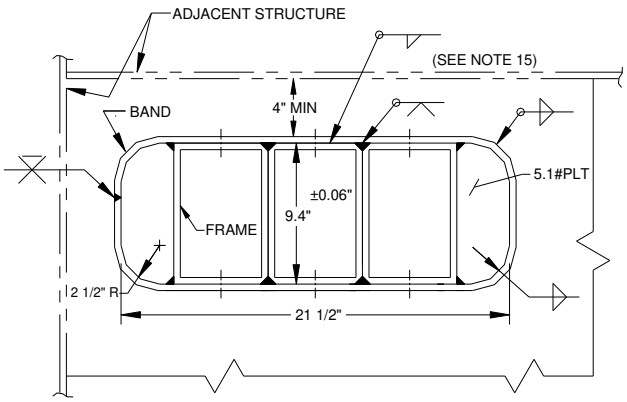


DOUBLE WALL 2 FRAME BANDED

Type II Size of band			
Wt of structure penetrated #	**Long'l strength member**		**BHDs and decks other than long'l strength members**
	BHDs	**Decks**	
7.65	5" x 5/16	12" x 5/16	5" x 5/16
10.2	5" x 5/16	12" x 5/16	5" x 5/16
12.75	5" x 5/16	12" x 5/16	5" x 5/16
15.3	5" x 3/8	12" x 3/8	5" x 5/16
17.85	5" x ½	12" x ½	5" x 5/16
20.4	5" x ½	12" x ½	5" x 5/16
25.5	5" x 5/8	12" x 5/8	5" x 3/8
30.6	5" x ¾	12" x ¾	5" x ½
35.7	5" x 7/8	12" x 7/8	5" x ½

FIGURE 3B26. Multiple cable penetrator details (types RGS and RGA) – Continued.

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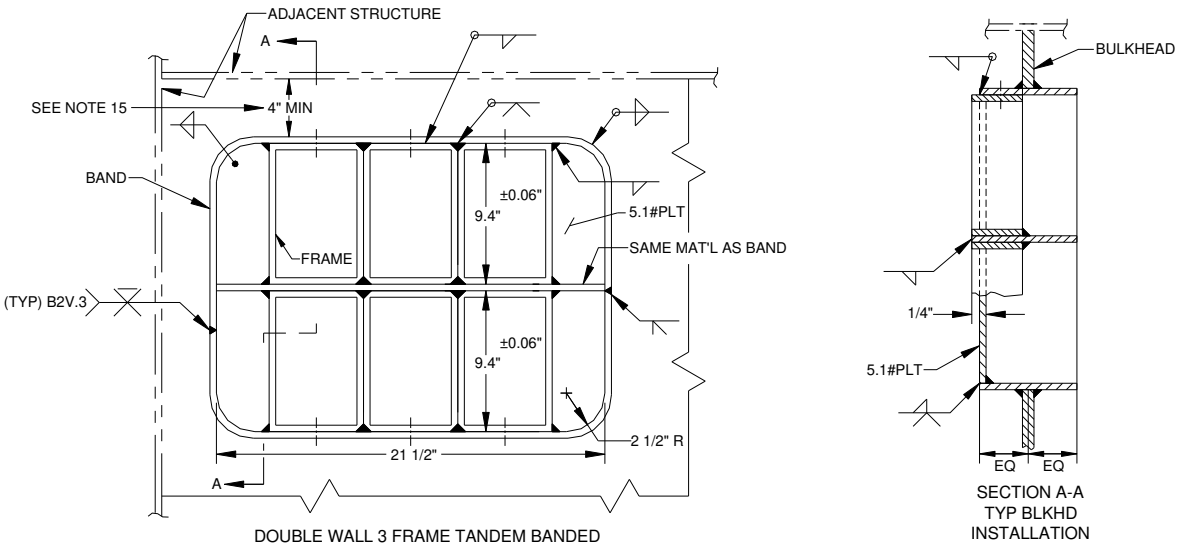


DOUBLE WALL 3 FRAME BANDED

Type III Size of band			
Wt of structure penetrated #	**Long'l strength member**		**BHDs and decks other than long'l strength members**
	BHDs	**Decks**	
7.65	6" x ⅜	12" x ⅜	5" x 5/16
10.2	6" x ⅜	12" x ⅜	5" x 5/16
12.75	6" x ⅜	12" x ⅜	5" x 5/16
15.3	6" x 7/16	12" x 7/16	5" x 5/16
17.85	6" x ½	12" x ½	5" x ⅜
20.4	6" x ⅝	12" x ⅝	5" x ⅜
25.5	6" x ¾	12" x ¾	5" x ½
30.6	7" x ¾	12" x ¾	5" x ½
35.7	7" x ⅞	12" x ⅞	5" x ¾

FIGURE 3B26. Multiple cable penetrator details (types RGS and RGA) – Continued.

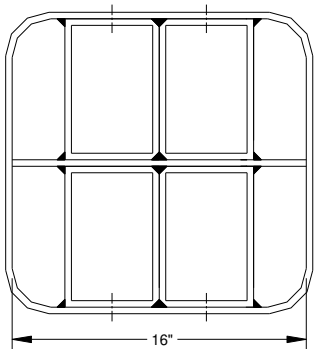
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Type IV Size of band			
Wt of structure penetrated #	**Long'l strength member**		**BHDs and decks other than long'l strength members**
	BHDs	**Decks**	
7.65	5" x 5/16	12" x5/16	5" x 5/16
10.2	5" x 3/8	12" x 3/8	5" x 5/16
12.75	5" x 3/8	12" x 3/8	5" x 5/16
15.3	5" x 3/8	12" x 3/8	5" x 5/16
17.85	5" x ½	12" x ½	5" x 3/8
20.4	5" x ½	12" x ½	5" x 3/8
25.5	5" x 5/8	12" x 5/8	5" x ½
30.6	5" x ¾	12" x ¾	5" x 5/8
35.7	6" x ¾	12" x ¾	5" x ¾

FIGURE 3B26. Multiple cable penetrator details (types RGS and RGA) – Continued.

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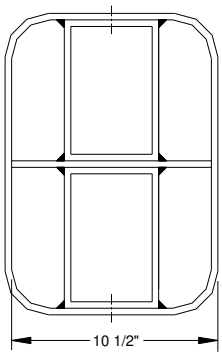


DOUBLE WALL 2 FRAME TANDEM BANDED

Type V Size of band			
Wt of structure penetrated #	**Long'l strength member**		**BHDs and decks other than long'l strength members**
	BHDs	**Decks**	
7.65	5" x 5/16	12" x 5/16	5" x 5/16
10.2	5" x 3/8	12" x 3/8	5" x 5/16
12.75	5" x 3/8	12" x 3/8	5" x 5/16
15.3	5" x 3/8	12" x 3/8	5" x 5/16
17.85	5" x ½	12" x ½	5" x 3/8
20.4	5" x ½	12" x ½	5" x 3/8
25.5	5" x 5/8	12" x 5/8	5" x ½
30.6	5" x ¾	12" x ¾	5" x 5/8
35.7	6" x ¾	12" x ¾	5" x ¾

FIGURE 3B26. Multiple cable penetrator details (types RGS and RGA) – Continued.

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SINGLE FRAME TANDEM BANDED

Type VI Size of band			
Wt of structure penetrated #	**Long'l strength member**		**BHDs and decks other than long'l strength members**
	BHDs	**Decks**	
7.65	5" x 5/16	12" x 5/16	5" x 5/16
10.2	5" x 3/8	12" x 3/8	5" x 5/16
12.75	5" x 3/8	12" x 3/8	5" x 5/16
15.3	5" x 3/8	12" x 3/8	5" x 5/16
17.85	5" x ½	12" x ½	5" x 3/8
20.4	5" x ½	12" x ½	5" x 3/8
25.5	5" x 5/8	12" x 5/8	5" x 3/8
30.6	5" x ¾	12" x ¾	5" x ½
35.7	5" x ¾	12" x ¾	5" x ¾

FIGURE 3B26. Multiple cable penetrator details (types RGS and RGA) – Continued.

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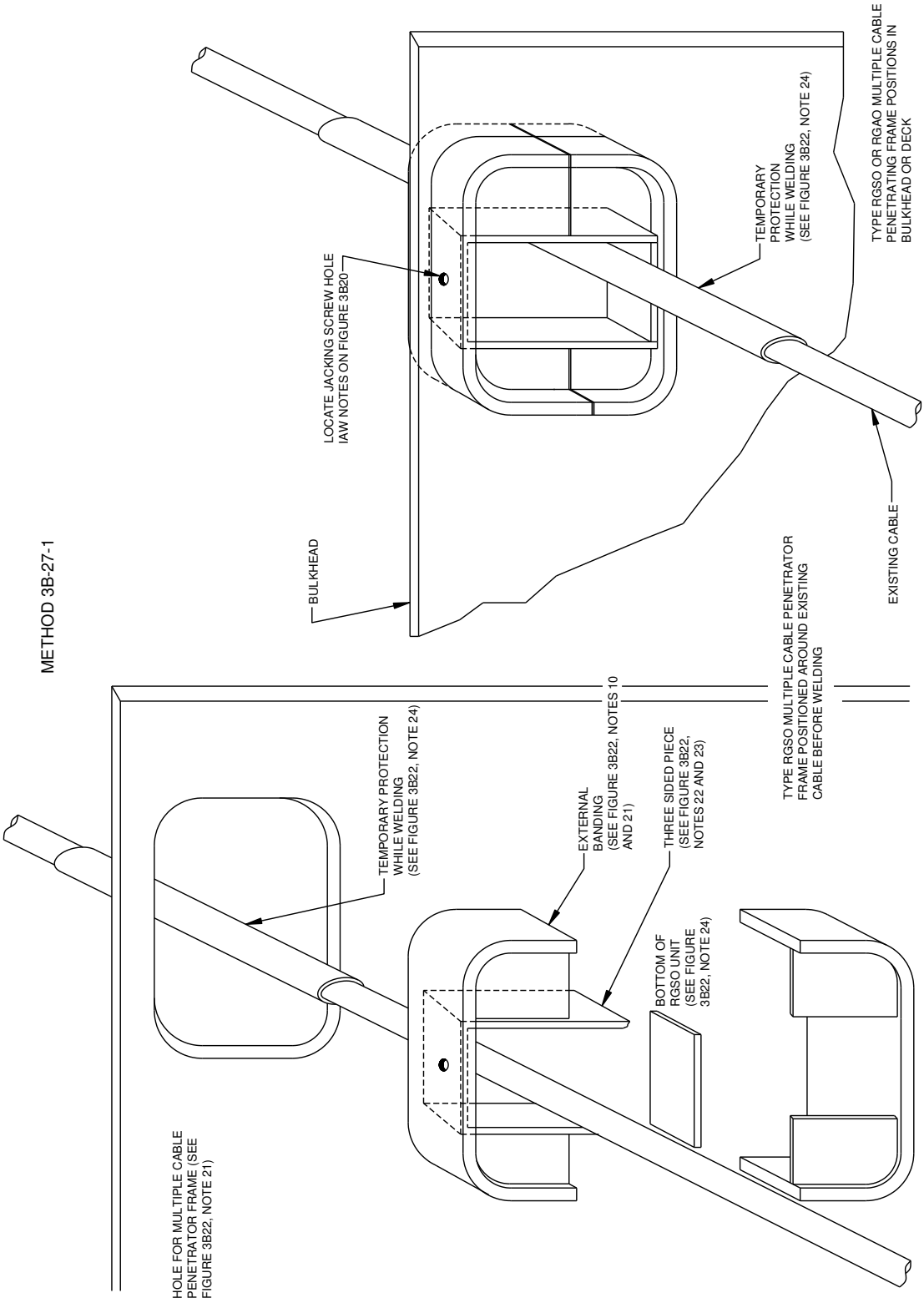


FIGURE 3B27. Multiple cable penetrator installation details (types RGS and RGA).

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NOTE:

1. See figure 3B22, notes 19 through 27 for additional information.

FIGURE 3B27. Multiple cable penetrator installation details (types RGS and RGA) – Continued.

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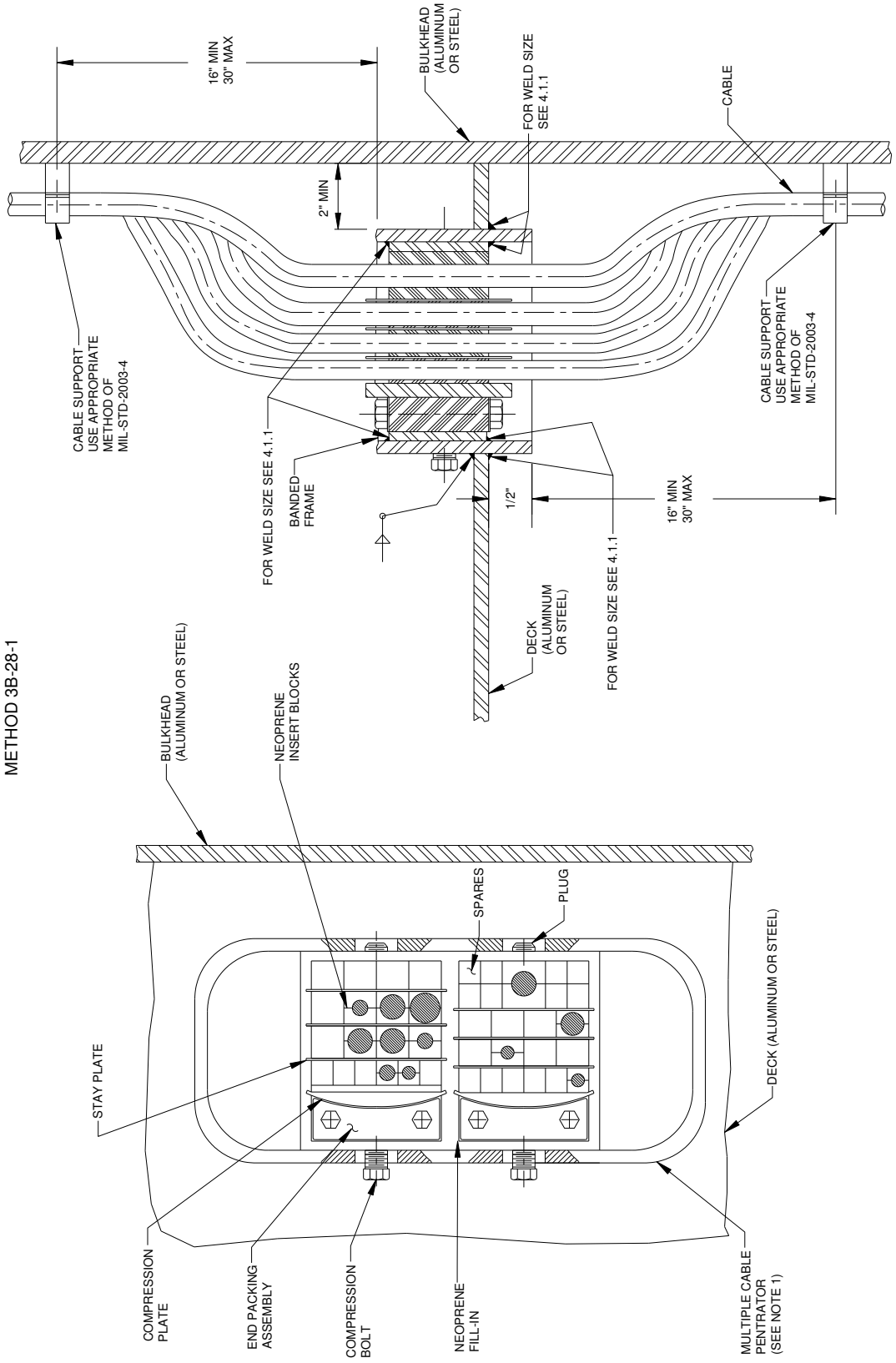


FIGURE 3B28. Multiple cable penetrator frame in decks (types RGS and RGA).

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NOTE:

1. Multiple cable penetrator frames are available in various sizes and arrangements. Shown is a two-frame penetrator.

FIGURE 3B28. Multiple cable penetrator frame in decks (types RGS and RGA) – Continued.

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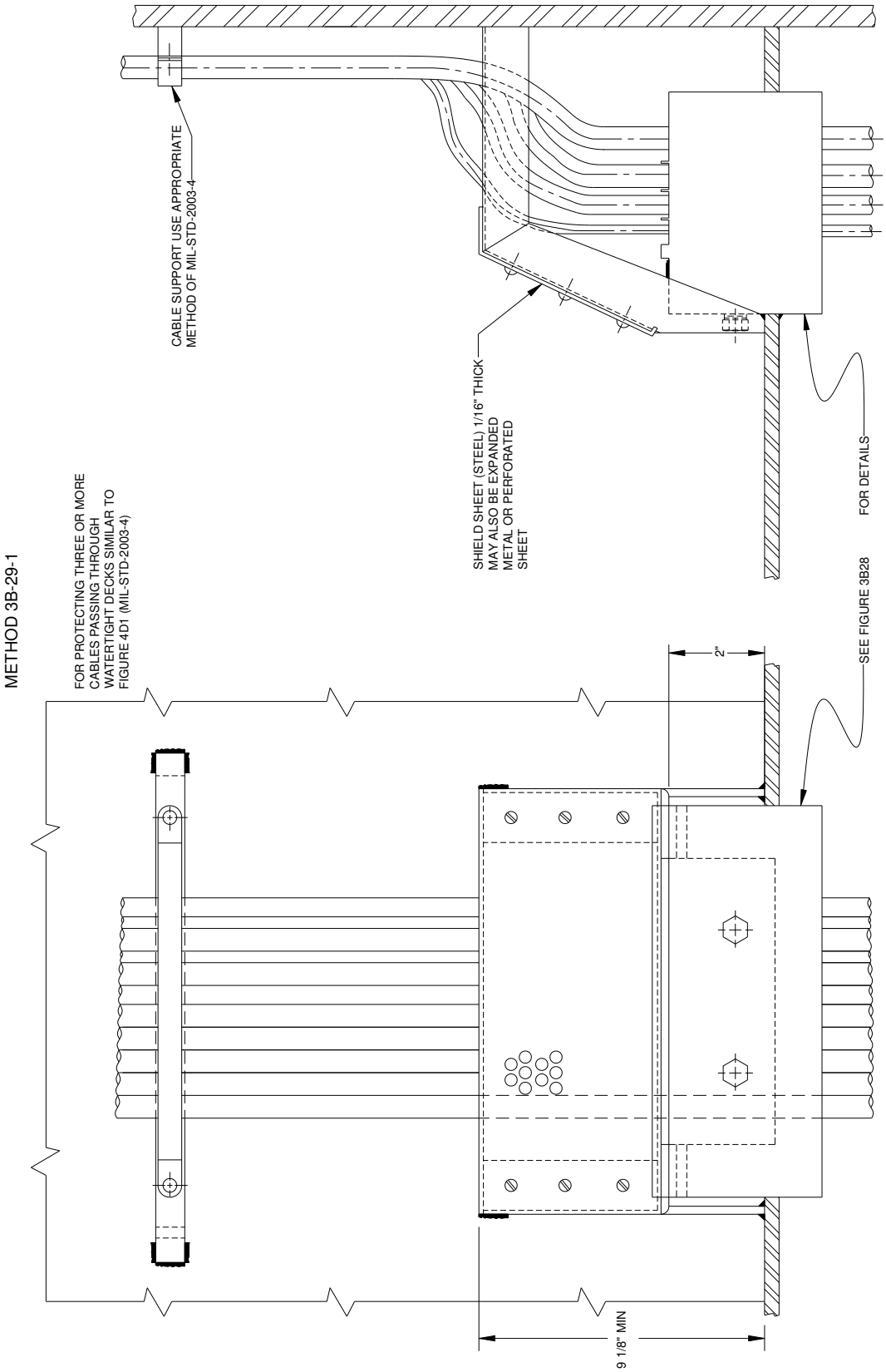


FIGURE 3B29. Multiple cable penetrator (type RGS and RGA) shield.

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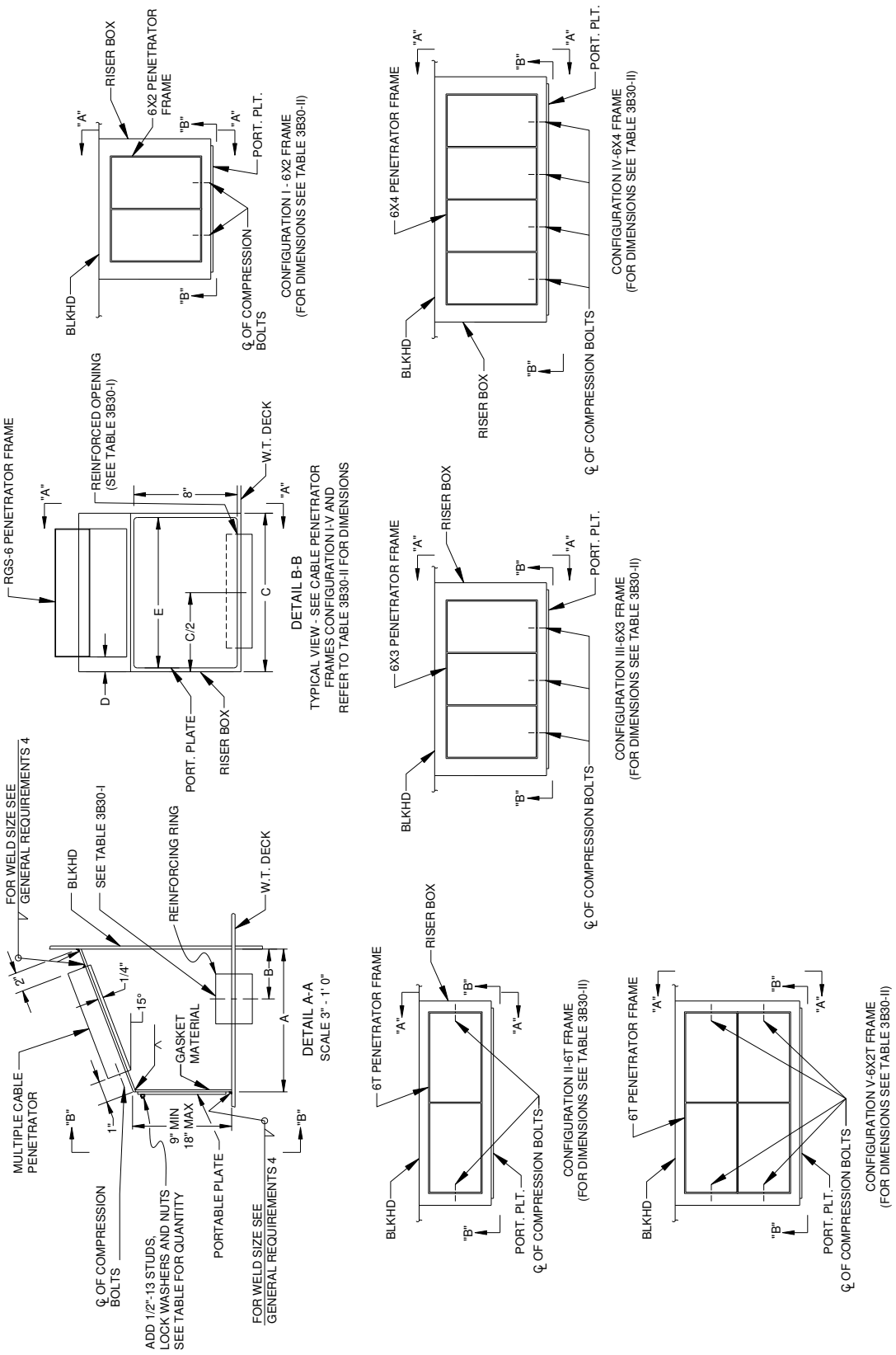


FIGURE 3B30. Multiple cable penetrator riser box.

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TABLE 3B30-I. Reinforced frame opening dimensions. [1/]

No. frames	Size of opening
6 x 2	10" x 4"
6 x 3	15" x 4"
6 x 4	20" x 4" or 16" x 5"

NOTE:

[1/] Min. of 20 sq. in. opening for each frame.

TABLE 3B30-II. Frame dimensions.

Dimensions						
Configuration number	**A**	**B**	**C**	**D**	**E**	**Bolts req'd.**
I	12½"	5"	14"	1¾"	13"	14
II	8"	4½"	21"	1⅛"	20"	16
III	12½"	5"	19"	1⅝"	18"	16
IV	12½"	5"	24"	1½"	23"	18
V	13½"	7"	21"	1⅛"	20"	16

FIGURE 3B30. Multiple cable penetrator riser box – Continued.

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METHOD 3B-31-1

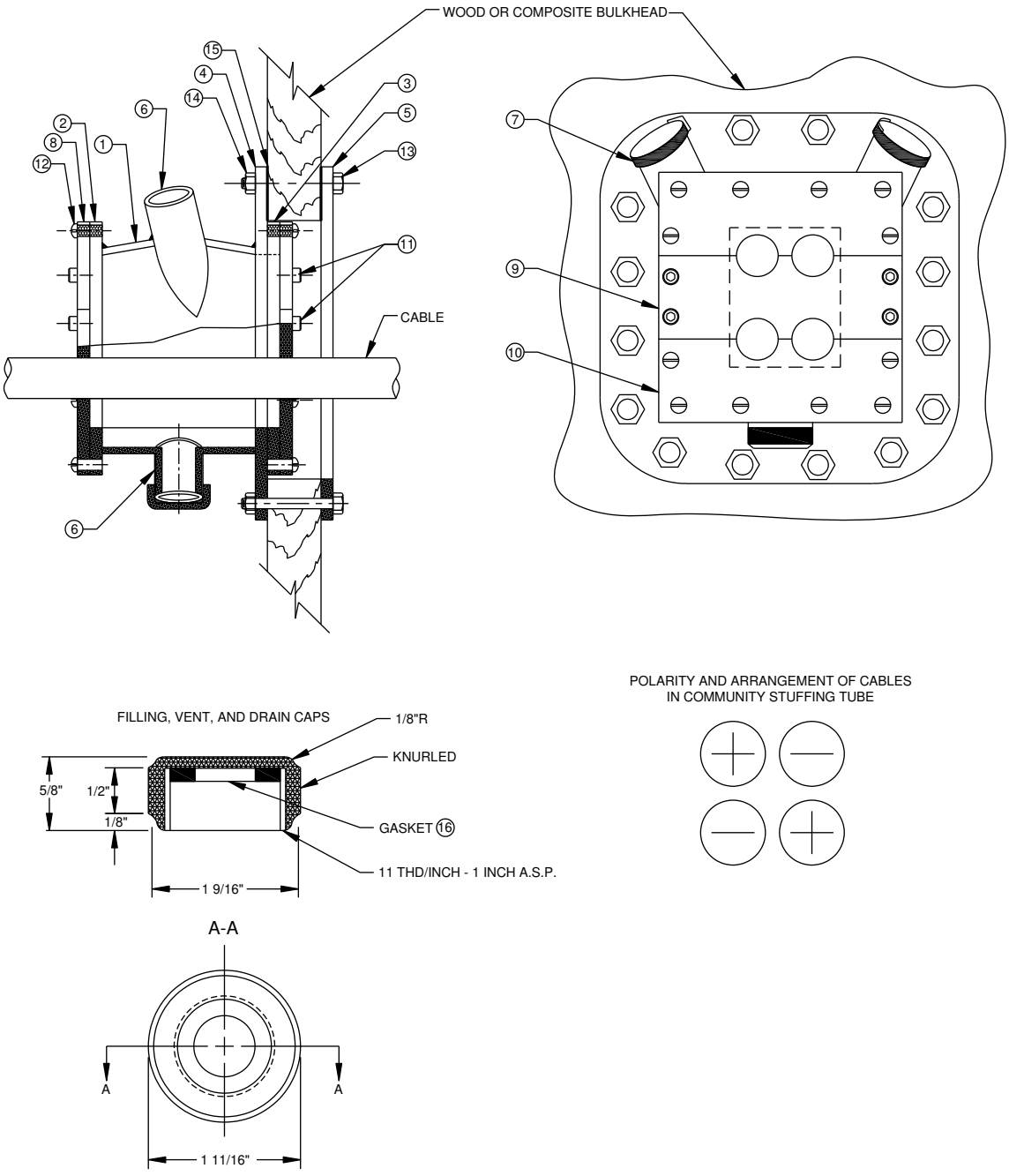


FIGURE 3B31. Community stuffing tube for minesweepers.

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TABLE 3B31-I. List of materials.

PC. no.	Qty.	Name	Material	Matl. spec. or std. dwg. no.
1	1	Gland body, 3/16" plate	Alum	ASTM B209
2	1	Front face plate, 5/16" pl.	Alum	ASTM B209
3	1	Back face plate, 3/8" pl.	Alum	ASTM B209
4	1	Flange, 1/4" pl.	Alum	ASTM B209
5	1	Backing plate, 1/4" pl.	Alum	ASTM B209
6	3	Pipe – 1", length to suit	Alum	ASTM B209
7	3	Cap – fill, drain and vent	Alum	ASTM B209
8	2	Slat, 2"	Alum	ASTM B209
9	2	Slat, 1 3/4"	Alum	ASTM B209
10	2	Slat, 2 1/4"	Alum	ASTM B209
11	32	1/4-20UNC socket head cap screw, 5/8" long	CRES 316	
12	32	1/4-20UNC socket head cap screw, 1/4" long	CRES 316	
13	16	3/8-16UNC hex head cap screw, length as reqd.	CRES 316	
14	16	3/8" nut	CRES 316	
15	2	Gasket – Flange, 0.090 thk.	Rubber	MIL-PRF-15624
16	3	Gasket Cap, 0.090 thk.	Rubber	MIL-PRF-15624
17	As req'd.	Filling compound	Rubber	
18	As req'd.	Plastic sealer		MIL-I-3064

Installation notes:

1. The gland is to be positioned on the bulkhead so that there is reasonable access to the filling, drain, and vent tubes, and so that the top of the gland is horizontal (maximum permissible tilt is 5 degrees).
2. An opening 7 3/8 inches high by 7 1/8 inches wide should be prepared in the bulkhead for receiving the gland.
3. Fit rubber gaskets and bolt gland to bulkhead.
4. Wipe the gland interior completely to remove traces of grease or oil.
5. Pull cables through gland, removing, where possible, all dirt and grease from cables.
6. The portions of the cables passing through the gland, especially at the slat assembly positions, shall be wiped clean with rags and cyclohexanone to remove all traces of grease or oil from the cables.
7. Coat the cable holes, butting edges, and butting side of the bottom slats with sealer, PC 18. Fit the slats into position and screw to gland body (one slat each side).
8. Position the bottom row of cables on the slats.
9. Proceed in a similar manner, to position the cables until all slats are in position.
10. After cables are in position, they should not be disturbed.

FIGURE 3B31. Community stuffing tube for minesweepers – Continued.

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Installation notes (continued):

11. Final gland cleaning operation – Remove the caps from the filling, drain, and vent tubes. Fit a hose to the drain tube to feed into a clean bucket and provide a clamp on the hose near the gland. Fill the gland with liquid degreasing agent.

    CAUTION – use adequate ventilation, avoid prolonged or repeated breathing of vapor, and avoid prolonged or repeated contact with skin. Do not take internally.

    After about 5 minutes, release the clamp to drain the gland, and blow compressed air through the gland from the top for about 5 minutes to ensure complete elimination of any fluid that might remain in the gland. Any leaks noted during the cleaning operation should be repaired.
12. The liquid filling compound, PC 17, consists of two separate liquids that are mixed together immediately prior to use. The success of the resultant compound depends on thorough mixing, and this should be a stirring action. The compound should be stirred for 5 minutes. Complete mixing instructions are printed on the filling compound container.
13. Pour the filling compound into the gland until the tubes are filled.
14. Both sides of the gland shall be inspected during the filling operation for leakage. It may be possible to repair slight leaks by external application of sealer. If leakage persists, the filling operation shall stop. Immediately drain the filling compound. After leak has been stopped, the filling operation may be restarted.
15. Fit filling and vent tube caps.
16. See 4.5 for fastener and fastener material requirements.

FIGURE 3B31. Community stuffing tube for minesweepers – Continued.

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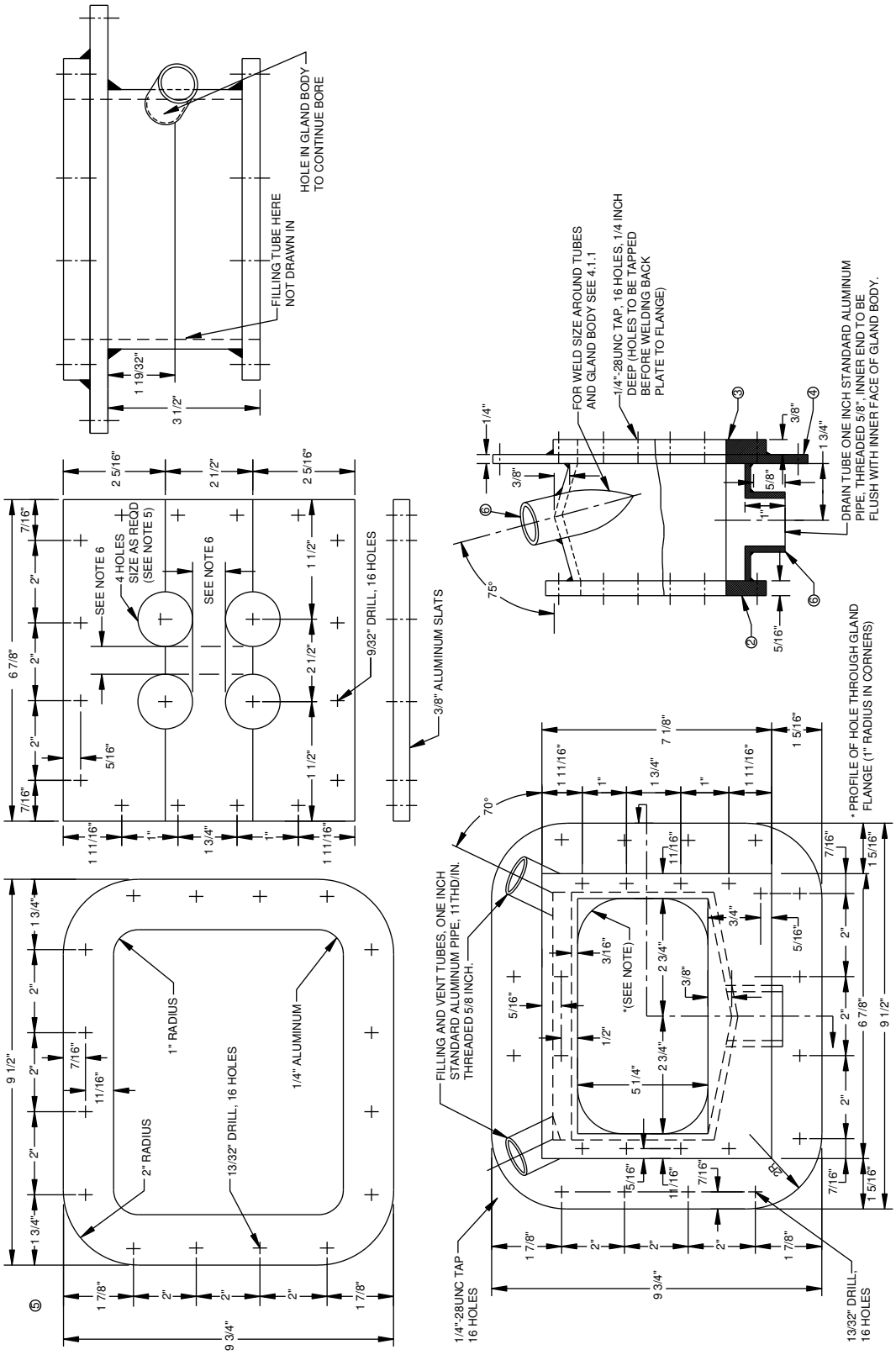


FIGURE 3B32. Community stuffing tubes for minesweepers (details).

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NOTES:

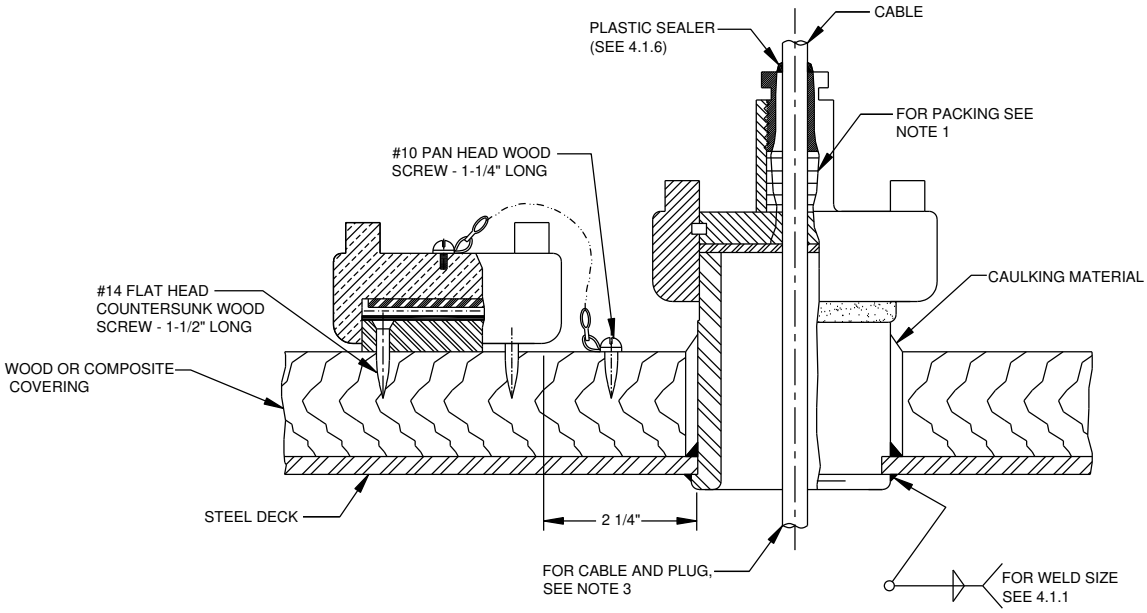
1. For a list of materials, see figure 3B31.
2. See figure 3B31 for fill, drain, and vent cap detail.
3. Drill front and back slats at the same time.
4. Painting shall be in accordance with approved methods.
5. The diameter of the holes should be same size (+1/16, -0) outside diameter of minesweep cable used.
6. The separation of these holes should provide a minimum spacing of 0.625 between cables and also provide a minimum spacing of 0.5 between cable and inner faces of gland body.

FIGURE 3B32. Community stuffing tubes for minesweepers (details) – Continued.

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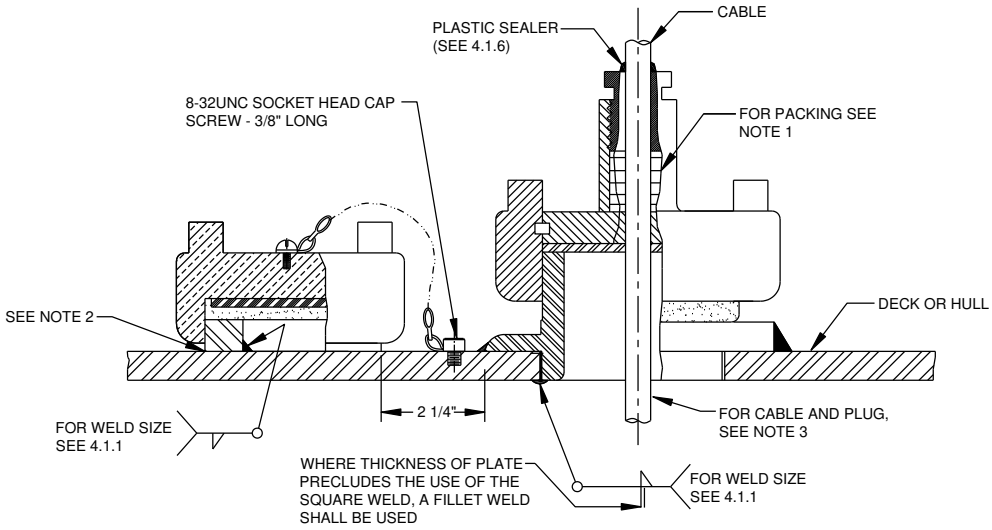
METHOD 3B-33-1

FOR PASSING CABLES THROUGH WEATHER DECK WITH WOOD OR COMPOSITE COVERING



METHOD 3B-33-2

FOR PASSING CABLES THROUGH WEATHER DECK AND HULL MIL-S-24235



NOTES:

1. Glands of outlets shall be packed with prefabricated packing type 2, size “B” in accordance with MIL-DTL-16685.
2. After welding dummy plug (MIL-S-24235) in place, an approved coal tar pitch emulsion shall be used to seal the cracks and fill the inside flush.
3. Outlets are designed to accommodate a single cable and type CCC-2 plug in accordance with MIL-DTL-2726/68 (see 4.3).
4. Plastic sealer is required on kickpipes and stuffing tubes when exposed to the elements (penetrates the skin of the ship).

FIGURE 3B33. Deck outlets for portable cables (surface ships).

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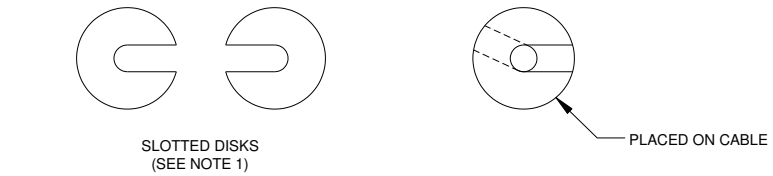
NOTES (continued):

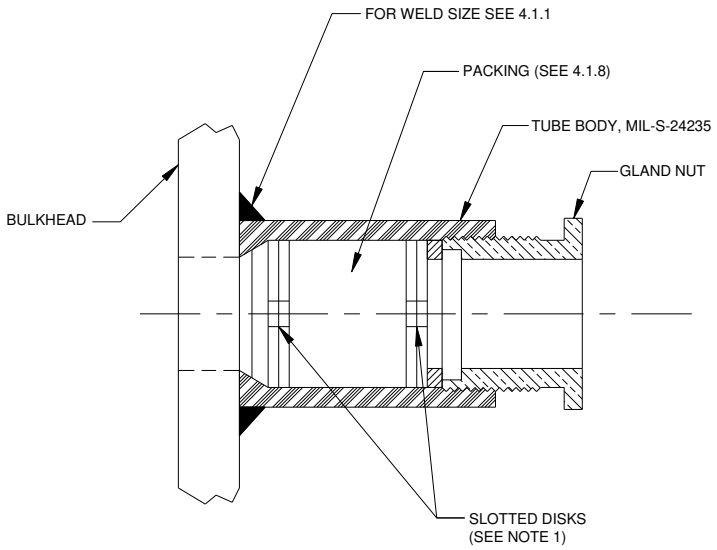
5. Wood screws shall be in accordance with table I of MIL-STD-2003-2.
6. See 4.8 for thread lubrication guidance.
7. See figure 3A1 for gland nut tightening methods.
8. See 4.5 for fastener and fastener material requirements.

FIGURE 3B33. Deck outlets for portable cables (surface ships) – Continued.

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METHOD 3B-34-1
(SEE NOTE 2)





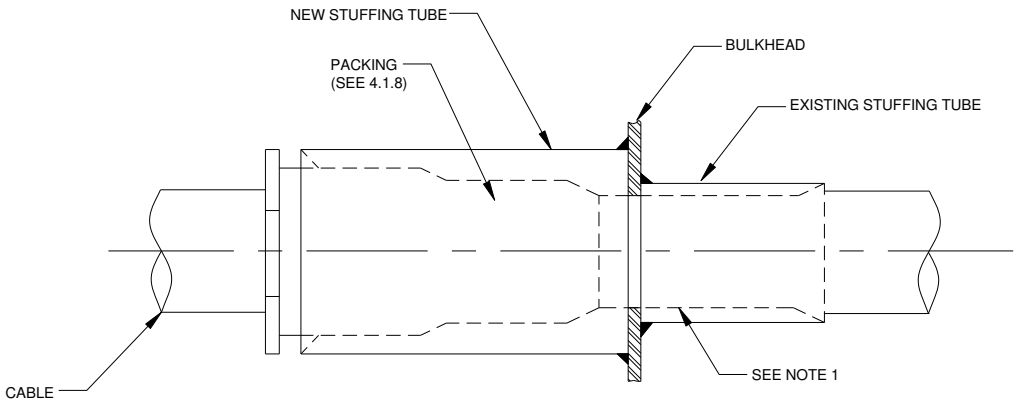
NOTES:

1. Cut a slot in a disk or washer on one side of the cable hole only. Two slotted disks or slotted washers are then slipped around the cable both in the gland bottom and under the gland nut to hold the packing securely in place.
2. See figure 3A1 for gland nut tightening methods.
3. See 4.8 for thread lubrication guidance.

FIGURE 3B34. MIL-S-24235 stuffing tubes for smaller size cables using slotted packing disks.

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METHOD 3B-35-1



TYPICAL INSTALLATION

TABLE 3B35-I. New stuffing tubes – MIL-S-24235/10. 1/

Existing stuffing tubes MIL-S-24235/10	Size	A	B	C	D	E	F	G	J	K	L	M	N	P	R	S	T	V	W	X	Y	Z	AA	BB
	A		X	X	X	X	X																	
	B			X	X	X	X	X																
	C				X	X	X	X	X															
	D					X	X	X	X	X														
	E						X	X	X	X														
	F							X	X	X	X													
	G								X	X	X	X												
	J									X	X	X	X	X										
	K										X	X	X	X	X									
	L											X	X	X	X									
	M												X	X	X	X								
	N													X	X	X	X							
	P														X	X	X	X						
	R															X	X	X	X					
	S																X	X	X	X	X			
	T																	X	X	X	X	X	X	
	V																		X	X	X	X	X	
	W																			X	X	X	X	
	X																				X	X	X	X
	Y																					X	X	X
	Z																						X	X
	AA																							X
	BB																							

NOTE:

1/ X indicates an acceptable combination.

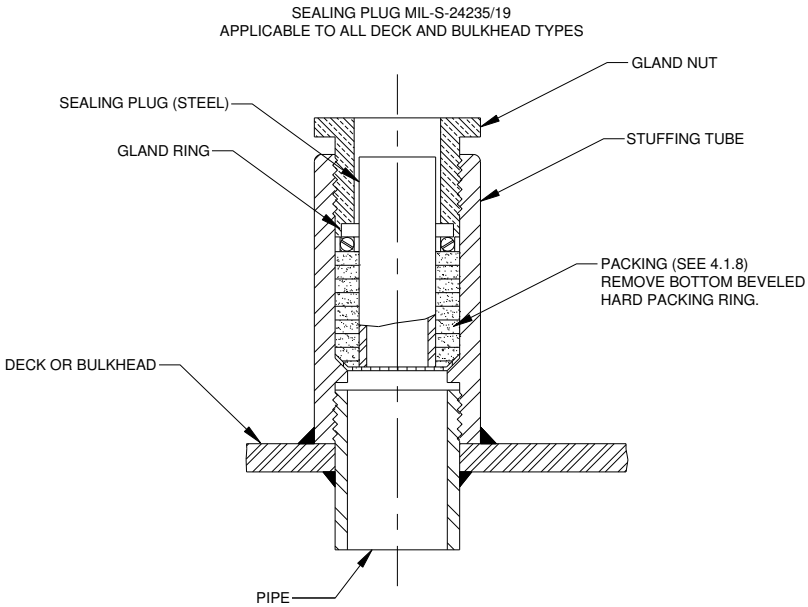
NOTES:

1. Table 3B35-I indicates new tube size limitations for adaptation to undersized non-reusable stuffing tubes.
2. Ream smooth and round off inside edge of existing tube to prevent cable chafing.
3. See figure 3A1 for gland nut tightening methods.
4. See 4.8 for thread lubrication guidance.

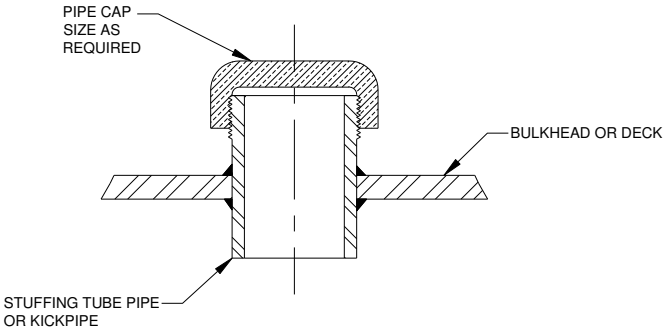
FIGURE 3B35. Changing stuffing tube sizes to accommodate larger cable.

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METHOD 3B-36-1
SEALING UNUSED STUFFING TUBES
IN DECKS AND BULKHEADS



METHOD 3B-36-2
SEALING UNUSED KICKPIPES AND
STUFFING TUBES WITH REMOVABLE
STUFFING TUBE BODY



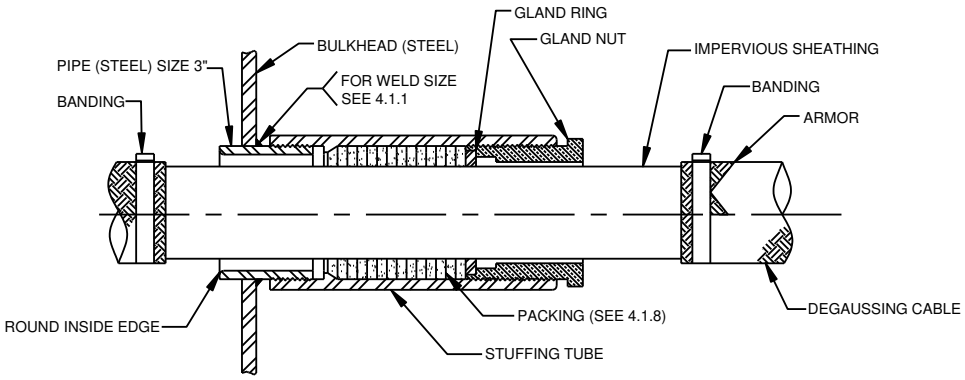
NOTES:

1. Application of the methods shown (3B-36-1 and 3B-36-2) are intended for existing installed stuffing tubes whose cables have been removed and not necessarily replaced.
2. See figure 3A1 for gland nut tightening methods.
3. See 4.8 for thread lubrication guidance.

FIGURE 3B36. Sealing unused stuffing tubes in medium steel bulkhead and decks.

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METHOD 3B-37-1



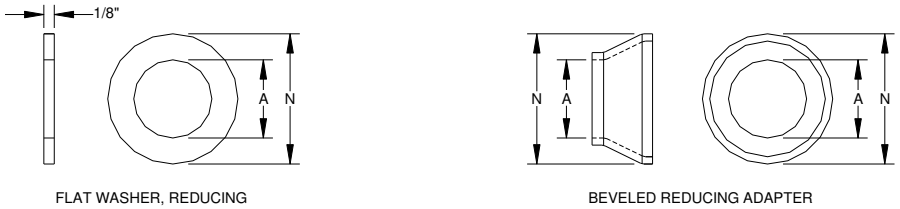
Instructions:

1. Cut away the outer armor, if installed, in the way of the stuffing tube and secure as shown.
2. Insert extra packing rings in stuffing tube.
3. Position the impervious sheathing area within the stuffing tube and tighten gland nut.
4. See figure 3A1 for gland nut tightening methods.
5. See 4.8 for thread lubrication guidance.

FIGURE 3B37. Sealing degaussing cable in bulkhead stuffing tubes.

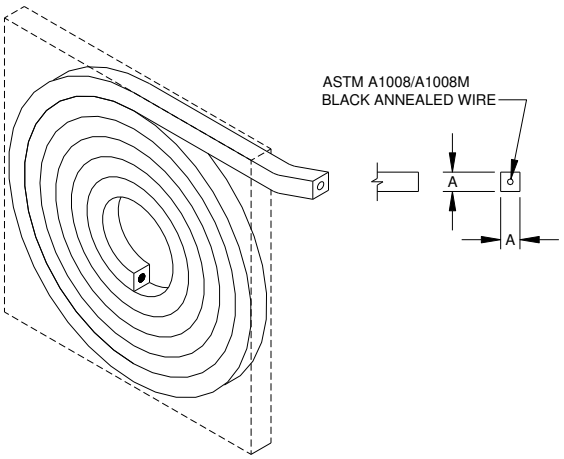
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METHOD 3B-38-1
STUFFING TUBES BEVEL REDUCING ADAPTER ASSEMBLIES
(MIL-S-24235/18)



FLAT WASHER, REDUCING

BEVELED REDUCING ADAPTER



REINFORCED NEOPRENE PACKING

TABLE 3B38-I. Packing wire diameter.

Part no.	"A" dim.
3B381	3/16"
3B382	1/4"
3B383	5/16"
3B384	3/8"
3B385	7/16"
3B386	1/2"
3B387	9/16"

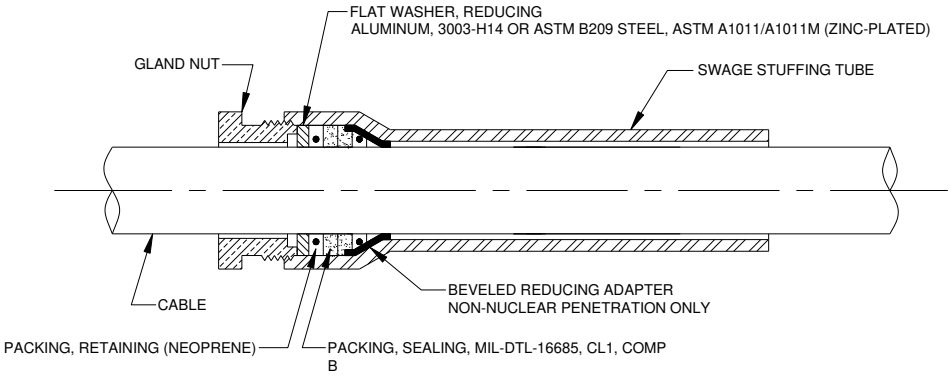


FIGURE 3B38. MIL-S-24235/18 stuffing tube bevel reducing adapter assemblies.

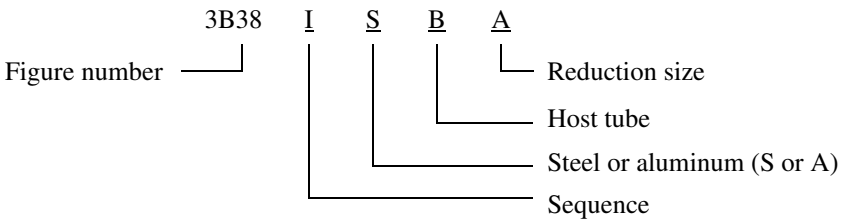
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TABLE 3B38-II. Stuffing tube adapter sets.

Adapter set (steel)	Adapter set (aluminum)	Existing tube	"N" dim. OD	Effective reduction size	"A" dim. ID
3B381SBA	3B381ABA	B	0.875	A	0.406
3B382SDA	3B382ADA	D	1.032	A	0.406
3B383SDB	3B383ADB	D	1.032	B	0.515
3B384SDC	3B384ADC	D	1.032	C	0.640
3B385SGB	3B385AGB	G	1.360	B	0.516
3B386SGC	3B386AGC	G	1.360	C	0.640
3B387SGD	3B387AGD	G	1.360	D	0.750
3B388SGE	3B388AGE	G	1.360	E	0.812
3B389SGF	3B389AGF	G	1.360	F	0.843
3B3810SLG	3B3810ALG	L	1.703	G	0.953
3B3811SLJ	3B3811ALJ	L	1.703	J	1.062
3B3812SLK	3B3812ALK	L	1.703	K	1.172
3B3813SNL	3B3813ANL	N	1.875	L	1.265
3B3814SNM	3B3814ANM	N	1.875	M	1.406
3B3815SSN	3B3815ASN	S	2.562	N	1.515
3B3816SSP	3B3816ASP	S	2.562	P	1.625
3B3817SSR	3B3817ASR	S	2.562	R	1.750
3B3818SWS	3B3818AWS	W	3.000	S	1.875
3B3819SWT	3B3819AWT	W	3.000	T	2.062
3B3820SWV	3B3820AWV	W	3.000	V	2.187
3B3821SAAX	3B3821AAAX	AA	3.625	X	2.500
3B3822SAAY	3B3822AAAX	AA	3.625	Y	2.609
3B3823SAAZ	3B3823AAAZ	AA	3.625	Z	2.781

NOTES:

1. Adapter set consists of one flat washer and one bevel reducing adapter.
2. Definition of adapter sets numbering system follows:



3. Neoprene shall be in accordance with MIL-PRF-15624, class I, grade 50.
4. See figure 3A1 for gland nut tightening methods.
5. See 4.8 for thread lubrication guidance.

FIGURE 3B38. MIL-S-24235/18 stuffing tube bevel reducing adapter assemblies – Continued.

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Installation notes:

1. Round multiple cable penetrator (RMCP) is a modular system that utilizes standard units and dimensions to allow maximum flexibility and compatibility with various numbers and sizes of cables. Final sealed units are designed to provide watertight, airtight, and fireproof bulkhead or deck penetrations. Round MCP components shall meet requirements of MIL-DTL-24705 and MIL-DTL-24705/3. Since there are more than one variant of RMCPs available, manufacturer's instructions should be consulted for installation details. The following general instructions apply. The dimension tables in figures 3B40 and 3B41 cover the smaller RMCPs shown just before step 3a. For dimensions for the larger RMCP, consult manufacturer's information.
2. Remove any dirt from inside the frame. Freely lubricate outside of the RMCP with cable lubricant (tallow) before inserting into pipe sleeve, but do not push RMCP completely into the sleeve at this time.
Pull all cables through the RMCP, bearing in mind that cables shall always approach parallel to each other and in a horizontal plane to the penetrator. This can be accomplished if the nearest cable hanger is 16 inches minimum from the bulkhead. Cable running up the bulkhead and through an RMCP shall have its full bending radius outside of the penetrator. While in the process of pulling cables, it is imperative that a reasonable amount of slack should be left to facilitate movement of cables within the RMCP to ease packing operation. See MIL-STD-2003-4 for cable slack requirements. Also, it is necessary that each cable be identified on the compression nut side of the multiple cable penetrator by lead number and cable size. All cables through a new round multiple cable penetrator shall be pulled prior to installing insert blocks. See 4.4 for specific restrictions and instructions for installation of fiber optic and blown fiber cables in electrical penetrations.
3. After pulling all cables through a particular round multiple cable penetrator, it can be packed. First, apply a liberal coat of cable lubricant (tallow) to all outside surfaces of the insert blocks and inside of the RMCP. When packing armored cable, apply Nelson Fire Stop Caulk, or equal (see 4.2), sealer on the surface of each block that contacts the cable (a ⅛-inch bead every ½ inch around the circumference.) Each RMCP shall be laid out (blocked) showing solid blocks (used as spares) and cable assignments prior to packing. A typical example of a packed RMCP unit with location assignments for packing is shown below.

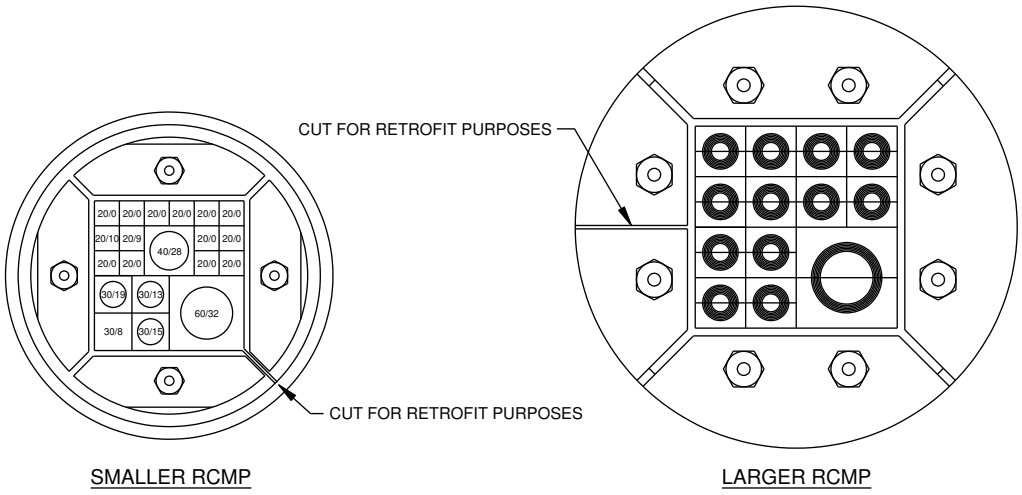


FIGURE 3B39. Round multiple cable penetrators installation notes.

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Installation notes (continued):

a. Select insert blocks and spares (see note 3) from tables 3B21-I and 3A10-I through 3A25-I (presented for information and guidance) and prepare as illustrated above (lubricant and sealer, where required). Blocks should be selected to avoid any gaps under compression. Fill-ins as explained in figure 3B20 and illustrated in figure 3B21 may be used if different size blocks are installed in the same row to ensure an even installation. Although not desirable, the fill-ins also may be used if soft cable would leave a gap under compression. For multi-diameter modules, the insert blocks should be adapted by peeling one layer at a time. The insert blocks should be adapted to achieve a 0.1- to 1-millimeter gap between the two halves when the block is held against the cable. See following illustration. The layers shall be peeled from both sides to achieve an even installation. Once a layer is peeled, the layer may not be reinserted into the block.



b. Starting with largest cable, place the lower half of the 60/32 insert block under the cable in the unit. Next, repeat the procedure with the 30/15, then add one 30/0 spare. Add the upper half of the 60/32 and the 30/15. Place lower half of the 30/13 and 30/19 under cable, then add upper halves.

CAUTION: For easiest packing, do not insert the blocks fully into the square hole until all cable holding and spare blocks are in their assigned locations.

c. On top of the previously installed blocks, place four 20/0 spares and the lower half of the 40/28 under its cable. In the next higher row, place two 20/0 spares and the lower half of the 20/9 and 20/11 under their respective cables. Then add upper halves of 40/28, 20/9, and 20/11 on top.
d. Add final row of six 20/0 spares.
e. All blocks and their RMCP can now be fully inserted simultaneously, i.e., the blocks pushed completely into the square hole of the plug and RMCP completely into the sleeve.
f. When the RMCP unit is completely filled and inserted, then the nuts are alternately tightened until the RMCP is secure in the hole. The installation is then complete.
g. If individual conductors of a three-phase power circuit are in different cables, the individual cables shall be installed in adjacent insert blocks. There shall be no ferrous material between the individual cables. The purpose of this requirement is to avoid heating due to electro-magnetic induction.
h. See SE000-01-IMB-010 for modifications to the MCPs to support the installation of waveguides.

4. To add a new cable to a packed RMCP unit, loosen the outside hex nuts and remove the desired spare insert modules. Pull the new cable and add matching cable holding blocks in accordance with previous instructions.
5. Template all work from ship.
6. All welding and inspection shall be in accordance with 4.1.1.
7. All painting shall be done in accordance with MIL-DTL-917.
8. This plan was developed for the reinforcement of openings in structure of surface ships other than in protective plating to permit installation of round multiple cable penetrator in watertight decks and bulkheads of surface ships.
9. Fillet weld reinforcement for T2V.1 weld joints in accordance with MIL-STD-22 shall be ⅛ inch for plating up to 15.3# and ¼ inch for plating above 17.85#.
10. The bulkheads and decks that are considered to be longitudinal strength members are defined as components in “strength envelope (i.e., in outside plating, uppermost strength deck, inner bottom, and their associated longitudinal framing) and for other strength decks within the midship three-fifths length”. If there is difficulty in definition, consult with structural design.

FIGURE 3B39. Round multiple cable penetrators installation notes – Continued.

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Installation notes (continued):

11. After installation is complete, pipe sleeve of penetrator may be painted to suit surrounding structure in accordance with applicable ship specifications. Wait 24 hours or longer (refer to manufacturer's instructions) after installation before exposing the cables/pipes to strain or pressure. The following wait time shall be observed prior to the post-installation testing.
12. Post-installation testing:
    a. After installation of a new RMCP or work on an existing RMCP, retest shall be conducted to confirm integrity of the electrical penetration.
    b. Perform a visual inspection on plastic or rubber inserts inside of the RMCP to ensure there are no visible or discernible openings.
    c. For airtight, oiltight, and watertight penetrations, conduct an air blast test or a vacuum box test as follows:
        (1) Air blast test: The air hose nozzle shall be held a maximum of 3 inches from the MCP area. The distance should be as close to 3 inches as practicable to avoid distortion of the MCP blocks due to high air flow. The pressure shall be directed at the structure under test in a manner most likely to disclose leaks. The minimum nozzle diameter shall be ⅜ inch and the nozzle pressure shall be 60 to 90 PSIG as monitored at the nozzle. Obtain a pressure differential in the same direction as would occur during a full compartment, tank, or void test using the air hose. The air pressure shall be maintained for at least 3 minutes. As soon as the pressure is applied, apply a soapy solution to the opposite side of the structure and inspect for leaks.
        (2) Vacuum box test: Apply a soapy solution to the joint to be tested. Install the vacuum box so the pressure differential will be in the same direction as a full compartment, tank, or void completion test. Draw 10.2"Hg vacuum (minimum). Maintain the vacuum for at least 3 minutes. Inspect the joint or fitting for leakage by observing the formation of bubbles.
    d. For penetrations through fume-tight bulkheads, conduct a visual/light test to ensure no light transmits from one side of the RMCP to the other.

Round multiple cable penetrators for existing cables:

13. Temporarily insulate cable for protection against welding and burning.
14. Remove old non-watertight structure.
15. Prepare hole in bulkhead to receive round multiple cable penetrator (designated "RMCPO"). For the smaller frames, one corner of RMCP is cut with a sharp tool to allow the frame to be opened and placed around the existing cables. Pipe sleeve shall also be placed around the cable. For the larger RMCP frames, the cut is between two of the screws that secure one of the compression plates. (See illustration prior to step 3a.)
16. Segregate and confine required cables for protection and installation.
17. Slide unit into prepared sleeve in bulkhead then proceed as outlined in notes 3-12.

NOTES:

1. The round multiple cable penetrator shall not be installed in bulkheads and decks that are exposed to the weather except for electro-magnetic compatibility (EMC) type penetrations covered in figure 3B63.
2. Insert block cable assignments are shown for guidance and information on figure 3B21.
3. Approved multi-diameter insert blocks may be used where available.

FIGURE 3B39. Round multiple cable penetrators installation notes – Continued.

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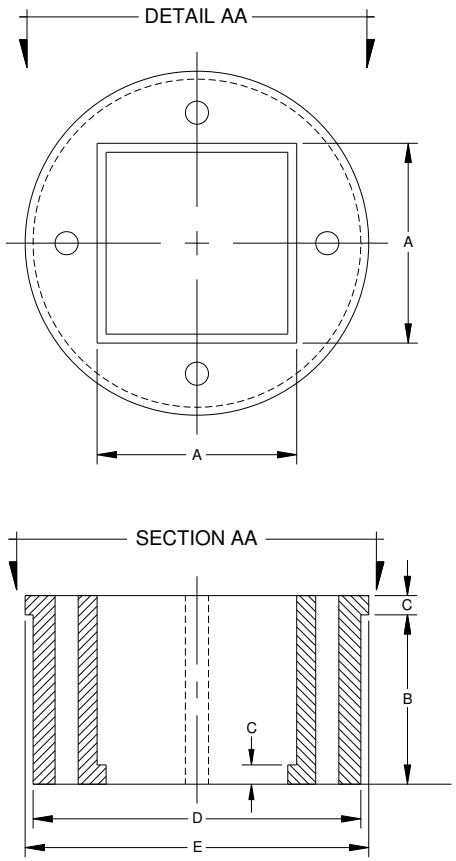


TABLE 3B40-I. Dimensions.

	A					
	IN	MM	B	C	D	E
RMCP2	1.181	30	2.559	0.197	2.000	2.232
RMCP25	1.220	30	2.559	0.197	2.302	2.634
RMCP3	1.57	40	2.559	0.197	3.000	3.340
RMCP4	2.362	60	2.559	0.197	4.000	4.282
RMCP5	3.15	80	2.559	0.197	5.000	5.339
RMCP6	3.543	90	2.559	0.197	6.000	6.375
RMCP8	4.72	120	2.559	0.197	8.000	8.312

For factory cut corner for retrofit applications add “O” to the nomenclature, i.e., “RMCPO”

Material: Special neoprene elastomer

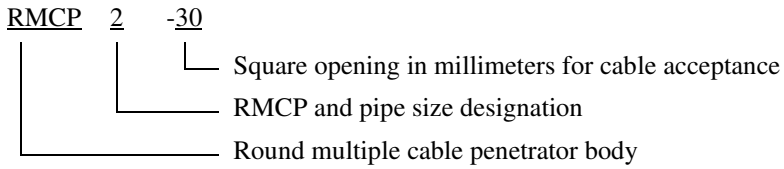
FIGURE 3B40. Round multiple cable penetrators (dimensions).

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Nomenclature:

Round multiple cable penetrators (RMCP): a system of passing cables through water and non-watertight bulkheads and decks designed to suit particular bulkhead and deck weights.

a. Physical arrangement and identification



b. RMCP consists of seven basic sizes: 2, 2.5, 3, 4, 5, 6, and 8 for insertion into steel pipe (table 3B41-I) or aluminum pipe (table 3B41-II).

c. Compression hardware for sealing is comprised of the following:
   (1) Back compression plate (detail 1) with press fitted compression screw (detail 2) (illustration 3) to make furnished assembly.
   (2) Front compression plate similar to back compression plate, only it is furnished without compression screw.
   (3) Hex nut (illustration 4) to be used with front compression plate and stud.

d. Insert blocks:
Twin half blocks are specially formulated of a neoprene elastomer with a centered semi-circular groove. When matched around a cable, these blocks form a single block with a tight fit. See note 2.

e. Spares:
Solid blocks are specially formulated of a neoprene elastomer and are used as fillers or spare provisions for addition of future cables. Sizes shown for information and guidance on figure 3B21. When installing new MCPs, at least 20 percent of the MCP usable area shall be reserved for future cable installations.

f. Fill-ins:
Same material as insert blocks and spares are used to fill thicknesses of 5 or 10 millimeters created by the presence of different size blocks in the same row. Also employed to increase the pressure in the penetrator unit when exceptionally soft cables are used. These fill-ins are available in two sizes: 24x5/0 or 12x10/0. See figure 3B21 for information and guidance. Serrations have been provided to permit slicing of the fill-ins to the required length. Example: 6x10/0, 8x5/0, etc.

g. Tallow:
Tallow is a natural lubricant that allows the blocks, stayplates, compression system and cables to move freely during the compression process. Without tallow, the friction during compression will create gaps in the system and an effective seal will not be obtained. Therefore, as stated in the installation instructions in figure 3B39, tallow must be applied for every insert block.

h. Sealer:
Liquid silicone rubber applied to cable side of each gauge block. Upon compression, this provides a seal between the armor of the cable and its impervious inner sheath.

i. Assembly steps:
   (1) Place RMCP into appropriate pipe sleeve and leave slightly extended.
   (2) Pass cable through RMCP.
   (3) Place modules around cable and inside RMCP.
   (4) Press complete assembly into pipe until flange is flush with end of pipe.
   (5) Then tighten hardware for compression and expansion of multi-plug to achieve air and water seal.

j. All dimensions are in inches unless otherwise stated.

FIGURE 3B41. Round multiple cable penetrators.

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TABLE 3B41-I. Round carbon steel tube.

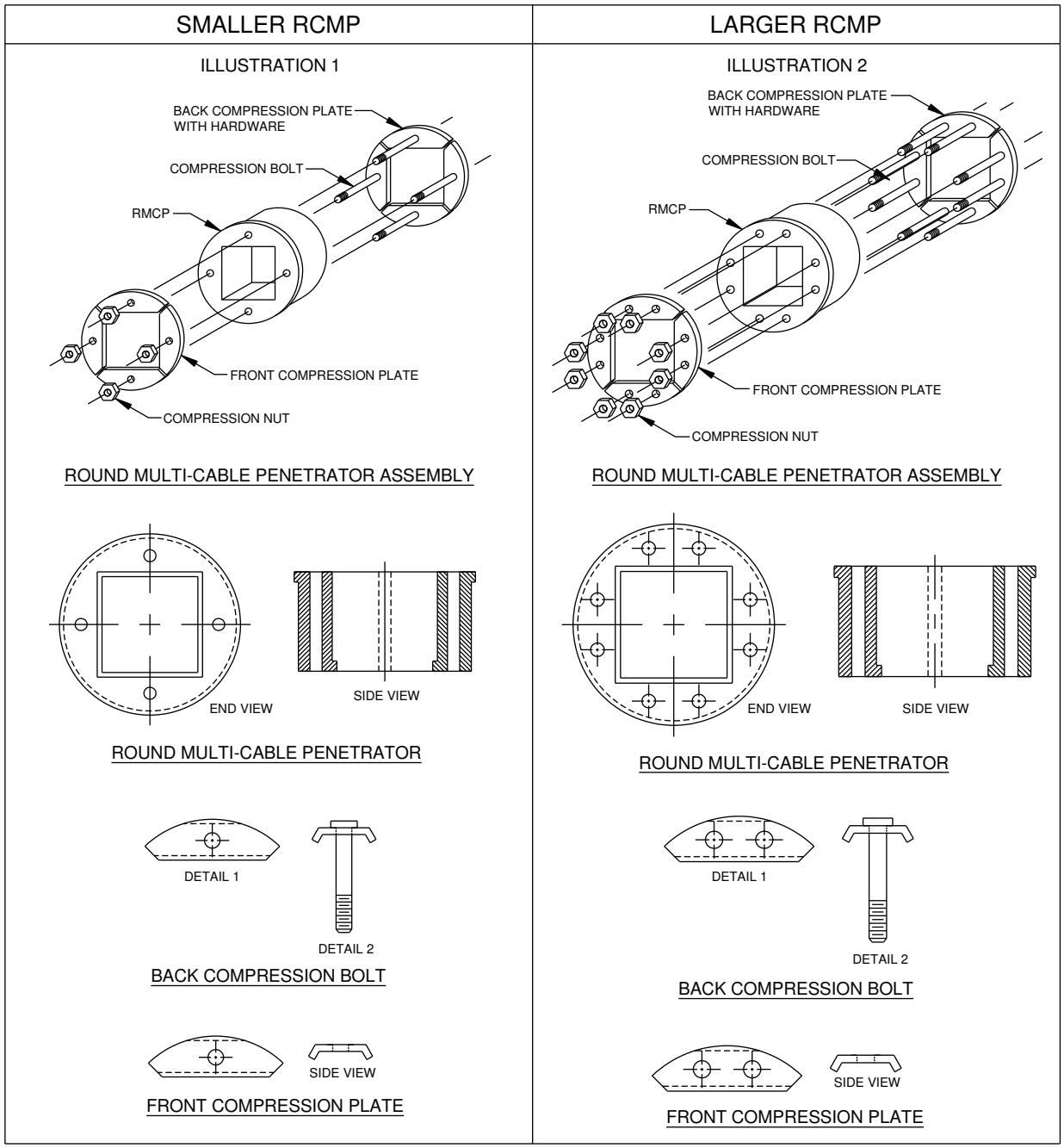
Ident. no.	OD	Wall	Theoretical ID	Type	Pounds per ft.	Length for blkhd	Length for deck
RMCP2	2⅜	0.154	2.067	SCH 40 PIPE	3.653	4	12
RMCP2.5	2⅞	0.203	2.469	SCH 40 PIPE	5.793	4	12
RMCP3	3½	0.216	3.068	SCH 40 PIPE	7.576	4	12
RMCP4	4½	0.237	4.026	SCH 40 PIPE	10.790	4	12
RMCP5	5$^{9}/_{16}$	0.258	5.047	SCH 40 PIPE	14.620	4	12
RMCP6	6⅝	0.280	6.065	SCH 40 PIPE	18.970	4	12
RMCP8	8⅝	0.322	7.981	SCH 40 PIPE	28.550	4	12

TABLE 3B41-II. Round aluminum (alloy 6061-T6).

Ident. no.	OD	Wall	Theoretical ID	Type	Pounds per ft.	Length for blkhd	Length for deck
RMCP2	2⅜	0.154	2.067	SCH 40 PIPE	1.264	4	12
RMCP2.5	2⅞	0.203	2.469	SCH 40 PIPE	2.004	4	12
RMCP3	3½	0.216	3.068	SCH 40 PIPE	2.621	4	12
RMCP4	4½	0.237	4.026	SCH 40 PIPE	3.733	4	12
RMCP5	5$^{9}/_{16}$	0.258	5.047	SCH 40 PIPE	5.057	4	12
RMCP6	6⅝	0.280	6.065	SCH 40 PIPE	6.564	4	12
RMCP8	8⅝	0.322	7.981	SCH 40 PIPE	9.878	4	12

FIGURE 3B41. Round multiple cable penetrators – Continued.

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NOTES:

1. Bolts and back compression plates are factory assembled into one unit.
2. Insert block cable assignments are shown for information and guidance on figure 3B21.

FIGURE 3B41. Round multiple cable penetrators – Continued.

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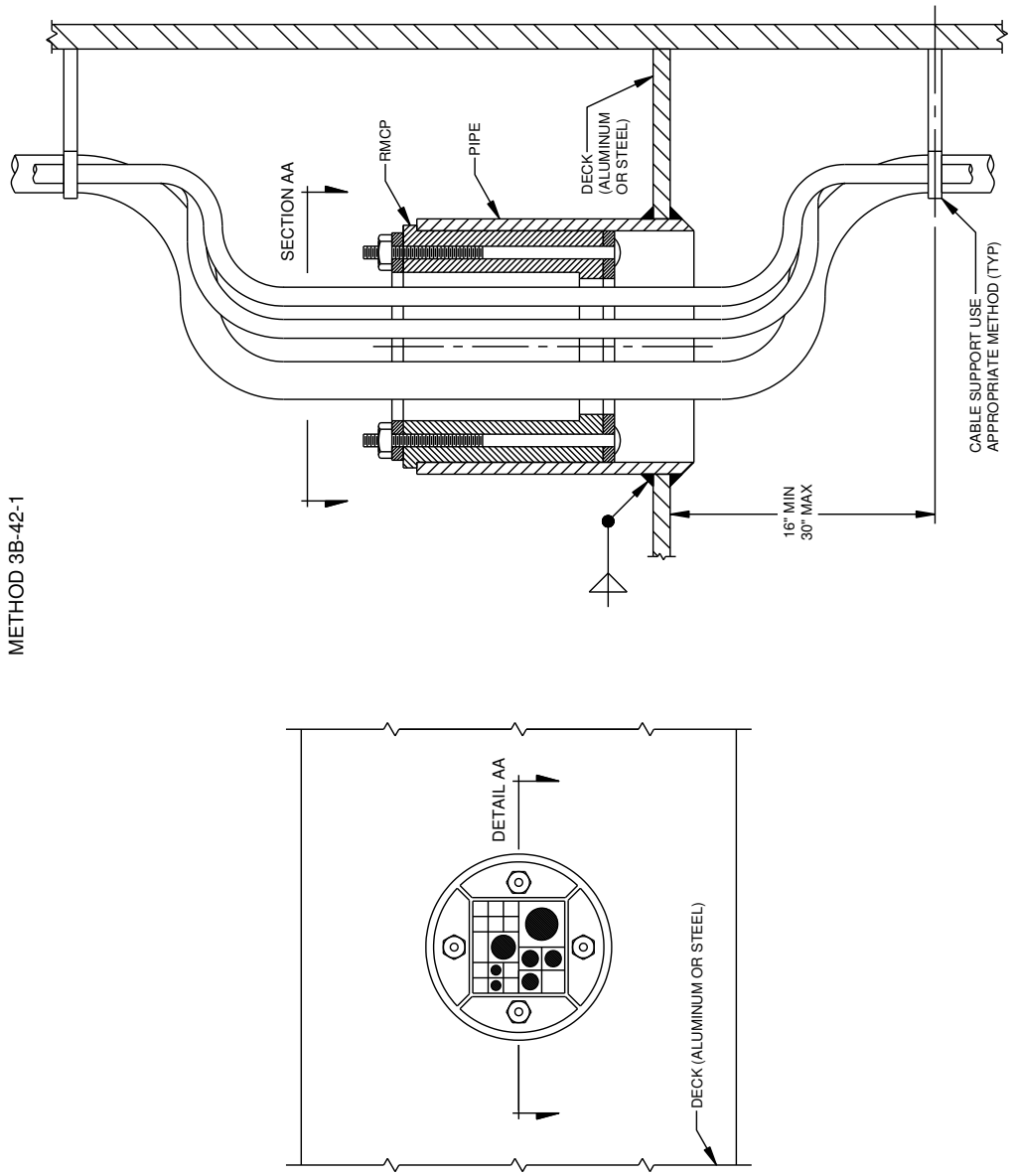


FIGURE 3B42. Round multiple cable penetrator installation in steel or aluminum decks adjacent to bulkheads.

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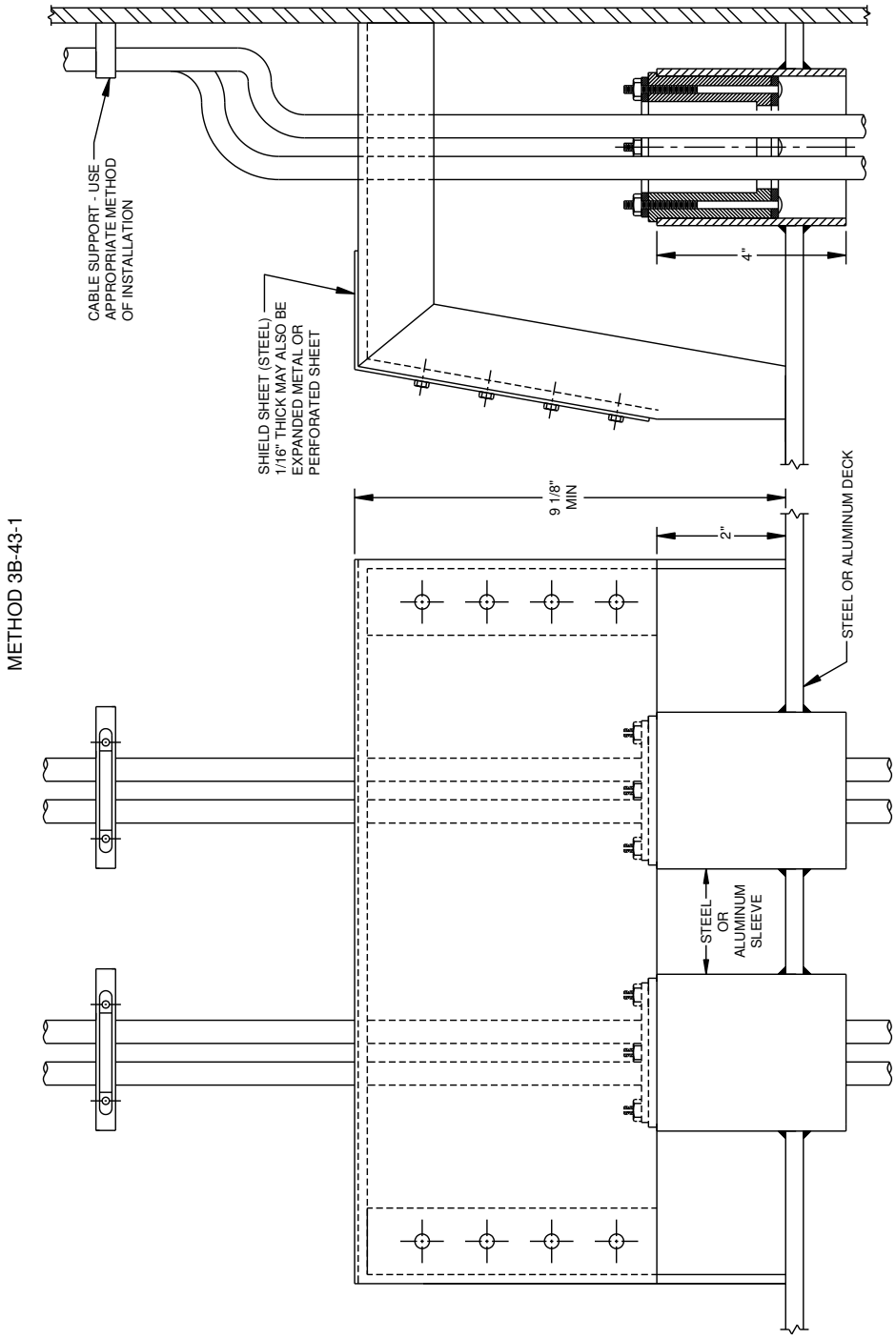


FIGURE 3B43. Round multiple cable penetrator shield.

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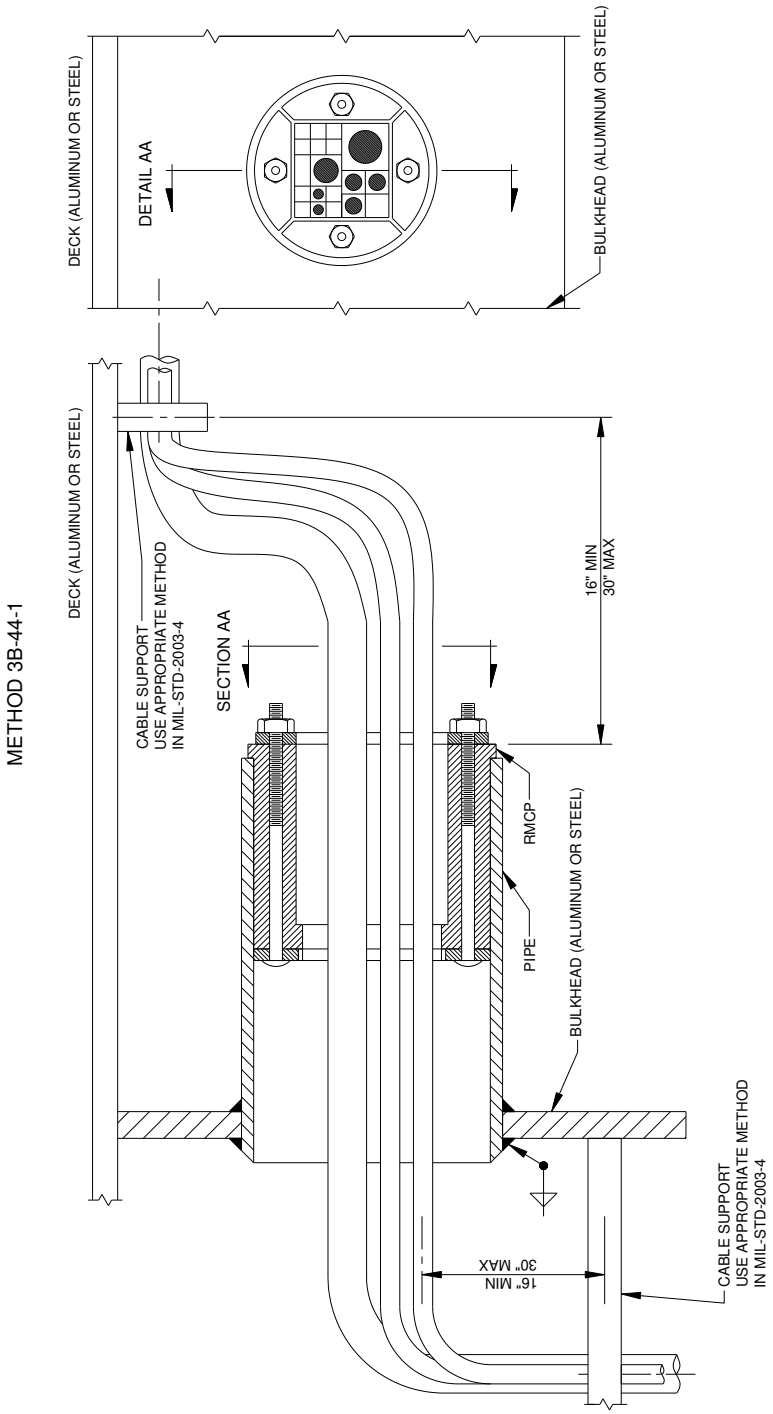


FIGURE 3B44. Round multiple cable penetrator installation in steel or aluminum bulkhead.

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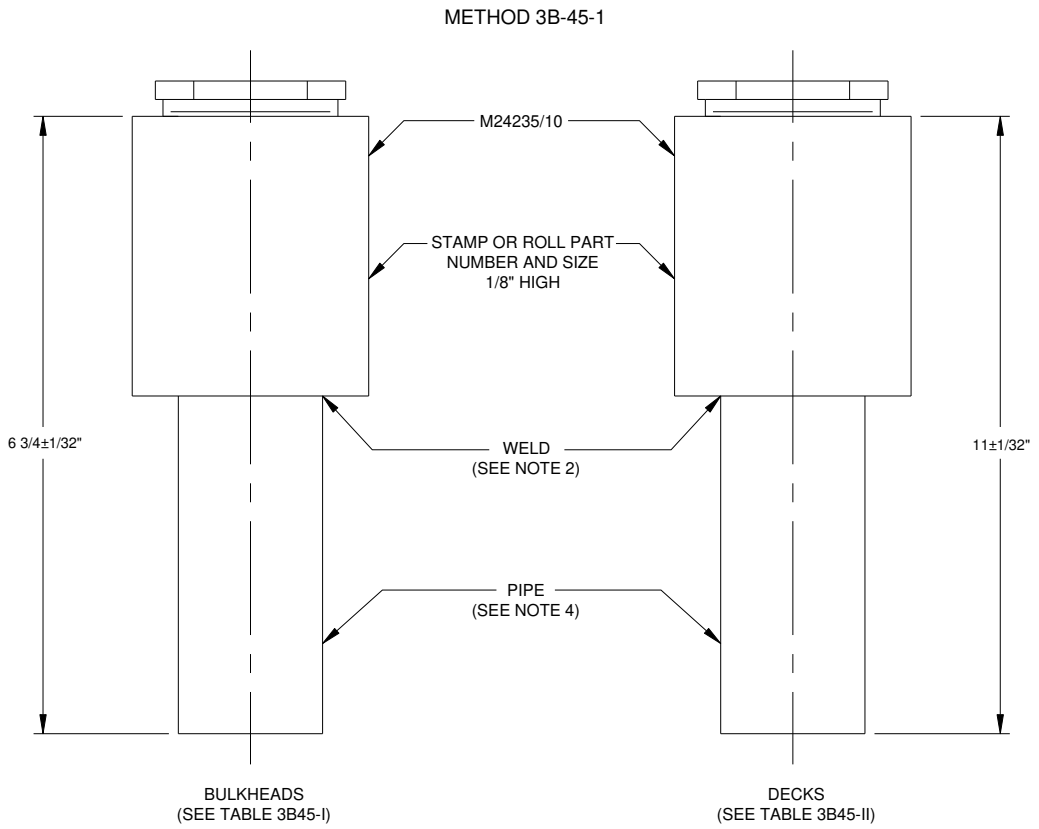


FIGURE 3B45. MIL-S-24235/10 stuffing tubes for steel decks and bulkheads with pipe protection (surface ships).

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TABLE 3B45-I. Bulkheads.

Part no.	Tube size	Stuffing tube	IPS pipe size	Pipe length [1/]
3B45B1	A	M24235/10-01	½ SCH 80-(27/32)	5
3B45B2	B	M24235/10-02	½ SCH 80-(27/32)	5
3B45B3	C	M24235/10-03	¾ SCH 40-(1 1/16)	5
3B45B4	D	M24235/10-04	¾ SCH 40-(1 1/16)	5
3B45B5	E	M24235/10-05	1 SCH 40-(1 21/64)	5
3B45B6	F	M24235/10-06	1 SCH 40-(1 21/64)	5
3B45B7	G	M24235/10-07	1 SCH 40-(1 21/64)	5 15/16
3B45B8	J	M24235/10-08	1¼ SCH 40-(1 21/32)	5 15/16
3B45B9	K	M24235/10-09	1¼ SCH 40-(1 21/32)	3⅝
3B45B10	L	M24235/10-10	1¼ SCH 40-(1 21/32)	3⅝
3B45B11	M	M24235/10-11	1½ SCH 40-(1 29/32)	3⅝
3B45B12	N	M24235/10-12	1½ SCH 40-(1 29/32)	3⅝
3B45B13	P	M24235/10-13	2 SCH 40-(1 5/16)	3⅝
3B45B14	R	M24235/10-14	2 SCH 40-(1 5/16)	3⅝
3B45B15	S	M24235/10-15	2 SCH 40-(1 5/16)	1⅝
3B45B16	T	M24235/10-16	2½ SCH 40-(2⅞)	1⅝
3B45B17	V	M24235/10-17	2½ SCH 40-(2⅞)	1⅝
3B45B18	W	M24235/10-18	2½ SCH 40-(2⅞)	1⅝
3B45B19	X	M24235/10-19	3 SCH 40-(3½)	1⅝
3B45B20	Y	M24235/10-20	3 SCH 40-(3½)	1⅝
3B45B21	Z	M24235/10-21	3 SCH 40-(3½)	1⅝
3B45B22	AA	M24235/10-22	-	-
3B45B23	BB	M24235/10-23	-	-

NOTE:
[1/] Pipe length shall be ±1/32.

FIGURE 3B45. MIL-S-24235/10 stuffing tubes for steel decks and bulkheads with pipe protection (surface ships) – Continued.

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TABLE 3B45-II. Decks.

Part no.	Tube size	Stuffing tube	IPS pipe size	Pipe length [1/]
3B45D31	A	M24235/10-01	½ SCH 80-(27/32)	9¼
3B45D32	B	M24235/10-02	½ SCH 80-(27/32)	9¼
3B45D33	C	M24235/10-03	¾ SCH 40-(1 1/16)	9¼
3B45D34	D	M24235/10-04	¾ SCH 40-(1 1/16)	9¼
3B45D35	E	M24235/10-05	1 SCH 40-(1 21/64)	9¼
3B45D36	F	M24235/10-06	1 SCH 40-(1 21/64)	9¼
3B45D37	G	M24235/10-07	1 SCH 40-(1 21/64)	9 3/16
3B45D38	J	M24235/10-08	1¼ SCH 40-(1 21/32)	9 3/16
3B45D39	K	M24235/10-09	1¼ SCH 40-(1 21/32)	7⅞
3B45D40	L	M24235/10-10	1¼ SCH 40-(1 21/32)	7⅞
3B45D41	M	M24235/10-11	1½ SCH 40-(1 29/32)	7⅞
3B45D42	N	M24235/10-12	1½ SCH 40-(1 29/32)	7⅞
3B45D43	P	M24235/10-13	2 SCH 40-(1 5/16)	7⅞
3B45D44	R	M24235/10-14	2 SCH 40-(1 5/16)	5⅞
3B45D45	S	M24235/10-15	2 SCH 40-(1 5/16)	5⅞
3B45D46	T	M24235/10-16	2½ SCH 40-(2⅞)	5⅞
3B45D47	V	M24235/10-17	2½ SCH 40-(2⅞)	5⅞
3B45D48	W	M24235/10-18	2½ SCH 40-(2⅞)	5⅞
3B45D49	X	M24235/10-19	3 SCH 40-(3½)	5⅞
3B45D50	Y	M24235/10-20	3 SCH 40-(3½)	5⅞
3B45D51	Z	M24235/10-21	3 SCH 40-(3½)	5⅞
3B45D52	AA	M24235/10-22	3 SCH 40-(3½)	4⅛
3B45D53	BB	M24235/10-23	3½ SCH 40-(4)	4⅛

NOTE:
[1/] Pipe length shall be ±1/32.

NOTES:

1. This method is equal to and is an acceptable replacement for M24235/17 and /18.
2. Welding shall be in accordance with 4.1.1.
3. Break all sharp edges.
4. Pipe shall be in accordance with ASTM A106/A106M, grade A.
5. See figure 3A1 for gland nut tightening methods.
6. See 4.8 for thread lubrication guidance.

FIGURE 3B45. MIL-S-24235/10 stuffing tubes for steel decks and bulkheads with pipe protection (surface ships) – Continued.

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General:

1. TW multiple cable penetrators (MCP) are devices utilizing standard units and dimensions that are compatible with various numbers and sizes of cables that may be used whenever there is a need to seal cables penetrating watertight, airtight, and fireproof bulkheads and decks. In accordance with 4.1, TW MCPs system shall meet all performance and qualification requirements of MIL-DTL-24705. See 4.4 for specific restrictions and instructions for installation of fiber optic and blown fiber cables in electrical penetrations.
2. Frame:
   a. One piece cast steel mounting frame to be welded directly into bulkhead or deck.
   b. Cast keyways in mounting frame. Align and position sealing block assemblies.
   c. Frames may be installed so that sealing block assemblies can be used inserted in either horizontal or vertical position.
3. Sealing block assembly:
   a. Elastomeric material between case-malleable pressure plates protects cable from mechanical damage, provides high pullout resistance, provides positive cable separation, and expands during fire to seal any voids left by burned cable insulation.
   b. Interchangeable sealing block assemblies fit all TW MCP mounting frames.
   c. Cast stops on front pressure plate prevents sealing block assemblies from slipping through frame during installation.
   d. Sealing block assemblies apply for all cable outside diameters from 0.250 inch to 3.500 inches (6.4 millimeters to 88.9 millimeters).
   e. Depending on opening size range, a standard sealing block assembly will seal from 2 to 11 cables. It is possible to increase cable density with double-sided sealing block assemblies (see table 3B47-I, figure 3B47) sandwiched between halves of a standard assembly.
   f. Reducers permit sealing block assemblies to accept cables with a smaller outside diameter than the specified range.
   g. Plugs are to fill unused openings in sealing block assemblies.
4. Blank sealing block assemblies are used to fill unused spaces in frames, providing for future expansion.
5. Template all work from ship.
6. All welding and inspection shall be in accordance with 4.1.1.
7. All painting shall be in accordance with MIL-DTL-917.
8. The structural reinforcement shown on this drawing does not apply to flight deck bents, flight deck support structures, or other similar type structures on other ships. Such cases shall be individually resolved with structural design.
9. Material or flat bar reinforcement is to be similar to material of bulkhead or deck.
10. This plan was developed to permit installation of the MCP frame in watertight bulkheads and decks on surface ships.
11. Fillet weld reinforcement for T-2 weld joint shall be ⅛ inch for plating up to 15.3# and ¼ inch for plating above 17.85#.
12. Post-installation testing:
    a. After installation of a new MCP or work on an existing MCP, retest shall be conducted to confirm integrity of the electrical penetration.
    b. Perform a visual inspection on plastic or rubber inserts inside of the MCP to ensure there are no visible or discernible openings.

FIGURE 3B46. Multiple cable penetrators installation notes (type TW).

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General (continued):

c. For airtight, oiltight, and watertight penetrations, conduct an air blast test or a vacuum box test as follows:
   (1) Air blast test: The air hose nozzle shall be held a maximum of 3 inches from the MCP area. The distance should be as close to 3 inches as practicable to avoid distortion of the MCP blocks due to high air flow. The pressure shall be directed at the structure under test in a manner most likely to disclose leaks. The minimum nozzle diameter shall be ⅜ inch and the nozzle pressure shall be 60 to 90 PSIG as monitored at the nozzle. Obtain a pressure differential in the same direction as would occur during a full compartment, tank, or void test using the air hose. The air pressure shall be maintained for at least 3 minutes. As soon as the pressure is applied, apply a soapy solution to the opposite side of the structure and inspect for leaks.
   (2) Vacuum box test: Apply a soapy solution to the joint to be tested. Install the vacuum box so the pressure differential will be in the same direction as a full compartment, tank, or void completion test. Draw 10.2"Hg vacuum (minimum). Maintain the vacuum for at least 3 minutes. Inspect the joint or fitting for leakage by observing the formation of bubbles.

d. For penetrations through fumetight bulkheads, conduct a visual/light test to ensure no light transmits from one side of the MCP to the other.

Selection of parts:

1. The selection of components is based on the quantity and size of cables going through the penetration. Once they are known, the sealing block assemblies and frames can be selected.
2. Group cables by outside diameter and rank from largest to smallest. Keeping in mind that sealing block assemblies are available in ¼-inch increments, group cables that fall within the same sealing block assembly outside diameter size range. Starting with the largest cable outside diameter, select the sealing block assemblies required. Specify reducers to accommodate smaller diameter cables and plugs to fill openings not used; all openings shall be filled.
3. Total the frame spaces required for the specified sealing block assemblies and select the appropriate frame(s). Keep “spare” requirements in mind when specifying frame. Specify blank sealing block assemblies to fill unused frame space.

Installation of sealing block assemblies:

1. Clean all inside surfaces of frame and edges of frame openings, removing all foreign materials.
2. Lubricate all inside surfaces of frames with a silicone-based lubricant.
3. Move cables from immediate area of frame where first sealing block assembly will be installed. Group cables according to their diameter and with respect to sizes of openings in the sealing block assemblies to be installed. The diameter of openings of sealing block assemblies are in ¼-inch increments (6.4-millimeter) and each is marked with their size range. Cables to be sealed in a given sealing block assembly need not all be of the same size. Reducers may be used with cables having an outside diameter up to ¼ inch (6.4 millimeters) smaller than the range of the opening in the sealing block assembly.
4. Slide the first section of sealing block assembly into frame keyways, beginning at one end or bottom of frame. Be certain that flat side of this assembly is against inside surface of frame.
5. Arrange cables in slots of sealing block assembly section. While holding these cables in place, slide the mating section of sealing block into frame.

   CAUTION: When individual conductors of a power circuit are carried through a frame, all conductors of that circuit should be installed in the same sealing block assembly to avoid heating of the pressure plate due to electro-magnetic induction. There shall be no ferrous metals between the cables that form a multi-phase power circuit.

   Repeat operations of steps 4 and 5 with appropriate sealing block assemblies until all sealing block assemblies are in place.
6. Install reducers where needed. Fill unused frame spaces with blank sealing block assemblies. Fill unused openings in the sealing block assemblies with correct size plugs. For holes one size larger than plug, use plug and reducer in combination. Be sure that plugs and reducers are fully inserted into openings of sealing block assemblies.
7. Assemble keeper bar over cast stops of front pressure plates.

FIGURE 3B46. Multiple cable penetrators installation notes (type TW) – Continued.

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Installation of sealing block assemblies (continued):

8. Tighten all nuts only a few turns at a time until sealing material "rolls" into spaces between pressure plates of sealing block assemblies and between pressure plates and frame. This step to be performed when ambient temperature is at least 40 °F (4.5 °C).
   Uniform tightening of compression hardware is important. It can be properly done by noting how far threaded studs extend through nuts. In general, when assemblies are all installed, extension of threaded studs will be nearly the same for those assemblies occupying the same number of frame spaces.
   CAUTION: Torque values on nuts should not exceed the values specified in S9086-CJ-STM-010/075. These torque levels are not ordinarily required to provide a seal. However, if nuts are torqued to these values and a tighter seal is required, retorque after 24 hours.
9. Check tightness of assembly by placing a bright light source on one side of the assembly and inspecting it from the opposite side to see if any light is visible through the assembly. Any visible light indicates that proper seal has not been formed. Tighten compression hardware further.

FIGURE 3B46. Multiple cable penetrators installation notes (type TW) – Continued.

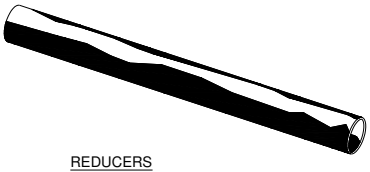
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TABLE 3B47-I. Type TW penetrator characteristics.

OPENING SIZE RANGE in. / mm	0.250-0.500 6.4-12.7		0.500-0.750 12.7-19.1		0.750-1.000 19.1-25.4	1.000-1.250 25.4-31.8	1.250-1.500 31.8-38.1	1.500-1.750 38.1-44.5	1.750-2.000 44.5-50.8	2.000-2.250 50.8-57.2	2.250-2.500 57.2-63.5	2.500-2.750 63.5-69.9	2.750-3.000 69.90-75.20	3.000-3.250 75.20-82.60	3.250-3.5000 82.60-88.90	BLANK-NO OPENINGS	
NO. OPENINGS IN BLOCK	11	11 ADDED	6	11 ADDED	6	5	4	3	3	3	2	2	2	2	2	NONE	NONE
SEALING BLOCK ASSEMBLY NUMBER	TWBS2111	TWBS1111	TWBS2062	TWBS2112	TWBS2063	TWBS3054	TWBS3045	TWBS30355	TWBS4036	TWBS40366	TWBS5027	TWBS50277	TWBS5028	TWBS60288	TWBS6029	TWBS1	TWBS3
FRAME SPACES REQUIRED	2	1	2	2	2	3	3	3	4	4	5	5	5	5	5	1	3
PLUG CAT. #	TWP1				TWP3		TWP5		TWP6		TWP7			TWP8		- -	
REDUCER CAT. #	- -		TWR2		TWR3	TWR4	TWR5	TWR55	TWR6	TWR66	TWR7	TWR77	TWR8	TWR88	TWR9	- -	



PLUGS
PLUGS WILL CLOSE ANY UNUSED OPENINGS IN SEALING BLOCK ASSEMBLIES. SEE TABLE ABOVE FOR PLUG NUMBERS WHICH MATCH SPECIFIC SEALING BLOCK ASSEMBLIES.



REDUCERS
REDUCERS WILL REDUCE OPENINGS BY 0.250" (6.4 MM) IN SEALING BLOCK ASSEMBLIES. SEE TABLE ABOVE FOR REDUCER NUMBERS WHICH MATCH SPECIFIC SEALING BLOCK ASSEMBLIES.

Example for specifying information:

Cables specified: 6 power cables – sizes ranging from 1.875 inches to 2.180 inches outside diameter (OD)
31 control cables – sizes ranging from 0.490 inch to 0.945 inch OD
Spare capacity required: 25 percent

Step 1. Group cables by OD and rank from largest to smallest.

	Cable qty.	Cable OD
	2	2.180
	2	2.140
	2	1.875
	1	0.945
	4	0.890
	5	0.700
	7	0.637
	6	0.550
	8	0.490
Total	37	

FIGURE 3B47. Multiple cable penetrators ordering/selection criteria (type TW).

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Step 2. Group cables that fall within the same sealing assembly OD range.

	Cable qty.	Sealing block OD range
	4	2.000-2.250
	2 (with reducers)	2.000-2.250
	5	0.750-1.000
	18	0.500-0.750
	8 (with reducers)	0.500-0.750
Total	37	

Step 3. Starting with the largest cable OD, select the quantity of sealing block assemblies required. Specify TWR reducers to accommodate smaller diameter cables where possible and TWP plugs to fill openings not used. (See example diagram.)

Sealing block assy cat. #	OD range	Number of openings	Cables to be sealed	Openings not used	Frame spaces req'd.
TWBS40366	2.000-2.250	3	3		4
TWBS40366	2.000-2.250	3	3		4
TWBS2063	0.750-1.000	6	5	1	2
TWBS2062	0.500-0.750	6	6		2
TWBS2112	0.500-0.750	11	11		2
TWBS2112	0.500-0.750	11	9	2	2
	Totals	40	37	3	16

NOTE: In this example, two TWBS2112 sealing block assemblies are sandwiched between two halves of a TWBS2062. This dramatically increases cable density in minimum frame space. One TWP3 plug is required for unused opening in TWBS2063 and two TWP1 plugs are required for unused openings in the TWBS2122. Ten TWR2 and two TWR66 reducers are required for sealing cables of smaller diameter and for use in conjunction with TWP1 in 0.750 holes.

Step 4. Total the frame spaces required for sealing block assemblies and select appropriate size mounting frame(s). Factor in spare capacity required for future expansion.

Total frame spaces required	16
Specification requires 25 percent	
Spare capacity	4
TOTAL	20 spaces

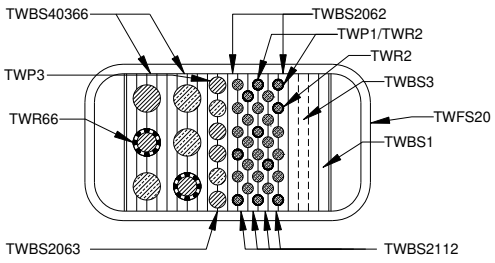
Selection: One TWFS20 mounting frame with total capacity of 20 spaces, one TWBS3, and one TWBS1 blank sealing block assembly to fill unused frame space. (Choice of frame could vary based on future expansion needs and/or specific cable arrangement.)

Step 5. Bill of materials for specification should read:

(1) TWFS 20	(1) TWBS2062	(2) TWP1	(10) TWR2	} For sealing smaller diameter cables
(2) TWBS40366	(2) TWBS2112	(1) TWBS3		
(1) TWBS2063	(1) TWP3	(1) TWBS1	(2) TWR66	

FIGURE 3B47. Multiple cable penetrators ordering/selection criteria (type TW) – Continued.

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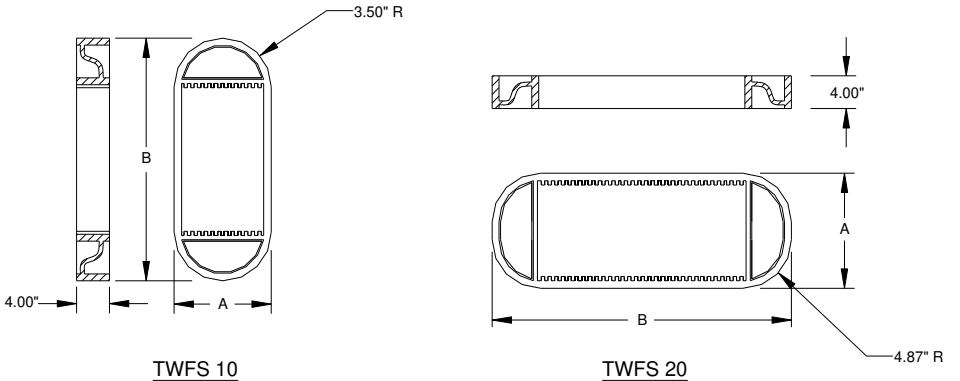
NOTES:

1. TWBS1111 and TWBS2112 sealing block assemblies are used to increase the density of the cable fill. Use these double-sided assemblies when the quantity of cables exceeds the number of openings in the TWBS2111 or TWBS2062 standard sealing block assemblies.
2. One or more of the double-sided assemblies can be sandwiched between halves of the standard assemblies. Make sure the TWBS1111 is matched with TWBS21111 and TWBS2112 with TWBS2062.

FIGURE 3B47. Multiple cable penetrators ordering/selection criteria (type TW) – Continued.

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METHOD 3B-48-1

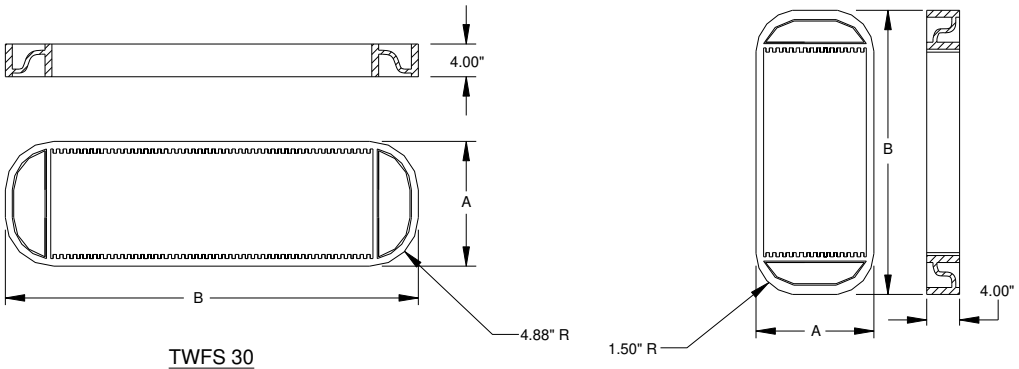


TWFS 10

BLKHD PL / LBS / TWFS10	7.65#	15.3#	25.5#
	10.2#	17.85#	30.6#
	12.75#	20.4#	
DIM "A"	9.10±0.098		
DIM "B"	14.50±0.105		

TWFS 20

BLKHD PL / LBS / TWFS20	7.65#	15.3#	25.5#
	10.2#	17.85#	
	12.75#	20.4#	
DIM "A"	11.00±0.112		
DIM "B"	23.72±0.126		



TWFS 30

BLKHD PL / LBS / TWFS30	7.65#	15.3#
	10.2#	17.85#
	12.75#	20.4#
DIM "A"	11.50±0.117	
DIM "B"	31.82±0.137	

TWFS 6

TWFS6		
"A"	5.86	0.087
"B"	13.00	0.015

FIGURE 3B48. Multiple cable penetrator (type TW) frame details.

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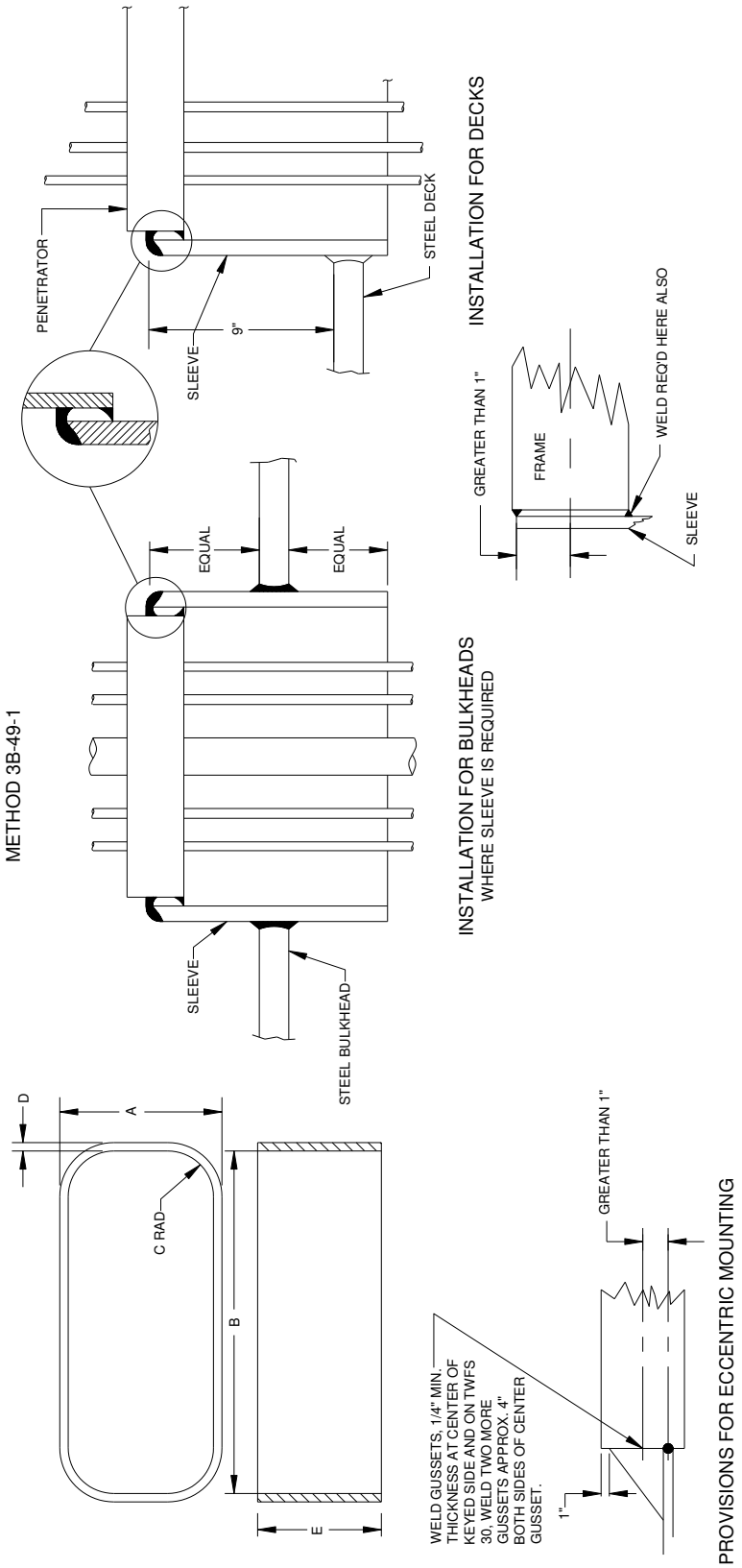


FIGURE 3B49. Multiple cable penetrator (type TW) sleeve installation.

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TABLE 3B49-I. Penetrator sleeve dimensions A, B, and C.

Sleeve			
Penetrator	Dimen. "A"	Dimen. "B"	Dimen. "C"
TWFS 6	6.11	12.25	1.62
TWFS 10	9.36	14.75	3.63
TWFS 20	11.25	23.98	5.00
TWFS 30	11.75	32.08	5.00

TABLE 3B49-II. Penetrator sleeve dimensions D and E for bulkhead installations.

Dk pl lbs / Penetrator	Dimen. "D"					Dimen. "E"				
	7.65# 10.2# 12.75# 15.3# 17.85# 20.4#	25.5#	30.6#	35.7#	40.8#	7.65# 10.2# 12.75# 15.3# 17.85# 20.4#	25.5#	30.6#	35.7#	40.8#
TWFS 6	N/R	N/R	N/R	N/R	0.625	N/R	N/R	N/R	N/R	3.75
TWFS 10	N/R	N/R	N/R	0.625	0.625	N/R	N/R	N/R	3.75	4.50
TWFS 20	N/R	N/R	0.625	0.625	0.75	N/R	N/R	4.25	5.00	4.75
TWFS 30	N/R	0.625	0.625	0.75	0.75	N/R	4.00	4.75	5.00	5.50

TABLE 3B49-III. Penetrator sleeve dimensions D and E for deck installations.

Dk pl lbs / Penetrator	Dimen. "D" / Dimen. "E"									
	7.65#	10.20#	12.75#	15.3#	17.85#	20.4#	25.5#	30.6#	35.7#	40.8#
TWFS 6	¼ / 10.0	¼ / 10.25	6/16 / 10.25	5/16 / 10.5	⅜ / 10.5	⅜ / 10.75	7/16 / 11.0	½ / 11.25	½ / 11.5	⅝ / 11.5
TWFS 10	¼ / 10.0	5/16 / 10.5	5/16 / 10.5	⅜ / 10.5	⅜ / 11.0	⅜ / 11.0	½ / 11.5	½ / 11.5	⅝ / 12.0	⅝ / 12.0
TWFS 20	5/16 / 10.5	5/16 / 10.5	⅜ / 10.75	⅜ / 10.75	7/16 / 11.0	½ / 11.0	½ / 11.5	⅝ / 11.5	⅝ / 12.0	¾ / 12.0
TWFS 30	5/16 / 10.25	⅜ / 10.75	⅜ / 10.75	7/16 / 11.0	½ / 11.25	½ / 11.25	⅝ / 11.75	⅝ / 11.75	¾ / 12.25	¾ / 12.25

FIGURE 3B49. Multiple cable penetrator (type TW) sleeve installation – Continued.

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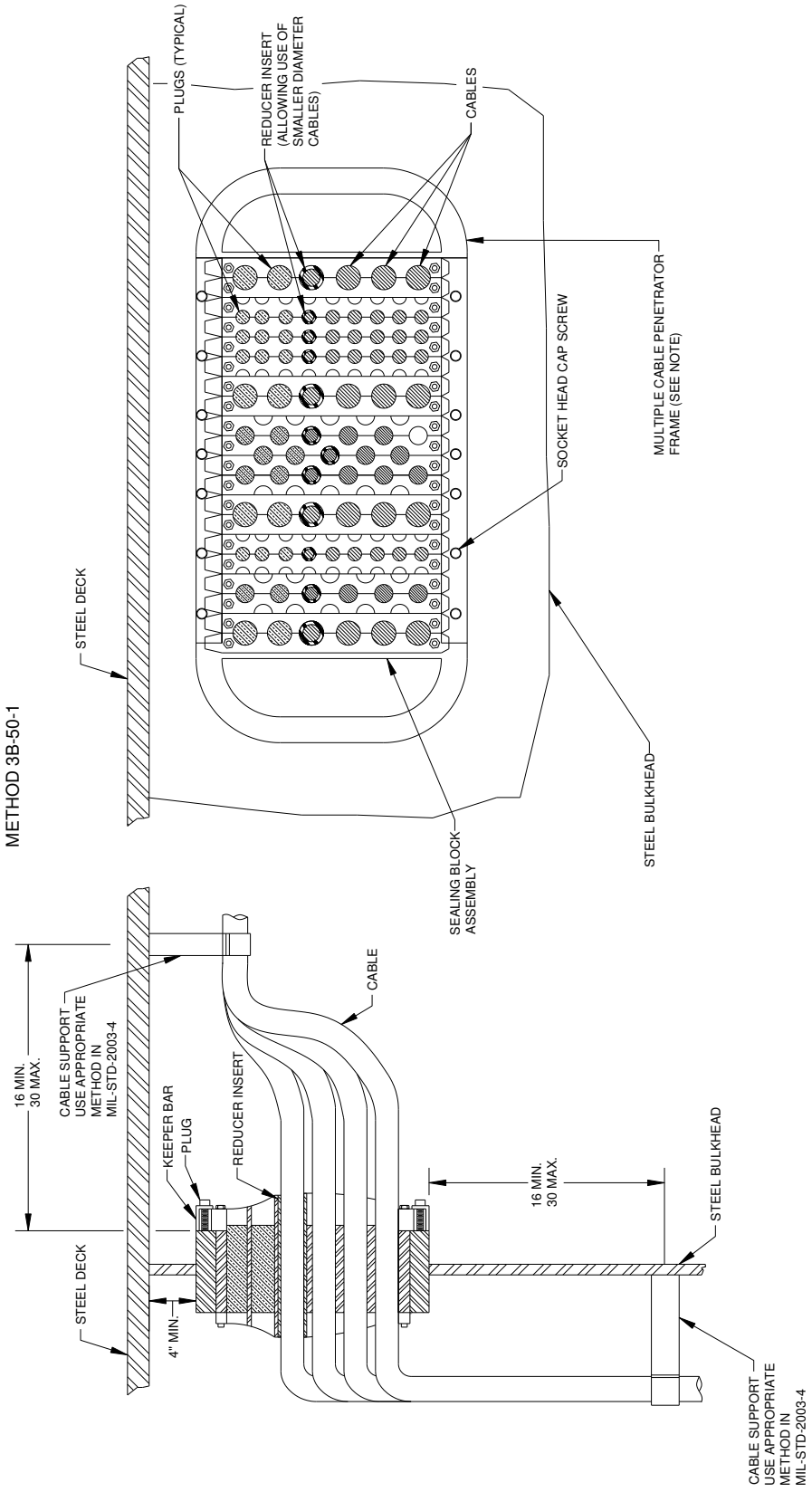


FIGURE 3B50. Multiple cable penetrator (type TW) typical installation in steel bulkhead.

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NOTE:

1. Multiple cable penetration penetrator frames are available in various sizes and arrangements; shown is a twenty-space penetrator.

FIGURE 3B50. Multiple cable penetrator (type TW) typical installation in steel bulkhead – Continued.

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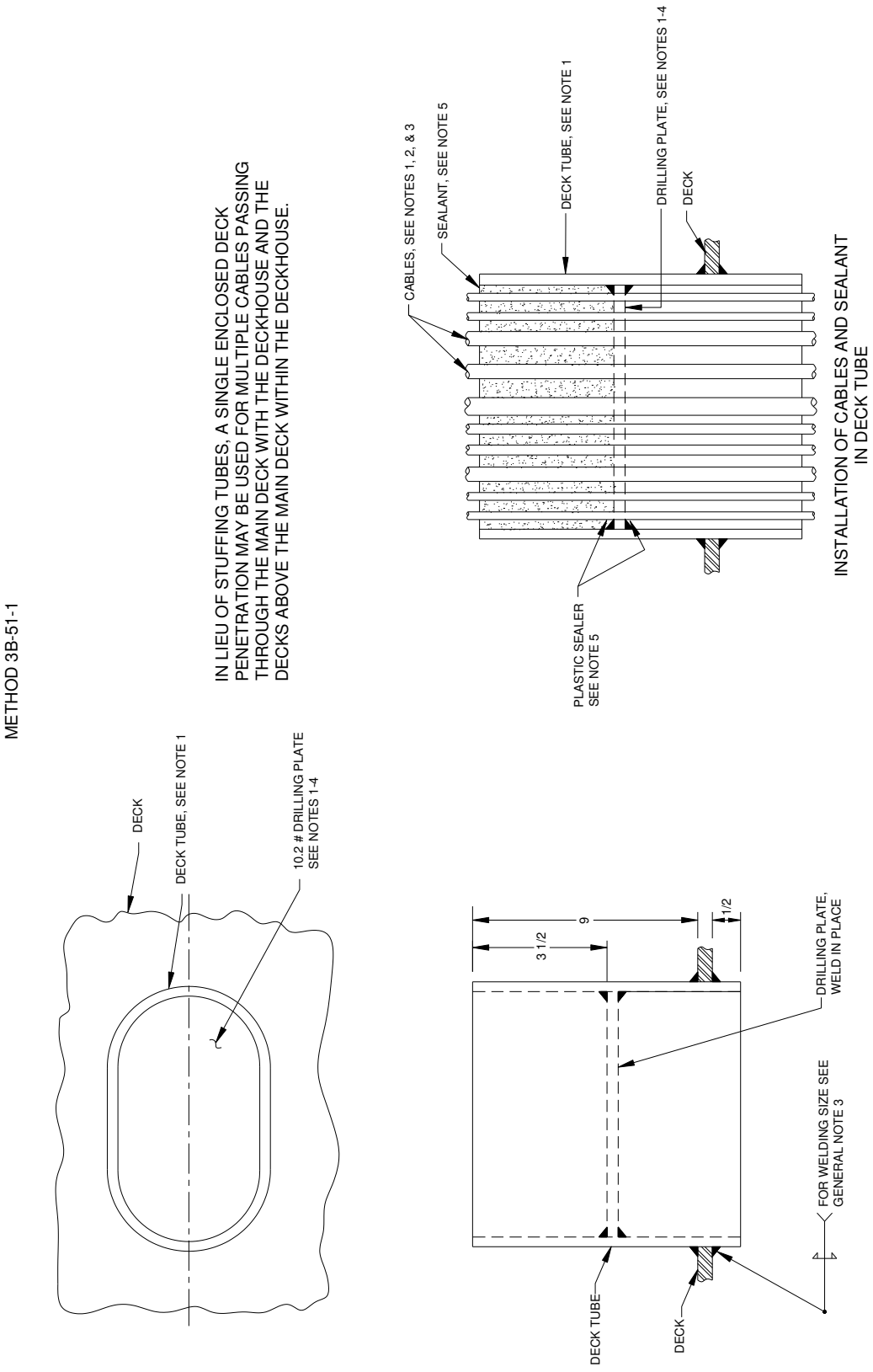


FIGURE 3B51. Community cable tube – watertight decks (poured seal).

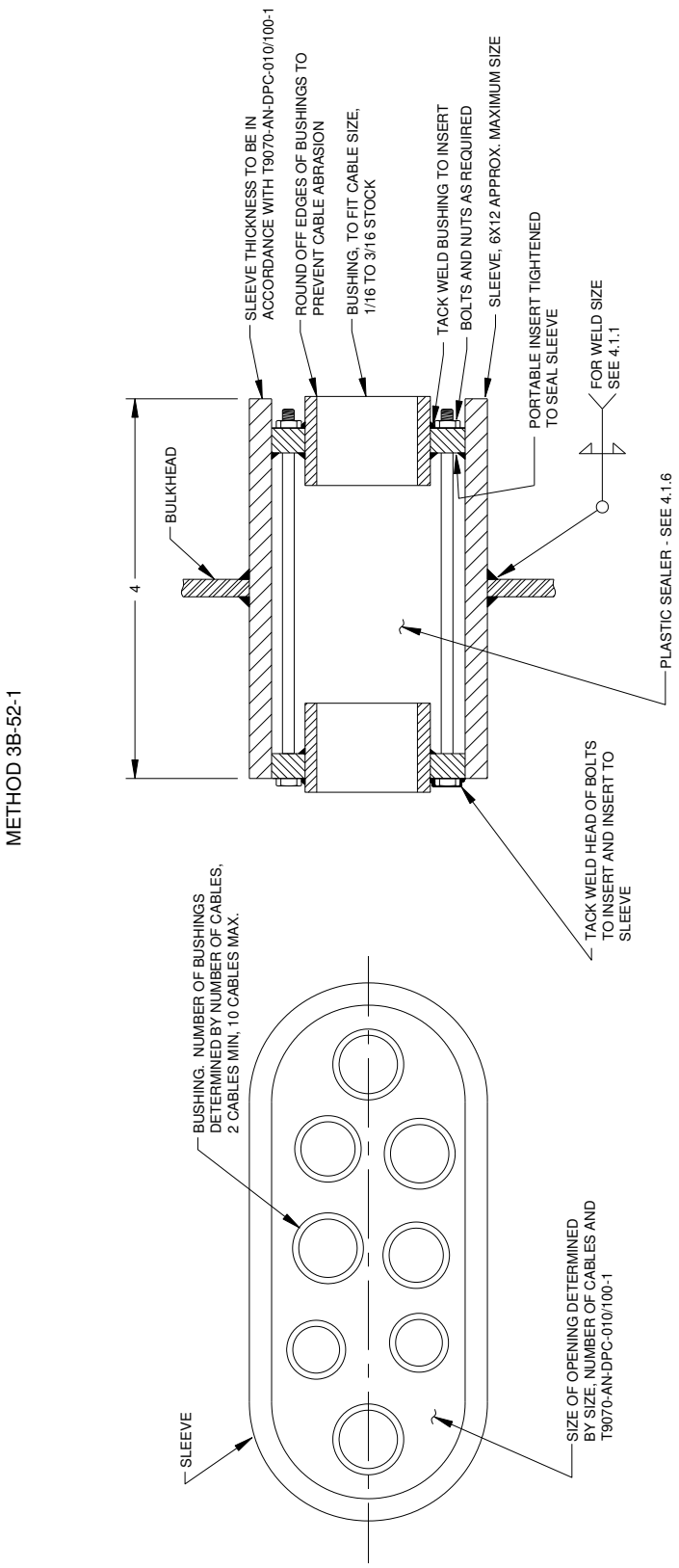
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NOTES:

1. Deck tube wall thickness, maximum size, and maximum quantity of cables through drilling plate shall be in accordance with T9070-AN-DPC-010/100-1.
2. Cable size and quantity up to the maximum allowed shall be determined by the shipyard.
3. Cable clearance holes shall be sized in accordance with standard shipyard practice. Holes to be rounded to prevent abrasion of cable.
4. Minimum clearance between holes in drilling plate and side of tube shall be ⅜ inch.
5. Sealant shall be in accordance with 4.1.6 with a minimum finished pour depth of 3 inches. The starting compound may be liquid to allow pouring into the mold. Prior to pouring sealant, gaps around the cables shall be packed with plastic sealer, MIL-I-3064, Type HF (see 4.1.6), to retain sealer when in the liquid state and serve as a permanent fire stop.
6. This method was canceled by ACN R240316Z dated July 1996. This method has been retained since activities may need to use the method in certain circumstances. NAVSEA approval is required to use this method.

FIGURE 3B51. Community cable tube – watertight decks (poured seal) – Continued.

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NOTE: This method was canceled by ACN R240316Z dated July 1996. This method has been retained since activities may need to use the method in certain circumstances. NAVSEA approval is required to use this method.

FIGURE 3B52. Community cable tube – watertight bulkheads (troweled seal).

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Method 3B-53-1
Repair of Deteriorated Stuffing tubes with CeramAlloy System (or equal [see 4.2])

1. Purpose:
   a. This procedure provides technical direction for conducting CeramAlloy System (or equal [see 4.2]) repairs for deteriorated stuffing tubes in surface ships only. This procedure shall not be used for submarines.
   b. This procedure may be used to restore watertight integrity of stuffing tubes in areas where removal of associated cables is undesirable due to cable material condition.
2. General:
   a. This procedure shall not be used for any penetrations that are part of Nuclear Primary or Secondary Containment Boundaries.
   b. This procedure shall only be used on HY-80 or HY-100 bulkheads or decks when the Repair Activity has the capability to perform the required pre-heat and post-heat welding procedures.
   c. This procedure may be used only for stuffing tubes that penetrate weather decks.
   d. This procedure shall be evaluated for use on a case basis and shall only be used where stuffing tubes are not broken and remain watertight through the gland nut. This procedure shall only be used where the deck around the stuffing tube will maintain structural integrity.
   e. Pitting and general corrosion in the area of the stuffing tubes should be evaluated to determine acceptability. CeramAlloy shall not be used to improve structural integrity of the deck around the stuffing tubes.
   f. Supervisor approval shall be obtained to use this procedure for stuffing tube repair.
3. Scope:
   a. In order to restore watertight integrity on deteriorated stuffing tubes, the Repair Activity shall apply MetalClad CeramAlloy (or equal [see 4.2]) to deteriorated stuffing tubes.
   b. The CeramAlloy shall be contained by a stainless steel cofferdam. The cofferdam will be referred to as the CeramAlloy Containment Box (CACB) and shall be custom fabricated liquid-tight seal capable of adequately containing the CeramAlloy during the curing phase.
4. Surface preparation:
   a. All surfaces that CeramAlloy (or equal [see 4.2]) will contact shall be cleaned and de-greased using acetone or isopropyl alcohol. No residue shall remain on the surface. These surfaces include the stuffing tubes, the surrounding weather deck area, and the steel plating used in the CACB.
   b. Surfaces shall be cleaned to bare metal and roughened using a carbide burr. Care shall be taken while cleaning to prevent damage of cables where they are exposed due to corrosion of stuffing tubes.
   c. Where there are holes in stuffing tubes due to corrosion, apply tape around the stuffing tube to seal the hole and prevent CeramAlloy from leaking into the stuffing tube. Tape such as duct tape with a shiny exterior should not be used since CeramAlloy will not bond well to this type of tape.
5. CACB construction and installation:
   a. The CACB shall be constructed of stainless steel plate.
   b. Steel plates shall be cleaned and roughened as detailed in paragraphs 4.a and 4.b above.
   c. For a four-sided CACB application, the CACB shall be installed as shown on the following illustration.
   d. If the location of the deteriorated stuffing tubes prevents use of a four-sided CACB due to proximity of the associated bulkhead, a three-sided CACB shall be used with the bulkhead serving as the fourth side.
   e. The overall size of the CACB shall be limited to the minimum required for the specific area, with special attention towards creation of trip hazards or operational obstructions.
   f. CACB depth shall be deep enough to contain a volume sufficient to allow application of the CeramAlloy to a level approximately ½ inch below the packing gland nut of shortest stuffing tube in the bundle.
   g. Four-sided CACBs shall have three sides assembled in the shop with the fourth side to be tack-welded in place when the CACB is placed around stuffing tubes.
   h. The CACB shall be tack-welded to the deck in two locations when the CACB is installed.
   i. VT shall be performed on all welded joints in accordance with 4.1.1.
   j. After CACB is assembled and CeramAlloy application is complete and has hardened, all new and disturbed painted surfaces shall be preserved.

FIGURE 3B53. Repair of deteriorated stuffing tubes.

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6. Application of CeramAlloy:
   a. The CeramAlloy System (or equal [see 4.2]) shall only be applied if the surface is completely dry and surface temperature is at least 5 °F above the dew point.
   b. Application of the CeramAlloy System (or equal [see 4.2]) shall not be undertaken at temperatures below 50 °F or above 90 °F.
   c. Mixing and application of CeramAlloy System (or equal [see 4.2]) shall be in accordance with product instructions.
   d. Apply CeramAlloy (or equal [see 4.2]) paste around stuffing tube to fill pitted, corroded areas. In the four-sided CACB application, use the paste to fair sides of plate that was tack-welded to the other three sides in order to seal the seams. Similarly, in the three-sided CACB application, use the paste to fair against the bulkhead and to seal the seams. Also use the paste to fair the outside seams around bottom as a seal (see the following illustration).
   e. After CeramAlloy (or equal [see 4.2]) paste has hardened, CACB shall be filled with CeramAlloy (or equal [see 4.2]) liquid to a level approximately ½ inch below the packing gland nut of shortest stuffing tube in the bundle in accordance with “Process Control Procedure - Kickpipes & Stuffing Tubes Repairing and Coating with Enecon CeramAlloy System.” After approximately 15 minutes, or when the liquid becomes like a paste but has not hardened, use some of the remaining paste to form the 15-degree slant on the top surface of the CeramAlloy, providing a runoff for water.
7. Post-installation test: Repaired stuffing tubes shall be tested to ensure watertight integrity prior to returning to service.

FIGURE 3B53. Repair of deteriorated stuffing tubes – Continued.

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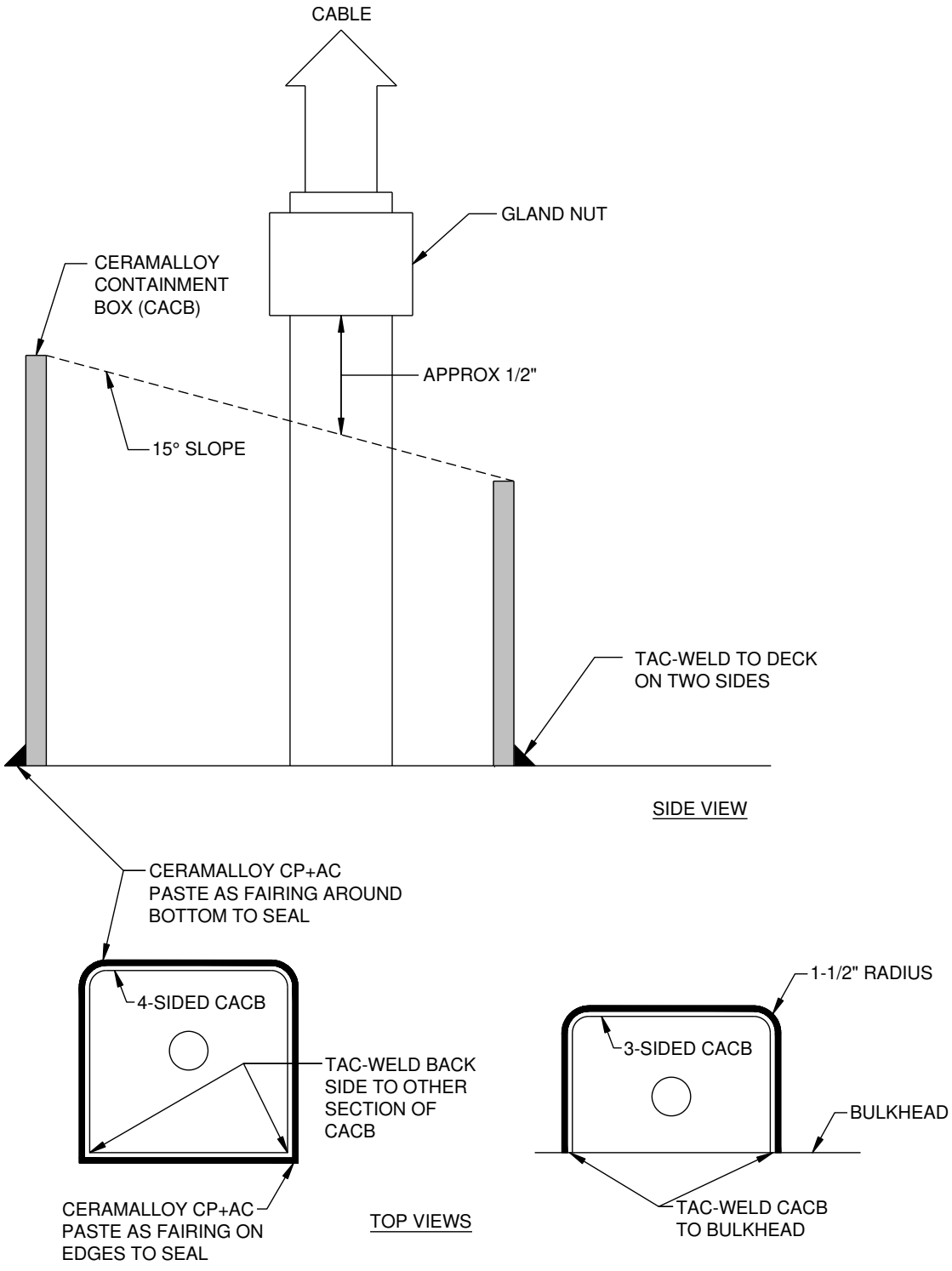


FIGURE 3B53. Repair of deteriorated stuffing tubes – Continued.

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Method 3B-53-2
Repair of Deteriorated Weather Deck Stuffing Tubes Using a Removable Dam and Epoxy Resin

1. Purpose:
   a. This procedure provides technical direction for conducting repairs of deteriorated weather deck stuffing tubes and deck areas around stuffing tubes in surface ships. This procedure shall not be used for submarines.
   b. This procedure may be used to restore watertight integrity of stuffing tubes in areas where removal of associated cables is undesirable due to cable, stuffing tube, or surrounding deck material condition.
2. General:
   a. This procedure shall not be used for any penetrations that are part of Nuclear Primary or Secondary Containment Boundaries.
   b. This procedure may be used only for stuffing tubes that penetrate weather decks.
   c. This procedure shall be evaluated for use on a case basis and shall only be used where stuffing tubes are not broken and remain watertight through the gland nut. This procedure shall only be used where the deck around the stuffing tube will maintain structural integrity once repaired.
   d. Pitting and general corrosion in the area of the stuffing tubes should be evaluated to determine acceptability.
   e. Supervisor approval shall be obtained to use this procedure for stuffing tube repair.
3. Scope:
   a. In order to restore watertight integrity on deteriorated stuffing tubes, the Repair Activity shall apply an epoxy resin to deteriorated stuffing tubes.
   b. The epoxy resin shall be contained in a removable 16 gauge steel cofferdam.
4. Surface preparation:
   a. All surfaces that the epoxy resin will contact shall be cleaned and de-greased using acetone or isopropyl alcohol. No residue shall remain on the surface. These surfaces include the stuffing tubes, the surrounding weather deck area, and the steel used for the removable steel cofferdam.
   b. Surfaces shall be cleaned to bare metal and roughened using a carbide burr. Care shall be taken while cleaning to prevent damage of cables where they are exposed due to corrosion of stuffing tubes.
   c. Where there are holes in stuffing tubes due to corrosion, tubes shall have cold patch applied to prevent flow of resin into tube and contacting cable.
   d. Wrap and seal holed stuffing tubes and exposed cabling with fiberglass.
5. Steel cofferdam construction and installation:
   a. Fabricate and install a temporary protective coaming enclosure around the group of stuffing tubes using 16-gauge steel.
   b. Template length, width and height of coaming enclosure, from existing stuffing tube group, allowing approximately one-inch clearance from stuffing tubes.
   c. Verify that the coaming placement provides a minimum of one inch of clearance from existing stuffing tube group and that the height of the coaming does not exceed 1.5 inches.
   d. The height of applied epoxy shall not protrude above the top of the stuffing tube body and gland nut. The cable stuffing tube gland nut must remain exposed and serviceable.
   e. Tack weld the steel coaming to the deck in accordance with 4.1.1.
6. Application of resin epoxy compound:
   a. Fill the coaming enclosure with resin in accordance with MIL-R-17882.
   b. The table below provides cure times for the resin. For temperatures below 60 °F, heat lamps shall be used for curing the epoxy resin. The pot life for the resin is 45 minutes, which means that the epoxy-resin compound should be used within 45 minutes after mixing. Shelf-life of the material before mixing is two years.

Temperature (°F)	Cure time (hours)
85	4
75	8
65	12

FIGURE 3B53. Repair of deteriorated stuffing tubes – Continued.

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7. Removal of steel coaming. Cut tack welds and remove the steel coaming installed in step 5 above.
8. Post-installation test: Repaired stuffing tubes shall be tested to ensure watertight integrity prior to returning to service. The test shall be a water hose test as follows.
   a. Use a 1½-inch hose with a minimum nozzle diameter of ½ inch.
   b. Pressure at the nozzle shall be 50 to 55 PSIG at a maximum distance of 10 feet from the surface being tested.
   c. The stream of water shall be directed against the structure in a manner most likely to disclose leaks.
   d. The opposite side of the structure shall be inspected to detect and locate leaks.
   e. Allowable leakage: None.

FIGURE 3B53. Repair of deteriorated stuffing tubes – Continued.

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General Requirements for Use of RISE Multiple Cable Penetrator (MCP) Sealant Installations:

1. RISE MCP sealant, or equal (see 4.2), is a surface ship penetration and structure sealing system that may be used as an alternative to new or existing penetrations in the following situations with the following restrictions:
   a. As an alternative to cable collars (steel) filled with MIL-I-3064 sealant to provide a watertight, airtight, fume-tight and fire-tight penetration for electric, fiber optic, and air blown fiber cables.
   b. In penetrations designed to accept MCPs built to MIL-DTL-24705 in applications that do not require withstanding a waterhead.
   c. As an alternate to new or existing stuffing tube areas in accordance with MIL-S-24235
   d. The RISE MCP penetration system shall be used with a circular or flat oval steel conduit sleeve (collar) with a maximum diameter of 8 inches or inside a circular or flat oval single or multiple standard MIL-DTL-24705 MCP frame with a maximum area of 50 square inches if the penetration is below the V-line for watertight applications and for applications at Collective Protective System boundaries. There is no size restriction for non-watertight cable penetrations above the V-line. See MIL-STD-2003-1 for the definition of V-line.
   e. RISE shall not be used aboard Submarines and In-Plant applications aboard Nuclear Aircraft Carriers.
   f. RISE shall not be used in ballistic and fragmentation plating, torpedo side protective systems, under bottom protective systems, any structures that have a blast loading requirement (gun, missile, or nuclear), and all major transverse watertight bulkheads.
   g. RISE shall not be used at magazine boundaries, or for spaces containing hazardous materials.
   h. See 4.4 for specific restrictions for installation of fiber optic and blown fiber cables in electrical penetrations.
2. The RISE MCP system is comprised of two parts; rubber insert sleeves and soft-seal sealant compound.
3. Any RISE MCP sealant installation for electro-magnetic compatibility (EMC) applications requires specific NAVSEA approval prior to installation. See figure 3B57 for specific procedures for installation of RISE for EMC applications.
4. The installation of RISE MCP sealant systems shall follow the requirements and methods in figures 3B54-3B58 of this standard and Drawing 302-7583331. If there is a conflict between these two documents, the requirements of this standard take precedence.
5. For new RISE MCP sealant installations, the cable packing density at the time of initial installation shall provide a minimum of 20 percent spare (free) surface required for future growth and to ensure the soft-seal can be effectively applied to produce a watertight seal.
6. When individual conductors of a power circuit are carried through a frame, all conductors of that circuit should be installed adjacent to each other to avoid heating due to electro-magnetic induction. There shall be no ferrous material between separate cables that form a set of three-phase conductors.
7. The minimum depth for the MCP for RISE application is 4 inches.
8. Prior to beginning installation of the sleeve system, the inside of the MCP and cable exterior surface shall be thoroughly cleaned so as to be free of dirt, dust, oils, or foreign residue of any kind. An all-purpose cleaner that dries rapidly and does not leave any residue or film shall be used. Ensure that the penetration and cables are fully dried and that the MCP and cables are completely clean before inserting sleeves or applying sealant. The adhesion of the sealant over time is essential for a lasting pressure boundary and depends on complete cleaning and drying of the MCP.
9. Curing: Curing of the sealant will depend on factors such as temperature and humidity. Colder temperature and higher humidity will cause longer curing times. At least one day of curing time shall be allowed for every 1 millimeter of sealant thickness; however, the curing time shall be at least five days.
10. Post-installation testing:
    a. For cable penetrations below the flooding water level II (see MIL-STD-2003-1) and for watertight, oiltight, and airtight penetrations, a pressure drop test shall be conducted as follows to confirm integrity of the electrical penetration:
       (1) The pressure drop test shall be conducted on the completed RISE MCP system after the sealant has completely cured. It is *essential* that the sealant be completely cured to prevent failure of the system. See step 9 for curing requirements.
       (2) The test pressure shall be no less than 15 psig. This test is not a compartment test and is not a substitute for a compartment test.

FIGURE 3B54. General requirements for RISE MCP sealant installations.

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General Requirements for Use of RISE Multiple Cable Penetrator (MCP) Sealant Installations (continued):

(3) Air or dry nitrogen shall be used to pressurize the RISE MCP. Options are available for applying and monitoring pressure. A ¼-inch threaded hole may be drilled into the MCP followed by installation of a threaded adapter (see illustration 1). As an alternative, a ¼-inch rubber or vinyl tube may be installed through one side of the MCP and sealed in the same manner as the cables through the MCP. The other end of the tube or threaded adapter will be attached to the pressure source.

(4) When the test configuration is complete, the MCP shall be pressurized to the specified pressure. When the test pressure is reached, the test valve shall be shut and the pressure shall be monitored for 15 minutes. The pressure shall not reduce by more than 0.1 psig in the 15-minute test time.

(5) If a hole was drilled to support the pressure test, the hole shall be sealed with a steel plug. If tubing was used, the tubing shall be removed and the sealant around the plug shall be repaired.

b. For fumetight penetrations that are not watertight, conduct a visual/light test to ensure no light transmits from one side of the MCP to the other.

11. Typical required tools to complete a RISE MCP installation:
    a. RISE sleeve system material
    b. Caulking gun
    c. Water spray bottle
    d. Plastic putty knife
    e. 0.25-inch plastic trowel
    f. Shop cloths
    g. All-purpose cleaner
    h. Cloth or latex gloves
    i. Plastic sleeve system re-entry tool for adding or deleting cables from an existing MCP

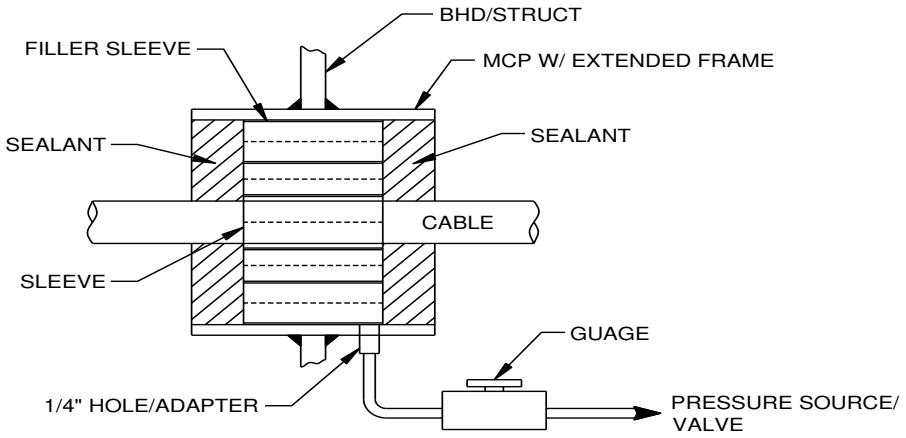


Illustration 1

FIGURE 3B54. General requirements for RISE MCP sealant installations – Continued.

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TABLE 3B54-I. RISE insert sleeve data.

Item No.	Description	NSN/MFR Part No.	Remarks
1	Sleeve, Rise, Rubber Insert 2.36″ LG Size 12/6	850RISE. 12/6. N	For CA Dia 0.19 - 0.27 for BHD WT pen
2	Sleeve, Rise, Rubber Insert 2.36″ LG Size 14/8	850RISE. 14/8. N	For CA Dia 0.27 - 0.35 for BHD WT pen
3	Sleeve, Rise, Rubber Insert 2.36″ LG Size 16/10	850RISE. 16/10. N	For CA Dia 0.35 - 0.43 for BHD WT pen
4	Sleeve, Rise, Rubber Insert 2.36″ LG Size 18/12	850RISE. 18/12. N	For CA Dia 0.43 - 0.51 for BHD WT pen
5	Sleeve, Rise, Rubber Insert 2.36″ LG Size 20/14	850RISE. 20/14. N	For CA Dia 0.51 - 0.59 for BHD WT pen
6	Sleeve, Rise, Rubber Insert 2.36″ LG Size 22/16	850RISE. 22/16. N	For CA Dia 0.59 - 0.67 for BHD WT pen
7	Sleeve, Rise, Rubber Insert 2.36″ LG Size 27/19	850RISE. 27/19. N	For CA Dia 0.67 - 0.82 for BHD WT pen
8	Sleeve, Rise, Rubber Insert 2.36″ LG Size 31/23	850RISE. 31/23. N	For CA Dia 0.82 - 1.00 for BHD WT pen
9	Sleeve, Rise, Rubber Insert 2.36″ LG Size 35/27	850RISE. 35/27. N	For CA Dia 1.00 - 1.14 for BHD WT pen
10	Sleeve, Rise, Rubber Insert 2.36″ LG Size 39/31	850RISE. 39/31. N	For CA Dia 1.14 - 1.30 for BHD WT pen
11	Sleeve, Rise, Rubber Insert 2.36″ LG Size 46/36	850RISE. 46/36. N	For CA Dia 1.30 - 1.53 for BHD WT pen
12	Sleeve, Rise, Rubber Insert 2.36″ LG Size 52/42	850RISE. 52/42. N	For CA Dia 1.53 - 1.77 for BHD WT pen
13	Sleeve, Rise, Rubber Insert 2.36″ LG Size 58/48	850RISE. 58/48. N	For CA Dia 1.77 - 2.00 for BHD WT pen
14	Sleeve, Rise, Rubber Insert 2.36″ LG Size 64/54	850RISE. 64/54. N	For CA Dia 2.00 - 2.25 for BHD WT pen
15	Sleeve, Rise, Rubber Insert 2.36″ LG Size 70/60	850RISE. 70/60. N	For CA Dia 2.25 - 2.50 for BHD WT pen
16	Sleeve, Rise, Rubber Insert 6.3″ LG Size 12/6	850RISE. 12/6	For CA DIA 0.19 - 0.27 for EMC penetrations
17	Sleeve, Rise, Rubber Insert 6.3″ LG Size 14/8	850RISE. 14/8	For CA Dia 0.27 - 0.35 for EMC penetrations
18	Sleeve, Rise, Rubber Insert 6.3″ LG Size 16/10	850RISE. 16/10	For CA Dia 0.35 - 0.43 for EMC penetrations
19	Sleeve, Rise, Rubber Insert 6.3″ LG Size 18/12	850RISE. 18/12	For CA Dia 0.43 - 0.51 for EMC penetrations

FIGURE 3B54. General requirements for RISE MCP sealant installations – Continued.

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TABLE 3B54-I. RISE insert sleeve data – Continued.

Item No.	Description	NSN/MFR Part No.	Remarks
20	Sleeve, Rise, Rubber Insert 6.3″ LG Size 20/14	850RISE. 20/14	For CA Dia 0.51 - 0.59 for EMC penetrations
21	Sleeve, Rise, Rubber Insert 6.3″ LG Size 22/16	850RISE. 22/16	For CA Dia 0.59 - 0.67 for EMC penetrations
22	Sleeve, Rise, Rubber Insert 6.3″ LG Size 27/19	850RISE. 27/19	For CA Dia 0.67 - 0.82 for EMC penetrations
23	Sleeve, Rise, Rubber Insert 6.3″ LG Size 31/23	850RISE. 31/23	For CA Dia 0.82 - 1.00 for EMC penetrations
24	Sleeve, Rise, Rubber Insert 6.3″ LG Size 35/27	850RISE. 35/27	For CA Dia 1.00 - 1.14 for EMC penetrations
25	Sleeve, Rise, Rubber Insert 6.3″ LG Size 39/31	850RISE. 39/31	For CA Dia 1.14 - 1.30 for EMC penetrations
26	Sleeve, Rise, Rubber Insert 6.3″ LG Size 46/36	850RISE. 46/36	For CA Dia 1.30 - 1.53 for EMC penetrations
27	Sleeve, Rise, Rubber Insert 6.3″ LG Size 52/42	850RISE. 52/42	For CA Dia 1.53 - 1.77 for EMC penetrations
28	Sleeve, Rise, Rubber Insert 6.3″ LG Size 58/48	850RISE. 58/48	For CA Dia 1.77 - 2.00 for EMC penetrations
29	Sleeve, Rise, Rubber Insert 6.3″ LG Size 64/54	850RISE. 64/54	For CA Dia 2.00 - 2.25 for EMC penetrations
30	Sleeve, Rise, Rubber Insert 6.3″ LG Size 70/60	850RISE. 70/60	For CA Dia 2.25 - 2.50 for EMC penetrations
31	Sleeve, Rise, Rubber Insert 8.27″ LG Size 12/6	850RISE210. 12/6	For CA Dia 0.19 - 0.27 for deck pen
32	Sleeve, Rise, Rubber Insert 8.27″ LG Size 14/8	850RISE210. 14/8	For CA Dia 0.27 - 0.35 for deck pen
33	Sleeve, Rise, Rubber Insert 8.27″ LG Size 16/10	850RISE210. 16/10	For CA Dia 0.35 - 0.43 for deck pen
34	Sleeve, Rise, Rubber Insert 8.27″ LG Size 18/12	850RISE210. 18/12	For CA Dia 0.43 - 0.51 for deck pen
35	Sleeve, Rise, Rubber Insert 8.27″ LG Size 20/14	850RISE210. 20/14	For CA Dia 0.51 - 0.59 for deck pen

FIGURE 3B54. General requirements for RISE MCP sealant installations – Continued.

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TABLE 3B54-I. RISE insert sleeve data – Continued.

Item No.	Description	NSN/MFR Part No.	Remarks
36	Sleeve, Rise, Rubber Insert 8.27″ LG Size 22/16	850RISE210. 22/16	For CA Dia 0.59 - 0.67 for deck pen
37	Sleeve, Rise, Rubber Insert 8.27″ LG Size 27/19	850RISE210. 27/19	For CA Dia 0.67 - 0.82 for deck pen
38	Sleeve, Rise, Rubber Insert 8.27″ LG Size 31/23	850RISE210. 31/23	For CA Dia 0.82 - 1.00 for deck pen
39	Sleeve, Rise, Rubber Insert 8.27″ LG Size 35/27	850RISE210. 35/27	For CA Dia 1.00 - 1.14 for deck pen
40	Sleeve, Rise, Rubber Insert 8.27″ LG Size 39/31	850RISE210. 39/31	For CA Dia 1.14 - 1.30 for deck pen
41	Sleeve, Rise, Rubber Insert 8.27″ LG Size 46/36	850RISE210. 46/36	For CA Dia 1.30 - 1.53 for deck pen
42	Sleeve, Rise, Rubber Insert 8.27″ LG Size 52/42	850RISE210. 52/42	For CA Dia 1.53 - 1.77 for deck pen
43	Sleeve, Rise, Rubber Insert 8.27″ LG Size 58/48	850RISE210. 58/48	For CA Dia 1.77 - 2.10 for deck pen
44	Sleeve, Rise, Rubber Insert 8.27″ LG Size 64/54	850RISE210. 64/54	For CA Dia 2.00 - 2.25 for deck pen
45	Sleeve, Rise, Rubber Insert 8.27″ LG Size 70/60	850RISE210. 70/60	For CA Dia 2.25 - 2.50 for deck pen
46	Sealant, FIWA 310 mL tube (10.3 oz)	850RISE-FIWA	Cartridge 6-month storage life
47	Flexible Rubber, Conductive, for EMC Instl 39″ X 6.36″ X 3/8″ roll	850RISE-CONDUCTION	For EMC installation See note 3 and figure 3B57.
48	Extender Frame for Size 4 MCP Frame BHD type	RISE80-0961	850EXTENDANAVYBULKHEAD 0.600
49	Extender Frame for Size 6 MCP Frame BHD type	RISE80-0962	850EXTENDANAVYBULKHEAD 0.700
50	Extender Frame for Size 8 MCP Frame BHD type	RISE80-0963	850EXTENDANAVYBULKHEAD 0.750

FIGURE 3B54. General requirements for RISE MCP sealant installations – Continued.

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Method 3B-55-1
RISE Multiple Cable Penetrator (MCP) Sealant Installation for Bulkhead and Deck Watertight and Non-Watertight Penetrations

1. Follow all general requirements and limitations of figure 3B54.
2. This figure provides instructions for configuring a RISE penetration using the RISE method.
3. Ensure the penetration can support a RISE MCP installation with at least a 4-inch depth. The minimum 4-inch depth is required to accommodate the minimum 2½-inch long insert sleeves plus the minimum ¾-inch depth of caulking sealant applied to both ends of the sleeves. Consequently, when 4-inch deep coamings are used (most bulkhead coamings), the insert sleeves shall be centered in the coaming. When longer coamings are used (most deck penetrations), centering the insert sleeves is not necessary as long as the minimum thickness of sealant is applied at both ends.
4. See illustrations 1-3 for guidance in configuring RISE MCPs.
5. Cables may be installed through the penetration in any order. Cables shall not be pulled so tight to prevent separation to apply sleeves and sealant.
6. After cables have been installed, a RISE insert sleeve shall be installed around each cable. The insert sleeves provide the necessary cable separation so that the sealant can flow between the cables to ensure an airtight seal. Each insert sleeve shall be adjusted in or out to be aligned with other sleeves and to form a clean edge recessed back approximately ¾ inch from the edge of the coaming.
   a. Select the sleeve for the applicable cable diameter from table 3B54-I.
   b. The insert sleeves are split lengthwise and, therefore, can be fitted around each cable prior to insertion inside the penetration.
   c. The RISE sleeve system does not require that the insert sleeve inside diameter exactly match the cable outside diameter; however, the closer the match, the more compact the installation will be.
   d. When packing deck MCPs, selecting insert sleeves with a smaller inside diameter than the cable is beneficial to help hold the sleeve in place over the cable during installation. When many small cables are passed through the MCP or when otherwise deemed appropriate by the installer, double sleeves may be used to provide adequate spacing around individual cables to permit/ensure the complete application of sealant.
   e. The sleeves may be overlapped or cut such that all parts of the cable length are covered. Gaps in the sleeving shall be avoided.
   f. Ensure that sufficient spacing exists around each cable to permit application of soft sealant, item 46 from table 3B54-I.
7. Once all cables have been sleeved, the cable bundle shall be maneuvered such that the free area inside the coaming will be in the most accessible position to facilitate future access for additional cables.
8. The remaining free space in the penetration shall be filled with RISE filler sleeves evenly split between items 4 and 7 for bulkhead penetrations and items 34 and 37 for deck penetrations from table 3B54-IFig_3B54.
9. Ensure the insert and filler sleeves fit tightly into the MCP. Cables shall not be bunched so tightly as to damage the insert sleeves or to preclude the injection of sealant between and around each cable. Insert sleeves and filler sleeves shall be adjusted in or out to form an aligned edge that is recessed ¾ inch from the edges of the MCP.
10. Clean and dry the entire penetration thoroughly. Remove any dirt, rust, or oil residues prior to applying the sealant.
11. Cables shall be supported by adjacent cable racks on each side of the MCP to ensure the cables do not sag during the sealant application and curing processes. Movement of the cable during the curing process can cause leaks. Cables are not permitted to contact the edge of the MCP. Ensure that cables are supported on both sides of the MCP such that, over time, the cables will not sag and contact the MCP edge.
12. Any cable bends shall be outside the RISE MCP. Cables within the penetration shall be parallel to each other. All cables shall enter the RISE MCP perpendicular to the MCP sealing surface. See MIL-STD-2003-4 for cable slack requirements for entrance into the MCP.

FIGURE 3B55. RISE MCP sealant installation instructions for bulkhead and deck watertight and non-watertight penetrations.

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13. Apply a layer of sealant, item 46 from table 3B54-IFig_3B54, at each end of the penetration.
    a. The completed installation shall have a minimum sealant depth of ¾ inch at each face.
    b. The penetration will need to be overfilled with sealant because some sealant will be pushed between and into the empty insert sleeves during the remainder of the RISE MCP installation process.
    c. It is advantageous to apply sealant starting at the center of the cable bundle and working toward the edges of the MCP.
    d. Sealant shall be applied completely around each cable. Moving the cable in a circular motion will enhance the flow of sealant around the cables. Long nozzles supplied with the sealant cartridges should be used to ensure sealant is completely applied around each cable.
    e. Loosening cable rack ties may be necessary while applying sealant to achieve sealant coverage. The cable rack ties shall be re-tightened after sealant application is complete, but before the sealant has time to begin curing.
    f. Application of the sealant shall be monitored such that there is not run or droop when the sealant is applied to the bottom face of a deck MCP.
14. Spray a cloth with water to prevent the sealant from sticking to the cloth.
15. While wearing gloves to prevent skin contact with the sealant, use the cloth to press down on the sealant layer. Then, press down on the sealant with a plastic putty knife, a paint stick, or by hand (wearing gloves). The smoothing action is essential to obtain effective fume resistance and watertightness.
16. Conduct flashlight tests on the completed penetration immediately after the work is completed. Perform a visual inspection of the sealant for holes in the sealant. Pay particular attention to the areas around the cables and the edge of the penetrator. If holes are detected, fill with additional sealant. Shine a one-million candlepower flashlight on the penetration. If any light is observed on the opposite side of the penetration, apply sealant to the areas that show light until there is no light shining through.
17. Once the curing requirements of figure 3B54 are met, conduct the post-installation testing required in figure 3B54.

FIGURE 3B55. RISE MCP sealant installation instructions for bulkhead and deck watertight and non-watertight penetrations – Continued.

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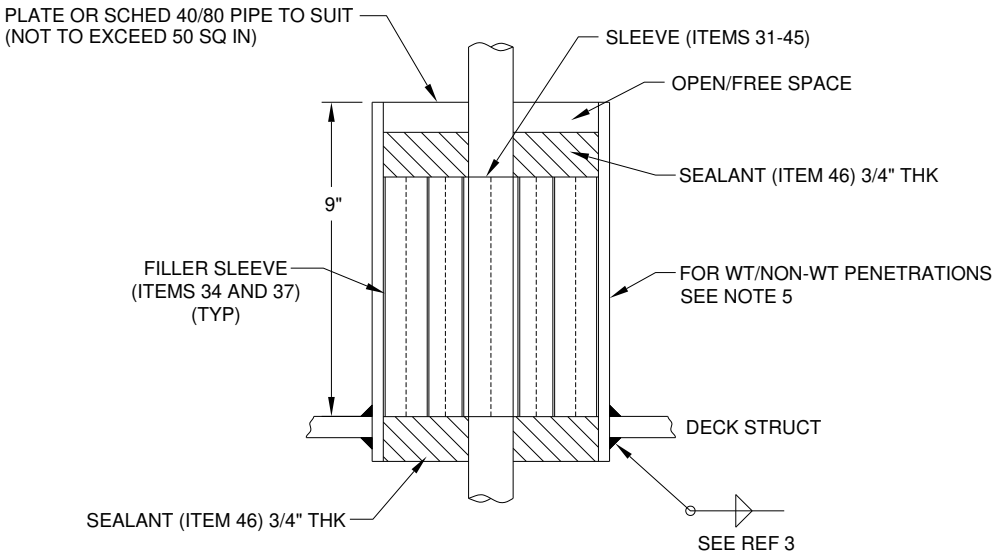


Illustration 1

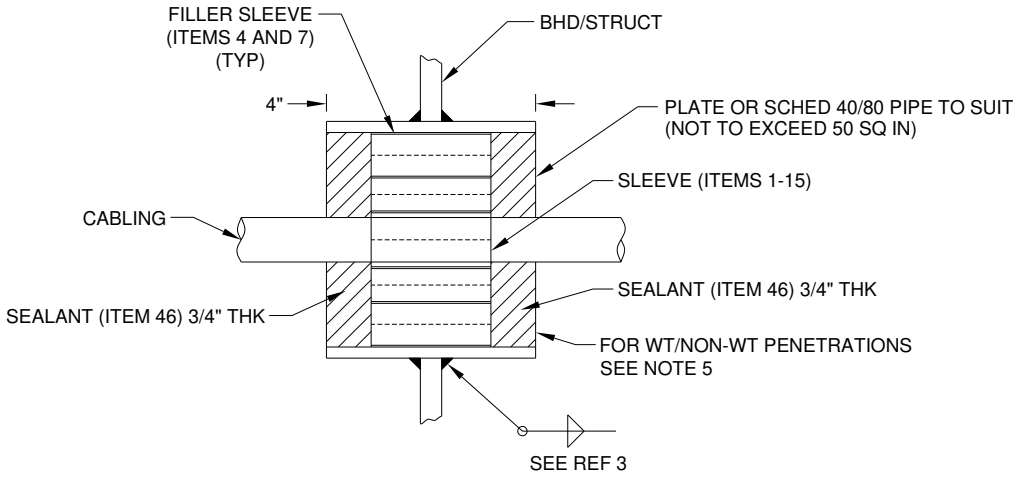


Illustration 2

FIGURE 3B55. RISE MCP sealant installation instructions for bulkhead and deck watertight and non-watertight penetrations – Continued.

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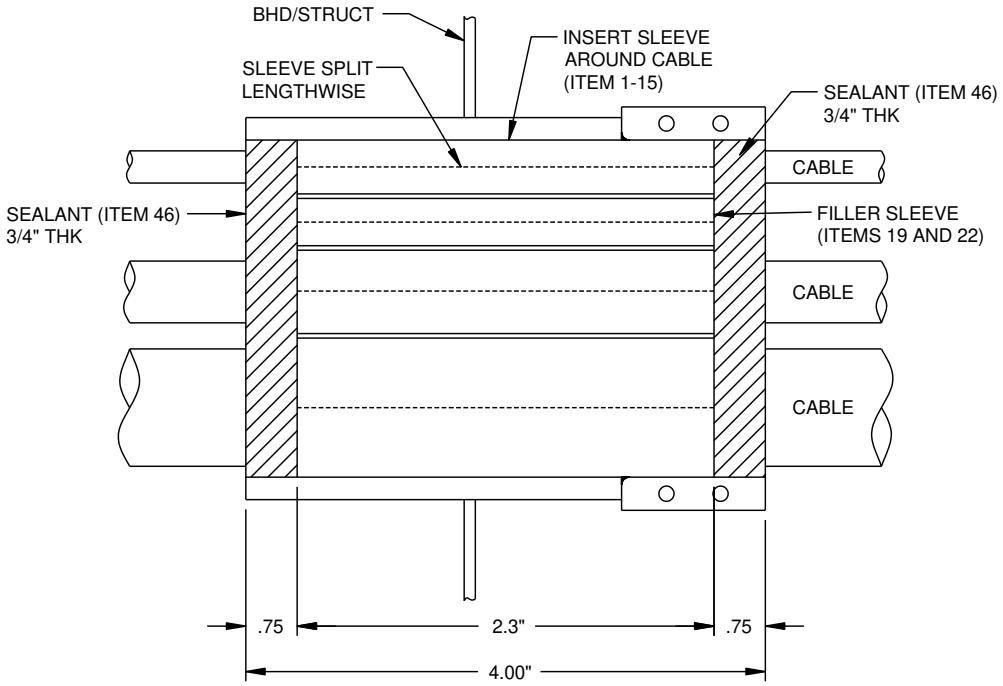


Illustration 3

FIGURE 3B55. RISE MCP sealant installation instructions for bulkhead and deck watertight and non-watertight penetrations – Continued.

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Method 3B-56-1
RISE Multiple Cable Penetrator (MCP) Sealant Installation Instructions for Replacing Existing Multiple Cable Penetrators

1. Follow all general requirements and limitations of figure 3B54.
2. This figure provides instructions for replacing a standard MCP with an MCP using the RISE method.
3. See illustrations 1-3 for guidance for replacing a standard MCP with a RISE MCP.
4. Remove the existing hardware from the existing MCP frame, leaving only the MCP frame in place.
5. Clean the area around the inside of the MCP, including existing cables. Clean surfaces will allow better bonding of the sealant.
6. Apply a thin layer of sealant, item 46 from table 3B54-I, inside the existing MCP prior to inserting the new extender plates to help seal the penetration.
7. Install the new MCP extension frames, items 48-50 from table 3B54-I, to achieve the minimum required penetration depth of 4 inches. The extension frames are shown in illustrations 2 and 3.
8. Install a bead of sealant around the perimeter of the new extender frames.
9. Separate the existing cables in the penetration as much as practicable. Pull any new cables to be added through the penetration. Ensure that, with the existing and added cables, the spare cable capacity requirement in step 5 of figure 3B54 is met.
10. After cables have been installed, a RISE insert sleeve shall be installed around each cable. The insert sleeves provide the necessary cable separation so that the sealant can flow between the cables to ensure an airtight seal. Each insert sleeve shall be adjusted in or out to be aligned with other sleeves and to form a clean edge recessed back approximately ¾ inch from the edge of the coaming.
    a. Select the sleeve for the applicable cable diameter from table 3B54-I.
    b. The insert sleeves are split lengthwise and, therefore, can be fitted around each cable prior to insertion inside the penetration.
    c. The RISE sleeve system does not require that the insert sleeve inside diameter exactly match the cable outside diameter; however, the closer the match, the more compact the installation will be.
    d. When packing deck MCPs, selecting insert sleeves with a smaller inside diameter than the cable is beneficial to help hold the sleeve in place over the cable during installation. When many small cables are passed through the MCP or when otherwise deemed appropriate by the installer, double sleeves may be used to provide adequate spacing around individual cables to permit/ensure the complete application of sealant.
    e. The sleeves may be overlapped or cut such that all parts of the cable length are covered. Gaps in the sleeving shall be avoided.
    f. Ensure that sufficient spacing exists around each cable to permit application of soft sealant, item 46 from table 3B54-I.
11. Once all cables have been sleeved, the cable bundle shall be maneuvered such that the free area inside the coaming of the MCP will be in the most accessible position to facilitate future access for additional cables.
12. The remaining free space in the penetration shall be filled with RISE filler sleeves evenly split between items 4 and 7 for bulkhead penetrations and items 34 and 37 for deck penetrations from table 3B54-I.
13. Ensure the insert and filler sleeves fit tightly into the MCP. Cables shall not be bunched so tightly as to damage the insert sleeves or to preclude the injection of sealant between and around each cable. Insert sleeves and filler sleeves shall be adjusted in or out to form an aligned edge that is recessed ¾ inch from the edges of the MCP.
14. Clean and dry the entire penetration thoroughly. Remove any dirt, rust, or oil residues prior to applying the sealant.
15. Cables shall be supported by adjacent cable racks on each side of the MCP to ensure the cables do not sag during the sealant application and curing processes. Movement of the cable during the curing process can cause leaks. Cables are not permitted to contact the edge of the MCP. Ensure that cables are supported on both sides of the MCP such that, over time, the cables will not sag and contact the MCP edge.
16. Any cable bends shall be outside the RISE MCP. Cables with the penetration shall be parallel to each other. All cables shall enter the RISE MCP perpendicular to the MCP sealing surface. See MIL-STD-2003-4 for cable slack requirements for entrance into the MCP.

FIGURE 3B56. RISE MCP sealant installation instructions for replacing existing multiple cable penetrations.

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17. Apply a layer of sealant, item 46 from table 3B54-I, at each end of the penetration.
    a. The completed installation shall have a minimum sealant depth of ¾ inch at each face.
    b. The penetration will need to be overfilled with sealant because some sealant will be pushed between and into the empty insert sleeves during the remainder of the RISE MCP installation process.
    c. It is advantageous to apply sealant starting at the center of the cable bundle and working toward the edges of the MCP.
    d. Sealant shall be applied completely around each cable. Moving the cable in a circular motion will enhance the flow of sealant around the cables. Long nozzles supplied with the sealant cartridges should be used to ensure sealant is completely applied around each cable.
    e. Loosening cable rack ties may be necessary while applying sealant to achieve sealant coverage. The cable rack ties shall be re-tightened after sealant application is complete, but before the sealant has time to begin curing.
    f. Application of the sealant shall be monitored such that there is not run or droop when the sealant is applied to the bottom face of a deck MCP.
18. Spray a cloth with water to prevent the sealant from sticking to the cloth.
19. While wearing gloves to prevent skin contact with the sealant, use the cloth to press down on the sealant layer. Then, press down on the sealant with a plastic putty knife, a paint stick, or by hand (wearing gloves). The smoothing action is essential to obtain effective fume resistance and watertightness.
20. Conduct flashlight tests on the completed penetration immediately after the work is completed. Perform a visual inspection of the sealant for holes in the sealant. Pay particular attention to the areas around the cables and the edge of the penetrator. If holes are detected, fill with additional sealant. Shine a one-million candlepower flashlight on the penetration. If any light is observed on the opposite side of the penetration, apply sealant to the areas that show light until there is no light shining through.
21. Once the curing requirements of figure 3B54 are met, conduct the post-installation testing required in figure 3B54.

FIGURE 3B56. RISE MCP sealant installation instructions for replacing existing multiple cable penetrations – Continued.

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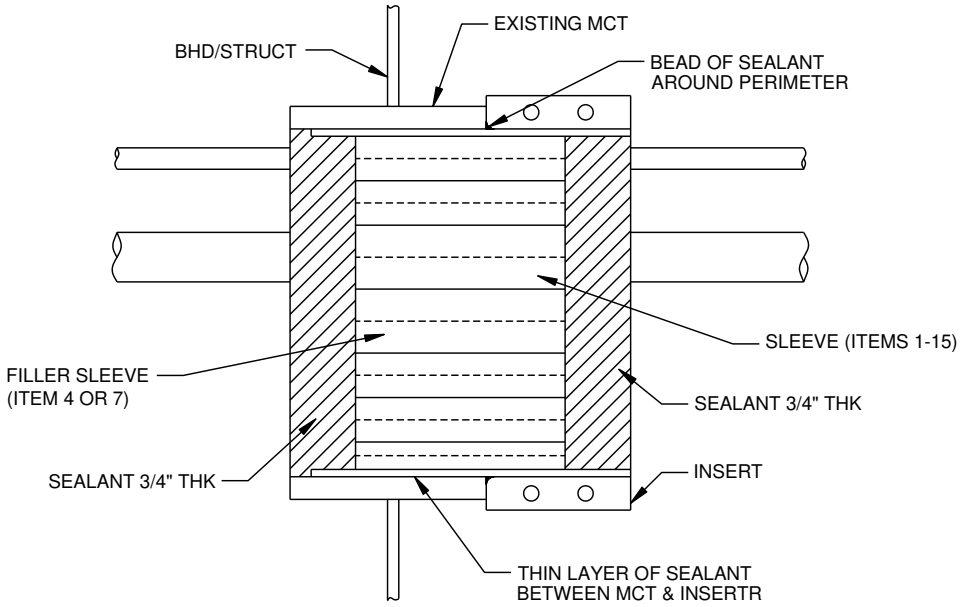


Illustration 1

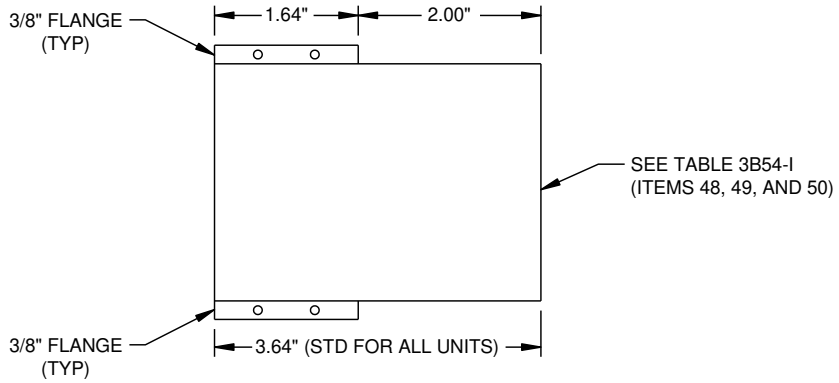


Illustration 2

FIGURE 3B56. RISE MCP sealant installation instructions for replacing existing multiple cable penetrations – Continued.

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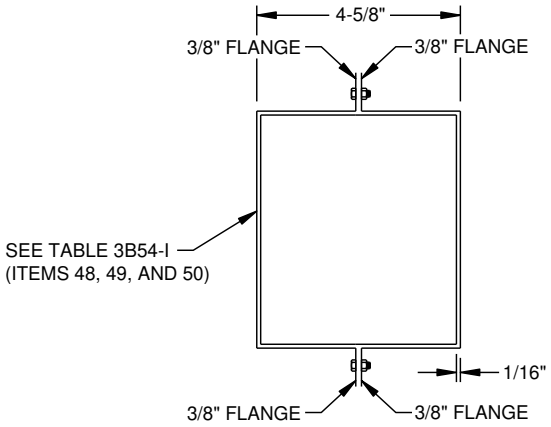


Illustration 3

FIGURE 3B56. RISE MCP sealant installation instructions for replacing existing multiple cable penetrations – Continued.

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Method 3B-57-1
RISE Multiple Cable Penetration (MCP) Sealant Installation Instructions for Electro-Magnetic Compatibility (EMC) Penetrations

1. Follow all general requirements and limitations of figure 3B54.
2. Any RISE MCP installation for EMC applications requires specific NAVSEA approval prior to installation.
3. This figure provides instructions for configuring a RISE penetration using the RISE method for a penetration used to provide EMC.
4. See illustrations 1 and 2 for guidance for configuring a RISE EMC penetration. Either illustration 1 or illustration 2 may be used.
5. At the place inside the penetration where the conduction flexible compound rubber, item 47 from table 3B54-I, will be applied, the penetration shall be bare steel without primer and thoroughly cleaned to ensure effective connection to earth.
6. If cables need to be installed, ensure that, with the existing and added cables, the spare cable capacity requirement in step 5 of figure 3B54 is met.
7. Clean the area around the inside of the MCP, including cables. Clean surfaces will allow better bonding of the sealant. Separate the cables in the penetration as much as practicable.
8. Remove the cable sheathing to expose the braid of the shielded cable for the 4-inch area where the conduction flexible compound rubber will be applied. For Entraco style cable conduit, the jacket shall be removed from ¾ inch inside the penetration to the opposite edge of the penetration. A waveguide with an outer layer shall have the outer layer removed for the same 4 inches of length that the conduction flexible rubber will be applied.
9. A RISE insert sleeve shall be installed around each cable and waveguide (if installed).
   a. Select the sleeve for the applicable cable diameter from table 3B54-I.
   b. The insert sleeves are split lengthwise and, therefore, can be fitted around each cable prior to insertion inside the penetration.
   c. When installing the sleeves on a rectangular or elliptical waveguide, the sleeve shall be molded around the waveguide for as tight a fit as possible. Overlapping or doubling the sleeve over the cable or waveguide is acceptable.
   d. Ensure that sufficient spacing exists around each cable to permit application of soft sealant, part 46 from table 3B54-I.
10. Once all cables have been sleeved, the cable bundle shall be maneuvered such that the free area of the MCP will be in the most accessible position to facilitate future access for additional cables.
11. The remaining free space in the penetration shall be filled with RISE filler sleeves evenly split between items 4 and 7 from table 3B54-I.
12. Ensure the insert and filler sleeves fit tightly into the MCP. For MCPs built in accordance with illustration 1, the insert and filler sleeves shall be pushed into the penetration such that there is 4.75 inches from the MCP edge to the sleeves. This should leave a ¾-inch recess on the opposite side of the MCP for application of sealant, item 46 from table 3B54-I. For MCPs built in accordance with illustration 2, the insert and filler sleeves shall be pushed into the penetration such that there is 5.5 inches from the MCP edge to the sleeves. The filler sleeves should be flush with the opposite side of the MCP. This should leave ¾ inch on either side of the conductive putty to apply the sealant.
13. Thoroughly clean the remaining inside wall of the penetration at both ends. Remove any dirt, rust, or oil residues prior to applying the sealant.

FIGURE 3B57. RISE MCP sealant installation instructions for electro-magnetic compatibility (EMC) penetrations.

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14. Cables shall be supported by adjacent cable racks on each side of the MCP to ensure the cables do not sag during the sealant application and curing processes. Movement of the cable during the curing process can cause leaks. Cables are not permitted to contact the edge of the MCP. Ensure that cables are supported on both sides of the MCP such that, over time, the cables will not sag and contact the MCP edge.
15. For MCPs built in accordance with illustration 1, apply ¾ inch of sealant, item 46 from table 3B54-I, at the edge of the MCP opposite of the side where the conduction flexible compound rubber will be applied. For MCPs built in accordance with illustration 2, apply ¾ inch of sealant, item 46 from table 3B54-I, inside the MCP at the inside edge of the insert and filler sleeves, adjacent to where the conductive putty will be applied.
    a. The completed installation shall have a minimum sealant depth of ¾ inch.
    b. The penetration will need to be overfilled with sealant because some sealant will be pushed between and into the empty insert sleeves during the remainder of the RISE MCP installation process.
    c. It is advantageous to apply sealant starting at the center of the cable bundle and working toward the edges of the MCP.
    d. Sealant shall be applied completely around each cable. Moving the cable in a circular motion will enhance the flow of sealant around the cables. Long nozzles supplied with the sealant cartridges should be used to ensure sealant is completely applied around each cable.
    e. Loosening cable rack ties may be necessary while applying sealant to achieve sealant coverage. The cable rack ties should be re-tightened after sealant application is complete, but before the sealant has time to begin curing.
16. Spray a cloth with water to prevent the sealant from sticking to the cloth.
17. While wearing gloves to prevent skin contact with the sealant, use the cloth to press down on the sealant layer. Then, press down on the sealant with a plastic putty knife, a paint stick, or by hand (wearing gloves). The smoothing action is essential to obtain effective fume resistance and watertightness.
18. Conduct a flashlight test on the penetration. Perform a visual inspection of the sealant for holes in the sealant. Pay particular attention to the areas around the cables and the edge of the penetrator. If holes are detected, fill with additional sealant. Shine a one-million candlepower flashlight on the penetration. If any light is observed on the opposite side of the penetration, apply sealant to the areas that show light until there is no light shining through.
19. Let the sealant cure for 24 hours if the ambient temperature is 50 °F or greater or 48 hours if the ambient temperature is less than 50 °F.
20. Pack conduction flexible compound rubber, item 47 from table 3B54-I, into the 4-inch long space in the penetration where the frame is bare metal and where the cable sheathing or waveguide covering has been removed in accordance with step 5 above. The conduction flexible compound rubber shall be tightly packed into place around the cable or waveguide (if applicable) and shall be packed in the entire 4-inch long space.
21. Clean the remaining inside wall of the penetration adjacent to the conductive putty in preparation for the final sealant application.
22. Repeat steps 14-18 above to apply sealant to the final ¾-inch long outside edge of the penetration adjacent to the conductive putty. This sealant layer may be crowned, as necessary, to allow water runoff. An excess of ¾-inch length of sealant is acceptable to achieve crowning.
23. Once the curing requirements of figure 3B54 are met, conduct the post-installation testing required in figure 3B54.

FIGURE 3B57. RISE MCP sealant installation instructions for electro-magnetic compatibility (EMC) penetrations – Continued.

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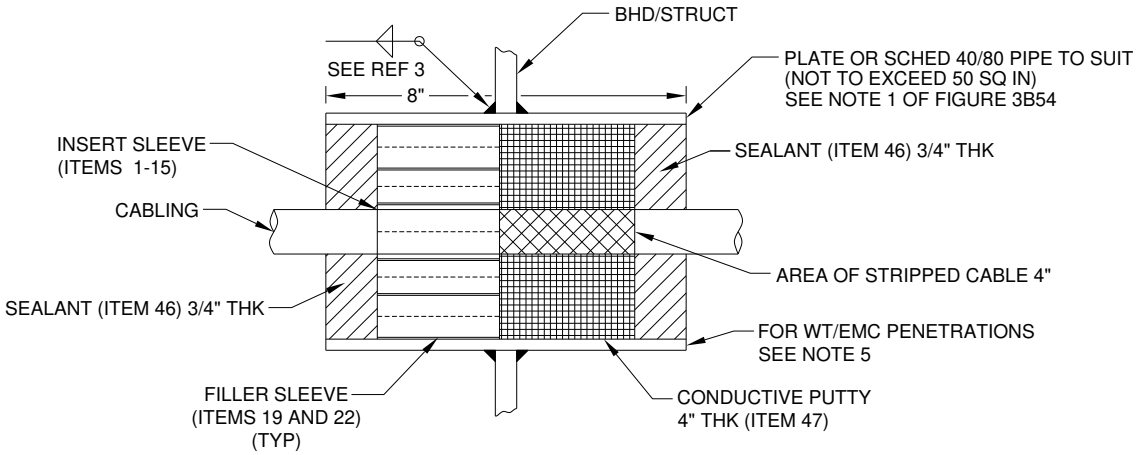


Illustration 1

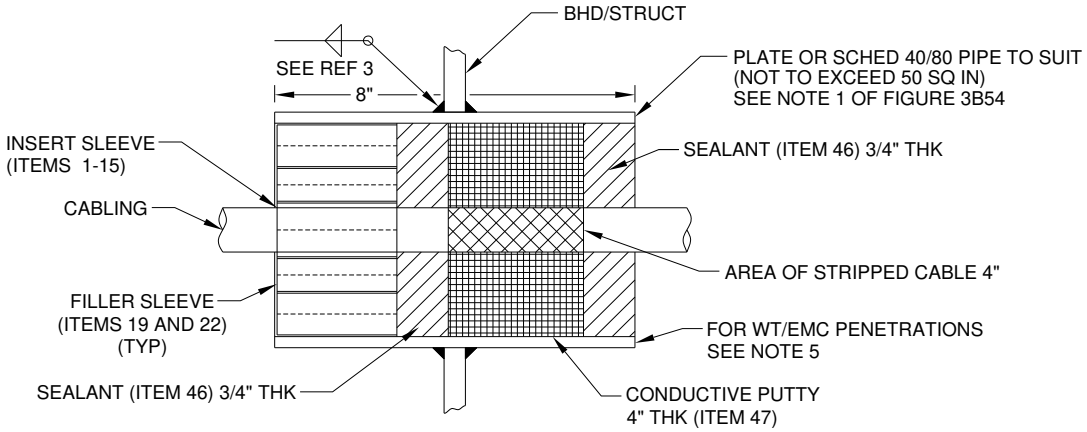


Illustration 2

FIGURE 3B57. RISE MCP sealant installation instructions for electro-magnetic compatibility (EMC) penetrations – Continued.

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Method 3B-58-1
Re-Entry/Repair of an Existing Expanding Sleeve Multiple Cable MCP System

1. Follow all general requirements and limitations of figure 3B54.
2. To add cables to a completed RISE MCP installation, carefully remove enough sealant from each face to expose sufficient filler sleeves to accommodate the added cables. Cut away the sealant layer at both sides of the penetration with a knife or a hollow punch in a tapering shape. Cutting the sealant at an angle of 45 degrees creates a good foundation for the sealant mass to be applied later. A non-metallic re-entry tool shall be used and the tools shall be non-conductive. Care shall be taken not to damage the insulation on existing cables in the MCP. Sharp edged tools shall not be used.
3. It may be possible to slip a small cable through an existing filler sleeve, whereas a larger cable may require removal of more than one filler sleeve. When a larger cable than will fit in an installed sleeve will be added, select a sleeve corresponding to the new cable from table 3B54-I. Place the used or new sleeves around the new cables and slide the sleeves adjacent to the other sleeves in place.
4. To remove cables from a completed RISE MCP, it is necessary to break the bond between the cables to be removed and the sealant. A small flat plastic bar may be slipped along the side of the cable to loosen the sealant from the cable on both ends of the MCP. No knives or sharp edge tools are allowed. Alternatively, the sealant may be carefully cut on both faces around the cable. The cables to be removed shall be pulled through and out of the MCP. The cable shall be replaced by filler sleeves, items 4 and 7, from table 3B54-I.
5. Sealant, item 46 from table 3B54-I, shall be applied to the affected area to ensure a completed sealant depth of ¾ inch for each MCP face.
6. Spray a cloth with water to prevent the sealant from sticking to the cloth.
7. While wearing gloves to prevent skin contact with the sealant, use the cloth to press down on the sealant layer. Then, press down on the sealant with a plastic putty knife, a paint stick, or by hand (wearing gloves). The smoothing action is essential to obtain effective fume resistance and watertightness.
8. Once the sealant is applied, perform a visual inspection of the sealant for holes in the sealant. Pay particular attention to the areas around the cables and the edge of the penetrator. If holes are detected, fill with additional sealant. Conduct a flashlight test on the completed penetration immediately after the work is completed. Shine a one-million candlepower flashlight on the penetration. If any light is observed on the opposite side of the penetration, apply sealant to the areas that show light until there is no light shining through.
9. Once the curing requirements of figure 3B54 are met, conduct the post-installation testing required in figure 3B54.

FIGURE 3B58. Re-entry/repair of an existing expanded sleeve multiple cable penetration system.

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Method 3B-59-1
Installation of Type A Square Frame Bulkhead Double Multi-Cable Transits (DMCTs)

General Notes:

1. This method is for installation of type A double multi-cable transits (DMCTs) through specific decks and bulkheads as required by 4.1.3. 4.1.3 requires that metal stuffing tubes or back-to-back multiple cable penetrations in accordance with MIL-DTL-24705 be installed in ballistic bulkheads and into spaces other than magazines that store volatile combustible or explosive material.
2. Compensation is required to restore the structural and protective integrity of the structure being penetrated.
3. Welding shall be in accordance with 4.1.1.
4. Observe the general requirements and installation instructions of figures 3B20 through 3B30. See 4.4 for specific restrictions and instructions for installation of fiber optic and blown fiber cables in electrical penetrations.
5. For bulkhead penetrations, the preferred installation is equal distance from both faces of the bulkhead penetrated. Compensation may be adjusted from either side to suit space arrangement. Sufficient clearance shall be maintained for welding.
6. The overall length of the compensation is determined by the shipboard location and the hole cut size (clear opening) in the structure penetrated. See figures 3B20 through 3B30 for standard assemblies.
7. Standard assemblies in accordance with this method are shown in figures 3B20 through 3B30 and shall meet the requirements of MIL-DTL-24705.
8. DMCT frames and filler plates shall be manufactured from steel plate and shall meet the dimensional requirements of MIL-DTL-24705 and figures 3B20 through 3B30.
9. Cable supports are not required internal to the DMCT, regardless of the coaming depth.
10. The maximum size of a DMCT shall not exceed a size 6, 4 frame tandem, as shown in figures 3B20 through 3B30 (a total of eight size 6 frames).

FIGURE 3B59. Installation of type A square frame bulkhead double multi-cable transits (DMCTs).

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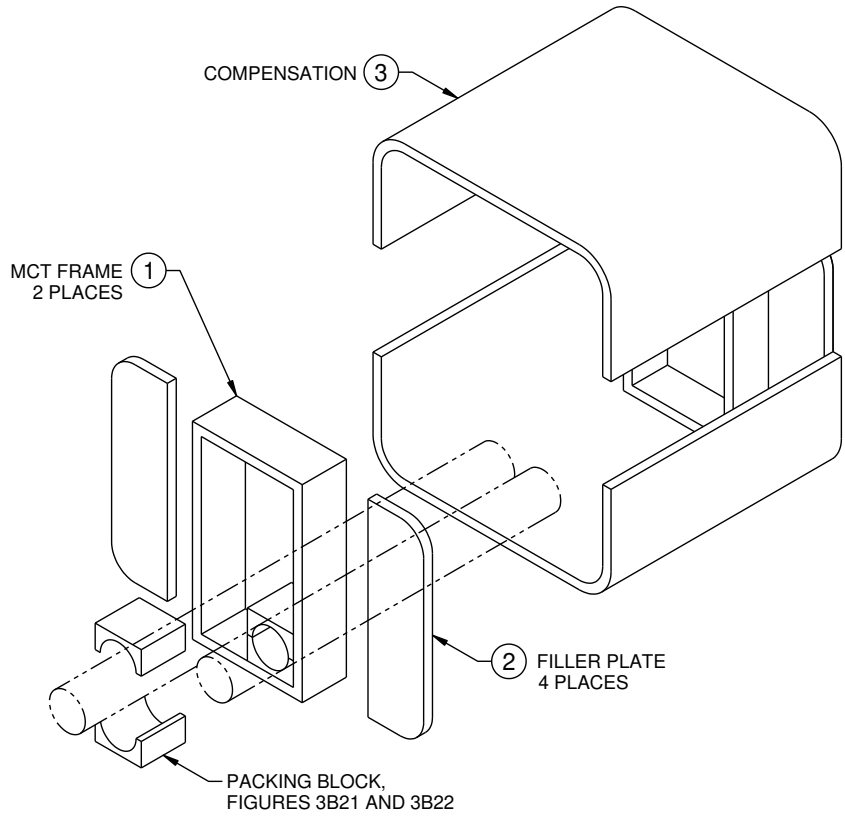


Illustration 1

TABLE 3B59-I. List of material for type A DMCT.

Item (see Illustrations)	Description
1	DMCT Frame, Steel Plate (see note 8)
2	Filler Plate, Steel Plate (see note 8)
3	Compensation, Steel Plate (see note 2)

FIGURE 3B59. Installation of type A square frame bulkhead double multi-cable transits (DMCTs) – Continued.

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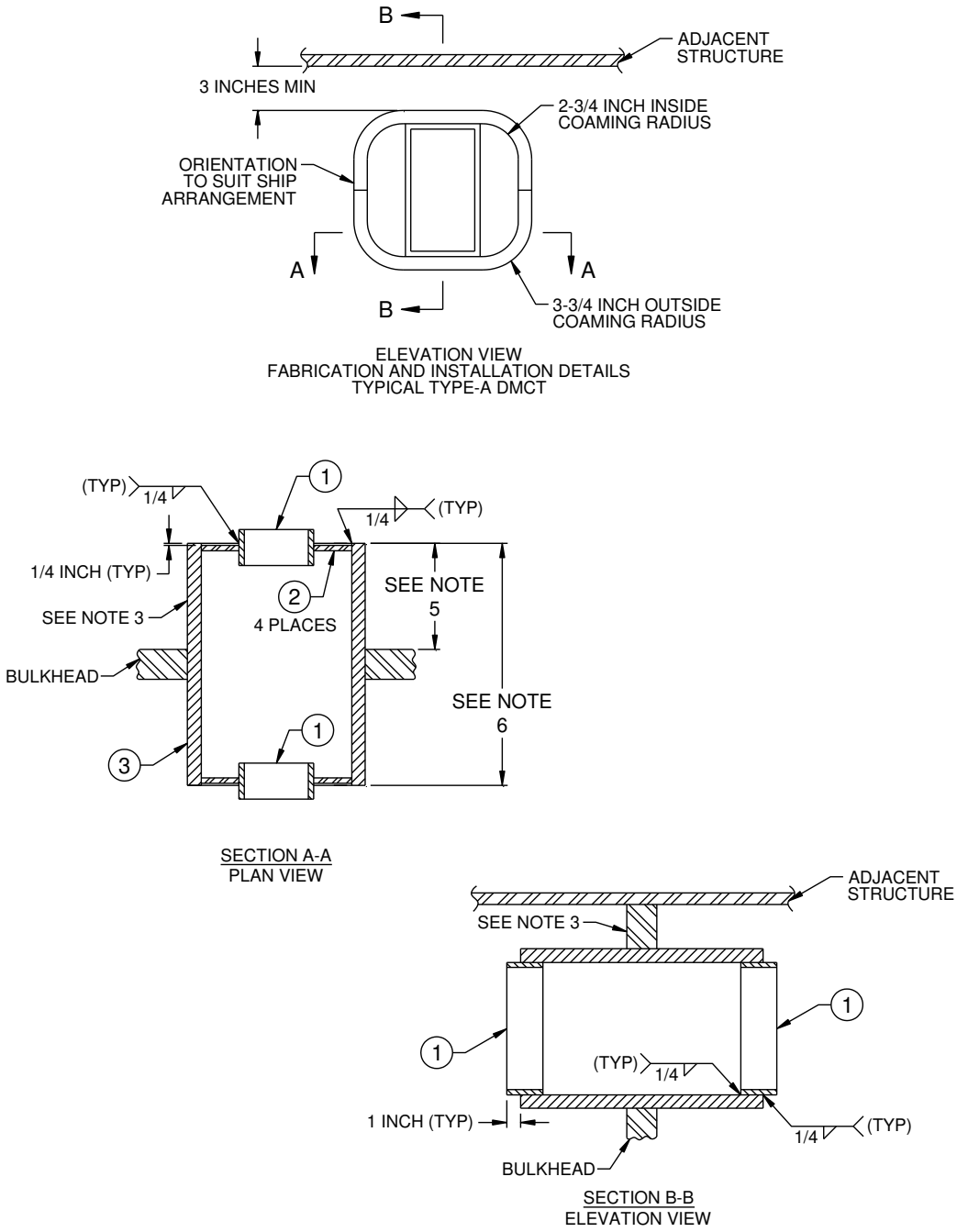


Illustration 2

FIGURE 3B59. Installation of type A square frame bulkhead double multi-cable transits (DMCTs) – Continued.

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General Requirements for Installation of Type B Double Multi-Cable Transits (DMCTs)

General Notes:

1. This method is for installation of type B double multi-cable transits (DMCTs) through specific decks and bulkheads as required by 4.1.3. 4.1.3 requires that metal stuffing tubes or back-to-back multiple cable penetrations in accordance with MIL-DTL-24705 be installed in ballistic bulkheads and into spaces other than magazine that store volatile combustible or explosive material.
2. Compensation is required to restore the structural and protective integrity of the structure being penetrated.
3. Welding shall be in accordance with 4.1.1.
4. Observe the general requirements and installation instructions of figures 3B20 through 3B30. See 4.4 for specific restrictions and instructions for installation of fiber optic and blown fiber cables in electrical penetrations.
5. For bulkhead penetrations, the preferred installation is equal distance from both faces of the bulkhead penetrated. Compensation may be adjusted from either side to suit space arrangement. Sufficient clearance shall be maintained for welding.
6. The overall length of the compensation is determined by the shipboard location and the hole cut size (clear opening) in the structure penetrated. See figures 3B20 through 3B30 for standard assemblies.
7. Standard assemblies in accordance with this method are shown in figures 3B20 through 3B30 and shall meet the requirements of MIL-DTL-24705.
8. DMCT frames and filler plates shall be manufactured from steel plate and shall meet the dimensional requirements of MIL-DTL-24705 and figures 3B20 through 3B30.
9. Cable supports are not required internal to the DMCT, regardless of the coaming depth.

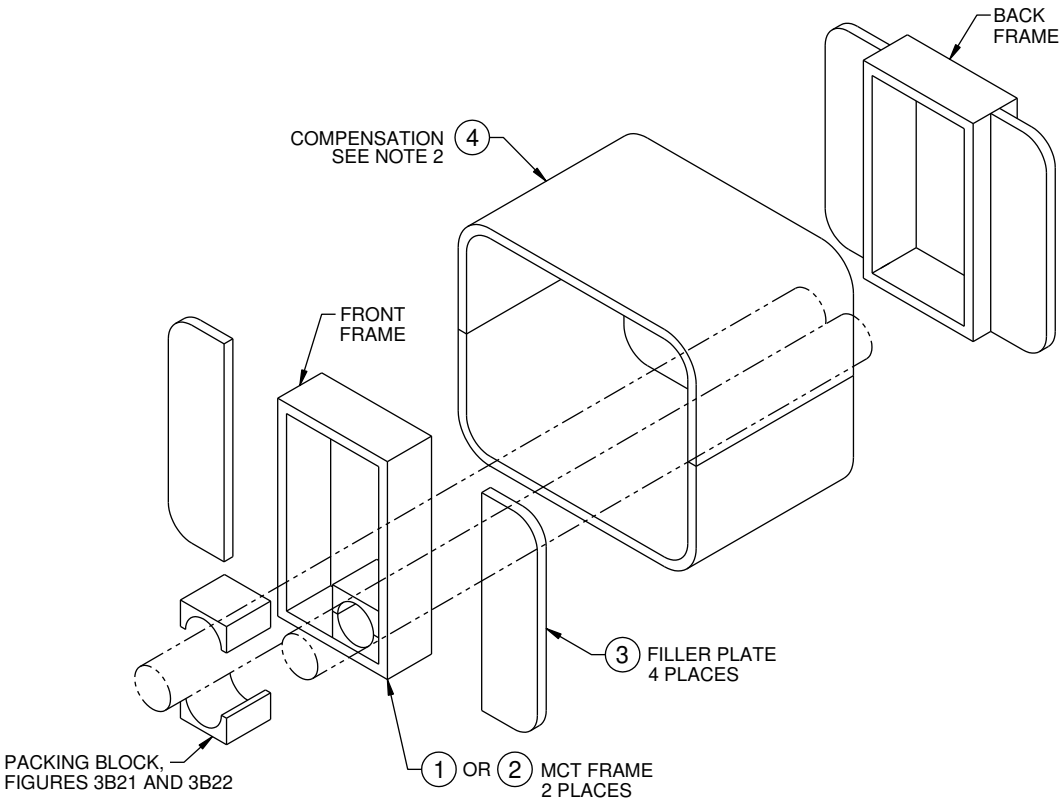


FIGURE 3B60. Installation of type B square frame double multi-cable transits (DMCTs).

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TABLE 3B60-I. List of material for type B DMCT.

Item (see Illustrations)	Description
1	DMCT Frame, Square, Steel Plate (see note 8)
2	DMCT Frame, Square Radius Corner, Steel Plate (see note 8)
3	Filler Plate, Steel Plate, ASTM A36/A36M (see note 8)
4	Compensation, Steel Plate (see note 2)

Method 3B-60-1
Installation of Type B Square Frame Double Multi-Cable Deck Transits

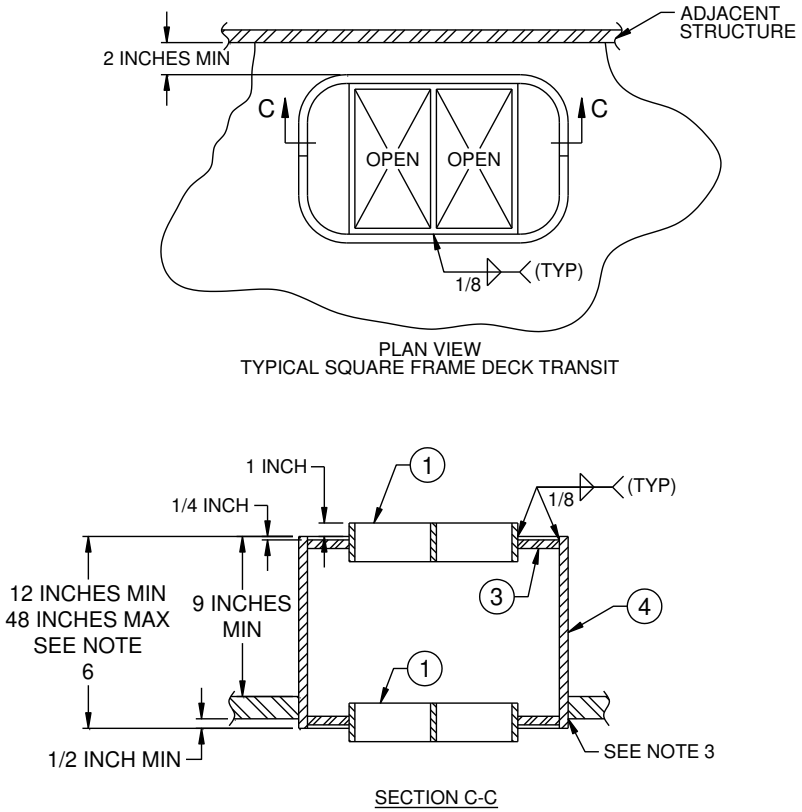


FIGURE 3B60. Installation of type B square frame double multi-cable transits (DMCTs) – Continued.

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Method 3B-60-2
Installation of Type B Square Frame Bulkhead Double Multi-Cable Transits

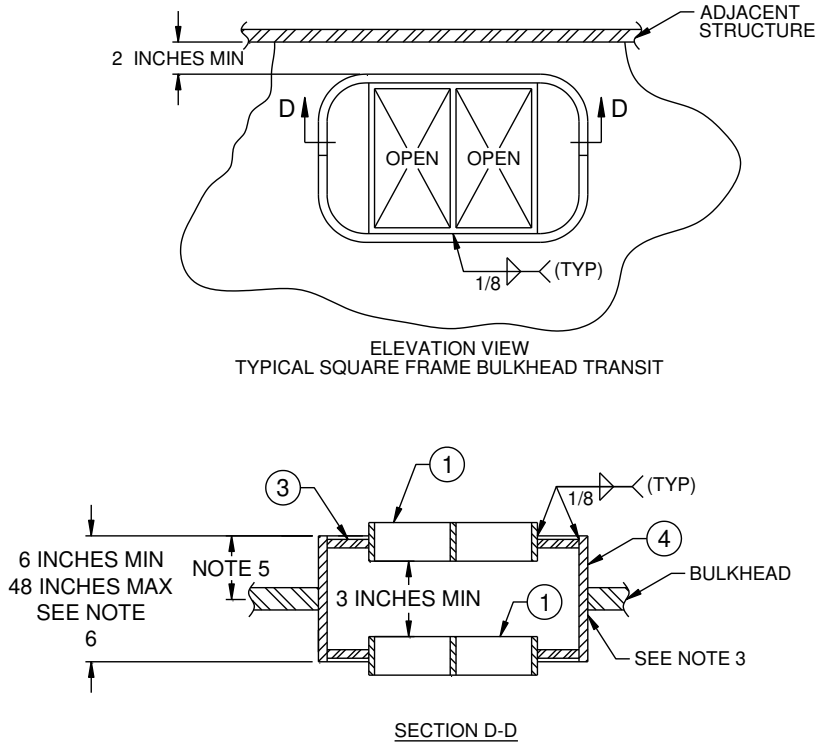


FIGURE 3B60. Installation of type B square frame double multi-cable transits (DMCTs) – Continued.

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Method 3B-60-3
Installation of Type B Radius Frame Double Multi Cable Deck Transits

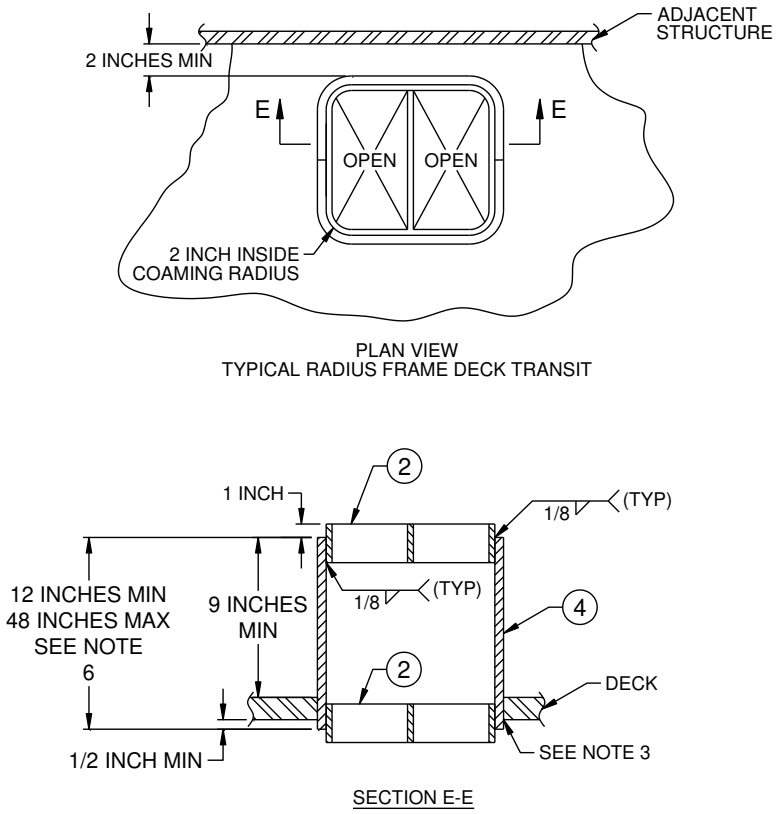
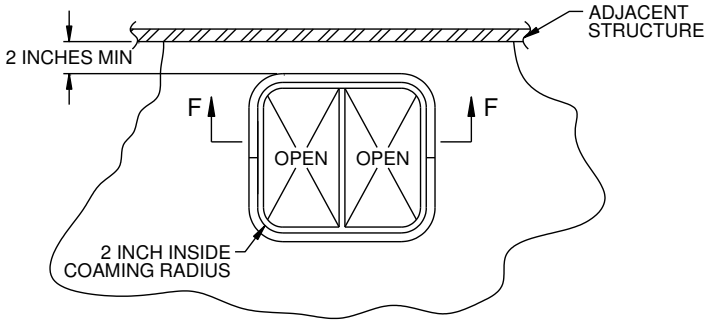


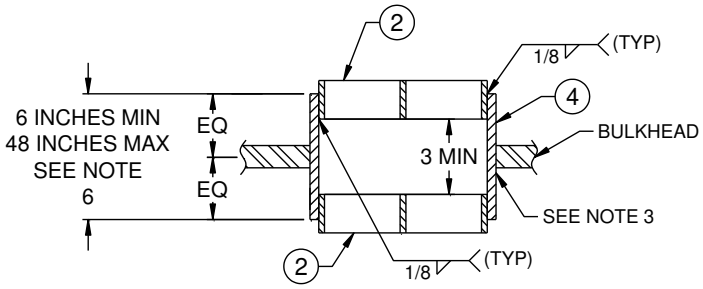
FIGURE 3B60. Installation of type B square frame double multi-cable transits (DMCTs) – Continued.

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Method 3B-60-4
Installation of Type B Radius Frame Double Multi Cable Bulkhead Transits



ELEVATION VIEW
TYPICAL RADIUS FRAME BULKHEAD TRANSIT



SECTION F-F

FIGURE 3B60. Installation of type B square frame double multi-cable transits (DMCTs) – Continued.

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General Requirements for Installation of Round Double Multi-Cable Transits (DMCTs)

General Notes:

1. This method is for installation of round double multi-cable transits (DMCTs) through specific decks and bulkheads as required by 4.1.3. 4.1.3 requires that metal stuffing tubes or back-to-back multiple cable penetrations in accordance with MIL-DTL-24705 be installed in ballistic bulkheads and into spaces other than magazine that store volatile combustible or explosive material.
2. Compensation is required to restore the structural and protective integrity of the structure being penetrated.
3. Welding shall be in accordance with 4.1.1.
4. Observe the general requirements and installation instructions of figures 3B39 through 3B44. See 4.4 for specific restrictions and instructions for installation of fiber optic and blown fiber cables in electrical penetrations.
5. For bulkhead penetrations, the preferred installation is equal distance from both faces of the bulkhead penetrated. Compensation may be adjusted from either side to suit space arrangement. Sufficient clearance shall be maintained for welding.
6. The overall length of the compensation is determined by the shipboard location and the hole cut size (clear opening) in the structure penetrated. See figures 3B39 through 3B44 for standard assemblies.
7. Standard assemblies in accordance with this method are shown in figures 3B39 through 3B44 and shall meet the requirements of MIL-DTL-24705.
8. DMCT frames and filler plates shall be manufactured from steel plate and shall meet the dimensional requirements of MIL-DTL-24705 and figures 3B39 through 3B44.
9. The round DMCTs may also use the larger diameter round frames shown in figure 3B39.

FIGURE 3B61. Installation of round double multi-cable transits (DMCTs).

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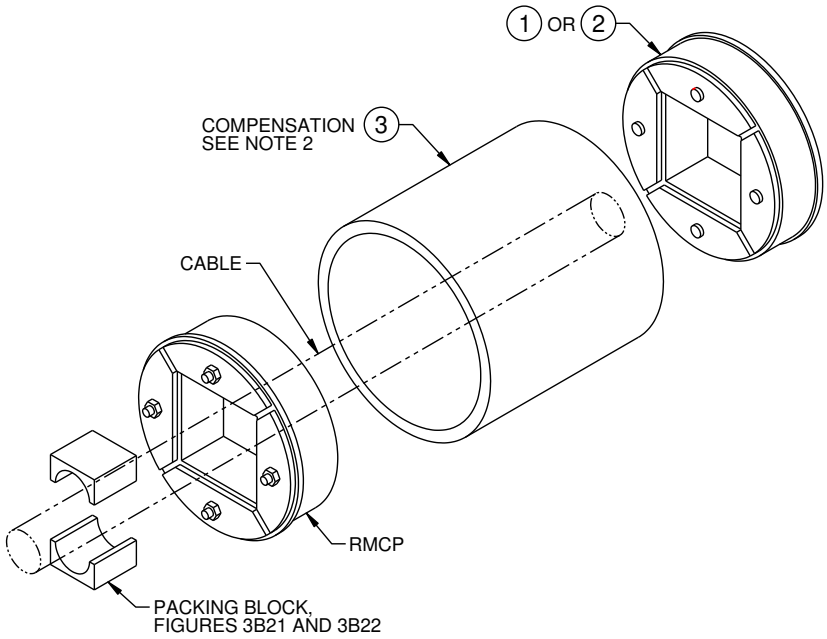


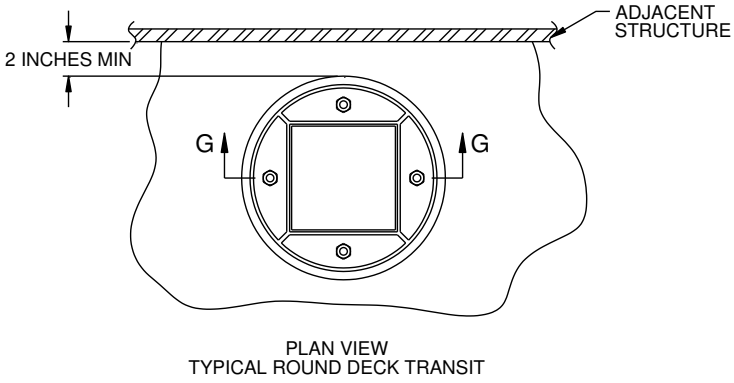
TABLE 3B61-I. List of material for round MCT.

Item, see Illustrations	Description
1	5" round MCT
2	8" round MCT
3	Compensation, steel plate, see note 2

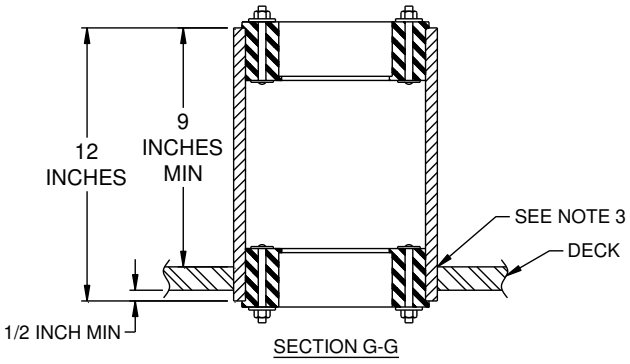
FIGURE 3B61. Installation of round double multi-cable transits (DMCTs) – Continued.

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Method 3B-61-1
Installation of Deck Round Double Multi-Cable Transits



PLAN VIEW
TYPICAL ROUND DECK TRANSIT



SECTION G-G

FIGURE 3B61. Installation of round double multi-cable transits (DMCTs) – Continued.

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Method 3B-61-2
Installation of Bulkhead Round Double Multi-Cable Transits (DMCTs)

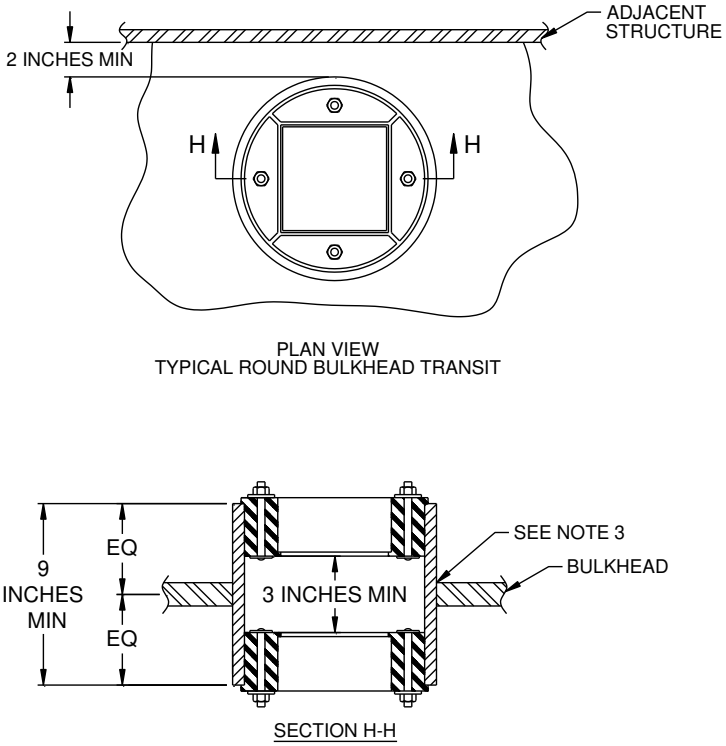


FIGURE 3B61. Installation of round double multi-cable transits (DMCTs) – Continued.

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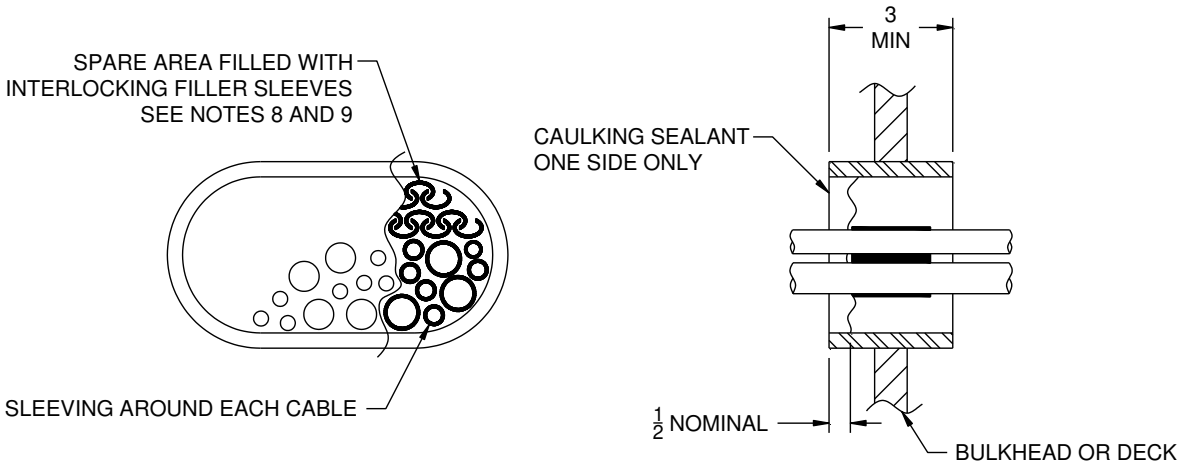
Method 3B-62-1
Instructions for RISE Multiple Cable Penetrator (MCP) Sealant Installation for Single Sided Air-Tight and Fire Zone Bulkhead and Deck Penetrations

1. Follow all general requirements and limitations of figure 3B54.
2. This figure provides instructions for configuring a RISE penetration using the RISE method for single-sided bulkhead and deck penetrations.
3. Ensure the penetration can support a RISE MCP installation with at least a 3-inch depth. The minimum 3-inch depth is required to accommodate the minimum 2½-inch long insert sleeves plus the minimum ½-inch depth of caulking sealant applied to one end of the sleeves.
4. Cables may be installed through the penetration in any order. Cables shall not be pulled so tight to prevent separation to apply sleeves and sealant.
5. After cables have been installed, a RISE insert sleeve shall be installed around each cable. The insert sleeves provide the necessary cable separation so that the sealant can flow between the cables to ensure an airtight seal. Each insert sleeve shall be adjusted in or out to be aligned with other sleeves and to form a clean edge recessed back approximately ½ inch from the edge of the coaming.
    a. Select the sleeve for the applicable cable diameter from items 1-15 for bulkhead penetrations or items 31-45 for deck penetrations from table 3B54-I.
    b. The insert sleeves are split lengthwise and, therefore, can be fitted around each cable prior to insertion inside the penetration.
    c. The RISE sleeve system does not require that the insert sleeve inside diameter exactly match the cable outside diameter; however, the closer the match, the more compact the installation will be.
    d. When packing deck MCPs, selecting insert sleeves with a smaller inside diameter than the cable is beneficial to help hold the sleeve in place over the cable during installation. Electrical tape wrapped around the cable and sleeve may be used to limit sleeve movement on the cable.
    e. The sleeves may be overlapped or cut such that all parts of the cable length are covered. Gaps in the sleeving shall be avoided.
    f. Ensure that sufficient spacing exists around each cable to permit application of soft sealant, item 46 from table 3B54-I.
6. Once all cables have been sleeved, the cable bundle shall be maneuvered such that the free area inside the coaming will be in the most accessible position to facilitate future access for additional cables.
7. The remaining free space in the penetration shall be filled with RISE filler sleeves evenly split between items 4 and 7 for bulkhead penetrations and items 34 and 37 for deck penetrations from table 3B54-I.
8. Ensure the insert and filler sleeves fit tightly into the MCP so the sleeves do not shift when sealant is applied. The filler sleeves should be contained such that the sleeves do not fall out prior to application of the sealant. Cables shall not be bunched so tightly as to damage the insert sleeves or to preclude the injection of sealant between and around each cable. Insert sleeves and filler sleeves shall be adjusted in or out to form an aligned edge that is recessed ½ inch from the edge on one side of the MCP. For deck penetrations, the ½-inch gap should typically be at the top face for ease of installation of the sealant.
9. Clean and dry the entire penetration thoroughly. Remove any dirt, rust, or oil residues prior to applying the sealant.
10. Cables shall be supported by adjacent cable racks on each side of the MCP to ensure the cables do not sag during the sealant application and curing processes. Movement of the cable during the curing process can cause leaks. Cables are not permitted to contact the edge of the MCP. Ensure that cables are supported on both sides of the MCP such that, over time, the cables will not sag and contact the MCP edge.

FIGURE 3B62. RISE MCP sealant installation instructions for single sided bulkhead and deck non-watertight penetrations.

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11. Apply a layer of sealant, item 46 from table 3B54-I, to the one end of the penetration made up of insert sleeve ends such that the completed installations will have approximately ½ inch of sealant. The penetration will need to be overfilled with sealant because some sealant will be pushed between and into the empty insert sleeves during the remainder of the RISE MCP installation process. It is acceptable if the sealant is applied over the edge of the coaming during installation.
12. Conduct a flashlight test on the penetration. Perform a visual inspection of the sealant for holes in the sealant. Pay particular attention to the areas around the cables and the edge of the penetrator. If holes are detected, fill with additional sealant. Shine a one-million candlepower flashlight on the penetration. If any light is observed on the opposite side of the penetration, apply sealant to the areas that show light until there is no light shining through.
13. Re-entry/repair process:
    a. To add cables to a completed single sided RISE installation, carefully remove enough sealant from the face with a non-metallic blunt instrument to expose sufficient filler sleeves to accommodate the added cable. A small flat plastic bar may be slipped along the side of the cable to loosen the sealant from the cable on both ends of the MCP. No knives or sharp edge tools are allowed and the tools shall be non-conductive. Slipping a small cable through an existing filler sleeve may be possible, whereas adding a larger cable may require removal of several filler sleeves. Where a larger cable is added, a filler sleeve should be sized and installed around the cable. Sealant should then be applied to the affected area to ensure a completed nominal sealant depth of ½ inch from the face of the MCP.
    b. To remove a cable, break the bond between the affected cable and the sealant using a dull screwdriver or small pry bar slipped along the side of the cable to pry the sealant loose. As an alternative, the sealant may be cut around the cable. The cable can then be pulled through the coaming. Insert sleeves and sealant can then be applied to the affected area.



FLAT OVAL COAMING INSTALLATION SHOWN
TYPICAL FOR ROUND COAMINGS OR SQUARE
RADIUS CORNER COAMINGS

FIGURE 3B62. RISE MCP sealant installation instructions for single sided bulkhead and deck non-watertight penetrations – Continued.

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Method 3B-63-1
Multiple Cable Penetrators Used for Electro-Magnetic Compatibility Applications

General Notes:

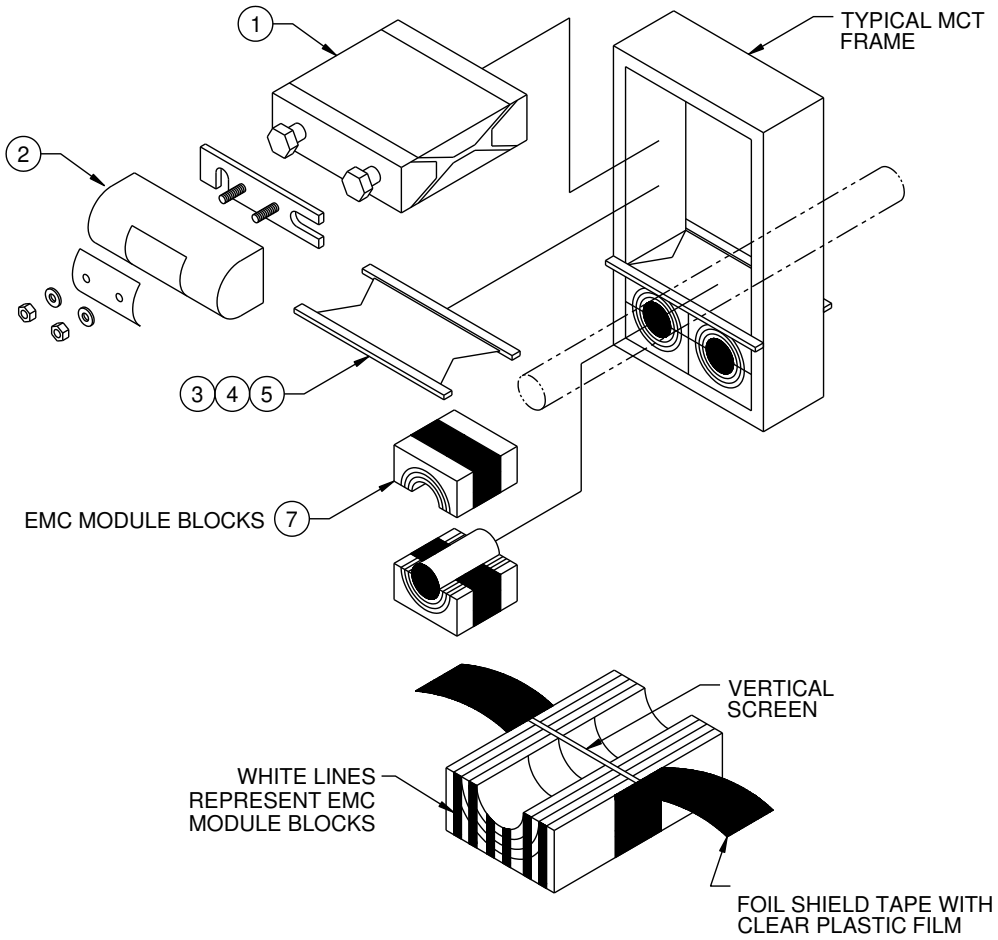
1. This method is applicable to installing and packing multiple cable penetrators (MCPs) that are used in electro-magnetic compatibility (EMC) applications. EMC MCPs are used on overall shielded type cables to provide a 360-degree shield ground at the point of cable penetration. EMC MCPs may be used for entry into electrical or electronic equipment, for bulkhead or deck penetrations, or for hull penetrations in topside locations. Shielded round frames may be used in place of rectangular frames. See 4.4 for specific restrictions and instructions for installation of fiber optic and blown fiber cables in electrical penetrations.
2. EMC MCPs shall meet EMC requirements of MIL-DTL-24705. EMC MCPs utilized for topside locations shall have frames and stay plates manufactured from CRES 316 material.
3. Packing requirements may vary for different suppliers of EMC MCP assemblies. Applicable supplier installation instructions should be followed.
4. See table 3B21-I for information and guidance to cross reference cable size with part number for EMC MCP modules.
5. See figure 3B20 for general MCP nomenclature.
6. A wedge designed for EMC applications may be substituted for the wedge cover assembly.

TABLE 3B63-I. Parts list.

Item	Description	Remarks
1	Wedge pack	
2	Wedge EMC cover assembly	Steel, see general note 6
3	Steel stay plate	Use with steel frame
4	Aluminum stay plate	Use with steel frame
5	CRES stay plate	See general note 2
6	Tallow	
7	EMC MCP module block	See general note 4

FIGURE 3B63. Multiple cable penetrators used for electro-magnetic compatibility applications.

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TYPICAL EMC MULTI-CABLE TRANSIT
EXPLODED VIEW

Illustration 1

Assembly Instructions:

1. To ensure good electrical conductivity for EMC grounding, clean all surfaces where the MCP frame will make electrical contact for the entire path to ground. Clean electrical contact surfaces to bright metal to ensure electrical contact.
2. On the cable, mark the module position where the outer sheath is to be removed (see illustration 2).
3. Cut cable outer sheath around the entire cable circumference to expose the shield. Be sure not to damage the cable shield.
4. Remove the cut outer sheath and make sure the exposed cable shield is clean to ensure good electrical contact.
5. Remove the plastic protective film from the foil shield tape on the EMC MCP modules and fold back the tape. Do not remove the protective tape from the spare modules.

FIGURE 3B63. Multiple cable penetrators used for electro-magnetic compatibility applications – Continued.

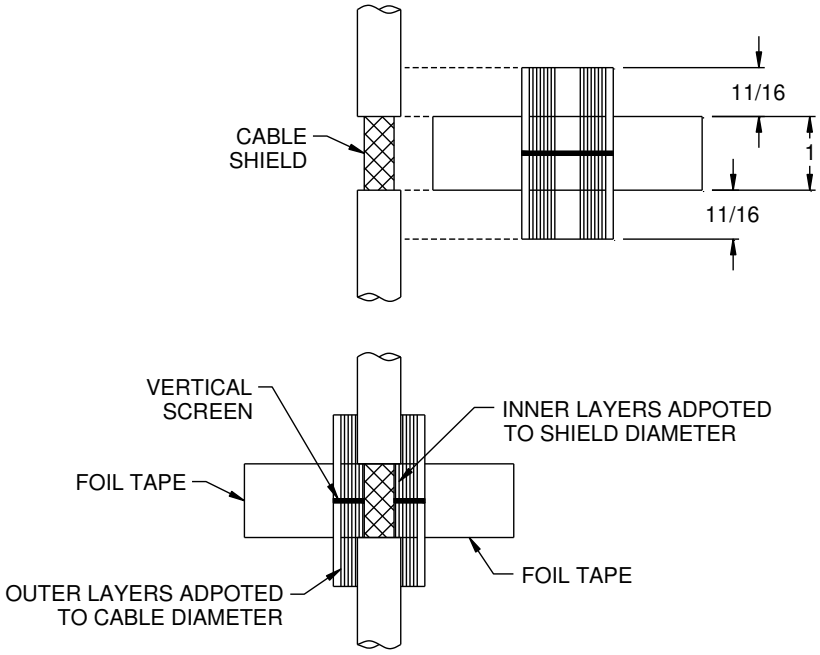
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Assembly Instructions (continued):

6. Adapting the EMC MCP module blocks:
   a. Peel the outer layers of the module block to the diameter of the cable insulation.
   b. Peel the inner layer of the module block to the diameter of the cable shielding.
   c. Cut the vertical screen in the module to fit the cable.
   d. Cable should fit inside the module block as shown in illustration 3.
7. Adapting the EMC MCP module block foil shield:
   a. Fold adhesive foil tape tightly inside the module half from one side along the inner layers.
   b. Lift the plastic film from the folded side of the adhesive foil.
   c. Fold the foil tape on the other side of the module tightly inside the module half. Ensure there are no air pockets.
   d. Fold back plastic film inside the module half.
8. Lubricate all modules sparsely with tallow, only the inside and outside rubber surfaces. Do not lubricate the frame or the plastic film.
9. Remove the plastic film while keeping the foil tape clean. Leave protective film on spare modules to protect foil for later use.
10. Starting with the largest module, insert the modules according to the packing plan. Insert module blocks directly on top of the cable.
11. Insert stay plates on top of every finished row of modules. Use steel plates with steel frames, aluminum plates with aluminum frames, and CRES plates with CRES frames.
12. Continue to fill the packing space with cable, modules, and stay plates. Do not slide in the modules to prevent damage to the conductive foil.
13. Before inserting the final row of modules, insert two stay plates together.
14. Separate the two stay plates and insert the final row of modules between the stay plates. Drop the upper stay plate on top of the modules. Optional: First insert a row of bottom module halves, then insert the cable or center plugs (spare modules), and then insert the top module halves.
15. Lubricate all sides of the wedge except the front and back face. Insert the wedge into the frame and tighten the wedge screws. Optional: The wedge may be inserted in the middle or bottom of the frame. Ensure the marked side of the wedge is against a stay plate.
16. Install the EMC wedge cover assembly, if provided.
17. Wait 24 hours or longer (refer to manufacturer's instructions) after installation before exposing the cables/pipes to strain or pressure. The following wait time shall be observed prior to the post-installation testing. Follow all retest instructions in figure 3B22.

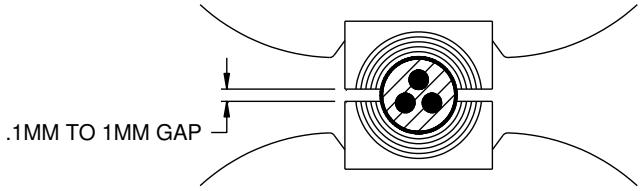
FIGURE 3B63. Multiple cable penetrators used for electro-magnetic compatibility applications – Continued.

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ADAPTING MODULE BLOCKS

Illustration 2



ADAPT MODULES BY PEELING OFF ONE LAYER AT A TIME TO ACHIEVE A 1 MM GAP BETWEEN THE TWO HALVES WHEN HELD AGAINST THE CABLE.

MODULE BLOCK SIZING

Illustration 3

FIGURE 3B63. Multiple cable penetrators used for electro-magnetic compatibility applications – Continued.

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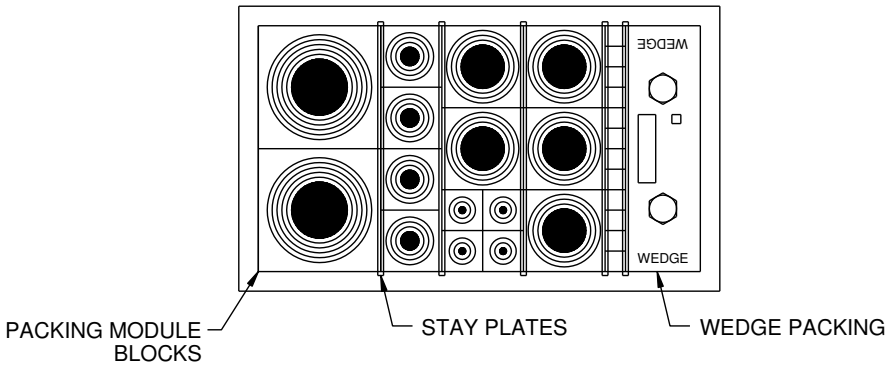
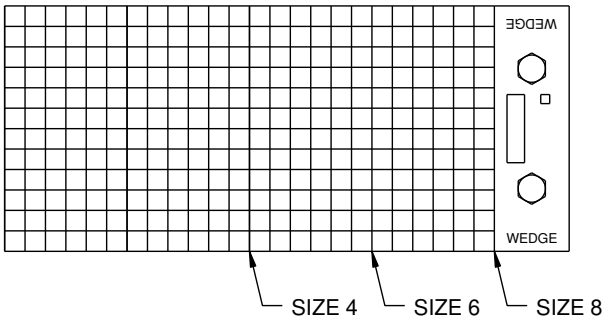


Illustration 4

Packing Plan:

1. List the outside dimension of all cables to be packed within the EMC MCP frame.
2. Select the proper size EMC MCP module block from table 3B21-I, shown for information and guidance.
3. Using the layout grid, arrange each selected block on the grid, outlining the location of the blocks in the grid. Indicate blank or spare modules as well as blocks with cables installed. Note: The grid layout is 10-millimeter squares.
4. A stay plate should be indicated and installed in between each row of module blocks.



LAYOUT GRID FOR PACKING PLAN

Illustration 5

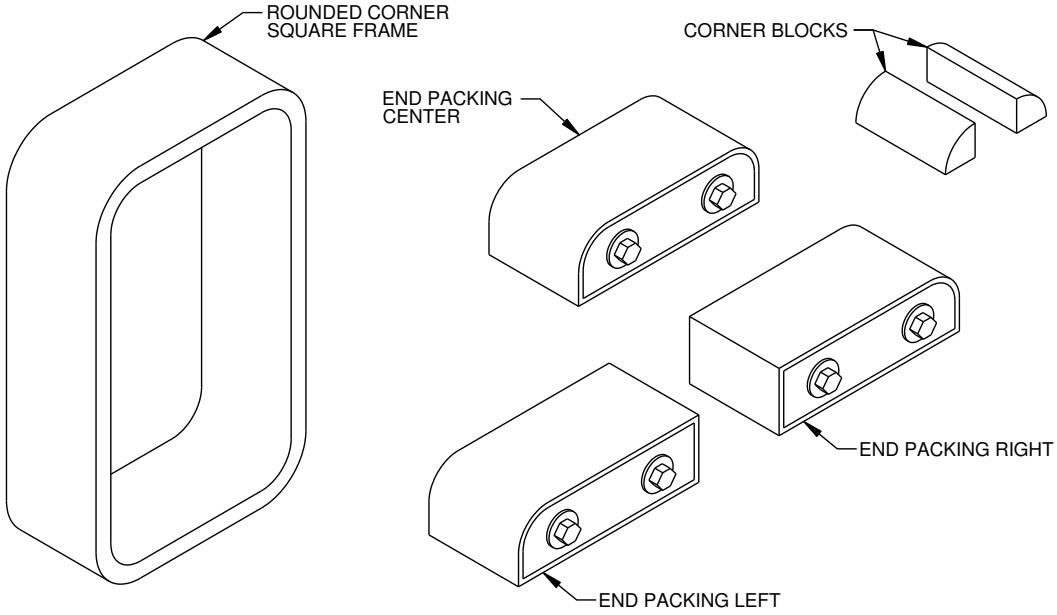
FIGURE 3B63. Multiple cable penetrators used for electro-magnetic compatibility applications – Continued.

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Method 3B-64-1
Installation of Rounded Corner Square Frame Multiple Cable Penetrators (RGSC)

The type Rounded Corner Square Frame (RGSC) Multiple Cable Penetrator (MCP) is a version of a rectangular MCP frame with rounded corners to reduce the risk of cracks forming in decks and bulkheads that are subjected to heavy loading. The RGSC metal frame is intended for load bearing structures. RGSC is designed with rounded corners to reduce stress concentration and the risk of cracks around the frame. Similar to the rectangular frames shown in figure 3B20, the RGSC frame is available in sizes 2, 4, 6, and 8. RGSC can also be installed as multiple frames. RGSC is available in mild steel, stainless steel, and aluminum. Special cornerblocks and endpackings with rounded corners are used when installing RGSC MCPs. The wedges included in figures 3B20 and 3B22 may also be used to secure the cable penetration.

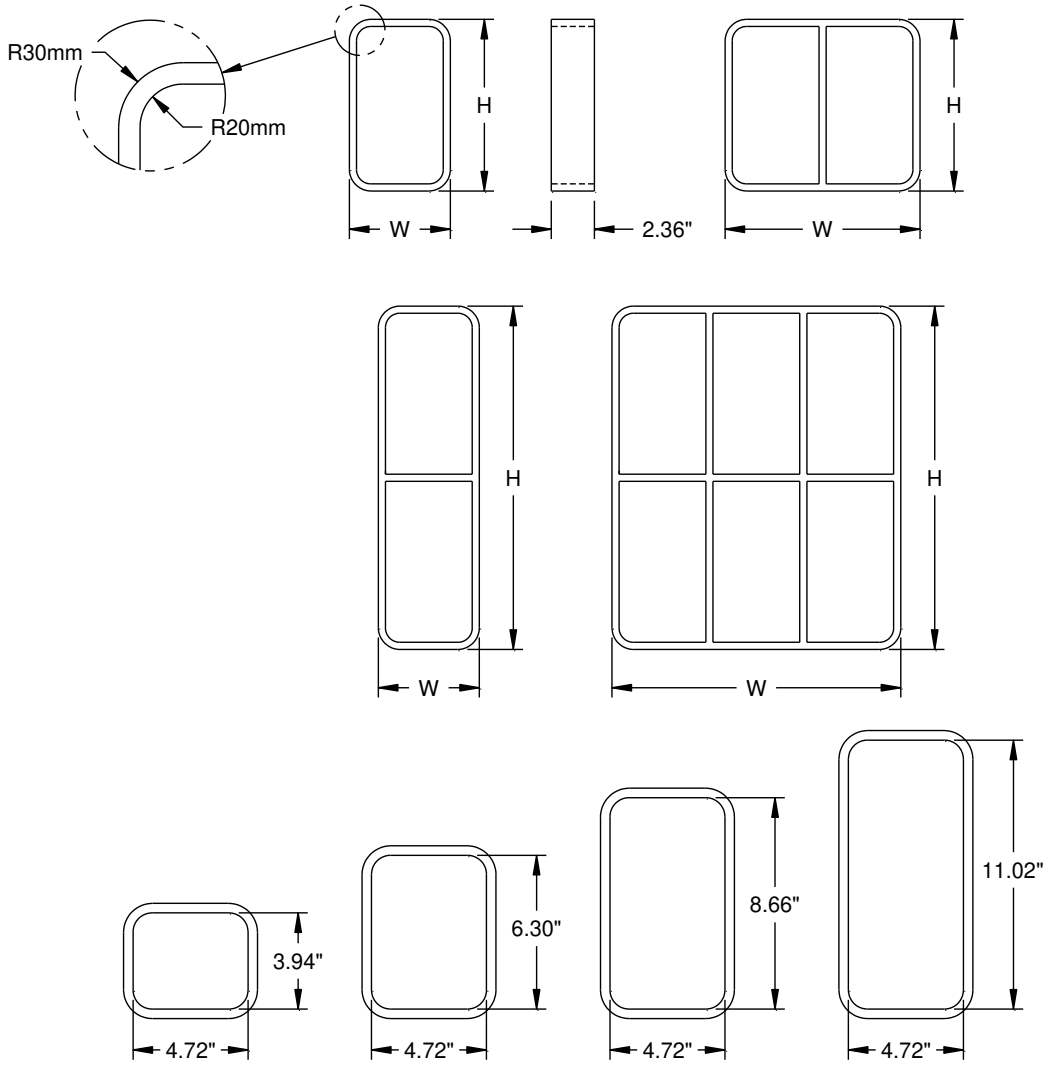
The following are illustrations of the basic parts of the RGSC frame. These are in addition to the wedge, compression plate, and insert blocks shown in figures 3B20 and 3B21. In addition, the following tables represent approximate size and weight values for the RGSC MCP frame variants. Manufacturer's data should be consulted if more precise information is needed.



Sample sizes for rounded corner square frame MCPs. Note that frames of different sizes can be combined to provide the desired cable penetrator. For example, a size four frame can be vertically attached to a size six frame.

FIGURE 3B64. Installation of rounded corner square frame multiple cable penetrators (RGSC).

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Frame sizes. See table 3B64-I for typical sizes of rounded corner square frames. Dimensions are in inches.

FIGURE 3B64. Installation of rounded corner square frame multiple cable penetrators (RGSC) – Continued.

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TABLE 3B64-I. Typical RGSC MCP frame sizes.

Frame Size	Height (H)	Width (W) Multiple Frames X1	Width (W) Multiple Frames X2	Width (W) Multiple Frames X3	Width (W) Multiple Frames X4	Width (W) Multiple Frames X5	Width (W) Multiple Frames X6
RGSC-2	4.76	5.53	10.67	15.81	20.94	26.08	31.22
RGSC-4	7.07	5.53	10.67	15.81	20.94	26.08	31.22
RGSC-6	9.37	5.53	10.67	15.81	20.94	26.08	31.22
RGSC-8	11.67	5.53	10.67	15.81	20.94	26.08	31.22
RGSC-2+2	9.53	5.53	10.67	15.81	20.94	26.08	31.22
RGSC-2+4	11.83	5.53	10.67	15.81	20.94	26.08	31.22
RGSC-2+6	14.13	5.53	10.67	15.81	20.94	26.08	31.22
RGSC-2+8	16.44	5.53	10.67	15.81	20.94	26.08	31.22
RGSC-4+4	13.74	5.53	10.67	15.81	20.94	26.08	31.22
RGSC-4+6	16.04	5.53	10.67	15.81	20.94	26.08	31.22
RGSC-4+8	18.35	5.53	10.67	15.81	20.94	26.08	31.22
RGSC-6+6	18.35	5.53	10.67	15.81	20.94	26.08	31.22
RGSC-6+8	20.65	5.53	10.67	15.81	20.94	26.08	31.22
RGSC-8+8	22.95	5.53	10.67	15.81	20.94	26.08	31.22
Frame weights. See table 3B64-II for typical weights of rounded corner square frames. Weight is in pounds.							

TABLE 3B64-II. Typical RGSC frame weights.

Material	Frame Size	Weight Multiple Frames X1	Weight Multiple Frames X2	Weight Multiple Frames X3	Weight Multiple Frames X4	Weight Multiple Frames X5	Weight Multiple Frames X6
Mild Steel	RGSC-2	4.9	8.6	12.6	16.3	20.3	24.0
Mild Steel	RGSC-4	6.0	10.1	14.3	18.5	22.7	26.9
Mild Steel	RGSC-6	7.1	11.9	16.8	21.6	26.5	31.3
Mild Steel	RGSC-8	8.4	13.9	19.6	25.1	30.9	36.4
Mild Steel	RGSC-2+2	7.9	17.9	26.2	34.6	43.0	51.4
Mild Steel	RGSC-2+4	9.3	19.4	28.2	36.8	45.6	54.2
Mild Steel	RGSC-2+6	10.6	20.9	30.0	39.2	48.3	57.3
Mild Steel	RGSC-2+8	12.1	22.7	32.4	42.1	51.8	61.5
Mild Steel	RGSC-4+4	10.6	20.9	30.0	39.2	48.3	57.3

FIGURE 3B64. Installation of rounded corner square frame multiple cable penetrators (RGSC) – Continued.

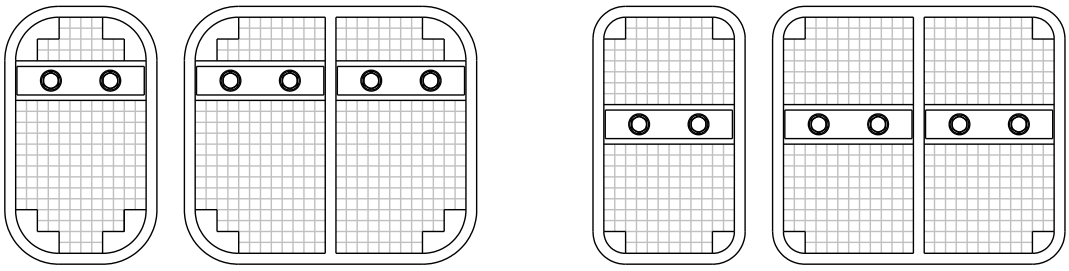
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TABLE 3B64-II. Typical RGSC frame weights – Continued.

Material	Frame Size	Weight Multiple Frames X1	Weight Multiple Frames X2	Weight Multiple Frames X3	Weight Multiple Frames X4	Weight Multiple Frames X5	Weight Multiple Frames X6
Mild Steel	RGSC-4+6	12.1	22.7	32.4	42.1	51.8	61.5
Mild Steel	RGSC-4+8	13.0	24.5	34.8	45.2	55.3	65.7
Mild Steel	RGSC-6+6	13.0	24.5	34.8	45.2	55.3	65.7
Mild Steel	RGSC-6+8	14.3	26.5	37.5	48.7	59.7	70.8
Mild Steel	RGSC-8+8	15.9	28.4	40.3	52.2	64.2	76.1
CRES	RGSC-2	4.9	8.8	12.8	16.8	20.7	24.7
CRES	RGSC-4	6.2	10.4	14.8	19.0	23.4	27.8
CRES	RGSC-6	7.3	12.1	17.2	22.0	27.1	31.7
CRES	RGSC-8	8.6	14.3	20.1	25.8	31.5	37.3
CRES	RGSC-2+2	8.2	18.3	26.9	35.5	44.1	52.7
CRES	RGSC-2+4	9.5	19.8	28.9	37.7	46.7	55.6
CRES	RGSC-2+6	10.8	21.4	30.9	40.1	49.6	58.9
CRES	RGSC-2+8	12.3	23.4	33.3	43.2	53.1	63.1
CRES	RGSC-4+4	10.8	21.4	30.9	40.1	49.6	58.9
CRES	RGSC-4+6	12.3	23.4	33.3	43.2	53.1	63.1
CRES	RGSC-4+8	13.2	25.1	35.7	46.3	56.9	67.5
CRES	RGSC-6+6	13.2	25.1	35.7	46.3	56.9	67.5
CRES	RGSC-6+8	14.8	27.1	38.6	49.8	61.3	72.5
CRES	RGSC-8+8	16.3	29.1	41.4	53.6	65.9	78.0
Aluminum	RGSC-2	1.8	3.1	4.4	5.7	7.1	8.4
Aluminum	RGSC-4	2.2	3.5	5.1	6.6	7.9	9.5
Aluminum	RGSC-6	2.4	4.2	6.0	7.5	9.3	11.0
Aluminum	RGSC-8	2.9	4.9	6.8	8.8	10.8	12.8
Aluminum	RGSC-2+2	2.9	6.2	9.3	12.1	15.2	18.1
Aluminum	RGSC-2+4	3.3	6.8	9.9	13.0	15.9	19.0
Aluminum	RGSC-2+6	3.7	7.3	10.6	13.7	17.0	20.1
Aluminum	RGSC-2+8	4.2	7.9	11.5	14.8	18.3	21.6
Aluminum	RGSC-4+4	3.7	7.3	10.6	13.7	17.0	20.1
Aluminum	RGSC-4+6	4.2	7.9	11.5	14.8	18.3	21.6
Aluminum	RGSC-4+8	4.6	8.6	12.1	15.9	19.4	22.9
Aluminum	RGSC-6+6	4.6	8.6	12.1	15.9	19.4	22.9
Aluminum	RGSC-6+8	5.1	9.3	13.2	17.0	20.9	24.7
Aluminum	RGSC-8+8	5.5	9.9	14.1	18.3	22.5	26.7
Packed RGSC MCPs. See the following illustrations. Note that the wedge (see figure 3B20) may be installed in any location within the MCP. Note that the corner blocks reduce the available packing area.							

FIGURE 3B64. Installation of rounded corner square frame multiple cable penetrators (RGSC) – Continued.

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Installation Requirements:

1. RGSC MCPs shall meet requirements of MIL-DTL-24705.
2. RGSC MCPs should be installed in accordance with manufacturer's instructions.
3. In general, the installation steps in figure 3B22 apply and should be observed.
4. After installation of the RGSC MCP, the retest shall be performed in accordance with figure 3B22. Wait 24 hours or longer (refer to manufacturer's instructions) after installation before exposing the cables/pipes to strain or pressure. The wait time shall be observed prior to the post-installation testing.

FIGURE 3B64. Installation of rounded corner square frame multiple cable penetrators (RGSC) – Continued.

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APPENDIX 3C – PENETRATIONS – STUFFING TUBES, GENERAL

C.1 SCOPE

C.1.1 Scope. This appendix describes the installation standard methods for stuffing tubes in general applications. This Appendix is a mandatory part of the standard. The information contained herein is intended for compliance.

C.2. APPLICABLE DOCUMENTS

C.2.1 General. The documents listed in this section are specified in this appendix. This section does not include documents cited in other sections of this standard or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in this appendix, whether or not they are listed.

C.2.2 Government documents.

C.2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-S-24235	-	Stuffing Tubes, Metal, and Packing Assemblies for Electric Cables, General Specification for
MIL-S-24235/3	-	Stuffing Tubes, Metal, and Packing Assemblies for Electric Cables, Bulkhead, for Ballast Tanks
MIL-S-24235/9	-	Stuffing Tubes, Metal, and Packing Assemblies for Electric Cables, Brass and Steel, for Decks and Bulkheads with Pipe Protection
MIL-S-24235/10	-	Stuffing Tubes, Metal, and Packing Assemblies for Electric Cables, Steel, for Decks and Bulkheads Without Pipe Protection
MIL-I-24391	-	Insulation Tape, Electrical, Plastic, Pressure-Sensitive
MIL-I-24768/1	-	Insulation, Plastic, Laminated, Thermosetting, Glass-Cloth, Melamine-Resin (GME)

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-2003-1	-	Electric Plant Installation Standard Methods for Surface Ships and Submarines (Cable)
MIL-STD-2003-4	-	Electric Plant Installation Standard Methods for Surface Ships and Submarines (Cableways)
MIL-STD-2003-5	-	Electric Plant Installation Standard Methods for Surface Ships and Submarines (Connectors)

(Copies of these documents are available online at https://quicksearch.dla.mil.)

C.2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

ASTM INTERNATIONAL

ASTM D710	-	Standard Specification for Vulcanized Fibre Sheets, Rolls, Rods, and Tubes Used for Electrical Insulation

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ASTM D4066 - Standard Classification System for Nylon Injection and Extrusion Materials (PA)

ASTM D5948 - Standard Specification for Molding Compounds, Thermosetting

ASTM D6779 - Standard Classification System for and Basis of Specification for Polyamide Molding and Extrusion Materials (PA)

ASTM F1836M - Standard Specification for Stuffing Tubes, Nylon, and Packing Assemblies (Metric)

(Copies of these documents are available online at www.astm.org.)

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

NEMA RE 2 - Electrical Insulating Varnish

(Copies of this document are available online at www.nema.org.)

SAE INTERNATIONAL

SAE-AMS-P-83461 - Packing, Preformed, Petroleum Hydraulic Fluid Resistant, Improved Performance at 275 °F (135 °C)

(Copies of this document are available online at www.sae.org.)

C.2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

C.3 REQUIRED EQUIPMENT AND MATERIALS

C.3.1 Required equipment and materials. The required equipment and materials for the proper installation of stuffing tubes in general applications are as shown in the Appendix C methods.

C.4 NOTES AND PROCEDURES

C.4.1 Dimensions. For figures and tables in this appendix, all dimensions are in inches unless otherwise noted.

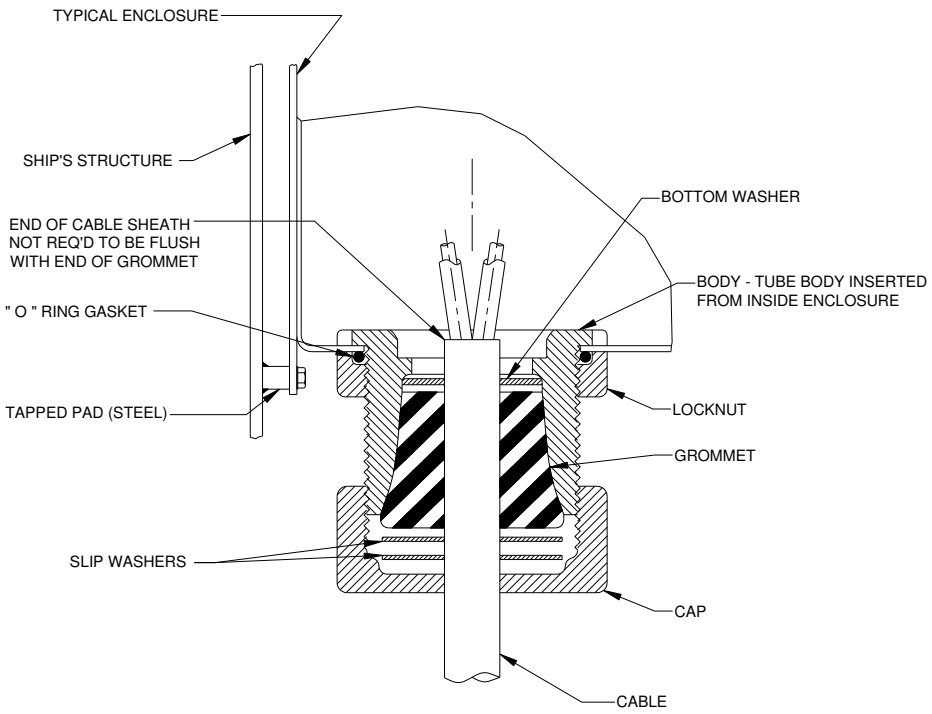
C.4.2 Figures. Table 3CI provides information for the figures in this appendix.

TABLE 3CI. Figures for general purpose stuffing tubes.

Figure number	Stuffing tube installation	Page
3C1	ASTM F1836M nylon stuffing tube typical installation	204
3C2	ASTM F1836M nylon stuffing tube assembly	206
3C3	ASTM F1836M nylon stuffing tube data	208
3C4	MIL-S-24235 stuffing tubes through refrigerated spaces (surface ships)	211
3C5	Stuffing tubes through refrigerated spaces	212
3C6	MIL-S-24235 stuffing tubes through refrigerated spaces (surface ships)	214
3C7	MIL-S-24235 stuffing tubes through refrigerated spaces (surface ships)	215
3C8	MIL-S-24235/3 community stuffing tubes for bulkheads (submarines or surface ships)	216
3C9	Stuffing tube inserts	217
3C10	Conduit transition fittings	219

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METHOD 3C-1-1
APPLIES ALSO TO ANGLE TUBES



Installation notes:

1. Inspect the hole in the enclosure for conformance with the clearance hole required, as shown on figure 3C3, and remove any burrs or irregularities.
2. For steel enclosures where roughness is greater than a 125 micro-inch finish (not required on aluminum enclosures), remove paint and roughen surface with emery paper ½ inch wide around the hole on the exterior of the enclosure. Apply one coat of approved shipboard primer and allow to dry. Dust primer-coated surface with talc (soapstone). If primer is thoroughly dry at time of tube installation, omit talc.
3. With straight tube type, insert the stuffing tube body in the hole from inside the enclosure. If necessary, remove the interior fitting from the enclosure.
4. With angle and "Y" tubes, insert the stuffing tube body in the hole from the outside of the enclosure. Excess length protruding into the enclosure may be removed.
5. Screw locknut onto body and tighten against "O" ring gasket sufficiently to obtain plastic-to-metal contact of the stuffing tube and enclosure. In cases where this plastic-to-metal contact cannot be obtained, tighten the locknut until the threads start to skip. This will be considered a satisfactory indication of tightness (see installation note 11).
6. Remove sufficient armor (when present) from the cable for connections within the enclosure plus the body length and trim to suit (see figure 3C2). Avoid cutting cable sheath. Secure armor (when present) on cable a minimum of 1 inch from plastic tube face with pressure sensitive vinyl tape, MIL-I-24391, or by a shrink fit plastic sleeve. Place cap and the two slip washers over end of cable.
7. Prepare conductors for making electrical connection.
8. Place bottom washer over the end of cable against the end of the grommet.
9. Insert end of cable through stuffing tube and into the enclosure so as to seat the grommet into tube body.

FIGURE 3C1. ASTM F1836M nylon stuffing tube typical installation.

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Installation notes (continued):

10. Screw on cap and tighten sufficiently to compress the grommet to form a tight seal between the cable and tube.
11. Hold tube body when tightening locknut to prevent turning. Also hold tube body when tightening cap to prevent breaking the watertight seal.
12. Sealing plugs in accordance with ASTM F1836M shall be used to seal nylon stuffing tubes from which the cables have been removed. When installing the plugs, the cable grommet shall be discarded, but the nylon washers shall be retained and left in the stuffing tube. Tube bonding agent is not required on plug.
13. "O" rings shall be furnished by the installing activity in accordance with SAE-AMS-P-83461.
14. When necessary to pass airtight test, apply RTV silicone rubber sealant to the grommet.

NOTES:

1. In general, the sizes of the stuffing tubes do not change when end sealing of cable is used as shown on Appendix 1A figures of MIL-STD-2003-1. In those instances where end sealing increases the diameter of the bunched individual leads larger than the throat diameter of the assigned tube, the next larger tube size is specified.
2. In those cases where the bunched conductor leads cannot be accommodated in a standard grommet, use the sealing plug assigned to the tube. Freeze the plug to -40 °F and drill a hole to the dimension given, thus making a new grommet.

FIGURE 3C1. ASTM F1836M nylon stuffing tube typical installation – Continued.

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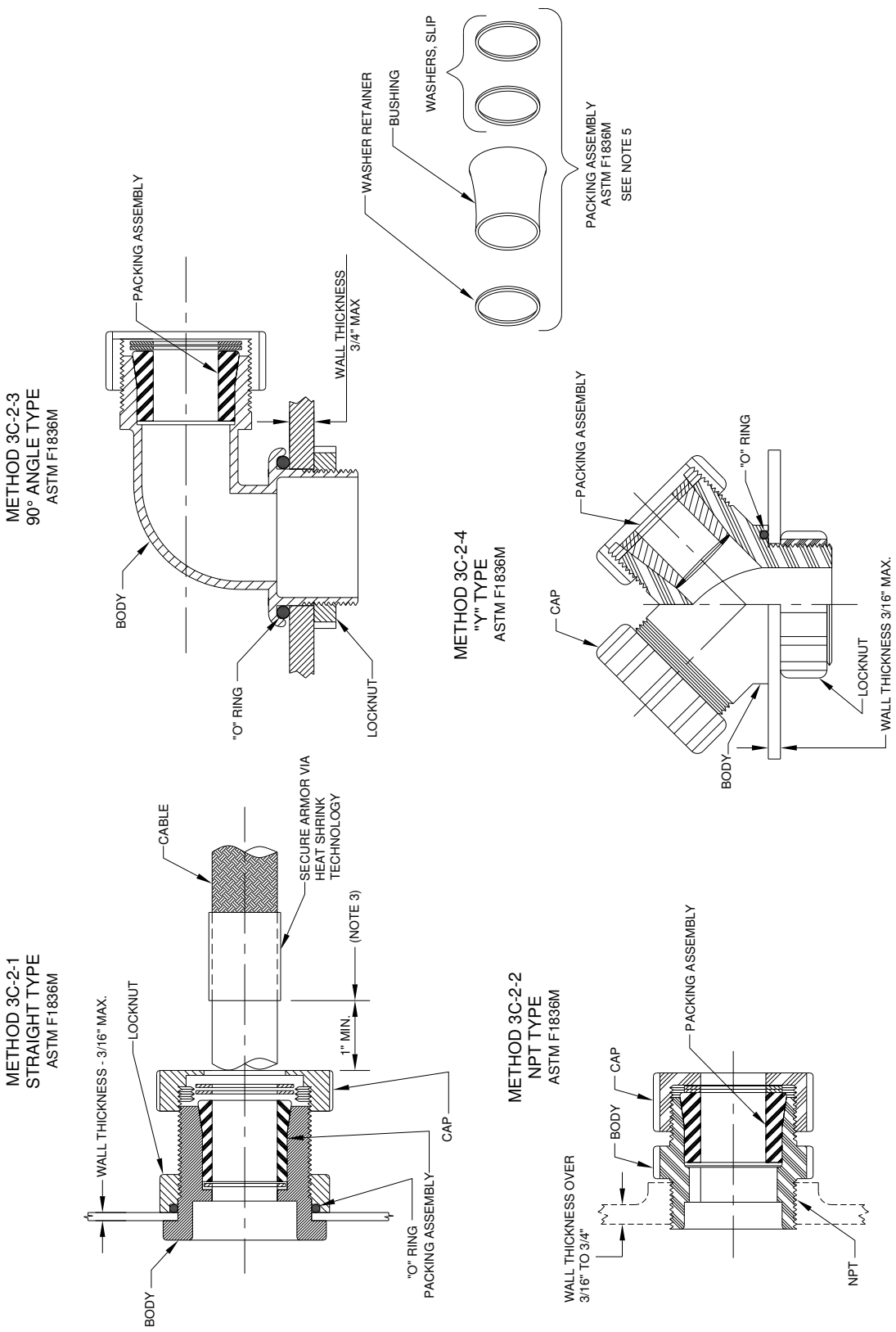


FIGURE 3C2. ASTM F1836M nylon stuffing tube assembly.

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NOTES:

1. Packing assemblies and “O” rings are not furnished with stuffing tubes. They shall be ordered separately by installing activity to suit installations.
2. “O” rings are not required for non-watertight installations.
3. Secure armor (when present) on cable a minimum of 1 inch from plastic tube face with shrink fit plastic sleeve.
4. See figure 3C1 for general installation notes.
5. See figures 3A10 through 3A25 for cable assignment.

FIGURE 3C2. ASTM F1836M nylon stuffing tube assembly – Continued.

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Stuffing Tube Sizes

TABLE 3C3-I. Straight tube sizes 1 through 9, ASTM F1836M.

Tube size	ASTM F1836M part number	National stock number (NSN)	O ring			Clearance hole
			Part number	Size	National stock number (NSN)	
1	1-001	5975-00-296-4092	M83461/1-212	212	5331-01-133-4718	0.885
2	1-002	5975-00-296-4093	M83461/1-214	214	5331-01-173-7962	1.010
3	1-003	5975-00-877-6957	M83461/1-216	216	5331-01-128-4873	1.135
4 See note 1	1-004	5975-00-296-4095	M83461/1-218	218	5331-01-123-3302	1.260
4T	1-005	5975-00-989-5046	M83461/1-220	220	5331-01-185-2570	1.385
5	1-006	5975-00-296-4096	M83461/1-226	226	5331-01-183-0972	2.010
6	1-007	5975-00-296-4097	M83461/1-230	230	5331-01-185-2574	2.510
7	1-008	5975-00-296-4098	M83461/1-232	232	5331-01-113-5624	2.760
8	1-009	5975-00-296-4099	M83461/1-236	236	5331-01-183-0971	3.260
9	1-010	5975-00-296-4100	M83461/1-242	242	5331-01-184-3906	4.010

FIGURE 3C3. ASTM F1836M nylon stuffing tube data.

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TABLE 3C3-II. Angle tube sizes 1 through 6, ASTM F1836M.

Tube size	ASTM F1836M part number	National stock number (NSN)	O ring			Clearance hole
			Part number	Size	National stock number (NSN)	
1	2-001	5975-00-503-4694	M83461/1-212	212	5330-00-187-3638	0.885
2	2-002	5975-00-503-4693	M83461/1-212	212	5330-00-187-3638	0.885
3	2-003	5975-00-877-6956	M83461/1-216	216	5330-00-641-0231	1.135
4T	2-005	5975-00-989-5045	M83461/1-226	220	5330-00-196-5385	1.260
5	2-006	5975-00-503-4695	M83461/1-226	226		2.010
6	2-007	5975-00-503-4697	M83461/1-230	230	5330-00-054-6866	2.510

FIGURE 3C3. ASTM F1836M nylon stuffing tube data – Continued.

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TABLE 3C3-III. NPT tube sizes 1 through 9, ASTM F1836M.

Tube size	ASTM F1836M part number	National stock number (NSN)	NPT tap
1	3-001	5975-00-808-4063	0.5
2	3-002	5975-00-808-4064	0.75
3	3-003	5975-00-877-6955	1.0
4T	3-004	5975-00-752-2945	1.0
5	3-005	5975-00-808-9272	1.5
6	3-006	5330-00-808-4066	2.0
7	3-007	5330-00-808-4067	2.5
8	3-008	5330-00-808-4068	3.0
9	3-009	5330-00-808-4069	3.5

TABLE 3C3-IV. "Y" type tube sizes 1 through 4, ASTM F1836M.

Tube size	ASTM F1836M part number	National stock number (NSN)	O ring			Clearance hole
			Part number	Size	National stock number (NSN)	
1	4-01	5975-00-782-6139	M83461/1-212	212	5331-01-133-4718	0.885
2	4-02	5975-00-782-6140	M83461/1-214	214	5331-01-173-7962	1.010
3	4-03	5975-00-782-6141	M83461/1-216	216	5331-01-128-4873	1.135
4T	4-04	5975-00-782-6142	M83461/1-220	220	5331-01-123-3302	1.385

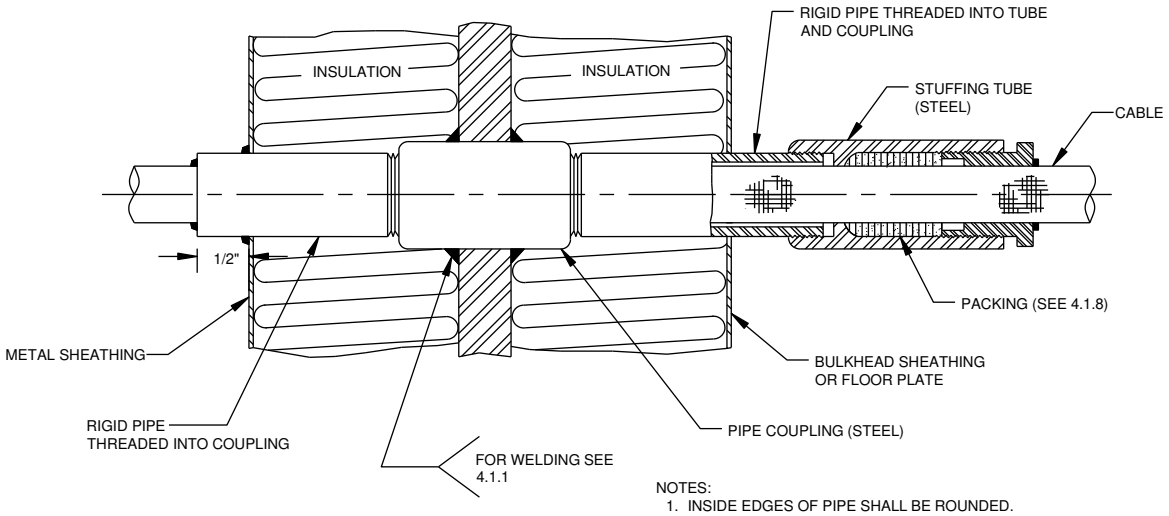
NOTE:

1. Size 4 straight and angle tube is for replacement only in existing installations where size 4T is not interchangeable.

FIGURE 3C3. ASTM F1836M nylon stuffing tube data – Continued.

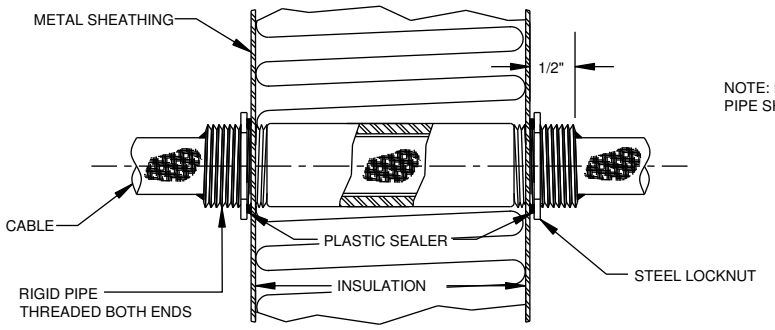
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METHOD 3C-4-1
CABLES THROUGH STEEL PARTITION BULKHEADS OR DECKS WITH INSULATION AND SHEATHING ON BOTH SIDES



NOTES:
1. INSIDE EDGES OF PIPE SHALL BE ROUNDED.
2. CHAFING NIPPLES USED TO PROTECT CABLES PASSING THROUGH BULKHEADS SHALL BE WELDED ON ONE SIDE OF BULKHEAD PENETRATED; EXCEPT FOR THE FOLLOWING TYPES OF BULKHEADS WHERE WELDS ON BOTH SIDES ARE REQUIRED:
   A. BALLISTIC BULKHEADS
   B. SHELL PLATING
   C. TORPEDO DEFENSE BULKHEADS

METHOD 3C-4-2
CABLES THROUGH PARTITION BULKHEADS OF FIBROUS GLASS AND SHEATHING



NOTES:

1. See figure 3A1 for gland nut tightening methods.
2. See 4.8 for thread lubrication guidance.

FIGURE 3C4. MIL-S-24235 stuffing tubes through refrigerated spaces (surface ships).

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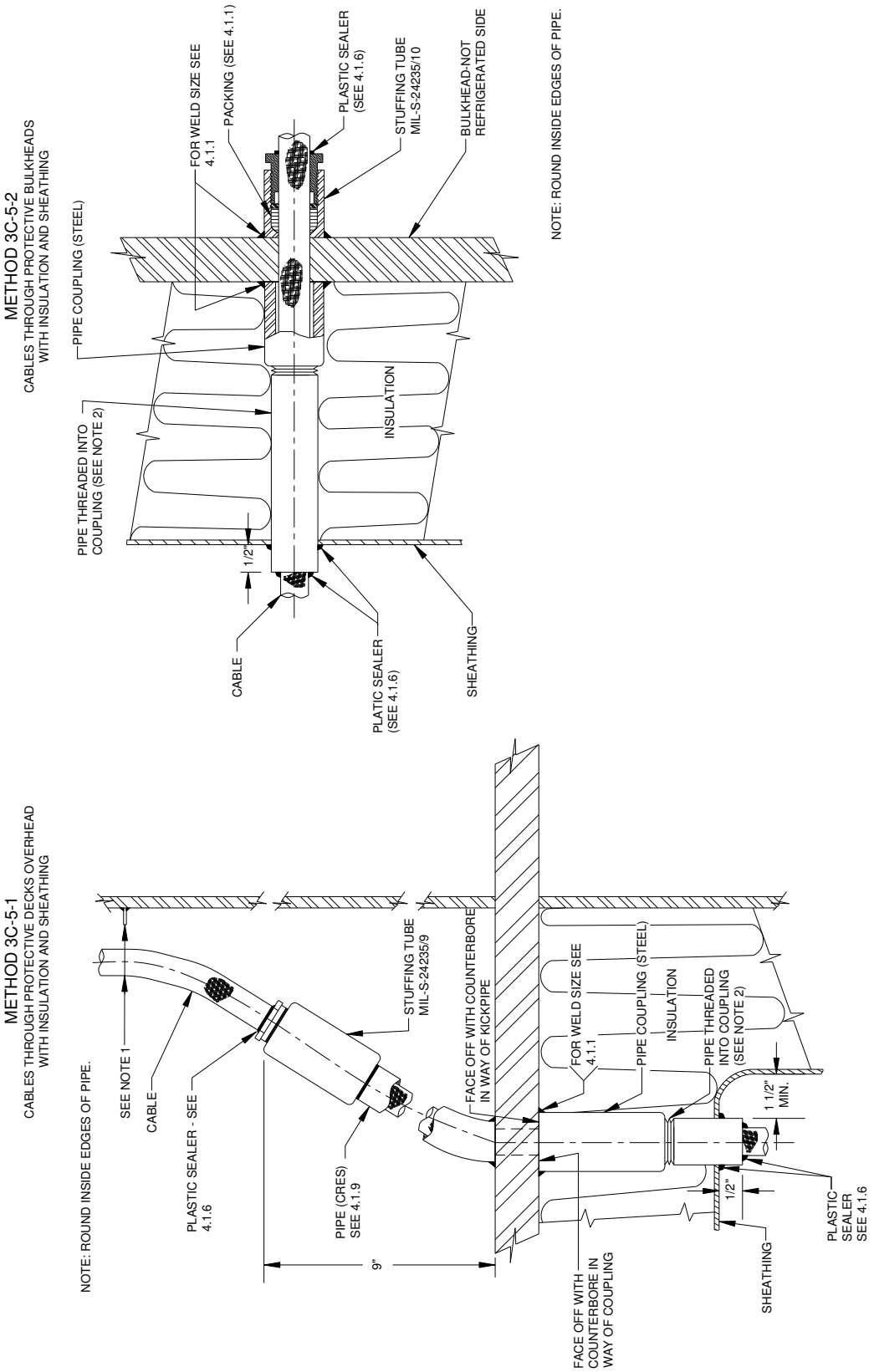


FIGURE 3C5. Stuffing tubes through refrigerated spaces.

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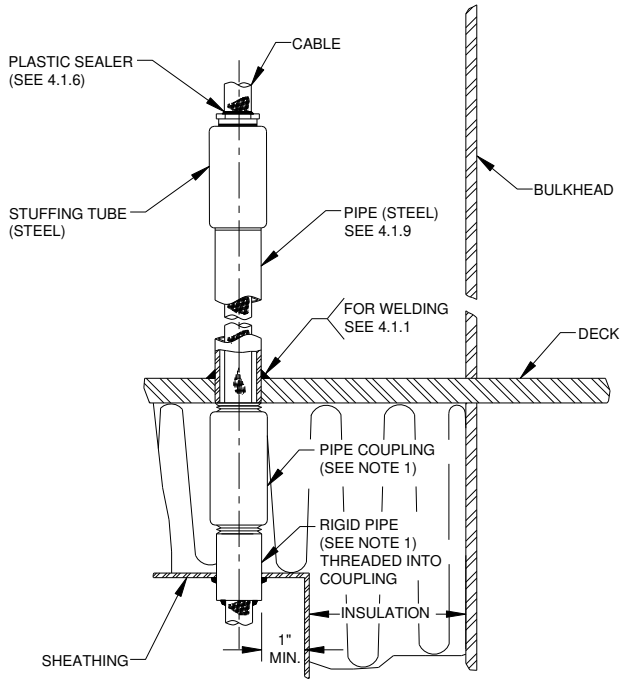
NOTES:

1. For suitable cable support, see MIL-STD-2003-4.
2. Plastic pipe and fitting shall be nylon.
3. See figure 3A1 for gland nut tightening methods.
4. See 4.8 for thread lubrication guidance.

FIGURE 3C5. Stuffing tubes through refrigerated spaces – Continued.

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METHOD 3C-6-1
CABLES THROUGH OVERHEAD STEEL DECKS
WITH INSULATION AND SHEATHING



NOTE: ROUND INSIDE EDGES OF PIPE.

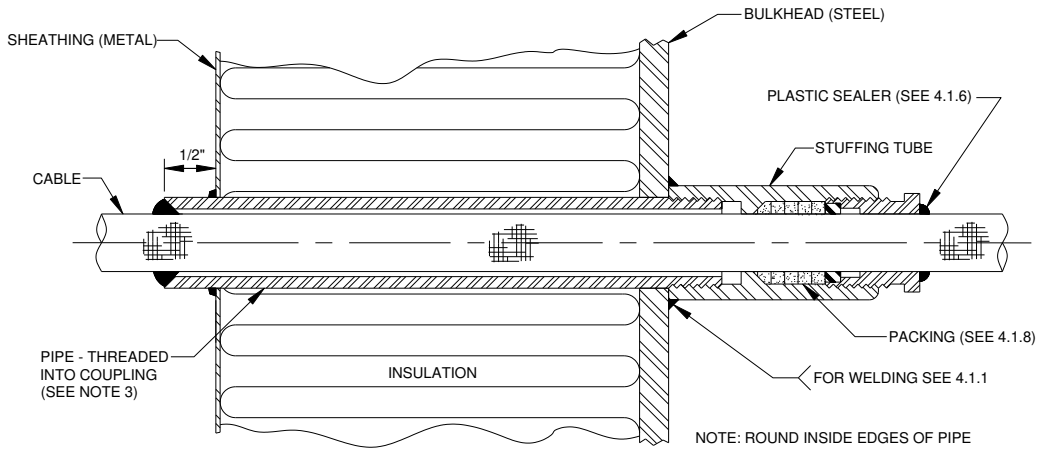
NOTES:

1. Plastic pipe and fittings shall be nylon.
2. See figure 3A1 for gland nut tightening methods.
3. See 4.8 for thread lubrication guidance.

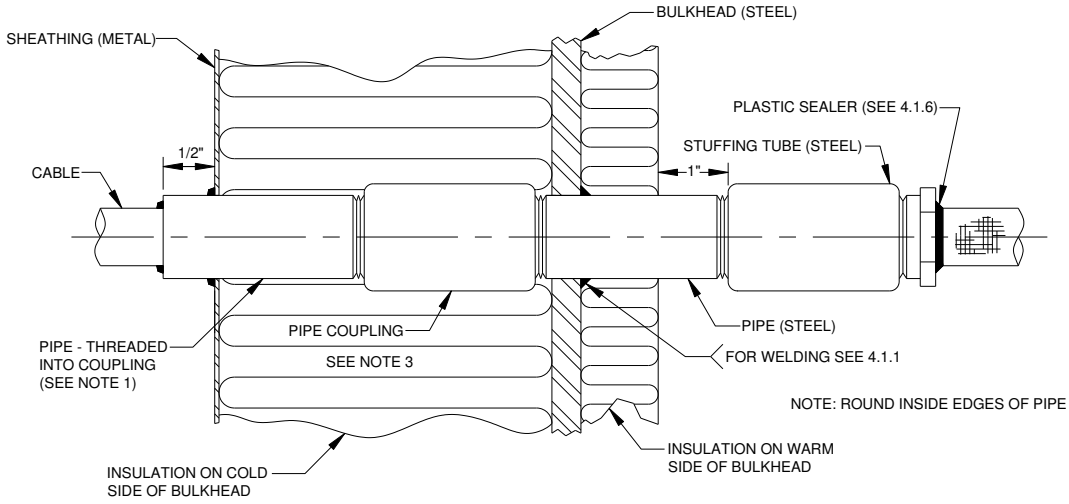
FIGURE 3C6. MIL-S-24235 stuffing tubes through refrigerated spaces (surface ships).

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METHOD 3C-7-1
CABLES THROUGH BOUNDRY BULKHEADS OR OVERHEAD DECKS WITH INSULATION AND SHEATHING ON THE COLD SIDE



METHOD 3C-7-2
CABLES THROUGH BOUNDRY BULKHEADS OR OVERHEAD DECKS WITH INSULATION AND SHEATHING ON THE COLD SIDE AND INSULATION WITHOUT SHEATHING ON THE WARM SIDE



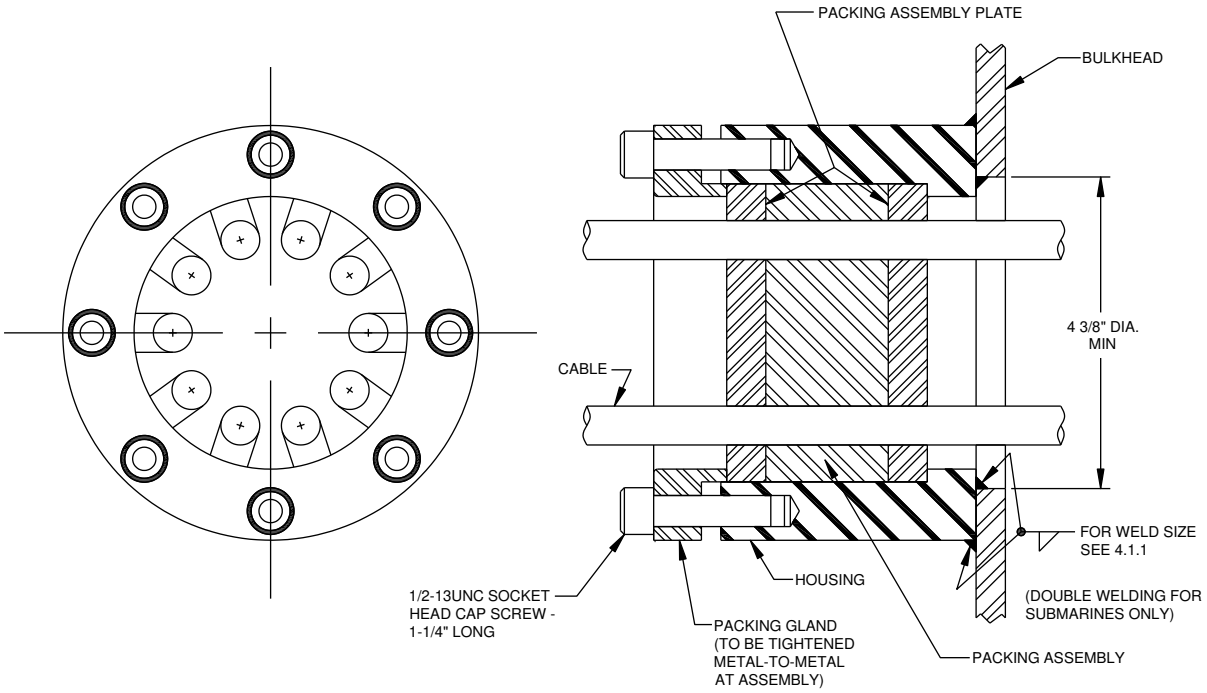
NOTES:

1. Cable entrance to refrigerated spaces preferably through bulkheads or overhead decks.
2. Stuffing tubes are located preferably on the warm side of boundary bulkheads or overhead decks.
3. Plastic pipe and fittings shall be nylon.
4. See figure 3A1 for gland nut tightening methods.
5. See 4.8 for thread lubrication guidance.

FIGURE 3C7. MIL-S-24235 stuffing tubes through refrigerated spaces (surface ships).

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METHOD 3C-8-1



NOTES:

1. The design shown hereon is for use for passing cables through ballast tank bulkheads for submarines and is a typical installation only. Bulkhead fittings for submarines shall withstand maximum pressure of 45 psi and conform to MIL-S-24235.
2. This method is also applicable to surface ships by varying size to suit the number and size of cables to be accommodated, and the specified material may be changed to suit the shipbuilder, except that grommet shall remain Neoprene of 40-45 durometer and gland ring shall be brass. Other changes may be made to lighten the construction, such as decrease of housing thickness and flange thickness, providing gland ring is threaded into housing using the same construction used for single cable stuffing tubes. Design for surface ships shall withstand minimum pressure of 15 psi.
3. For wood construction, housing shall be designed with a suitable flange for securing to wood bulkhead.
4. Radial staggering of cables to attain maximum acceptance of cables in minimum space is satisfactory. The original bulkhead strength and tightness shall be maintained.
5. This sheet showing method 3C-8-1 is considered a working plan. The submission or preparation of plans showing details of accomplishing the method or the permissive notes on this sheet is not required.
6. Position stuffing tube so that packing gland is on the most accessible side of ballast tank bulkhead.
7. See 4.5 for fastener and fastener material requirements.

FIGURE 3C8. MIL-S-24235/3 community stuffing tubes for bulkheads (submarines or surface ships).

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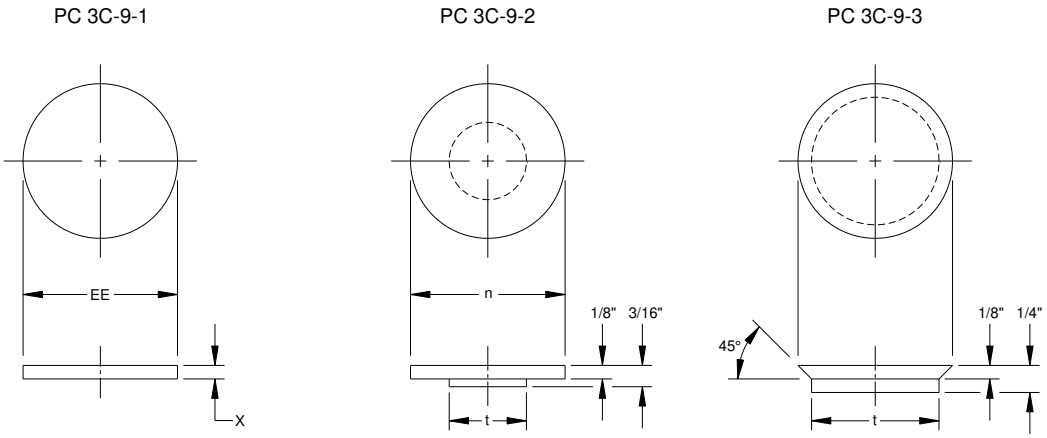


TABLE 3C9-I. Inserts PC 3C-9-1.

Size	Dimensions	
	EE	X
1	0.550	3/32
2	0.615	
3	0.755	
4	0.896	
5	1.383	1/8
6	1.740	
7	1.927	
8	2.302	
9	3.240	

FIGURE 3C9. Stuffing tube inserts.

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TABLE 3C9-II. Inserts PCs 3C-9-2 and 3C-9-3.

Size	Dimensions	
	n	t
A	3/4	3/8
B	7/8	15/32
C	1	19/32
D	1 1/8	23/32
E	1 1/8	25/32
F	1 3/16	13/16
G	1 3/8	29/32
J	1 1/2	1 1/32
K	1 5/8	1 1/8
L	1 11/16	1 7/32
M	1 7/8	1 3/8
N	1 15/16	1 15/32
P	2 1/16	1 19/32
R	2 3/16	1 23/32
S	2 9/16	1 27/32
T	2 3/4	2 1/32
V	2 7/8	2 5/32
W	3	2 9/32
X	3 1/8	2 15/32
Y	3 1/4	2 9/16
Z	3 7/16	2 3/4
AA	3 5/8	2 27/32

NOTES:

1. For method of sealing cable ends, see 4.1.6.
2. For cable assignment, see figure 3C3.
3. Insert PC 3C-9-1 used with nylon stuffing tubes, ASTM F1836M.
4. Insert PCs 3C-9-2 and 3C-9-3 used with steel stuffing tubes, MIL-S-24235, drawn and machined respectively.
5. Insert PCs 3C-9-1, 3C-9-2, and 3C-9-3 may be made of the following materials:
   a. Polyamide (nylon) plastic, ASTM D4066 or ASTM D6779.
   b. Laminated plastic, MIL-I-24768/1. The laminations shall be parallel to the faces of the insert.
   c. Molded phenolic, Type MAJ-60, ASTM D5948.
   d. Fiber, ASTM D710, grade “CH”, form “B” or “S”. The fiber shall be given a coat of insulating varnish in accordance with NEMA RE 2.

FIGURE 3C9. Stuffing tube inserts – Continued.

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Method 3C-10-1
Brazed Conduit Transition Fittings

NOTES:

1. This method covers using adapters and brazing to connect stuffing tubes, coaxial connectors, and connectors with flexible conduit.
2. Adapter and transition body may be supplied separately or as a unit.
3. Due to tight braid construction, conduit should be formed into the approximate contour of the intended run before conduit cutting and brazing of the final assembly. This will ensure the flexibility desired for installation.
4. Prior to cutting the conduit, the wire braid should be lightly silver soldered at the place the cut will be made.
5. The conduit cut shall be square and shall be made with a high-speed knife-edge saw. The cut ends shall be deburred with a file such that no sharp edges remain.
6. Conduit shall be cleaned by blowing out with air or nitrogen until the conduit is free of any moisture or foreign particles.
7. For attaching adapters to stuffing tubes:
   a. Place end fitting over flexible conduit and silver solder (braze) in place. When soldering, hold the end fitting and flexible conduit vertically to prevent excess solder from running back onto the conduit and introducing unwanted stiffness.
   b. Slide the coupling nut, the tapered ferrule, and the transition body, in that order, over the end fitting and onto the flexible conduit.
   c. Complete the cable installation through the stuffing tube by normal procedure, except the stuffing tube gland nut is to be replaced by the adapter.
   d. Install the adapter onto the stuffing tube by the same procedure that would have been used for installing the gland nut. Apply the same torque as would be required for torqueing the gland nut. Connect the transition body to the adapter and tighten.
8. For attaching coaxial connectors or connectors in accordance with MIL-STD-2003-5:
   a. Place end fitting over flexible conduit and silver solder (braze) in place. When soldering, hold the end fitting and flexible conduit vertically to prevent excess solder from running back onto the conduit and introducing unwanted stiffness.
   b. Slide the coupling nut, the tapered ferrule, and the transition body, in that order, over the end fitting and onto the flexible conduit.
   c. Complete the cable installation to the connector by normal procedure, except the cable grip is to be replaced by the adapter.
   d. Install the adapter onto the connector plug by the same procedure that would have been used for installing the cable grip. Connect the transition body to the adapter and tighten.
9. For conduit termination at equipment panels:
   a. Slide the coupling nut, the tapered ferrule, and the transition body, in that order, over the flexible conduit.
   b. Place panel fitting over flexible conduit and silver solder (braze) in place. When soldering, hold the panel fitting and flexible conduit vertically to prevent excess solder from running back onto the conduit and introducing unwanted stiffness.
   c. Complete cable installation to panel termination by the normal manner. Place the threaded end of the panel fitting into the panel. Install the lock nut and tighten.

FIGURE 3C10. Conduit transition fittings.

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10. Hold the transition body in one hand, pull back on the transition body, and pull back on the flexible conduit until the back end of the end fitting extends approximately ½ inch beyond the male threaded end of the transition body. Slide the tapered ferrule into place and tighten coupling nut into the transition body.
    a. Tighten the coupling nut by hand until snug.
    b. Using a strap wrench, continue to advance the coupling nut approximately ⅛ turn.
    c. Check for tightness by pulling on the conduit.
    d. If the coupling is not sufficiently tight to hold the conduit, loosen the coupling nut approximately ⅛ turn past the point to which the coupling nut was previously tightened.
    e. If necessary, repeat step 10.d, increasing in ⅓ turn increments until the conduit is sufficiently tightened.
11. To make flexible conduit transition fittings watertight, install O-rings as required.

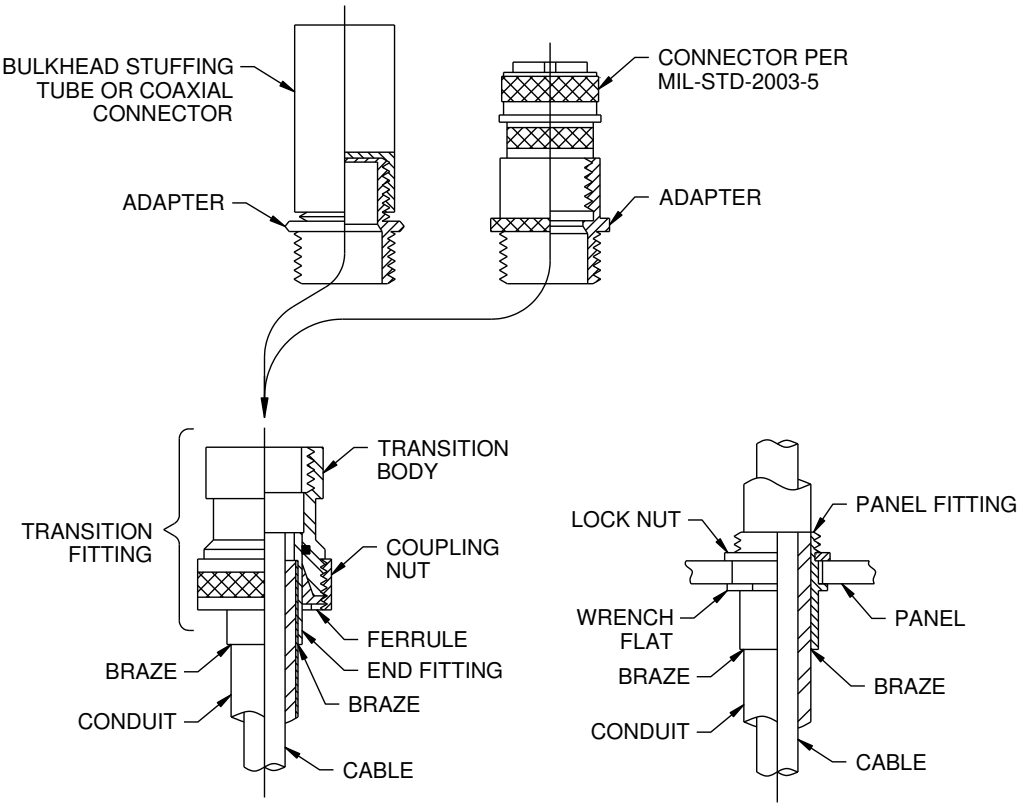


FIGURE 3C10. Conduit transition fittings – Continued.

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Method 3C-10-2
Reusable Conduit Transition Fittings

NOTES:

1. Adapter and transition body may be supplied separately or as a unit.
2. Due to tight braid construction, conduit should be formed into the approximate contour of the intended run before conduit cutting and brazing of the final assembly. This will ensure the flexibility desired for installation.
3. Prior to cutting the conduit, the wire braid should be lightly silver soldered at the place the cut will be made.
4. The conduit cut shall be square and shall be made with a high-speed knife-edge saw. The cut ends shall be deburred with a file such that no sharp edges remain.
5. Conduit shall be cleaned by blowing out with air or nitrogen until the conduit is free of any moisture or foreign particles.
6. For attaching adapters to stuffing tubes:
   a. Slide the coupling nut, the tapered ferrule, and the transition body, in that order, over the end fitting and onto the flexible conduit.
   b. Complete the cable installation through the stuffing tube by normal procedure, except the stuffing tube gland nut is to be replaced by the adapter.
   c. Install the adapter onto the stuffing tube by the same procedure that would have been used for installing the gland nut. Apply the same torque as would be required for torqueing the gland nut. Connect the transition body to the adapter and tighten.
7. For attaching coaxial connectors or connectors in accordance with MIL-STD-2003-5:
   a. Slide the coupling nut, the tapered ferrule, and the transition body, in that order, over the end fitting and onto the flexible conduit.
   b. Complete the cable installation to the connector by normal procedure, except the cable grip is to be replaced by the adapter.
   c. Install the adapter onto the connector plug by the same procedure that would have been used for installing the cable grip. Connect the transition body to the adapter and tighten.
8. For conduit termination at equipment panels:
   a. Slide the coupling nut, the tapered ferrule, and the transition body, in that order, over the flexible conduit.
   b. Complete cable installation to the panel termination by the normal manner. Place the threaded end of the panel fitting into the panel. Install locknut and tighten.
9. Hold the transition body in one hand, pull back on the transition body, and pull back on the flexible conduit until the back end of the end fitting extends approximately ½ inch beyond the male threaded end of the transition body. Slide the tapered ferrule into place and tighten coupling nut into the transition body.
   a. Tighten the coupling nut by hand until snug.
   b. Using a strap wrench, continue to advance the coupling nut approximately ⅛ turn.
   c. Check for tightness by pulling on the conduit.
   d. If the coupling is not sufficiently tight to hold the conduit, loosen the coupling nut approximately ⅛ turn past the point to which the coupling nut was previously tightened.
   e. If necessary, repeat step 9.d, increasing in ⅓ turn increments until the conduit is sufficiently tightened.

FIGURE 3C10. Conduit transition fittings – Continued.

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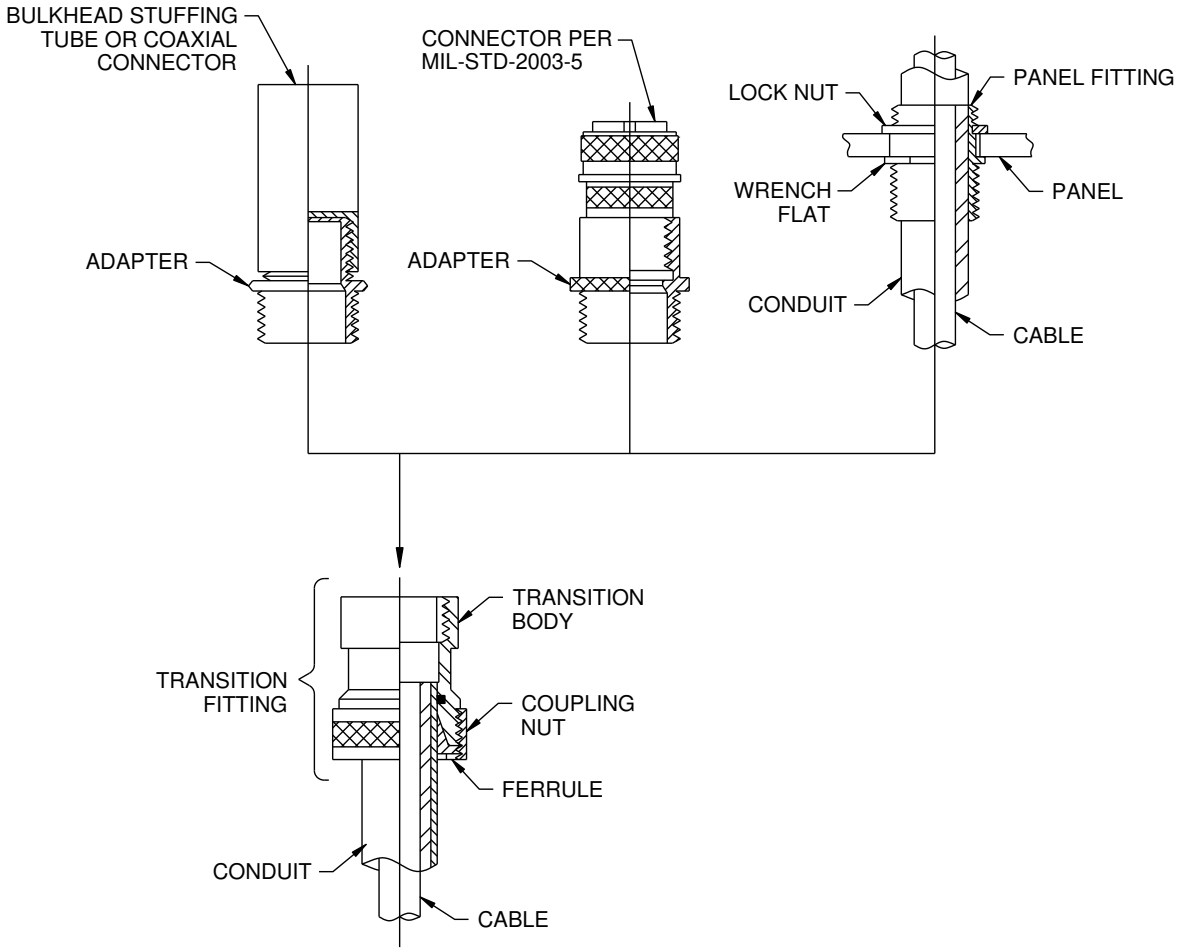


FIGURE 3C10. Conduit transition fittings – Continued.

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APPENDIX 3D – PENETRATIONS – KICKPIPES

D.1 SCOPE

D.1.1 Scope. This appendix describes the installation standard methods for kickpipes. This Appendix is a mandatory part of the standard. The information contained herein is intended for compliance.

D.2. APPLICABLE DOCUMENTS

D.2.1 General. The documents listed in this section are specified in in this appendix. This section does not include documents cited in other sections of this standard or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in this appendix, whether or not they are listed.

D.2.2 Government documents.

D.2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-S-24235	-	Stuffing Tubes, Metal, and Packing Assemblies for Electric Cables, General Specification for
MIL-S-24235/9	-	Stuffing Tubes, Metal, and Packing Assemblies for Electric Cables, Brass and Steel, for Decks and Bulkheads with Pipe Protection
MIL-S-24235/18	-	Stuffing Tubes, Metal, and Packing Assemblies for Electric Cables, Swage Type, Reduced Diameter, Steel and Aluminum, for Deck and Bulkheads with Pipe Protection

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-2003-4	-	Electric Plant Installation Standard Methods for Surface Ships and Submarines (Cableways)

(Copies of these documents are available online at https://quicksearch.dla.mil.)

D.2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

ASTM INTERNATIONAL

ASTM F1836M	-	Standard Specification for Stuffing Tubes, Nylon, and Packing Assemblies (Metric)

(Copies of this document are available online at www.astm.org.)

SAE INTERNATIONAL

SAE-AMS-WW-T-700/6	-	Tube, Aluminum Alloy, Drawn, Seamless, 6061

(Copies of this document are available online at www.sae.org.)

D.2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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D.3 REQUIRED EQUIPMENT AND MATERIALS

D.3.1 Required equipment and materials. The required equipment and materials for the proper installation of kickpipes are as shown in the Appendix D methods.

D.4 NOTES AND PROCEDURES

D.4.1 Dimensions. For figures and tables in this appendix, all dimensions are in inches unless otherwise noted.

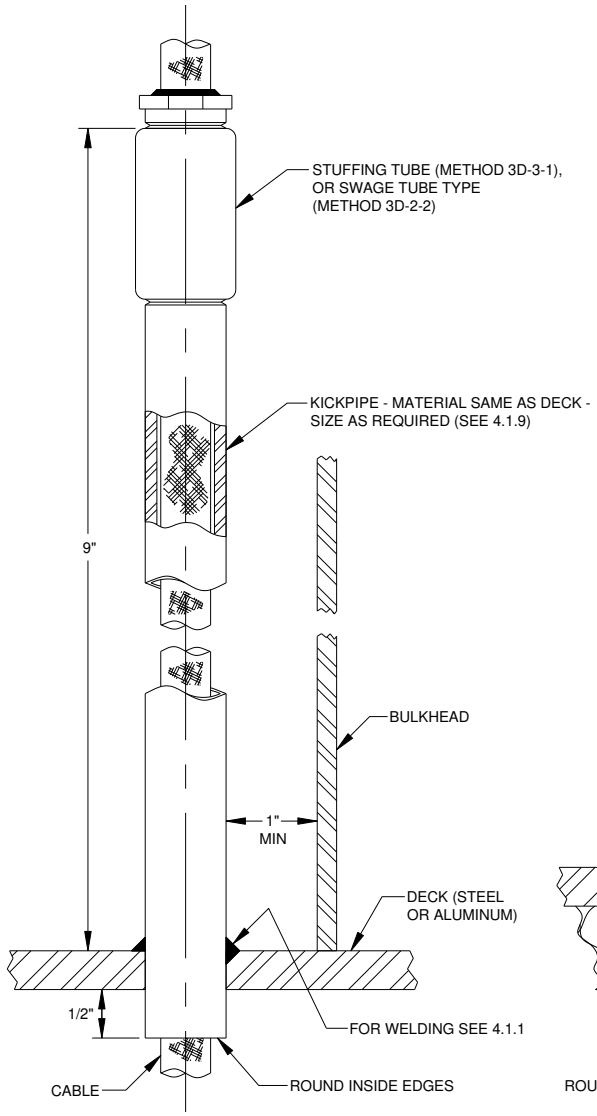
D.4.2 Figures. Table 3DI provides information for the figures in this appendix.

TABLE 3DI. Figures for kickpipes.

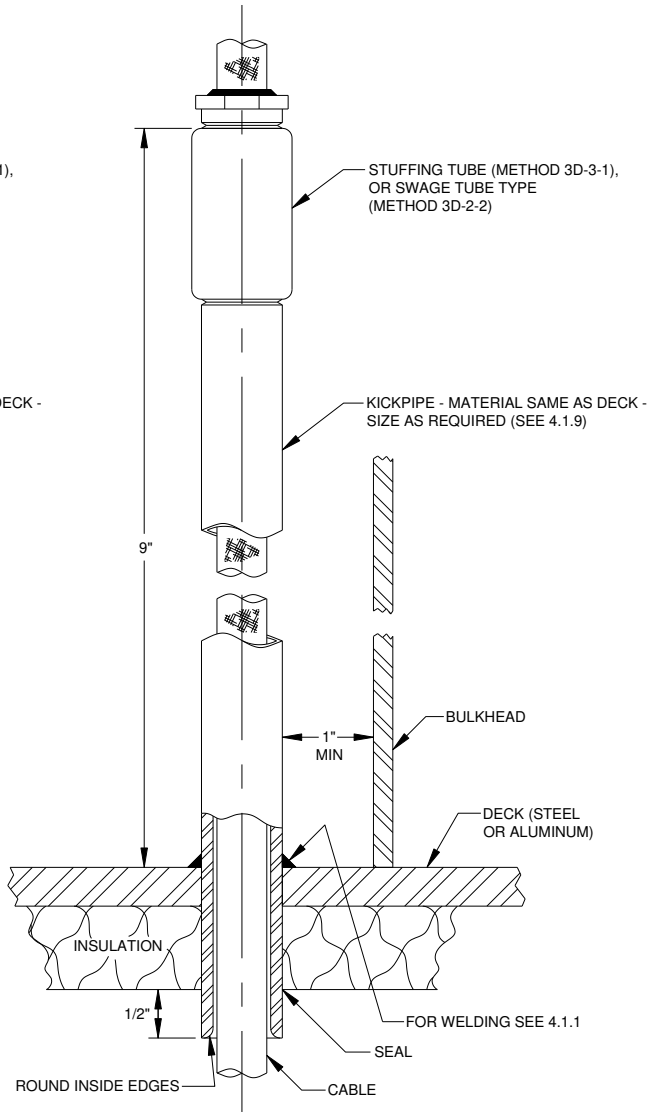
Figure number	Installation of kickpipes	Page
3D1	Kickpipes through steel or aluminum decks (surface ships)	225
3D2	Kickpipes through steel or aluminum decks (surface ships)	226
3D3	Kickpipes through steel or aluminum decks (surface ships)	227
3D4	Kickpipes through non-watertight decks (surface ships)	228
3D5	Kickpipes through ballistic plating (surface ships)	229
3D6	Kickpipes through ballistic plating (surface ships)	231
3D7	Kickpipes through wooden decks	232
3D8	Kickpipes with unions (surface ships)	234

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METHOD 3D-1-1
KICKPIPES
THROUGH WATERTIGHT DECK WITHOUT INSULATION



METHOD 3D-1-2
KICKPIPES
THROUGH WATERTIGHT DECK WITH INSULATION

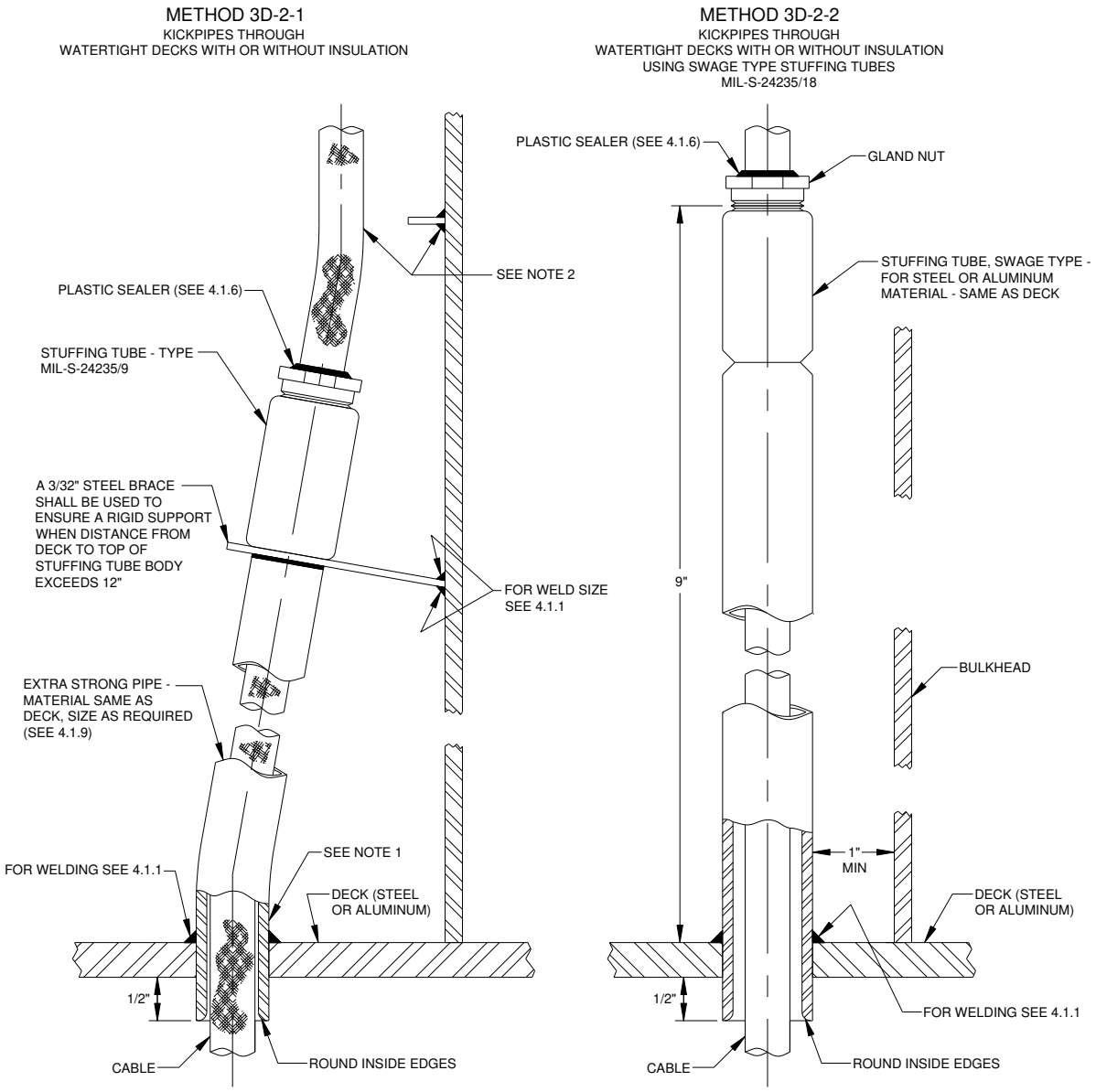


NOTES:

1. Use bent kickpipe to conserve space, if needed.
2. For suitable cable support, see MIL-STD-2003-4.
3. Pipes used in conjunction with stuffing tubes shall be welded on one side of the bulkhead or deck penetrated, except for the following types of bulkheads or decks where welds on both sides are required:
   a. Ballistic bulkheads
   b. Shell plating
   c. Torpedo defense system bulkheads
   d. Penetrations exposed to weather or wet spaces
4. See figure 3A1 for gland nut tightening methods.
5. See 4.8 for thread lubrication guidance.

FIGURE 3D1. Kickpipes through steel or aluminum decks (surface ships).

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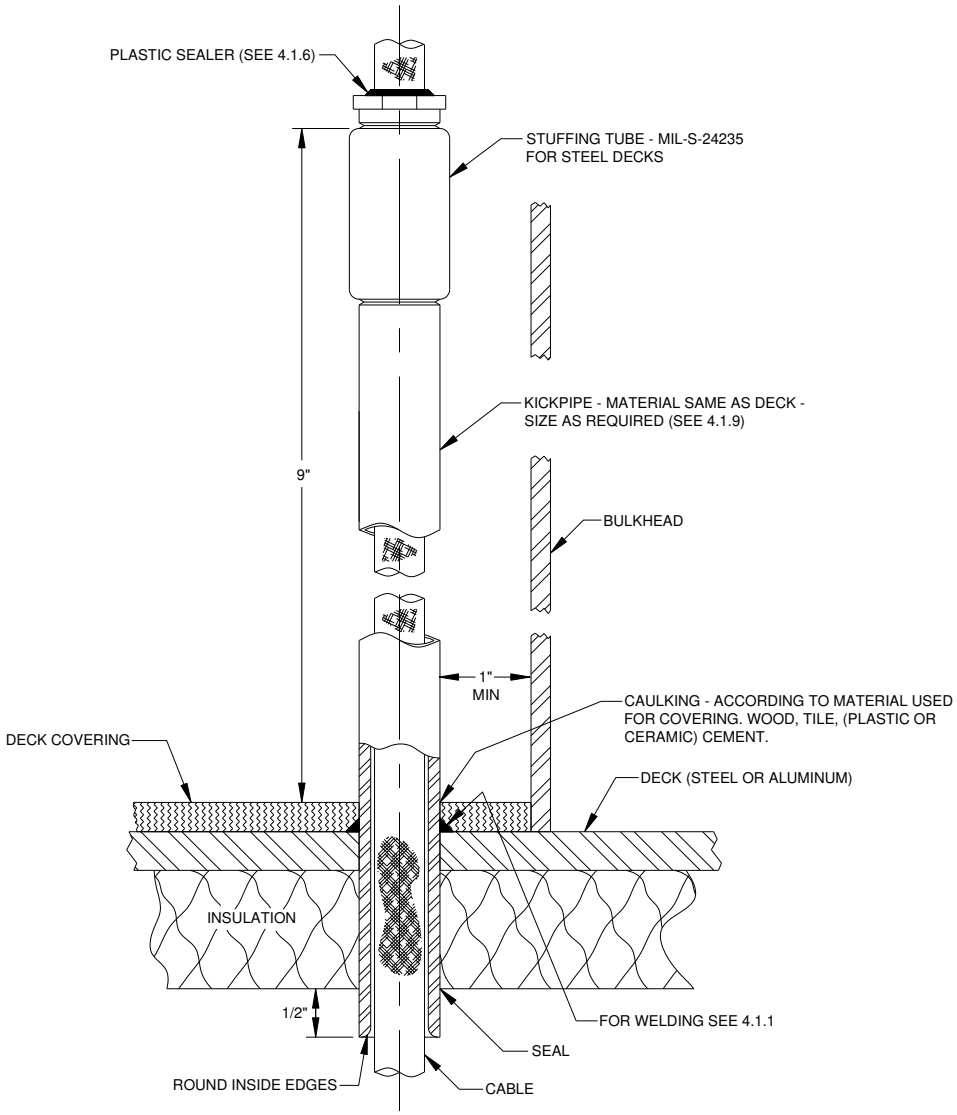
NOTES:

1. Use bent kickpipe to conserve space, if needed.
2. For suitable cable support, see MIL-STD-2003-4.
3. Pipes used in conjunction with stuffing tubes shall be welded on one side of the bulkhead or deck penetrated, except for the following types of bulkheads or decks where welds on both sides are required:
   a. Ballistic bulkheads
   b. Shell plating
   c. Torpedo defense system bulkheads
   d. Penetrations exposed to weather or wet spaces
4. Plastic sealer is only required on kickpipes and stuffing tubes when exposed to the elements (penetrates the skin of the ship).
5. See figure 3A1 for gland nut tightening methods.
6. See 4.8 for thread lubrication guidance.

FIGURE 3D2. Kickpipes through steel or aluminum decks (surface ships).

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METHOD 3D-3-1
KICKPIPES THROUGH
WATERTIGHT DECKS WITH COVERING AND
WITH OR WITHOUT INSULATION

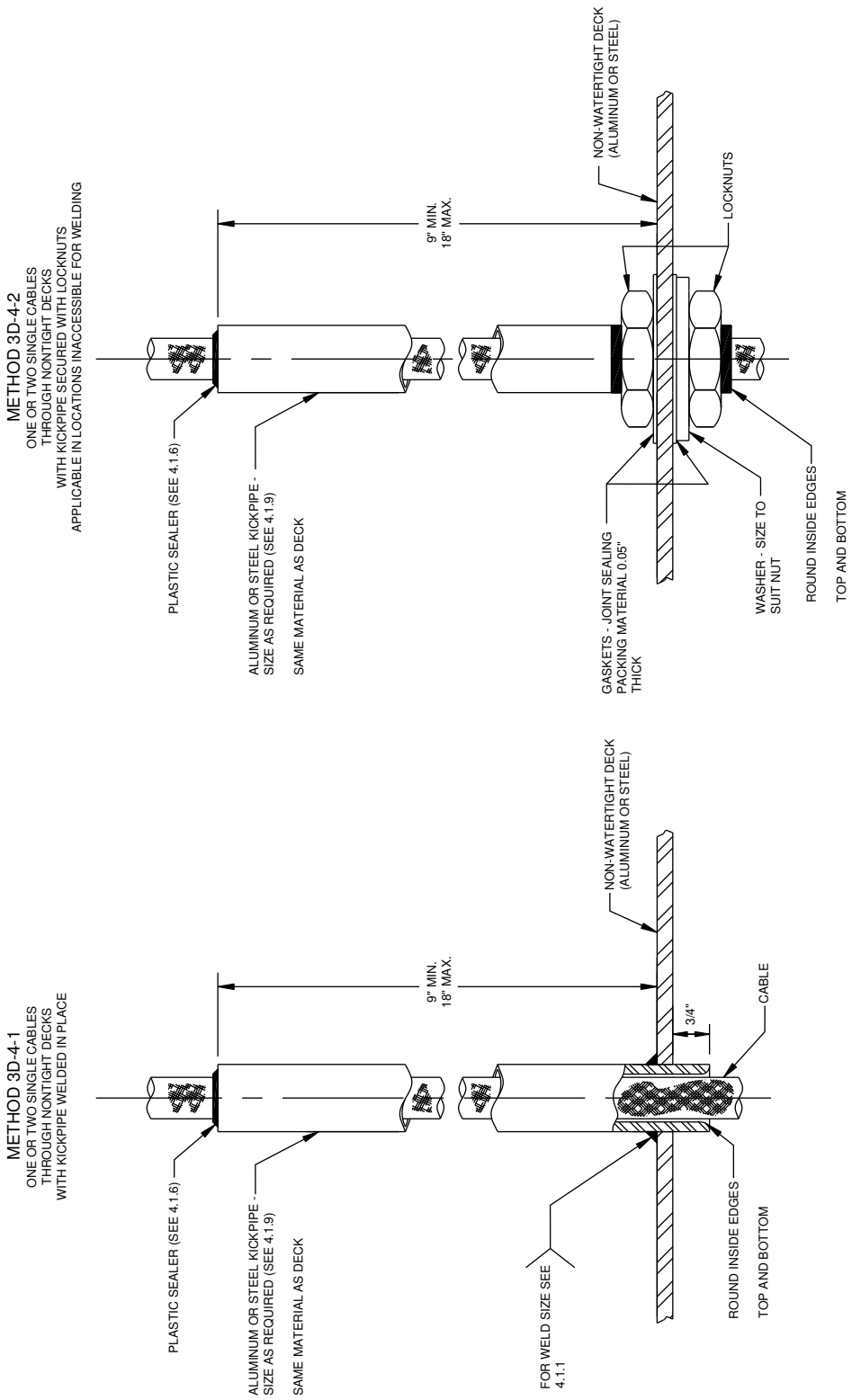


NOTES:

1. Pipes used in conjunction with stuffing tubes shall be welded on one side of the bulkhead or deck penetrated, except for the following types of bulkheads or decks where welds on both sides are required:
   a. Ballistic bulkheads
   b. Shell plating
   c. Torpedo defense system bulkheads
   d. Penetrations exposed to weather or wet spaces
2. See figure 3A1 for gland nut tightening methods.
3. See 4.8 for thread lubrication guidance.

FIGURE 3D3. Kickpipes through steel or aluminum decks (surface ships).

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NOTE:

1. See 4.5 for fastener and fastener material requirements.

FIGURE 3D4. Kickpipes through non-watertight decks (surface ships).

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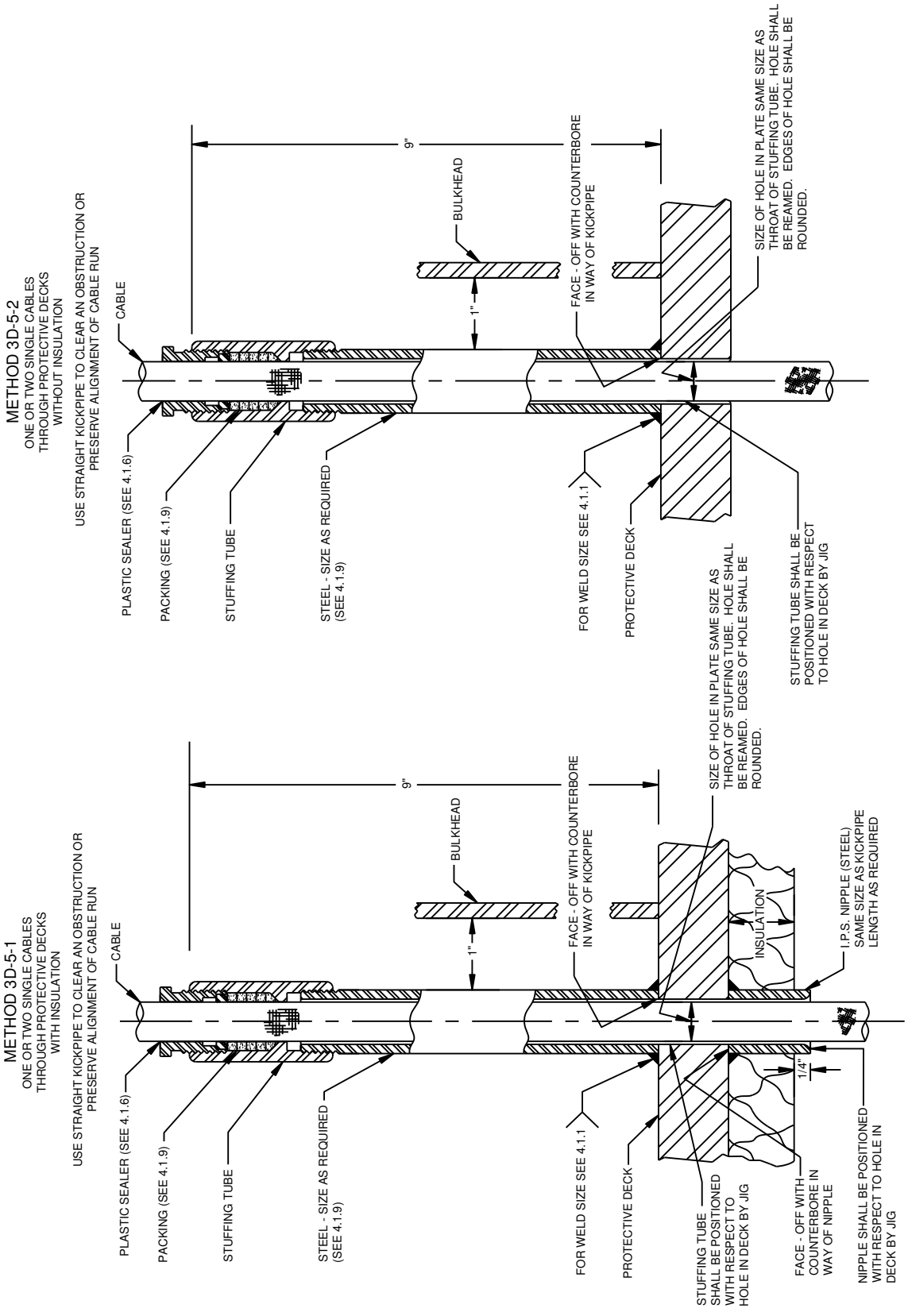


FIGURE 3D5. Kickpipes through ballistic plating (surface ships).

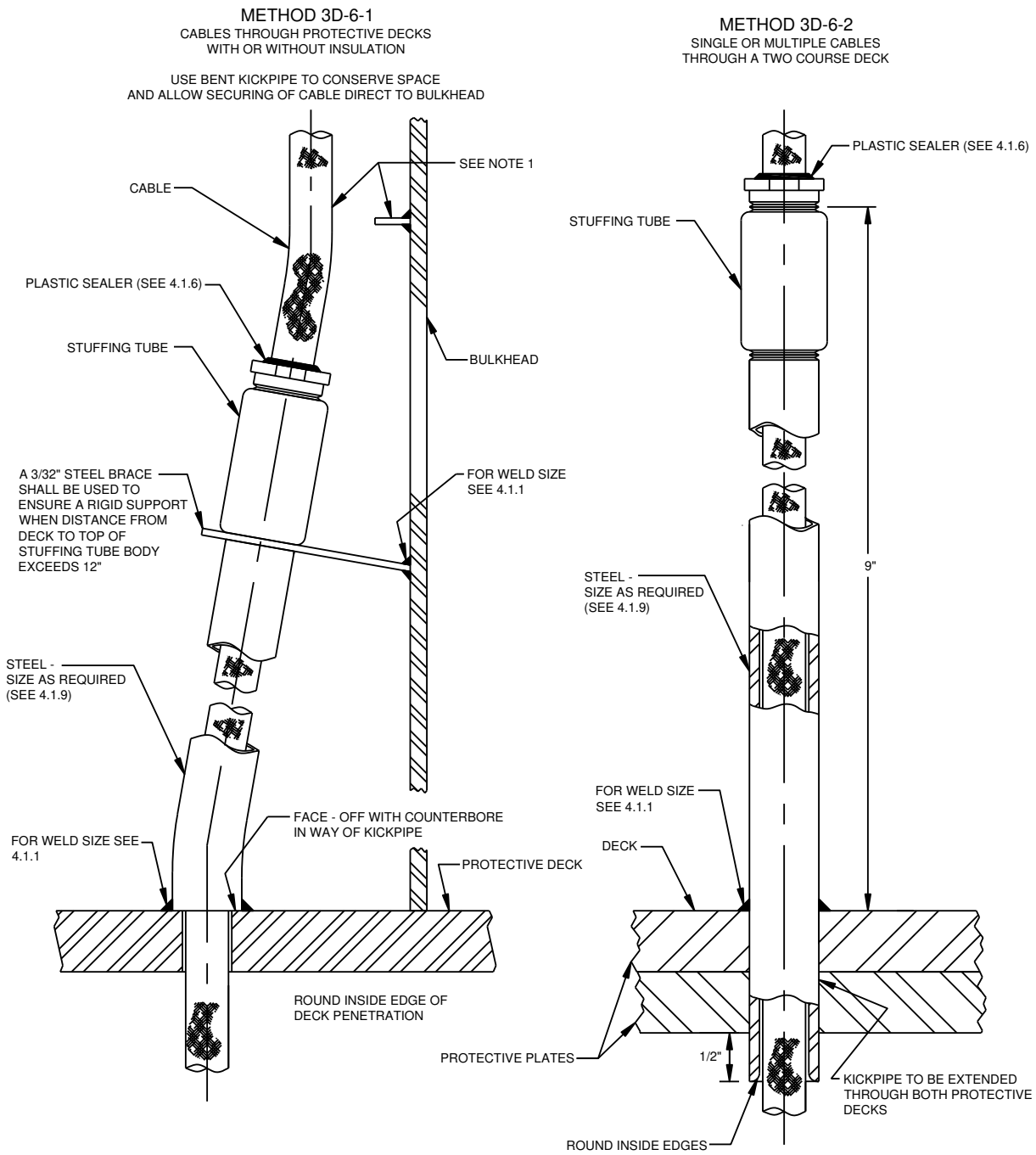
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NOTES:

1. See figure 3A1 for gland nut tightening methods.
2. See 4.8 for thread lubrication guidance.

FIGURE 3D5. Kickpipes through ballistic plating (surface ships) – Continued.

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NOTES:

1. For suitable cable support, see MIL-STD-2003-4.
2. See figure 3A1 for gland nut tightening methods.
3. See 4.8 for thread lubrication guidance.

FIGURE 3D6. Kickpipes through ballistic plating (surface ships).

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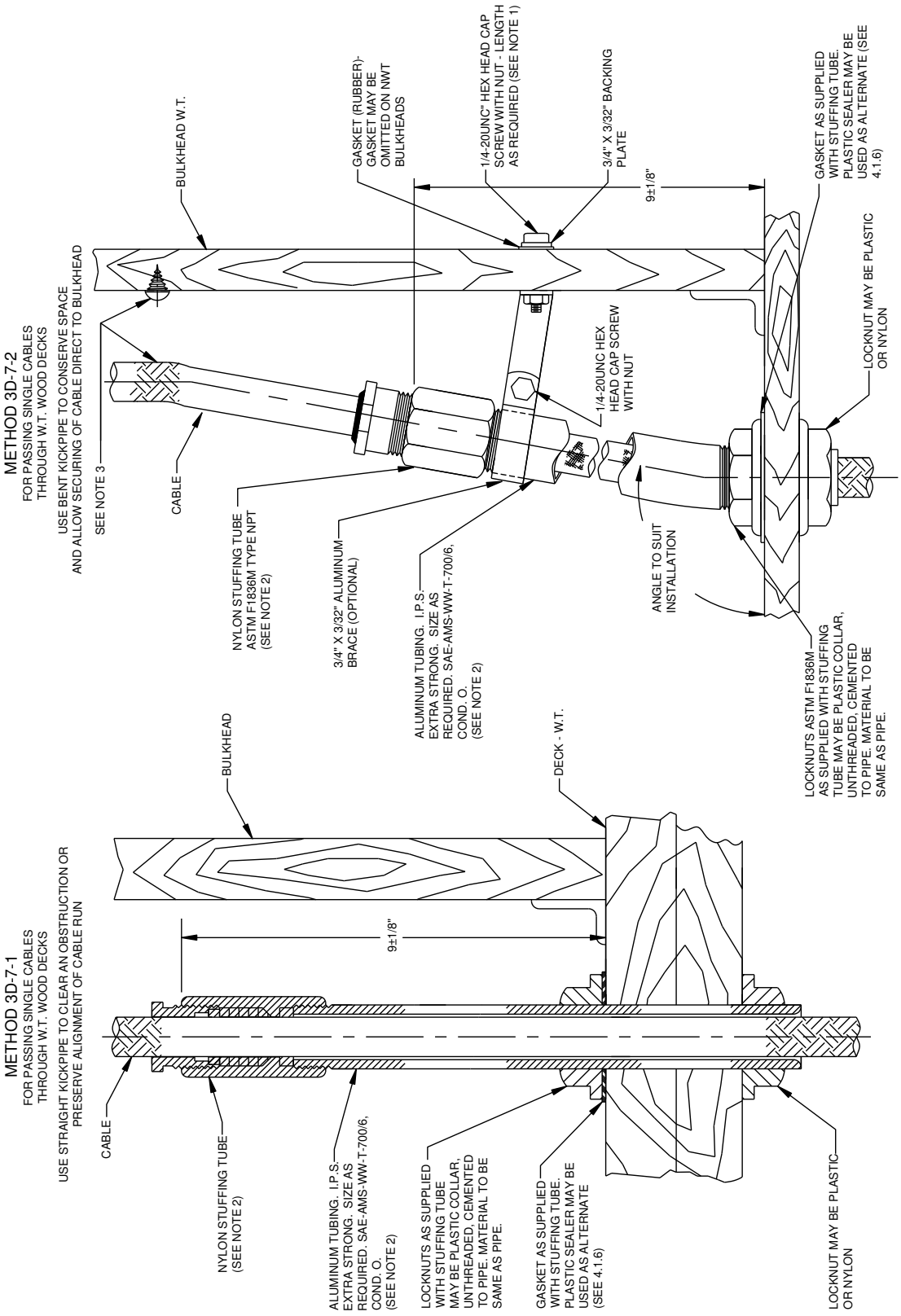


FIGURE 3D7. Kickpipes through wooden decks.

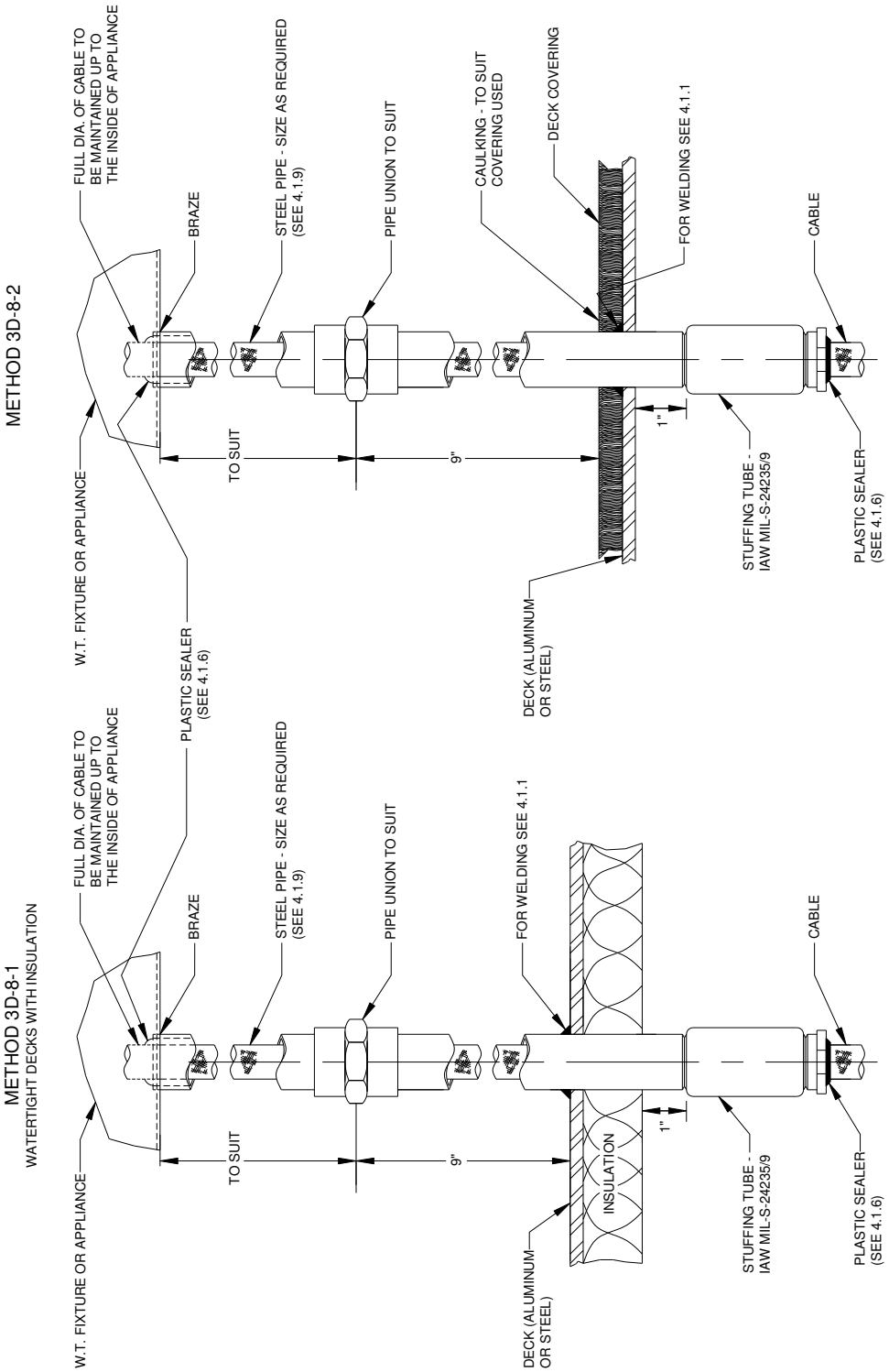
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NOTES:

1. Lag or wood screws (CRES) not less than 1 inch long may be used in lieu of hex head cap screws on bulkheads over 1 inch thick.
2. Plastic pipe fittings and stuffing tube may be used in lieu of aluminum shown. Stuffing tube should be nylon, ASTM F1836M, NPT type, sized to fit.
3. For suitable cable support, see MIL-STD-2003-4.
4. See figure 3A1 for gland nut tightening methods.
5. See 4.8 for thread lubrication guidance.
6. See 4.5 for fastener and fastener material requirements.

FIGURE 3D7. Kickpipes through wooden decks – Continued.

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NOTES:

1. See figure 3A1 for gland nut tightening requirements.
2. See 4.8 for stuffing tube lubrication requirements.

FIGURE 3D8. Kickpipes with unions (surface ships).

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APPENDIX 3E – PENETRATIONS – PRESSURE HULLS, SUBMARINES

E.1 SCOPE

E.1.1 Scope. This appendix describes the installation standard methods for penetration through pressure hulls of submarines. This Appendix is a mandatory part of the standard. The information contained herein is intended for compliance.

E.2. APPLICABLE DOCUMENTS

E.2.1 General. The documents listed in this section are specified in this appendix. This section does not include documents cited in other sections of this standard or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in this appendix, whether or not they are listed.

E.2.2 Government documents.

E.2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

FEDERAL SPECIFICATIONS

QQ-N-281 - Nickel-Copper Alloy Bar, Rod, Plate, Sheet, Strip, Wire, Forgings, and Structural and Special Shaped Sections

COMMERCIAL ITEM DESCRIPTIONS

A-A-59588 - Rubber, Silicone

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-915 - Cable, Electrical, for Shipboard Use, General Specification for

MIL-DTL-24231 - Connectors, Plugs, Receptacles, Adapters, Hull Inserts, and Hull Insert Plugs, Pressure-Proof, General Specification for

MIL-DTL-24231/1 - Connector, Plug, Type I, Molded, Three-, Four-, and Five-Conductor

MIL-DTL-24231/3 - Connector, Plug, Type I, Molded, Seven- and Nine-Conductor (Straight and 90 Degrees)

MIL-DTL-24231/4 - Connector, Plug, Type I, Molded, 14-, 24-, 30-, or 40-Conductor (Straight and 90 Degrees)

MIL-DTL-24231/18 - Connector, Plug, Type I, Molded, 65-Conductor

MIL-S-24235/16 - Stuffing Tubes, Metal, and Packing Assemblies for Electric Cables, Cable Shearing Valve, Hull, Pressureproof, Symbols 512, 512.1, 513, and 513.1

(Copies of these documents are available online at https://quicksearch.dla.mil.)

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E.2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

NAVAL SEA SYSTEMS COMMAND (NAVSEA) DRAWINGS

803-1197218 - Hull Fitting for AN/BQQ-1, Sonar, Sym No. 538

815-1197212 - Electrode - 4000 SBM, Plug In Type Symbols 3012 & 3012-1

(Copies of these documents are available from the applicable repositories listed in S0005-AE-PRO-010/EDM, which can be obtained online via Technical Data Management Information System (TDMIS) at https://mercury.tdmis.navy.mil/. Copies of these documents may also be obtained from the Naval Ships Engineering Drawing Repository (NSEDR) online at https://199.208.213.105/webjedmics/index.jsp. To request an NSEDR account for drawing access, send an email to NNSY_JEDMICS_NSEDR_HELP_DESK@navy.mil.)

NAVAL SEA SYSTEMS COMMAND (NAVSEA) PUBLICATIONS

T9074-AD-GIB-010/1688 - Requirements for Fabrication, Welding, and Inspection of Submarine Structure

(Copies of this document are available online via Technical Data Management Information System (TDMIS) at https://mercury.tdmis.navy.mil/ by searching for the document number without the suffix. Refer questions, inquiries, or problems to: DSN 296-0669, Commercial (805) 228-0669. This document is available for ordering (hard copy) via the Naval Logistics Library (NLL) at https://nll.navsup.navy.mil. For questions regarding the NLL, contact the NLL Customer Service at nllhelpdesk@navy.mil, (866) 817-3130, or (215) 697-2626/DSN 442-2626.)

E.2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

ASTM INTERNATIONAL

ASTM F468 - Standard Specification for Nonferrous Bolts, Hex Cap Screws, Socket Head Cap Screws, and Studs for General Use

(Copies of this document are available online at www.astm.org.)

SAE INTERNATIONAL

SAE-AMS-G-4343 - Grease, Pneumatic System

SAE-AS568 - Aerospace Size Standard for O-Rings

SAE-AS8660 - Silicone Compound NATO Code Number S-736

SAE-AS23053/5 - Insulation Sleeving, Electrical, Heat Shrinkable, Polyolefin, Flexible Crosslinked

SAE-AS23053/18 - Insulation Sleeving, Electrical, Heat Shrinkable, Modified Fluoropolymer, Crosslinked

(Copies of these documents are available online at www.sae.org.)

E.2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

E.3 REQUIRED EQUIPMENT AND MATERIALS

E.3.1 Required equipment and materials. The required equipment and materials for the proper installation of penetrations through pressure hulls of submarines are as shown in the Appendix E methods.

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E.4 NOTES AND PROCEDURES

E.4.1 Dimensions. For figures and tables in this appendix, all dimensions are in inches unless otherwise noted.

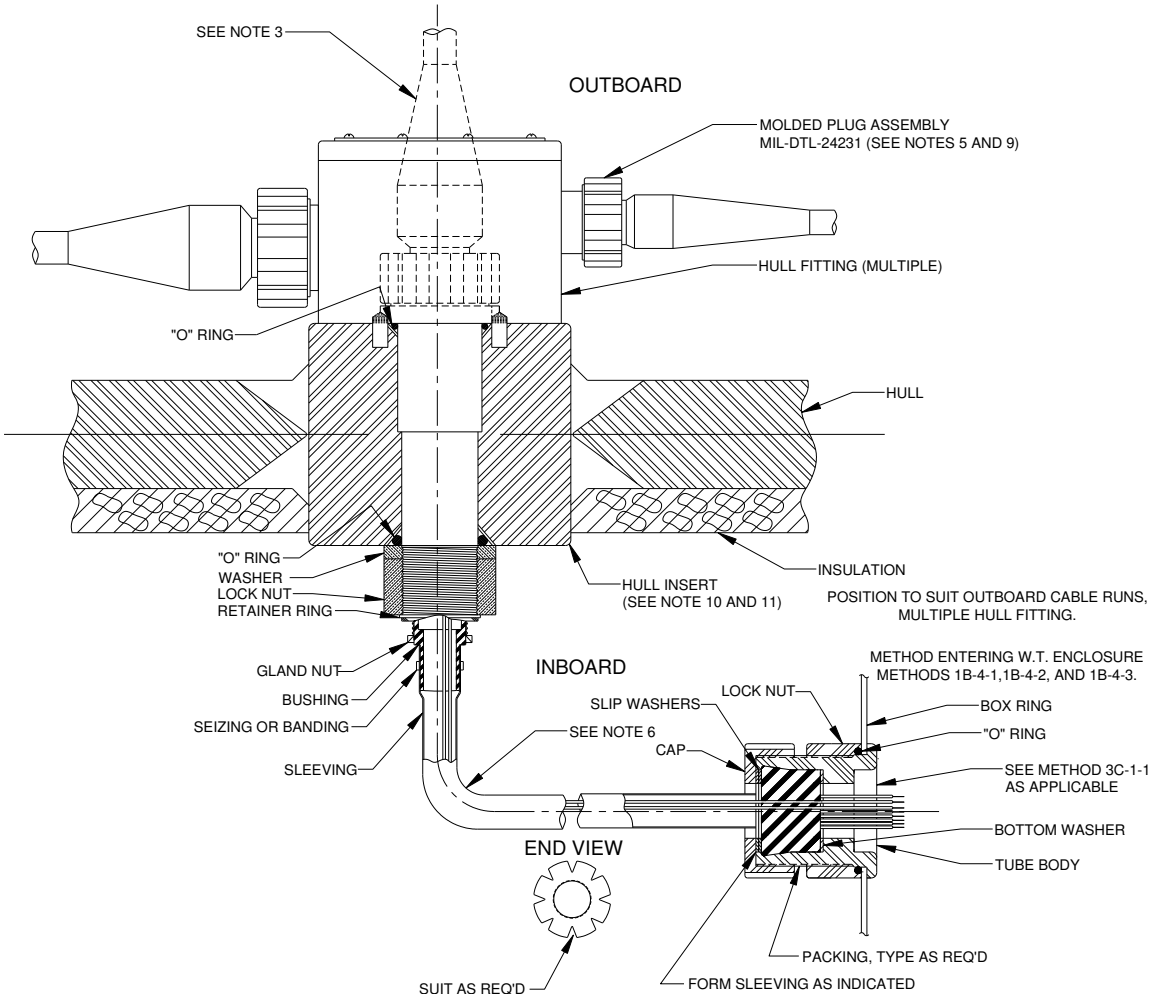
E.4.2 Figures. Table 3EI provides information for the figures in this appendix.

TABLE 3EI. Figures for penetrations of submarine pressure hulls.

Figure number	Submarine pressure hull penetrations	Page
3E1	Cable connections through pressure hulls of submarines	238
3E2	Sealing plugs for hull inserts on submarines	239
3E3	Passing cable through pressure hull of submarines using cable shearing valve	240
3E4	Hull fitting installation for sonar sphere	242
3E5	Mounting pressure-proof electrodes on submarines	244
3E6	Temporary sealing and securing pressure-proof molded plug assemblies	245

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METHOD 3E-1-1



NOTES:

1. Single and multiple cable connectors shall be installed and inspected in accordance with the applicable Submarine Maintenance Standard.
2. “O” ring surfaces on hull inserts shall be suitably protected from damage or weld splatter at all times prior to installation of hull fitting.
3. Dotted line indicates outline for single cable hull fitting.
4. Coat “O” rings and “O” ring surfaces with grease conforming to SAE-AS8660.
5. Remove receptacle caps prior to installation of molded plug assemblies (see note 9).
6. Secure inboard cable in place with cable supports as required.
7. Methods shown herein have been tested and approved for deep diving submarines.
8. Methods for temporary sealing hull inserts and pressure-proof plug assemblies are shown on figure 3E2.
9. When molded plug assemblies are temporarily disconnected, reinstall pressure-proof caps (see note 5 above).
10. Hull insert to be furnished by shipbuilder of a steel compatible with applicable hull. For further detailed requirements, see MIL-DTL-24231.
11. Position hull insert to outboard cable runs (multiple hull fittings). Hull insert is to be installed with dowel pins at zero degrees forward (or parallel to the ship’s centerline) unless otherwise directed by ship’s drawings.
12. See 4.7 for installations and servicing within the SUBSAFE boundary.

FIGURE 3E1. Cable connections through pressure hulls of submarines.

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METHOD 3E-2-1

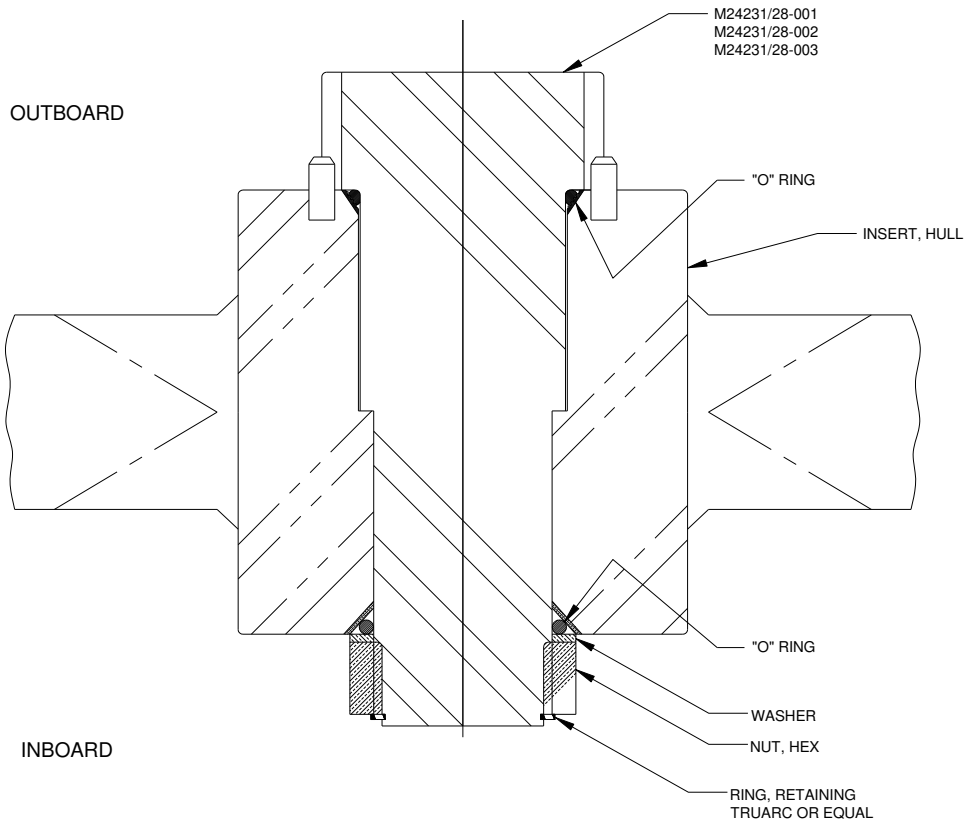


TABLE 3E2-I. Quantities for one sealing plug assembly.

Insert plug assy.	Number required - one						
	Insert plug	Retaining ring	Nut	Washer	"O" ring 1/ 2/ - outboard	"O" ring 1/ 2/ - inboard	For hull insert
M24321/28-001	M24231/28-010	M24231/5-076	M24231/5-074	M24231/5-075	-329	-327	M24231/25-001
M24231/28-002	M24231/28-020	M24231/10-045	M24231/10-042	M24231/10-043	-334	-332	M24231/25-002
M24231/28-003	M24231/28-030	M24231/19-014	M24231/19-024	M24231/19-025	-342	-339	M24231/25-003

NOTES:

1/ Silicone rubber A-A-59588, class III.

2 "O" ring sizes shall be in accordance with SAE-AS568.

NOTES:

1. Install hull insert sealing plug in accordance with applicable Submarine Maintenance Standard.
2. Hull inserts shall be furnished with sealing plugs in place to protect "O" ring surfaces from damage. Sealing plugs shall remain in place during welding and shall not be removed until insertion of single or multiple cable hull fitting.
3. Each sealing plug shall be checked with "go" and "no go" gages to ensure interchangeability with hull inserts.
4. For installation of hull fittings and hull inserts, see figure 3E1.
5. Silicone "O" rings are used only when welding in hull insert. Buna type "O" rings in accordance with table 3E2-I are required for service installation.
6. See 4.7 for installations and servicing within the SUBSAFE boundary.

FIGURE 3E2. Sealing plugs for hull inserts on submarines.

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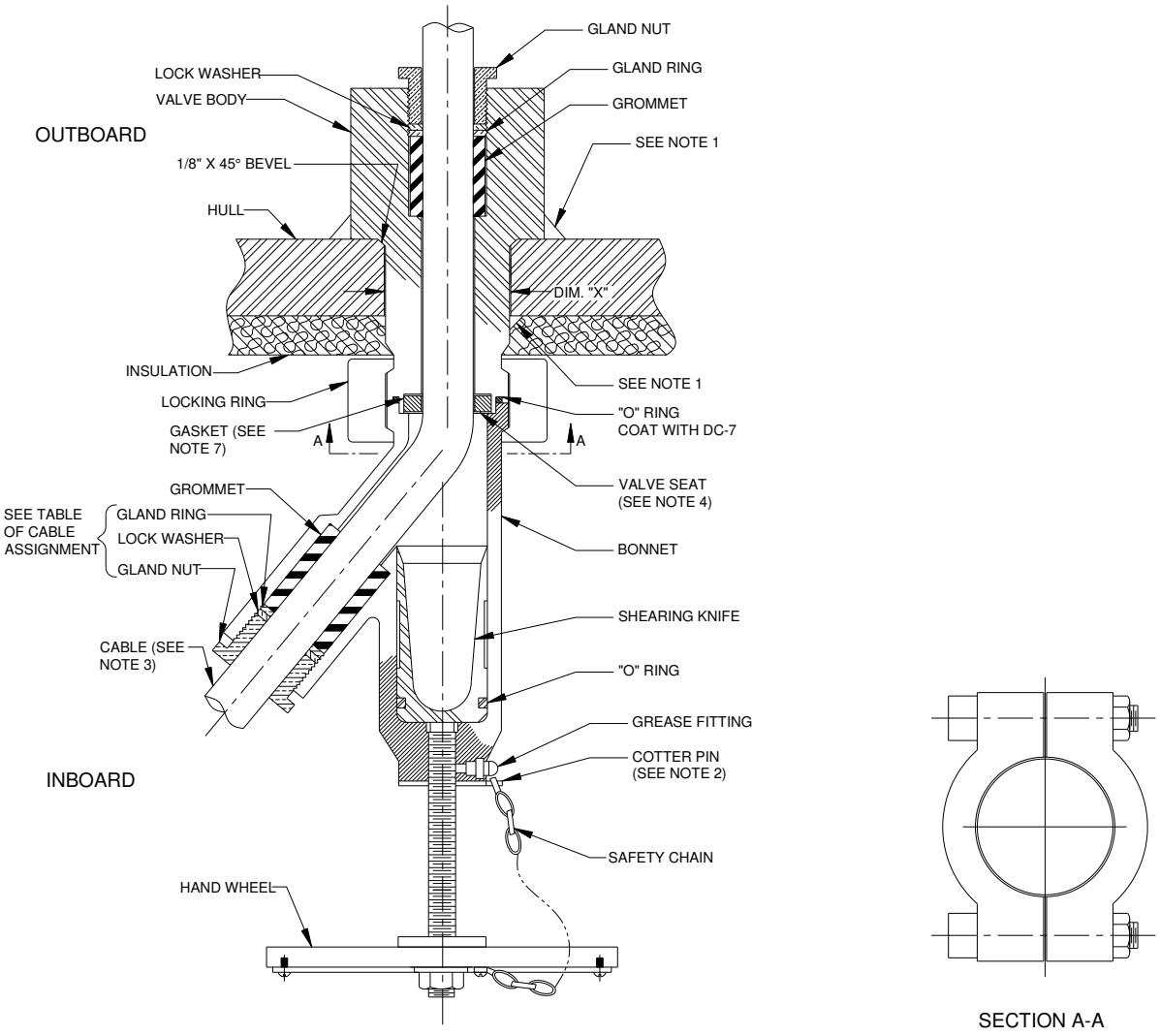


FIGURE 3E3. Passing cable through pressure hull of submarines using cable shearing valve.

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TABLE 3E3-I. Cable assignment.

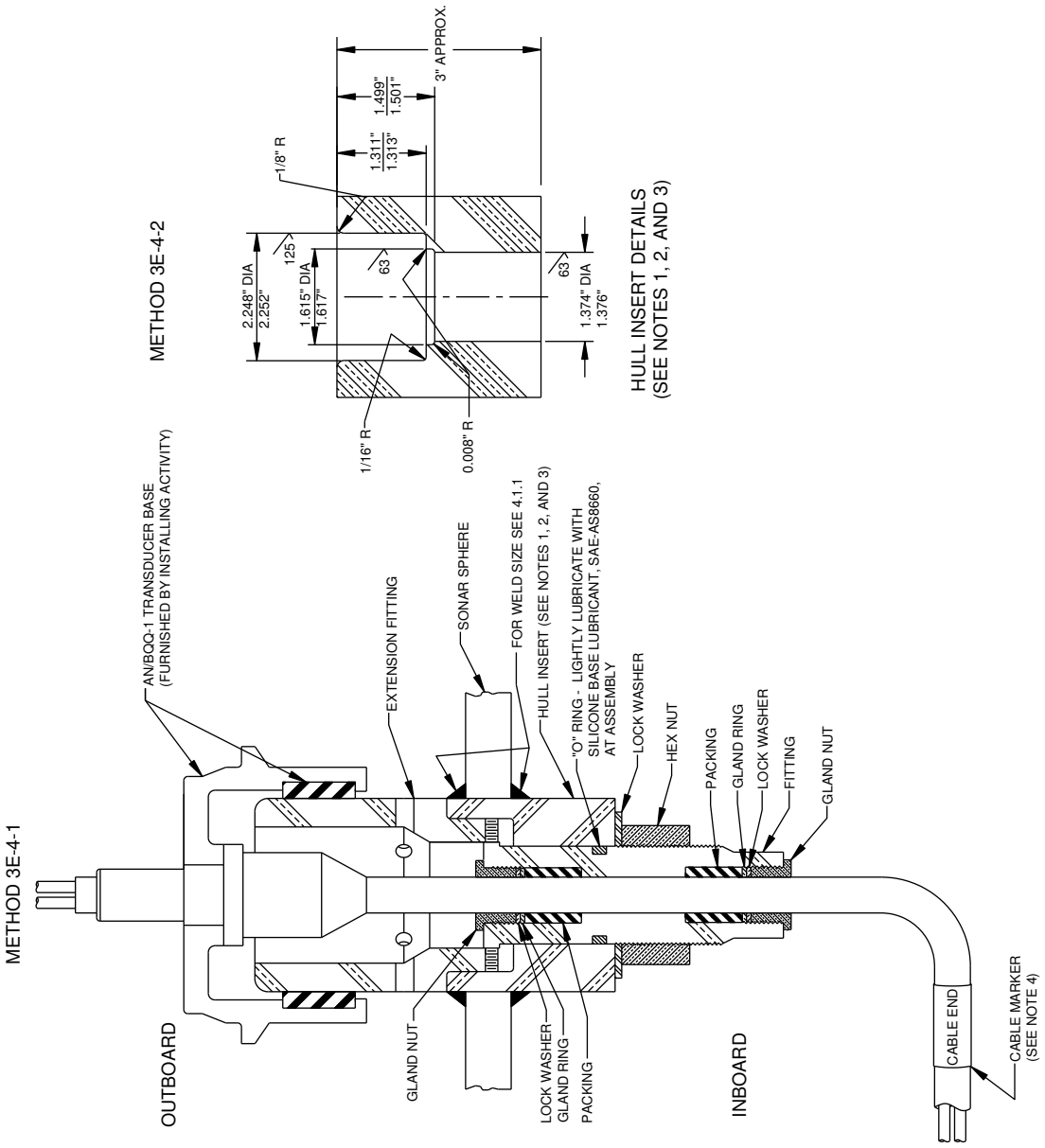
Symbol no.	Cable OD	Grommet PC no.	Gland ring PC no.	Dim "X"
512	0.475	46	59	2.062
	0.510	47	60	
	0.520			
	0.545	48	61	
512.1	0.627	49	62	2.062
	0.632			
	0.720	50	63	
	0.750	51	64	
513	0.795	52	65	2.562
	0.805			
	0.830	53	66	
	0.844			
	0.870	54	67	
513.1	0.995	55	68	2.562
	1.020	56	69	
	1.115	57	70	
	1.120			
	1.150	58	71	
	1.156			

NOTES:

1. Welding shall comply with T9074-AD-GIB-010/1688.
2. Before assembly, drill hole for cotter pin with valve in full open position. Pin shall be easily removed. Bend only enough to prevent falling out.
3. The cable shearing valves are designed for cables in accordance with MIL-DTL-915, MIL-DTL-24643, and MIL-DTL-24640.
4. If valve seat becomes damaged, it can be reversed and reinstalled with a new valve seat gasket. Removal of valve seat is accomplished by use of a ⅞-12 stud for symbol no. 512 or a 1⅜-12 stud for symbol no. 513.
5. Terminate inboard cable in a junction box, which will be located in the vicinity of the shear valve.
6. For specifications, see MIL-S-24235/16.
7. Cement gasket to valve seat and body.
8. See 4.7 for installations and servicing within the SUBSAFE boundary.

FIGURE 3E3. Passing cable through pressure hull of submarines using cable shearing valve – Continued.

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NOTES:

1. Install hull fitting in accordance with applicable Submarine Maintenance Standard and the applicable drawing for the ship class.
2. Hull insert shall be machined to finished dimensions as shown under method 3E-4-2 after all welding has been completed on sonar sphere.

FIGURE 3E4. Hull fitting installation for sonar sphere.

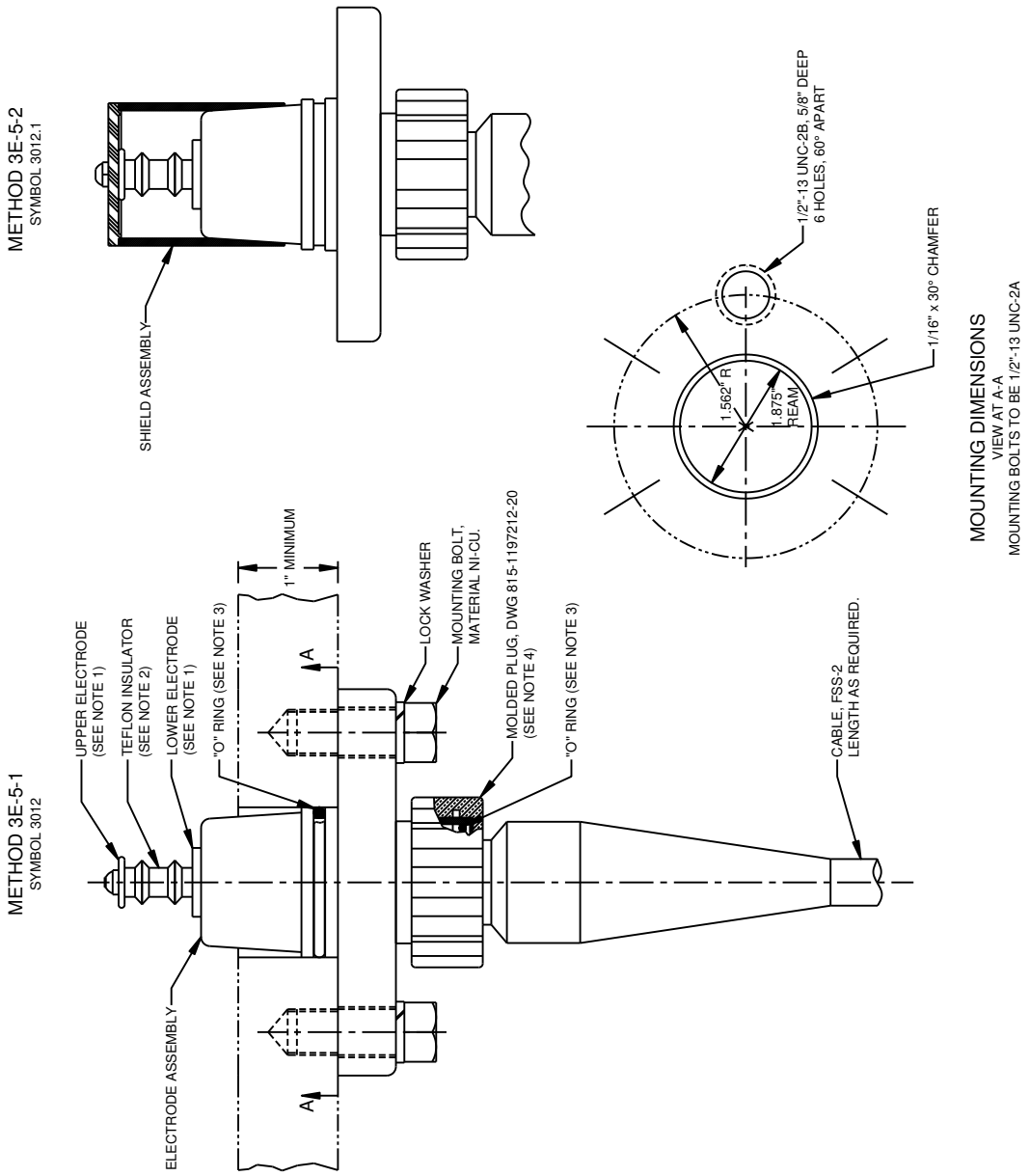
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NOTES (continued):

3. Hull insert material shall be Ni-CU Monel, QQ-N-281, class A, hot rolled. Dimensions shown are applicable after welding in sonar sphere and after machining to accept PCs 1, 3, and 9 of Drawing 803-1197218. Outer diameter shall be such as to provide adequate compensation. Hull inserts are to be furnished by the shipbuilder.
4. Cable marker shall consist of heat shrinkable tubing-white in accordance with SAE-AS23053/5, class 1, with cable designation stamped as required and then covered by a clear heat shrinkable tubing in accordance with SAE-AS23053/18. This method may be used as an alternate to the standard method of marking sonar cables.
5. See the ship drawings for configuration.
6. See 4.7 for installations and servicing within the SUBSAFE boundary.

FIGURE 3E4. Hull fitting installation for sonar sphere – Continued.

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NOTES:

1. Upper and lower electrode surfaces shall be free from contaminants to ensure circuit when immersed in water.
2. Surfaces of Teflon insulator shall be protected during installation to prevent damage and/or contamination by dirt, grease, etc.
3. Coat “O” rings and “O” ring surfaces with grease conforming to SAE-AMS-G-4343.
4. Remove receptacle cap furnished with electrode assembly prior to installation of molded plug assembly.
5. This electrode is designed to withstand 2000 psi hydrostatic pressure against either the electrode or the receptacle side.
6. This drawing was developed from Drawing 815-1197212.
7. Mounting hardware in method 3E-5-1 shall be in accordance with ASTM F468.
8. See 4.7 for installations and servicing within the SUBSAFE boundary.

FIGURE 3E5. Mounting pressure-proof electrodes on submarines.

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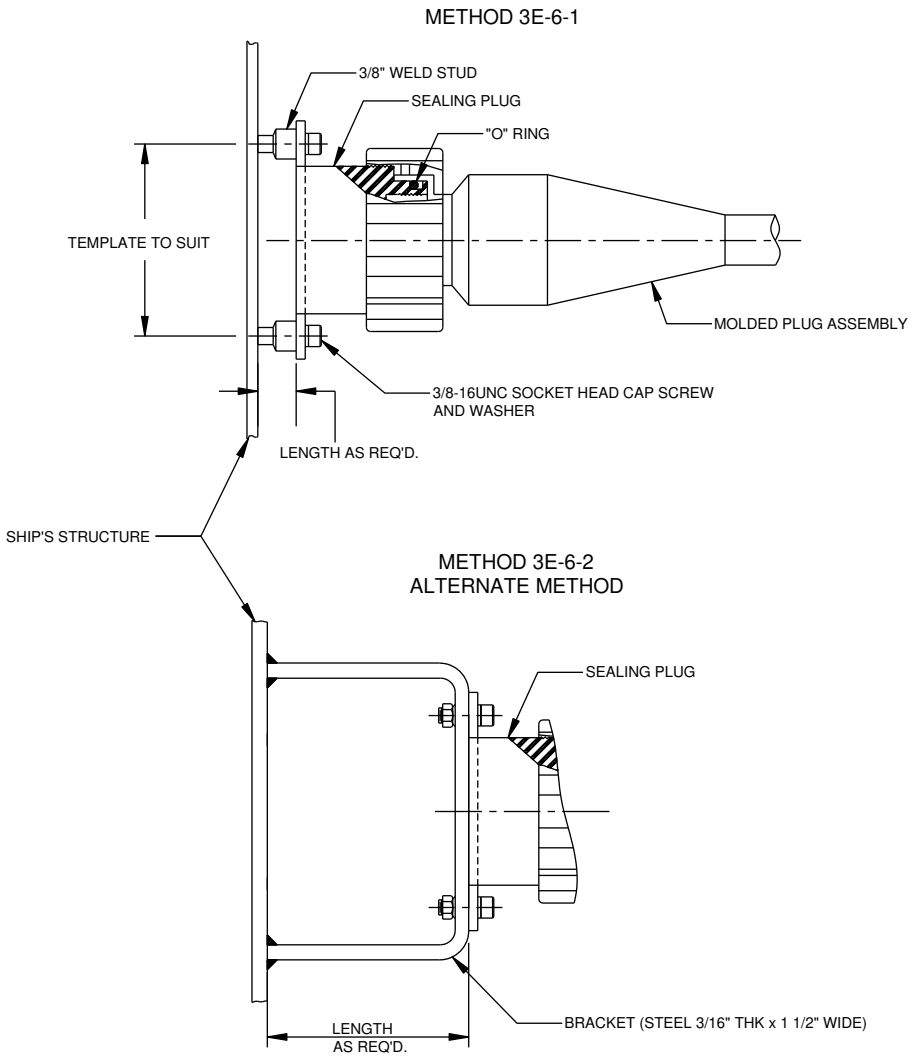


TABLE 3E6-I. Sealing plug assembly.

Molded plug specification	Symbol number	Sealing plug assembly
MIL-DTL-24231	713	M24231/1-020
	713.5	
	713.1	M24231/3-020
	713.2	M24231/4-020
	713.3	
	713.4	
	1139	M24231/18-020
	1118	

FIGURE 3E6. Temporary sealing and securing pressure-proof molded plug assemblies.

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NOTES:

1. Typical methods shown hereon are for information and guidance to installing activities for sealing and/or securing of pressure-proof pin type molded connectors that have been temporarily disconnected from their associated receptacle.
2. Temporary sealing of pin connector hull fittings or receptacles are shown on figures 3E1 and 3E2.
3. Cables shall be properly secured or protected to prevent damage from chafing, weld splatter, and/or other hazards.
4. See 4.7 for installations and servicing within the SUBSAFE boundary.
5. See 4.5 for fastener and fastener material requirements.

FIGURE 3E6. Temporary sealing and securing pressure-proof molded plug assemblies – Continued.

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CONCLUDING MATERIAL

Preparing activity:
Navy – SH
(Project SESS-2015-029)

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